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Measurement of $R(D^{(*)})$ using semileptonic tagging and leptonic tau decays

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Abstract

We describe the measurement of the ratio of branching fractions of semi-tauonic to semi-leptonic B decays into $D^{(*)}$ mesons, , using a semileptonic tagging method and leptonic τ decays. Including the full amount of data from the Long Shutdown 1 dataset (365 fb^{-1}), this will be the first semileptonically tagged result for $\mathcal{R}(D^{(*)})$ from Belle II.

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1. OUTLINE OF THE STRUCTURE OF THIS NOTE

The main part of this note describes the analysis strategy and the studies performed before unboxing (Sections 3-10).

The unboxing procedure following the strategy described in Section 2 is described in Sections 11.1-11.4.

For the final results after some slight changes in the fit strategy during CWR1 see Section 12.

Initially, our approach involved conducting a straightforward yield fit that did not incorporate any nuisance parameters. To estimate the magnitude of systematic uncertainties, we relied on toys. However, in order to have a more refined method to accurately quantify these uncertainties, we decided to transition to a bin-wise nuisance parameter implementation, which allowed us to obtain a more precise and detailed understanding of the size and impact of systematic uncertainties. Additionally, to further enhance our ability to capture the nuances in the shapes of the fitted distributions, we revised our binning strategy. We replaced the initially symmetric, two-dimensional binning with an asymmetric one, allowing for a more accurate representation of the distribution shapes. We repeated steps 3-4 of the unboxing procedure (see Section 11.5).

The ability to directly extract $\mathcal{R}(D)$ and $\mathcal{R}(D^*)$ along with their correlations from the fit, coupled with an improved capacity to evaluate the contribution of each systematic source to the total uncertainty, led us to adopt a different fit implementation using `pyhf`, which allows for using source-wise nuisance parameters. We re-blinded and performed several studies described in Sections 11.6-11.7. Finally we unboxed again with results presented in Section 11.8.

After that, several cross-checks were performed (Sections 11.9 - 11.15).

These were repeated with the final fit setup. Section 12 summarises the final results, their uncertainties and a summary of the re-evaluated cross-checks.

Section 13 provides a conclusion.

105 2. UNBOXING STRATEGY

106 The unboxing plan for this analysis is as follows:

- 107 0. check data/MC agreement in all selection variables and classifier inputs
- 108 1. if (0) looks good, check for good agreement in the classifiers themselves
- 109 2. if (1) looks good, run simultaneous signal extraction fit on $\mathcal{R}(D)_\mu$ and $\mathcal{R}(D^*)_\mu$
- 110 3. if (2) looks good, do same with electron mode
- 111 4. final unboxing of the result for $\mathcal{R}(D)$ and $\mathcal{R}(D^*)$ if everything else looks good

112 In steps 2-3, the result on $\mathcal{R}(D)$ and $\mathcal{R}(D^*)$ are kept blinded. The results of each step
113 are presented in Section11.

114 3. INTRODUCTION

115 3.1. Motivation

116 In the past few years, several measurements provide evidence that the conservation of
 117 lepton flavour universality (LFU), as predicted by the Standard Model of Particle Physics
 118 (SM), is an issue to be questioned. One of these deviations that have been detected
 119 involves the $b \rightarrow c\tau\nu$ transition, by measuring the ratios of the branching fractions (BFs)
 120 of the semi-tauonic decay of a B meson into a $D^{(*)}$ meson to the same decay but with a
 121 lighter lepton (e, μ):

$$\mathcal{R}(D^{(*)}) = \frac{\mathcal{B}(B \rightarrow D^{(*)}\tau\nu)}{\mathcal{B}(B \rightarrow D^{(*)}l\nu)} \quad (1)$$

122 with $l = e, \mu$. In the SM, these decays proceed via couplings to a $W^\pm$ boson, which are
 123 supposed to be universal for all three lepton generations. Investigating decays including
 124 τ leptons provide the opportunity to test this universality since the heaviest of the three
 125 lepton generations might couple stronger to certain hypothetical non-SM particle. Com-
 126 pared to purely leptonic B decays these semi-leptonic (-tauonic) B decays are convenient
 127 to analyse due to their higher branching fractions, since they are not Cabbibo-suppressed.
 128 Moreover, the hadron that is produced in addition to the τ lepton provides access to ad-
 129 ditional observables such as its momentum, which are useful for the analysis of these
 130 decays.

131 From the theoretical point of view, the ratio of these decay can be well-described due
 132 to their independence of $|V_{cb}|$ and to most part also of the $B \rightarrow D^{(*)}$ form factor model.
 133 According to Ref. [5], the theoretical values for the ratios are the following:

- 134 • $\mathcal{R}(D) = 0.299 \pm 0.003$ (average computed from the theoretical values of Refs. [15],
 135 [12] and [53])
- 136 • $\mathcal{R}(D^*) = 0.258 \pm 0.005$ (averaging the theoretical values of Refs.[12], [53] and [20])

137 Deviations in this ratio from the SM expectation are seen independently by several
 138 experiments. Table I summarises the results from previous measurements in comparison
 139 to the value predicted by the SM.

TABLE I: Previous results on $\mathcal{R}(D^{(*)})$

experiment	$\mathcal{R}(D)$	$\mathcal{R}(D^*)$	Ref.
BaBar 2013	$0.440 \pm 0.058 \pm 0.042$	$0.332 \pm 0.024 \pm 0.018$	[37] [38]
Belle 2015	$0.375 \pm 0.064 \pm 0.026$	$0.293 \pm 0.038 \pm 0.015$	[33]
LHCb 2015	-	$0.336 \pm 0.027 \pm 0.03$	[49]
Belle 2017	-	$0.270 \pm 0.035^{+0.028}_{-0.025}$	[52]
LHCb 2018	-	$0.280 \pm 0.018 \pm 0.029$	[22] [21]
Belle 2019	$0.307 \pm 0.037 \pm 0.016$	$0.283 \pm 0.018 \pm 0.014$	[25]
Average	$0.340 \pm 0.027 \pm 0.013$	$0.295 \pm 0.011 \pm 0.008$	[5]
(HFLAV Summer 2018)			
Standard Model	0.299 ± 0.003	0.258 ± 0.005	[15], [12],[53], [20]

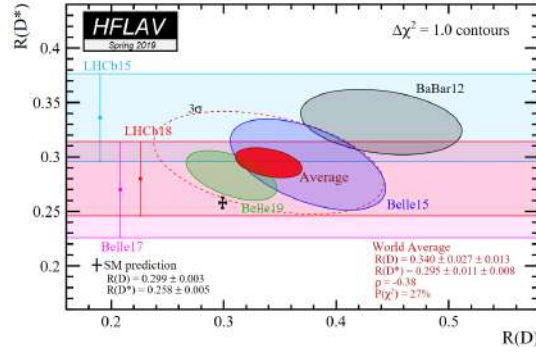


FIG. 1: Two-dimensional average of results for $\mathcal{R}(D)$ and $\mathcal{R}(D^*)$ from various experiments as well as theoretical expectations according to the SM [5]

The current combined world average of these ratios deviates more than 3σ from the SM expectation, which is visualised in fig.1. This suggests that these semi-tauonic B decays take place more often than we expect, which could hint at new Physics Beyond the Standard Model (BSM). This could, for example, involve a charged Higgs boson or a leptoquark being mediated in the $b \rightarrow c\tau\nu$ transition. The majority of the past measurements of $\mathcal{R}(D^{(*)})$ uses a hadronic tagging technique. The only semileptonically tagged result listed in table I is from Belle I (2019) [25]. Our aim is to provide the first semileptonically tagged result from Belle II using the Long Shutdown 1 (LS1) dataset.

3.2. Analysis overview

The basic idea of this analysis is to reconstruct signal $B^0 \rightarrow D^{(*)+}\tau^-\nu_\tau$ events as well as normalisation $B^0 \rightarrow D^{(*)+}\ell^-\nu_\ell$ events (as well as their charge conjugates) exactly the same way by defining appropriate selection criteria, all of which are described in detail in Section 6. This is possible since we are using only leptonic τ decays, such that the final state particles for both modes are the same ones. It also leads to the cancellation of several systematic uncertainties. Both signal and normalisation yields can then be extracted from the same fit, which is outlined in Section 8. It is a two-dimensional binned negative log-likelihood fit (cf. Section 8.8.3), simultaneously extracting $\mathcal{R}(D)$ and $\mathcal{R}(D^*)$. Thus, it is aimed to separate six categories of events, i.e. events belonging to the signal and normalisation categories in either the D or D^* modes as well as the remaining backgrounds. A component of true $B_{\text{sig}}^0 \rightarrow D^*(\ell/\tau)\nu_{\ell/\tau}$ events that is misreconstructed in the wrong mode (i.e. D mode) shares a yield with the correctly reconstructed D^* events. These two are the only shared parameters between both modes. Both fit variables are classifier outputs: A multiclassification algorithm is trained on several variable distributions in order to separate the three event categories and to assign a confidence score to each event of belonging to one of the three categories, respectively (see Section 8.8.1. Before performing the fit, several efficiency corrections are applied, which are presented in Section 7. In the 2D plane, the three event categories of signal, normalisation and backgrounds are concentrated in different areas, allowing us to define certain sidebands to check the data-MC agreement in both the fit variables as well as the BDT input variables (see Section 9). Using the uncertainties on the efficiency weights, we can estimate the size of the systematic uncertainties in Section 10. In addition, we check for additional systematic effects that might arise in this analysis (cf. Sections 10.1.1-10.1.5).

173 4. USED DATA AND SIMULATED SAMPLES

174 The data used for the studies presented in this note correspond to an integrated
175 luminosity of 364.4 fb^{-1} , corresponding to proc13 and buckets 26-36.

176 In addition to the data taken at a centre-of-mass energy of 10.58 GeV , i.e. at the
177 energy of the $\Upsilon(4S)$ resonance, data collected at a slightly lower centre-of-mass energy
178 of 10.52 GeV (“off-resonance”) is used to estimate the background from $q\bar{q}$ ($q = u, d, s, c$)
179 production in a data-driven way. This background will be labelled as “continuum” in the
180 following, due to the uniform shape of its cross-section as a function of the centre-of-mass
181 energy. In total, 42.32 fb^{-1} of off-resonant data is available in the current dataset (proc12,
182 bucket 20, bucket 24). Accordingly, the amount of these events has to be scaled to match
183 the amount of on-resonant data, accounting for the ratio of the two luminosities as well
184 as the inverse ratio of the different beam energies squared:

$$c_{\text{offres}} = \frac{\mathcal{L}_{\Upsilon(4S)}}{\mathcal{L}_{\text{offres}}} \cdot \left(\frac{E_{\text{offres}}^*}{E_{\Upsilon(4S)}^*} \right)^2 \quad (2)$$

185 The latter factor originates from the fact that the cross-section is inversely proportional
186 to the centre-of-mass energy squared. Since the amount of off-resonant data is much
187 smaller than those collected at the $\Upsilon(4S)$ resonance, and the event selection (see Section
188 6) already suppresses a large amount of those events, the statistical uncertainty in bins
189 of certain distributions containing off-resonant data is relatively high.

190 The total number of $B\bar{B}$ events in the used dataset can be estimated from a B counting
191 study to $N_{B\bar{B}} = (387 \pm 6) \times 10^6$. [17].

192 The Monte Carlo (MC) simulated events used for the plots in this note correspond to
193 the MC15rib production. First of all, they comprise a sample of 2800 fb^{-1} of $B\bar{B}$ events
194 containing equal fractions of neutral (“mixed”, i.e. $e^+e^- \rightarrow \Upsilon(4S) \rightarrow B^0\bar{B}^0$ events) and
195 charged B mesons produced by the MC generator EvtGen [51]. While 800 fb^{-1} of this
196 dataset is used for a BDT training, we use 2000 fb^{-1} for the generation of the MC fit
197 templates in the asimov fits as well as the final signal extraction fit. The total number
198 of simulated events in a sub-sample of 100 fb^{-1} can be estimated to $100.64 \cdot 10^6$. These
199 generic $B\bar{B}$ events already contain signal events as well as various types of backgrounds.
200 However, as the amount of $D^{(*)}\tau\nu$ events in these samples is comparatively small due to
201 the small BF of these decays, a dedicated sample of $500 \cdot 10^6$ signal events is used in the
202 BDT training in order to provide more statistics to train on.

203 Furthermore, the generic MC sample simulates semi-leptonic B decays into non-
204 resonant final states as $B \rightarrow D^{(*)}(\pi\pi/\eta)l\nu$ with an equal distribution in phase space
205 in order to fill the so-called “gap” between the overall B decay width and exclusive B
206 decays of $B \rightarrow D^{(*)}l\nu$, the latter of which could be measured with sufficient accuracy.

207 In this analysis, this is replaced by dedicated signal MC which simulates these “gap”
208 events via a first three-body semi-leptonic decay into a broad D^{**} resonance which sub-
209 sequently decays into the previously mentioned final states. The same signal MC is also
210 produced for $B \rightarrow D^{**}\tau\nu$, which has not been simulated at all in the existing generic
211 MC, and is implemented in this analysis.

212 See also ref. [8] for a more detailed description of the modelling of these gap events in
213 our MC and the differences to the signal MC to replace it.

214 In addition to the $B\bar{B}$ MC, simulated events for $\Upsilon(4S) \rightarrow q\bar{q}$ also exist, which have
215 been simulated via the KKMC software. But as the off-resonance data is expected to
216 model the continuum background better, this simulated dataset is replaced by the previ-
217 ously described off-resonance data for data-MC comparisons and is aimed to be used in

the signal extraction fit.

The used samples are processed in the Belle-II-specific software framework `basf2` [36] using the release `light-2212-foldex`.

5. TRUTHMATCHING

In simulation, it is possible to check whether a decay was reconstructed correctly by accessing truth-level information. In particular, we need to identify the following two components in order to construct the MC templates in the fit with the aim to extract their yields:

- signal events, i.e. the signal B decays as $B \rightarrow D^{(*)}\tau\nu_\tau$
- normalisation events, i.e. the signal B decays as $B \rightarrow D^{(*)}\ell\nu_\ell$

The true signal and normalisation samples are identified in MC in the following way:

- The reconstructed signal lepton has been identified correctly (`mcPDG == PDG`)
- The lepton mother (normalisation case) or grandmother (signal case) is the reconstructed signal B (checking `genParticleID` and `mcPDG`)
- The first lepton sister (normalisation case) or lepton aunt (signal case) is the reconstructed $D^{(*)}$
- The second daughter of the signal B is the light lepton in the normalisation case (checking `genParticleID`). It has to be the τ lepton in the signal case (checking `genParticleID==genMotherID` of the light lepton and `mcPDG`). Events selected like this do not include decays with further hadronic sisters of the lepton.[41]

This definition is slightly looser than the definition via `isSignalAcceptMissingNeutrino` by not requiring a correct reconstruction of the $D^{(*)}$ and increases the number of correctly reconstructed events in both the signal and normalisation categories by a factor of about 1.25. It also leads to an enhancement of the fit significance in the signal extraction fit which is further described in Sect. 8.

Note that a correct reconstruction of the tag-side B meson is not required for the truthmatching.

6. SELECTION

In the following, various cuts on different variables are used for the event selection (or subsequently for the signal/normalisation yield extraction). Thus, for the purpose of clarifying how exactly these are defined, some definitions are given below.

- $\cos\theta_{BY}^{\text{sig/tag}}$: angle between the momentum of the B candidate and the vector sum of the momenta of its visible decay products (Y); should be between -1 and 1 for correctly reconstructed normalisation events. The quantity is calculated as:

$$\cos\theta_{BY}^{\text{sig/tag}} = \frac{2E_{\text{beam}}E_Y - m_{B\text{sig/tag}}^2 - m_Y^2}{2|p_{B\text{sig/tag}}|\vec{p}_Y|},$$

where $m_{B\text{sig/tag}}$ is the invariant mass of the signal or tag-side B meson and E_Y , m_Y and $\vec{p}_Y$ describe the energy, invariant mass and momentum of the Y system.

• DMID: tag-side decay mode ID indicating which decay mode has been reconstructed by the SL FEI on the tag side. There are eight different SL decay modes in total (see Sec.6.6.1).

• $\mathcal{P}$: FEI signal probability; final classifier output of the FEI related to the probability of the B^{tag} meson being reconstructed correctly

• E_{ECL} : the extra energy in the calorimeter from unused neutral clusters in the event. The clusters are selected by the criteria defining the so-called Rest Of Event (ROE) which will be described in detail later in this Section.

• $\cos^2 \Phi_B$: establishes a relation between both tag- and signal-side $\cos^2 \theta_{\text{BY}}$ values in the following way:

$$\cos^2 \Phi_B = \frac{\cos^2 \theta_{\text{BY}}^{\text{sig}} + \cos^2 \theta_{\text{BY}}^{\text{tag}} + 2 \cos \theta_{\text{BY}}^{\text{sig}} \cos \theta_{\text{BY}}^{\text{tag}} \cos \gamma}{\sin^2 \gamma},$$

where γ denotes the angle between the two B decay planes. The value of $\cos^2 \Phi_B$ is expected to be ≤ 1 for correctly reconstructed $\Upsilon(4S)$ in the normalisation case. See also ref.[29]

• q^2 : momentum transfer squared using the so-called “Diamond Frame” calculation in the CMS frame as:

$$q^2 = (p_B - p_h)^2,$$

where p_h corresponds to the CMS momentum of all hadrons in the decay. In this case, the variable is used for the signal side B meson in order to define a low- q^2 sideband. For the computation of this variable, weighted average of the B meson around the reconstructed B meson cone is employed. See also ref. [27]. The variable is only a valid quantity for true normalisation events; in case of $\cos^2 \theta_{\text{BY}}^{\text{sig}} > 1$, a more simple calculation is used under the assumption that the B meson momentum in CMS is negligibly small (i.e. setting it to 0).

• m_{miss}^2 : missing mass squared of event calculated as:

$$m_{\text{miss}}^2 = E_{\text{miss}}^{*2} - p_{\text{miss}}^{*2}$$

with E_{miss}^{*2} being calculated as the difference between the CMS energy and the energy corresponding to those clusters in the event that fulfil $E > 0.1$ and $0.296706 < \theta < 2.61799$. p_{miss}^{*2} is computed by considering the momentum associated with all tracks that are required to have $|z_0| < 2$, $|d_0| < 0.5$ and $p_T > 0.1$.

• $\cos \theta_{\vec{x}\vec{p}}$: angle between momentum and vertex vector of a given particle, in this case used for the selection of K_S^0

1.	5.	9.
2.	6.	10.
3.	7.	11.
4.	8.	

Tables II-III present the cuts applied to select events with $B_{\text{sig}}^0 \rightarrow D^{(*)}l/\tau\nu$ decays. It comprises the selection cuts defined in the SL FEI skims, an additional event-level and tag-side selection as well as a selection to identify the exclusive decays on the signal side.

TABLE II: Synopsis of the selection cuts applied on different variables; the grey selection criteria are those applied in the skimming procedure (part 1)

variable	cut
event selection	
$n(\text{cleaned tracks})$	> 2
$n(\text{cleaned clusters})$	> 2
E_{vis}^*	$> 4 \text{ GeV}$
2nd-order Fox-Wolfram moment	< 0.4
tag-side selection	
DMID	< 4
$\log_{10}(\mathcal{P})$	> -1.0
p_l^*	$> 1 \text{ GeV}$
$\cos(\theta_{\text{BY}}^{\text{tag}})$	$-1.75 \text{ to } +1.1$
$\mathcal{P}$	choose best candidate (highest value)
signal-side selection	
$\cos(\theta_{\text{BY}}^{\text{sig}})$	$-15 \text{ to } +1.1$
E_{ECL}	$< 1.2 \text{ GeV}$
tracks	
$ d_0 $ (except slow $\pi^\pm$)	$< 0.5 \text{ cm}$
$ z_0 $ (except slow $\pi^\pm$)	$< 2.0 \text{ cm}$
θ	$0.297 \text{ to } 2.618$ (CDC acceptance)
$n(\text{CDChits})$ (except $\pi^\pm$)	> 0
electrons (in addition to track selection)	
electronIDnoSVDnoTOP	> 0.5
p	$> 0.2 \text{ GeV}$
brems corrections	
muons (in addition to track selection)	
p	$> 0.4 \text{ GeV}$
muonIDnoSVD	> 0.9
D mesons	
m_D	$m_D^{\text{PDG}} \pm 2.5\sigma$
vertex fit	via TreeFit

298 The measure applied for choosing the cut values used in this selection is the signifi-
 299 cance, defined as $S/\sqrt{S+B}$. In this definition S stands for the number of $D^*\tau\nu$ events
 300 and B represents the number of background events. The number of $D^*\ell\nu_\ell$ events, with ℓ
 301 being an electron or muon, is omitted for this measure, since the signal decays is much
 302 rarer than the number of normalisation events, such that it is especially important to keep
 303 as many of these events as possible when aiming at rejecting the backgrounds, since we
 304 have a much poorer understanding of the latter. Reconstructing signal events in any case
 305 yields a substantial amount of normalisation events. As these, in contrast to the other

TABLE III: Synopsis of the selection cuts applied on different variables (part 2)

variable	cut
D* mesons	
$\Delta m(D^* - D)$	0.13 GeV to 0.15 GeV
vertex fit	via TreeFit
π^0 s	
photon daughters	pi0:eff40May2020
 clusterTiming 	< 200 ns
$(\frac{\text{minC2TDist}}{40\text{cm}})^2 + (\frac{\text{clusterZernikeMVA}}{0.4})^2$	> 1
K_S^0 s	
particle list	stdKshorts
	(incl. TreeFit + $0.45 < m < 0.55$)
flight distance	> 0
significance of distance	> 0.5
$\cos \theta_{\vec{x}\vec{p}}$	> 0.8
B mesons	
p value corr. to χ^2 from D TreeFit	choose best candidate (highest value)
Rest of Event	
$ d_0 $	< 0.5 cm
$ z_0 $	< 2.0 cm
θ	0.297 to 2.618 (CDC acceptance)
photons	all
 clusterTiming 	< 200 ns
HadronicSplitOffSuppressionMVA	> 0.2
$n_{\text{ROEtracks}}$	= 0

306 background events, correspond to the component of events which is the best understood
 307 one from the ones that are present in this analysis, such that they do not necessarily have
 308 to be considered in the calculation of the significance.

TABLE IV: Reconstructed D modes

D^0	BF (acc. to PDG)	D^+	BF (acc. to PDG)
$D^0 \rightarrow K^- \pi^+ \pi^0$	14.40%	$D^+ \rightarrow K^- \pi^+ \pi^+$	9.38%
$D^0 \rightarrow K^- \pi^+ \pi^+ \pi^-$	8.23%	$D^+ \rightarrow K_S^0 \pi^+ \pi^0$	7.36%
$D^0 \rightarrow K_S^0 K^+ K^-$	4.42%	$D^+ \rightarrow K_S^0 \pi^+ \pi^+ \pi^-$	3.10%
$D^0 \rightarrow K^+ K^-$	4.08%	$D^+ \rightarrow K_S^0 \pi^+$	1.562 %
$D^0 \rightarrow K^- \pi^+$	3.95%	$D^+ \rightarrow K^- K^+ \pi^+$	0.968%
$D^0 \rightarrow K_S^0 \pi^+ \pi^-$	2.80%	$D^+ \rightarrow K_S^0 K^+$	—
$D^0 \rightarrow \pi^- \pi^+$	0.15%		

6.1. Event-level and tag side

The selection starts by an event-level preselection, then applying the SL FEI and a tag-side preselection as defined in the SL FEI skimming procedure. [50]

Beyond the cuts applied in the skimming procedure, a cut on the second-order Fox-Wolfram moment at 0.4 is used to further decrease the continuum background. The limit of 0.4 is chosen because previous studies report some mismodelling when applying tighter cuts. Above 0.4, one hardly finds any $B\bar{B}$ events, which makes this cut very efficient for continuum suppression.

Furthermore, in order to have a cleaner tag side, which becomes important when applying the completeness constraint, the cuts on $\cos \theta_{BY}^{\text{tag}}$ and on the FEI classifier output $\mathcal{P}$ are tightened. Beyond that, the $D^* \pi l \nu$ channels do not contribute any events when tightening the $\log \mathcal{P}$ cut to -1, such that only six tag-side decay modes are left (DMID < 6). Further, since the DMIDs 4 and 5 do not contribute many events and hardly signal, these two are excluded from the selection as well (see also Section 11.1.1.2). The DMIDs correspond to the following tag-side decay modes, from which the **first four** are used in this analysis:

$$0. \ B_{\text{tag}}^0 \rightarrow D^+ e^- \nu_e$$

$$1. \ B_{\text{tag}}^0 \rightarrow D^+ \mu^- \nu_\mu$$

$$2. \ B_{\text{tag}}^0 \rightarrow D^{*+} e^- \nu_e$$

$$3. \ B_{\text{tag}}^0 \rightarrow D^{*+} \mu^- \nu_\mu$$

$$4. \ B_{\text{tag}}^0 \rightarrow D^+ \pi e^- \nu_e$$

$$5. \ B_{\text{tag}}^0 \rightarrow D^+ \pi \mu^- \nu_\mu$$

$$6. \ B_{\text{tag}}^0 \rightarrow D^{*+} \pi e^- \nu_e$$

$$7. \ B_{\text{tag}}^0 \rightarrow D^{*+} \pi \mu^- \nu_\mu$$

6.2. Signal side

After the tag-side selection, there can in principle be multiple tag-side B candidates, from which only one should be kept in order to not introduce any kind of weights to the

signal side like this when using the same event multiple times. Therefore, only the B^{tag} candidate with the highest FEI classifier output is selected.

For selecting the tracks, recommended cuts on the impact parameters are applied ($d_0 < 0.5 \text{ cm}$, $z_0 < 2.0 \text{ cm}$) and it is required that the track θ is inside the acceptance of the CDC while demanding at least one CDC hit. This is because the CDC information is used to calculate the PID likelihood, such that there should be some CDC information to evaluate. One exception here is the slow pion from the decay of the D^* meson, for which the impact parameter and CDC hits cuts are not applied in order to increase the slow pion efficiency. Although the impact parameter variables are modelled poorly, the cuts introduced here do not introduce a bias, because the distribution is concentrated within a much narrower range (see also Section 11.1.1).

For the final state leptons, there are additional cuts targeting the magnitude of their momentum, which should be $> 0.2 \text{ GeV}$ for electrons and $> 0.4 \text{ GeV}$ for muons. These cuts are aimed at suppressing low-momentum lepton fakes which did not make it into the acceptance limits of the ECL or KLM, respectively. However, as light leptons originating from a τ decay tend to have lower momenta as well, these cuts are not further tightened. The respective cut values are chosen according to the minimum values of lepton momenta for which the lepton ID corrections are available. The availability of these corrections also limits the possible working points for lepton ID to 0.5, 0.9, 0.95 and 0.99. In order to suppress the background from charged pion tracks resembling those of muons, the PID likelihood threshold is set to 0.9 here, which increases the significance. For electrons which are mostly not affected by this kind of fakes the looser working point of 0.5 is chosen. In addition, brems photons are recovered and corrections are applied for electrons as described in Ref.[8]. For charged pions, no PID criteria are applied since they correspond to the majority of reconstructed tracks. For charged Kaons, it has been tested to use a PID threshold as well. However, although it increases the significance as defined above, we could decrease the uncertainty on the signal yield in the signal extraction fit when leaving out this cut. For this reason and due to the potential of introducing a further uncertainty in the analysis attributed to the PID likelihoods, we do not cut on a Kaon ID variable.

The identification of neutral final state particles follows the recommendations of [35], who performed detailed studies on how to maximise the reconstruction efficiency of these particles. All of the cuts taken from [35] are validated for this analysis, ensuring they maximise the significance with respect to $D^*\tau\nu$. The photon daughters of neutral pions are required to be outside an ellipsis in the two dimensional distribution of `clusterZernikeMVA` vs. `minC2TDist`, which denote an MVA classifier using eleven Zernike moments of the respective photon cluster and the minimum cluster-to-track-distance, respectively. This ellipsis contains the region where background events are concentrated. Further selection criteria for neutral pions are taken from the recommended standard list of `gamma:pi0eff40May2020`, additionally cutting at `|cluster timing| > 200 ns`. For neutral Kaons, the standard list is used which includes a vertex fit using TreeFit and a mass cut of $0.45 \text{ GeV} < m_{K_S^0} < 0.55 \text{ GeV}$. Each of the cuts recommended by [35] on the flight distance, significance of distance and the cosine of the angle between momentum and vertex vector maximises the significance with respect to $D^{(*)}\tau\nu$ and is therefore adopted in this analysis.

The selected D^0 and D^+ modes are listed in table IV. Note that in case of the $B^0 \rightarrow D^{*+}\ell^-/\tau^-\nu$ decay, we only use the D^0 modes, while in the $B^0 \rightarrow D^{*+}\ell^-/\tau^-\nu$ case, we employ solely the D^+ modes. Yet, the selection of the D mesons follows the same criteria. They are selected according to their signal purity and the width of the respective D mass peak. Concerning the latter, an additional cut on the mass of the D meson is applied. Due

to the worse detector resolution in case of neutral pion decay products, it is differentiated between those modes including neutral pions and those only comprising of charged pions, the latter of which are chosen with a tighter mass window. To define these mass windows, a Gaussian is fitted to the mass peak and the fitted σ s are averaged for the π^0 and non- π^0 modes, respectively. Beyond that, a Gaussian fit is performed to data in a sideband of $0 < q^2 < 3 \text{ GeV}^2$ in order to check for a shift of the peak in data vs MC. The purpose of this was the following: If there is a shift of the D mass peak in data vs MC, and only the MC width would be considered for the selection, we might obtain a bias due to this cut. Instead, this potential shift should be taken care of by choosing slightly wider mass windows, such that we can assure that the entire peak is covered. Since the fitted means in data vs MC were not compatible within their uncertainties, a rather loose cut on the D mass of $2.5\sigma_m^D$ is chosen, with σ_m^D being either the fitted width of the D modes including π^0 s or the one of the D modes with only charged tracks. We also checked that the D mass peak widths are compatible in data vs MC within their uncertainties in the above mentioned sideband, such that we can approximate the D mass peak widths by the ones obtained from the Gaussian fits to the entire MC, in order to be more precise than when relying on the fits to just the sideband data.

After applying these selection criteria, a vertex fit on the D meson is performed using TreeFit and the corresponding χ^2 probability is saved in order to use it for the purpose of selecting B^{sig} candidates in case of multiple of those candidates per event.

In the $B^0 \rightarrow D^{*+}\ell^-/\tau^-\nu$ case, D^0 mesons are combined with a charged pion to reconstruct a D^{*+} meson. The D^{*+} decay modes with a neutral pion or photon were also investigated but dropped again due to the large candidate multiplicity impeding the selection of a signal side candidate. A mass window of $0.13 \text{ GeV} < \Delta m_{D^{*+}-D} < 0.15 \text{ GeV}$ is chosen loosely comprising the entire $\Delta m_{D^{*+}-D}$ peak and a cross-check in the low- q^2 sideband is used to ensure that the peak in data is not significantly shifted with respect to the peak in MC. In addition, after applying this cut window, a vertex fit with TreeFit is performed in order to refit the slow pion track to the D^{*+} decay vertex.

After that, a signal-side B meson is reconstructed from the previously reconstructed intermediate particles. A cut on $\cos\theta_{\text{BY}}^{\text{sig}}$ of $< +1.1$ further limits the reconstructed particles to a physically meaningful range. While true $D^{(*)}\ell\nu$ events are expected to peak between -1 and +1 in $\cos\theta_{\text{BY}}^{\text{sig}}$, true $D^{(*)}\tau\nu$ events are spread out towards more negative values due to the missing energy associated with the additional neutrinos. A lower cut on $\cos\theta_{\text{BY}}^{\text{sig}}$ is nevertheless possible and reduces a considerably large amount of backgrounds: Below -15, only a small fraction of $< 5\%$ of signal events are found, making a cut at this value reasonable.

6.3. Rest Of Event

When combining the signal and tag B mesons to form the $\Upsilon(4S)$ candidate, in general not all of the supplied detector information is used. Instead, there is a considerable amount of unused tracks and clusters in the event. To select tracks produced by physical particles, the same selection criteria as for the final state particles except for the CDC hits cut are applied. Since in case of a correctly reconstructed $\Upsilon(4S)$, all tracks in the event should be used for the reconstruction, a cut on the number of remaining tracks at exactly 0 is a powerful tool to considerably reduce the amount of backgrounds. The additional energy deposited in the calorimeter, by contrast, may help to distinguish signal/normalisation events from backgrounds via their shapes, such that here only a loose cut at $> 1.2 \text{ GeV}$ will be applied for the signal extraction fit. The choice of neutral

clusters in the rest of event (ROE) is therefore crucial for the analysis, which is why a separate section is dedicated to their selection (see 6.6.4).

With this selection, we count on average 1.08 $B\bar{B}$ candidates and 1.03 $q\bar{q}$ candidates per event in MC for the D^* mode and 1.05 $B\bar{B}$ and $q\bar{q}$ candidates per event for the D mode. If there are multiple $\Upsilon(4S)$ candidates after applying all of the named selection criteria, they are ranked according to the χ^2 probability of the performed TreeFit on the D vertex and only the highest-ranked $\Upsilon(4S)$ candidate is chosen.

6.4. Choice of ROE mask defining the extra energy in the calorimeter

Given the complication of achieving a good modelling in the additional energy in the calorimeter, this section will be dedicated to the studies performed in order to find the optimal ROE definition as a foundation of the E_{ECL} distribution, a variable with potentially high discrimination power between the $D^{(*)}(l/\tau)\nu$ and background templates, which thus requires a certain definition of the neutral unused detector information. Ultimately, a cut of $E_{\text{ECL}} < 1.2 \text{ GeV}$ will be applied, since the backgrounds tend to come with a larger amount of unused energy in the calorimeter due to mis-reconstruction.

From theory, one would expect the signal as well as normalisation events to peak at zero in this variable, because ideally, all tracks and clusters in the event should have been used for the reconstruction of the $\Upsilon(4S)$, given that the invisible neutrinos are the only non-reconstructed particles in the event. If not everything in the reconstruction was correct, however, some additional clusters might be leftover, such that the distribution of background events is expected to be rather flat across a larger range in E_{ECL} . There are, however, two types of backgrounds leading to a less discriminating shape of signal/normalisation and backgrounds in E_{ECL} : beam background photons and clusters originally belonging to a hadronic cluster, but considered as separate clusters by the detector.

Possibilities of cleaning up the E_{ECL} distribution from these backgrounds include cuts on two MVAs intrinsically developed for the purpose of rejecting either of the two background contributions.

To obtain a first idea of where to cut on the classifier outputs, these are plotted on ECL cluster level, differentiating between the different types of backgrounds (see Figs.2-3).

For these plots, 800 fb^{-1} of generic MC is used. The used data corresponds to proc13 which was processed with release 6. Continuum is modelled by off-resonant data from proc13. The Beam Background MVA distribution shows reasonable agreement and suggests a cut on this classifier at 0.5, thus being less sensitive to mismodelling errors towards values of 0 or 1. When applying this cut, also the agreement in the Hadronic Split-Off Suppression MVA seems appropriate (cf. Fig.3).

In principle, one could now also cut on the Hadronic Split-Off Suppression MVA at 0.5. In order to test the data-MC agreement when cutting on the MVAs, we can avail ourselves of a sideband in the reconstructed q^2 value of $0 < q^2 < 3 \text{ GeV}^2$, since signal events are expected from theory to have a $q^2 > 3 \text{ GeV}^2$. In order to test the agreement despite potential missing normalisation factors due to a missing FEI calibration, we are only looking at normalised distributions. Doing so, a slight sculpting of the background distribution is observed when cutting on both MVAs at 0.5. They tend to peak towards 0 and thus resemble the normalisation (as well as signal) events. This should be avoided, since E_{ECL} is expected to be the most important variable in this analysis to distinguish the background events from the other two event categories. Thus, a slightly looser cut on the Hadronic Split Off Suppression MVA might be a better choice. To make a final decision,

cuts between 0.1 and 0.4 are tested, each time calculating a χ^2 and corresponding p value that quantify the agreement between data and MC in the q^2 sideband. Fig. 5 shows these p values for the cases of $B \rightarrow D(\ell/\tau)\nu$ (blue) and $B \rightarrow D^*(\ell/\tau)\nu$ (violet) in depending on the cut on the Hadronic Split-Off Suppression MVA. It suggests that cutting at 0.2 may be a good choice since the p values are similarly high for both cases. Figure 4 reveals the data-MC agreement in the q^2 sideband using the aforementioned cuts on the MVAs. The agreement appears reasonable. When applying the chosen ROE mask and cutting on E_{ECL} at 1.2 GeV, the following values for efficiency and purity (defined as $\frac{N_{\text{sig}}}{N_{\text{sig}} + N_{\text{backgr}}}$ omitting N_{norm}) are obtained:

- $B \rightarrow D(\ell/\tau)\nu$: efficiency 0.96, purity 0.021 (compared to a purity of 0.013 before the selection)
- $B \rightarrow D^*(\ell/\tau)\nu$: efficiency 0.98, purity 0.146 (compared to a purity of 0.085 before the selection).

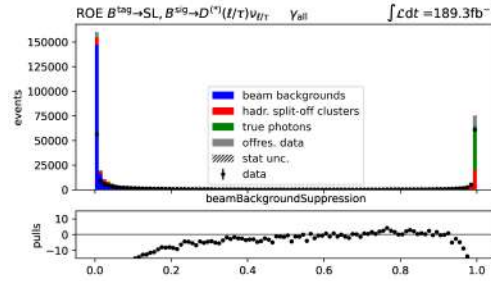
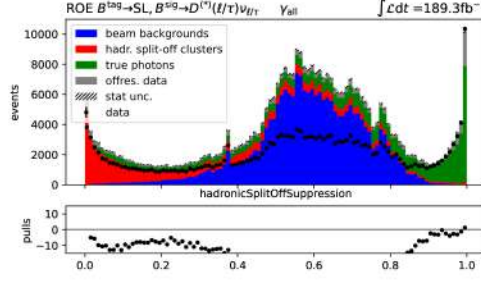
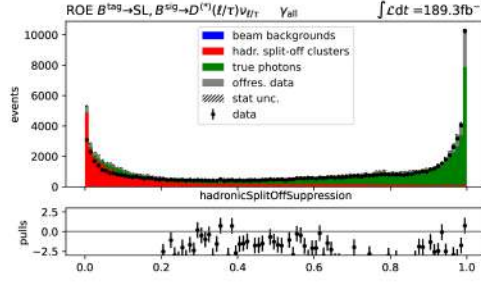


FIG. 2: Data (proc13) and MC distributions of the Beam Background MVA at ECL cluster level, differentiating between different sources of backgrounds.



(a)



(b)

FIG. 3: (a) Same as Fig.2 but for the HadronicSplitOffSuppressionMVA.
(b) Same distribution with a cut on the Beam Background MVA at 0.5

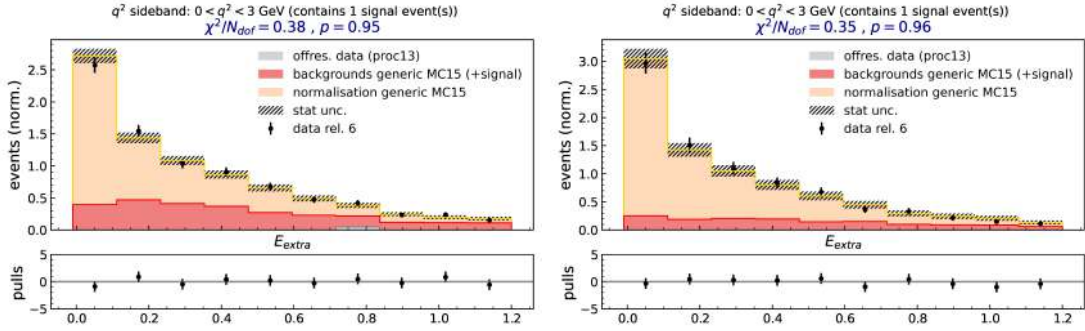


FIG. 4: E_{ECL} in the q^2 sideband for a cut on the Beam Background MVA at 0.5 and on the Hadronic Split-Off Suppression MVA at 0.2 for $B \rightarrow D(\ell/\tau)\nu$ (left) and $B \rightarrow D^*(\ell/\tau)\nu$ (right)

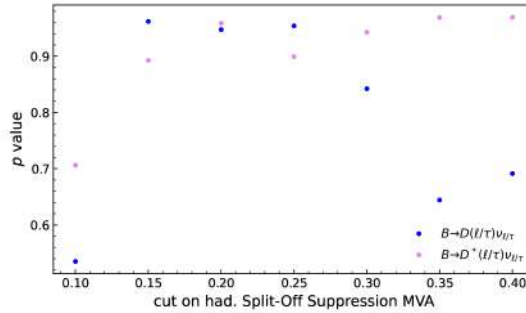


FIG. 5: p values calculated from a χ^2 of the data and MC distributions in the q^2 sideband using different cuts on the Hadronic Split-Off Suppression MVA.

7. CORRECTIONS APPLIED TO THE DATASETS

We apply several event weights to correct the signal efficiency. Those are related to:

- Branching fractions (BFs)
- Form factors (FFs)
- Slow pion efficiency (for the $D^{*+} \rightarrow D^0\pi^+$ decay)
- Lepton ID
- Track momentum scale
- Photon energy bias

In addition, the used signal MC samples ($D^*\tau\nu$, gap modes) as well as off-resonance data receive event weights in order to correct for the luminosity differences (cf. Section 4). We describe the applied corrections in further detail in the following section.

Firstly, branching fractions used for event generation [2] might differ from the most recently measured values. Thus, the BF of signal and normalisation modes as well as all relevant background contributions are updated, meaning that the BF of $B \rightarrow X(\ell/\tau)\nu_{\ell/\tau}$ are corrected. The values of the generated and updated BF can be seen in table V [5]. The correction is done by introducing a weight for all events belonging to a certain decay mode, being calculated as the ratio of the updated and generated BF. The BF of $\tau \rightarrow \ell\nu_{\ell}\nu_{\tau}$ is not corrected, as it divides out in the ratio of $\mathcal{R}(D^{(*)})$.

Secondly, the form factor (FF) model for the MC generated events is updated to the most recent predictions using the BLPRXP form-factors of Ref. [11], with values determined from a fit of the measurements of Refs. [28, 55] and by using **HAMMER** [9]. Table VI summarises the FF models employed in the MC generation as well as the most up-to-date FF models after the reweighting with **Hammer** for different B^0 decay modes. For the gap events, the same FF model is used as for the resonant broad D^{**} events, as the gap-mode signal MC simulates an intermediate broad D^{**} resonance.

Thirdly, the efficiency of reconstructing the slow pion from the $D^{*+} \rightarrow D^0\pi^+$ decay may differ in data vs MC. Thus, we use efficiency weights derived in Ref.[18] to correct for these differences. The referenced study provides momentum-dependent correction weights which are normalised to the highest $p^{\pi^\pm}$ bin, together with a their statistical and systematic uncertainties, which are presented in table VII.

Moreover, as we use the likelihood-based PID variables **electronID** and **muonID**, our selection is sensitive to differences in efficiency between data and simulation. This difference in efficiencies is accounted for by reweighting the MC datasets in bins of lepton momentum p and θ [30]. Corrections are applied to both electron and muon identification efficiency and to the rates of pions or kaons faking leptons. The former are derived from $J/\Psi \rightarrow l^+l^-$, $e^+e^- \rightarrow e^+e^-l^+l^-$, $e^+e^- \rightarrow \mu^+\mu^-\gamma$ and $e^+e^- \rightarrow e^+e^-(\gamma)$ decays in 62.8fb^{-1} of data while the pion fake rates are determined from $K_S \rightarrow \pi\pi$, $D^* \rightarrow D^0(K\pi)\pi$ and $ee \rightarrow \tau(1p)\tau(3p)$ and the kaon fake rates are deduced from $D^* \rightarrow D^0(K\pi)\pi$.

TABLE V: B decay modes together with the value to which the BF is updated as well as the one used for MC generation [5]

decay mode	BF B^0 [10^{-3}]	gen BF [10^{-3}]
$B \rightarrow D\ell^+\nu_\ell$	22.4 ± 0.7	21.4
$B \rightarrow D^*\ell^+\nu_\ell$	51.1 ± 1.1	51.1
$B \rightarrow D_1\ell^+\nu_\ell$	6.1 ± 1.0	7.04
$B \rightarrow D_0^*\ell^+\nu_\ell$	3.9 ± 0.7	3.62
$B \rightarrow D_1'\ell^+\nu_\ell$	3.9 ± 0.8	4.01
$B \rightarrow D_2^*\ell^+\nu_\ell$	2.73 ± 0.3	3.47
$B \rightarrow D\pi\ell^+\nu_\ell$	0.0	1.38
$B \rightarrow D^*\pi\ell^+\nu_\ell$	0.0	1.38
$B \rightarrow D\pi\pi\ell^+\nu_\ell$	0.6 ± 0.9	0.49
$B \rightarrow D^*\pi\pi\ell^+\nu_\ell$	2 ± 1.0	2.45
$B \rightarrow D\eta\ell^+\nu_\ell$	4.09 ± 4.09	2.17
$B \rightarrow D^*\eta\ell^+\nu_\ell$	4.09 ± 4.09	2.17
$B \rightarrow D\tau^+\nu_\tau$	6.698 ± 0.22	6.3986
$B \rightarrow D^*\tau^+\nu_\tau$	13.133 ± 0.322	13.1838
$B \rightarrow D_1\tau^+\nu_\tau$	0.62 ± 0.118	1.3
$B \rightarrow D_0^*\tau^+\nu_\tau$	0.312 ± 0.13	1.3
$B \rightarrow D_1'\tau^+\nu_\tau$	0.234 ± 0.092	2.0
$B \rightarrow D_2^*\tau^+\nu_\tau$	0.191 ± 0.034	2.0
$B \rightarrow D\pi\pi\tau^+\nu_\tau$	0.0665 ± 0.094	0.0665
$B \rightarrow D^*\pi\pi\tau^+\nu_\tau$	0.23 ± 0.109	0.23
$B \rightarrow D\eta\tau^+\nu_\tau$	0.469 ± 0.469	0.469
$B \rightarrow D^*\eta\tau^+\nu_\tau$	0.469 ± 0.469	0.469

TABLE VI: FF model used for MC generation and updated FF models

B decay modes	generator FF model	new FF model	number parameters
$D\ell\nu_\ell$	BGL	BLPRXP	9
$D\tau\nu_\tau$	CLN	BLPRXP	9
$D^*\ell\nu_\ell$	BGL	BLPRXP	9
$D^*\tau\nu_\tau$	CLN	BLPRXP	9
$(D_1^-/D_2^{*-})(\ell/\tau)^+\nu_{\ell/\tau}$	BLR	BLR	4
(narrow D^{**} resonances)			
$(D_0^{*-}/D_1'^-)(\ell/\tau)^+\nu_{\ell/\tau}$	BLR	BLR	3
(broad D^{**} resonances)			

TABLE VII: Slow pion efficiency corrections together with their statistical and systematic uncertainties [18]

momentum bin	0.05-0.12 GeV	0.12-0.16 GeV	0.16-0.2120 GeV	0.2-0.32 GeV
weight	0.996	0.990	0.987	1
stat.unc. uncorr.	0.022	0.017	0.019	1
stat. unc. corr.	0.015	0.015	0.015	1
syst. unc.	0.003	0.003	0.003	1

531 8. DEVELOPING THE SIGNAL AND NORMALISATION YIELD EXTRAC- 532 TION FIT

533 To ultimately obtain $R(D^*)$, a binned negative log-likelihood (NLL) fit can be per-
534 formed to the measured data in order to extract the yields of signal and normalisation
535 modes. The challenging aspect of this fit is its desired ability to accurately distinguish
536 between three categories of events: Apart from separating $D^{(*)}\tau\nu$ from $D^{(*)}l\nu$, the fit
537 should be capable of distinguishing both components against the backgrounds. Thus, we
538 require one or more variables in which two or all three components have discriminating
539 shapes. Different variables fulfilling this condition can be thought of, all of which are pre-
540 sented in Section 8.8.2. Utilising the discriminating variables, a multiclassifier training
541 is performed (cf. Subsec.8.8.1), and using its outputs in a signal extraction fit is com-
542 pared to several different other fit configurations with combinations of the input physics
543 variables (via asimov fits). Ultimately, a combination of the classifier outputs is chosen
544 (cf. Subsec. 8.8.2). The signal extraction fit is implemented in two different versions,
545 firstly accounting for systematic uncertainties using bin-wise nuisance parameters (see
546 Subsec.8.8.3) and secondly with source-wise nuisance parameters (see Subsec. 11.11.6).
547 The two approaches are compared and the latter one is chosen for obtaining the final
548 results (see Subsec.11.11.8).

549 8.1. Multiclassifier training

550 For the signal extraction fit, we use the output of a multiclassification algorithm in
551 order to distinguish the three event categories of signal, normalisation and background
552 events. Specifically, we use the **Gradient Boosting classifier** belonging to the python
553 package **sklearn**. For each event, it outputs three probabilities of belonging to either of
554 the categories. Each individual probability is in the range $[0, 1]$ while the sum of all three
555 probabilities amounts to one:

$$p_\tau + p_l + p_{\text{backr}} = 1. \quad (3)$$

556 The variable distributions to be used in the training are chosen according to the ability
557 to separate between the three categories of events via their shapes as well as on the data-
558 MC agreement. To test the latter, we again consider the sideband in $q^2 < 3 \text{ GeV}^2$, where
559 we also compute a χ^2 probability belonging to the consistency of the shapes of simulation
560 and data. Fig. 6 shows this variable in data compared to simulation. Figs. 7-8 show the
561 data-MC agreement in this sideband for the final choice of training features. Note that in
562 contrast to the previous version of the training using MC14 and rel.5, the variable m_{miss}^2 ,
563 describing the missing mass squared of the event, is dropped from the list of training
564 features, as we discovered some mismodelling in the lower m_{miss}^2 bins, which has been
565 seen in other analyses as well. Due to the potential systematic uncertainty introduced by it
566 as well as the low feature importance, a training on that variable is not worthwhile.

567 It is further ensured that no large correlations between the training features occur, i.e.
568 by removing one of the variables in question. We make sure that all remaining correlations
569 are not larger than 0.5. Fig. 9 reveals the final importance of the so-selected BDT input
570 variables as well as the correlations of the remaining variables, which are sufficiently low.
571 For the importance of each variable in relation to the most important feature, see table
572 VIII.

573 In order to not use the same events in the training as for the signal extraction, we
574 use half of the available $D^{(*)}\tau\nu$ signal MC for the training of the multiclassifier. The
575 other half of the signal MC will be employed in the fit for the yield extraction later. This

576 ensures that the classifier is not applied to the exact same events that it was trained on,
577 which would lead to an overestimation of its performance.

As the amount of $D^{(*)}\tau\nu$ signal MC events is the smallest of the available MC samples, we limit the number of events for each of the other samples by this number.

The normalisation events are drawn from 800fb^{-1} of generic MC, which is independent from the 2000fb^{-1} of generic MC used for the plots and asimov fits presented in the remaining sections of this note. Also the background events stem from this sample; but in addition, the dedicated signal MC for semileptonic / semitauonic gap modes is used (cf. Sect.4). To all of the samples, the same selection criteria are applied as those used for the templates in the signal extraction fit in order for the BDT to learn only those features which it will encounter in the fit templates.

The resulting ROC curves for each of the classifiers can be seen in Fig. 10. Those
 588 portray the misclassification rate vs. the true positive rate for independent training and
 589 testing samples. , The area under the ROC curves (annotated to the plots) is sufficiently
 590 large.

Finally, Fig.11 reveals the resulting classifier outputs for the same training and testing samples. The performance of the classifiers is still optimisable; especially due to the diverse origin of the backgrounds, these appear to be difficult to identify for the background classifier. Using more background events than signal events, on the contrary, would lead to a large degradation of the $D^{(*)}\tau\nu$ classifier performance. What is more, we prioritise the fact that there are as few background events in the region where signal peaks, meaning that we in the first place require an accurate performance of the signal classifier. Thus the backgrounds will not interfere as much with the accurate extraction of the signal yield. So in that respect, the ability to correctly classify backgrounds is the property of the multiclassifier that may be most dispensable while its ability to accurately identify signal events is of highest importance.

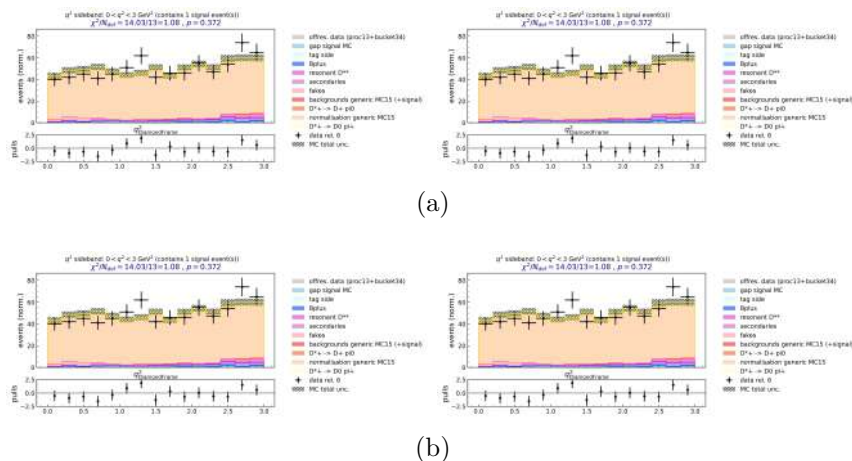


FIG. 6: $q^2 < 3 \text{ GeV}^2$ sideband used for blinding

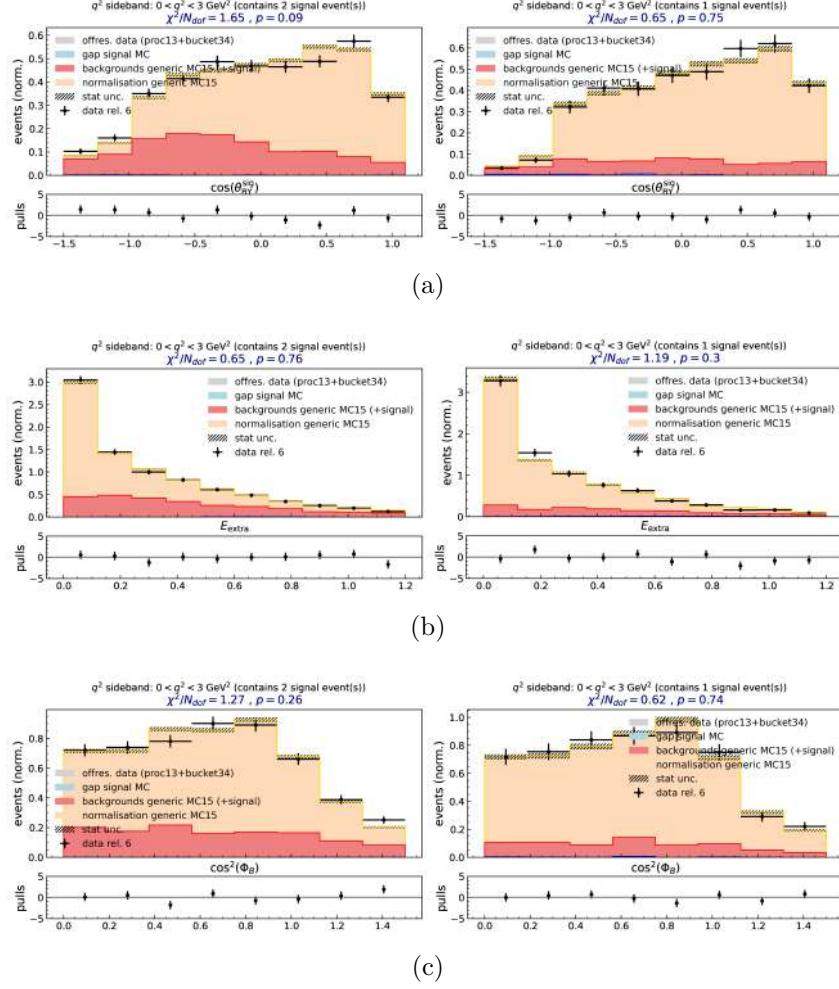


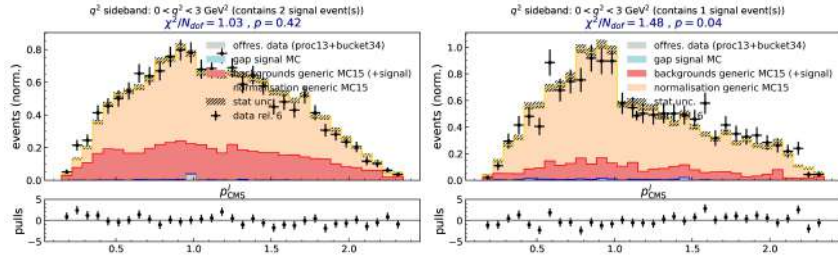
FIG. 7: BDT input variables in the $q^2 < 3 \text{ GeV}^2$ sideband, once for the D mode (left) and once for the D^* mode (right):

- (a) $\cos \theta_{BY}$
- (b) E_{extra}
- (c) $\cos^2 \Phi_B$

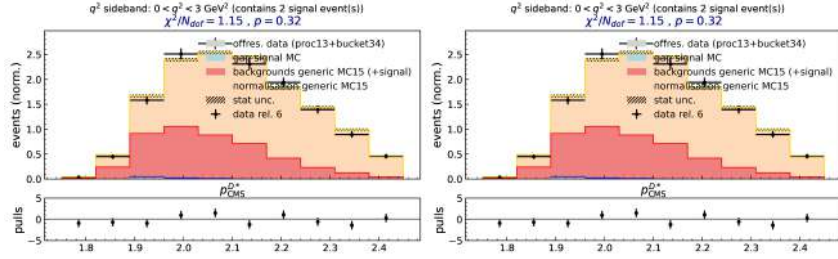
The MC components are scaled such that the MC yield matches the data yield, in order to better visualise the shape differences. Each plot includes a χ^2 and corresponding p value, which refer to the compatibility of the normalised data and MC distributions.

TABLE VIII: Importance of the BDT input variables relative to the most important one.

rank	variable name	importance rel. to most important [%]
1	cosBY	100.0
2	EextraR0Emask50hadronicSplitOffSuppression025	51.8
3	cos2PhiB	15.9
4	pDCMS	8.7
5	plCMS	8.4



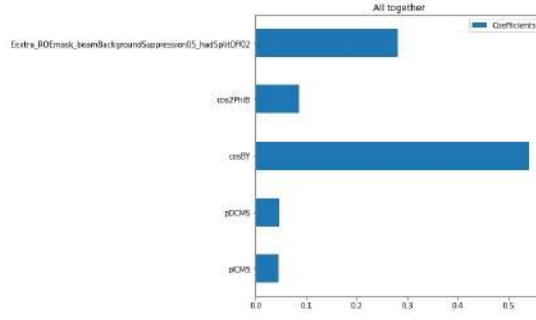
(a)



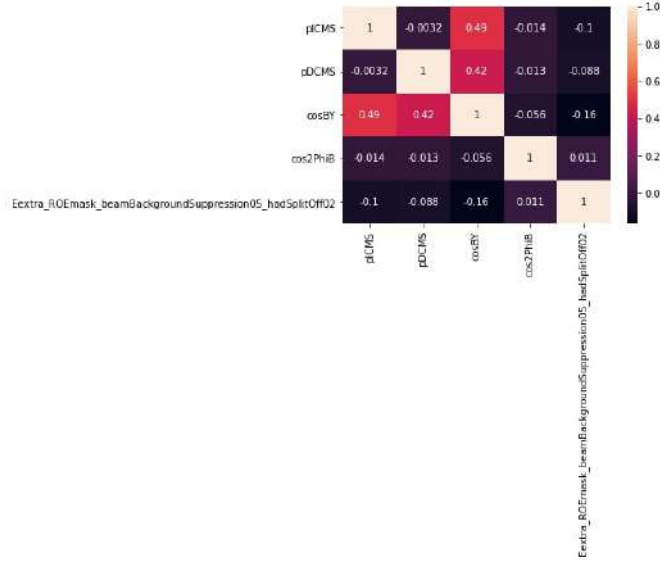
(b)

FIG. 8: Same as Fig.7, but for:

- (a) p_l^*
- (b) p_{D^*}



(a)



(b)

FIG. 9: Importance of the feature variables for the classification into signal, normalisation and background categories (a) and visualisation of the correlation matrix between the input variables (b). Cf. table VIII for the importance of each variable rel. to the most important feature.

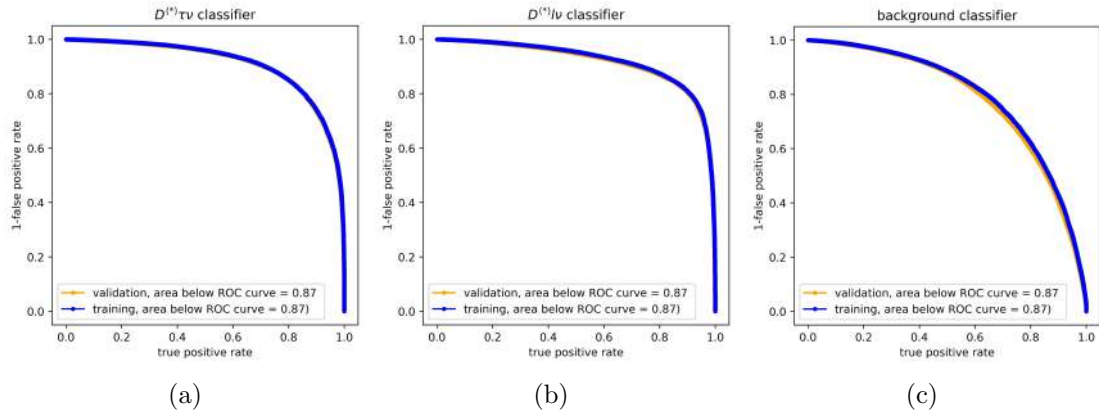


FIG. 10: ROC curves for the three classifiers:

(a) $D^{(*)}\tau\nu$

(b) $D^{(*)}l\nu$

(c) backgrounds

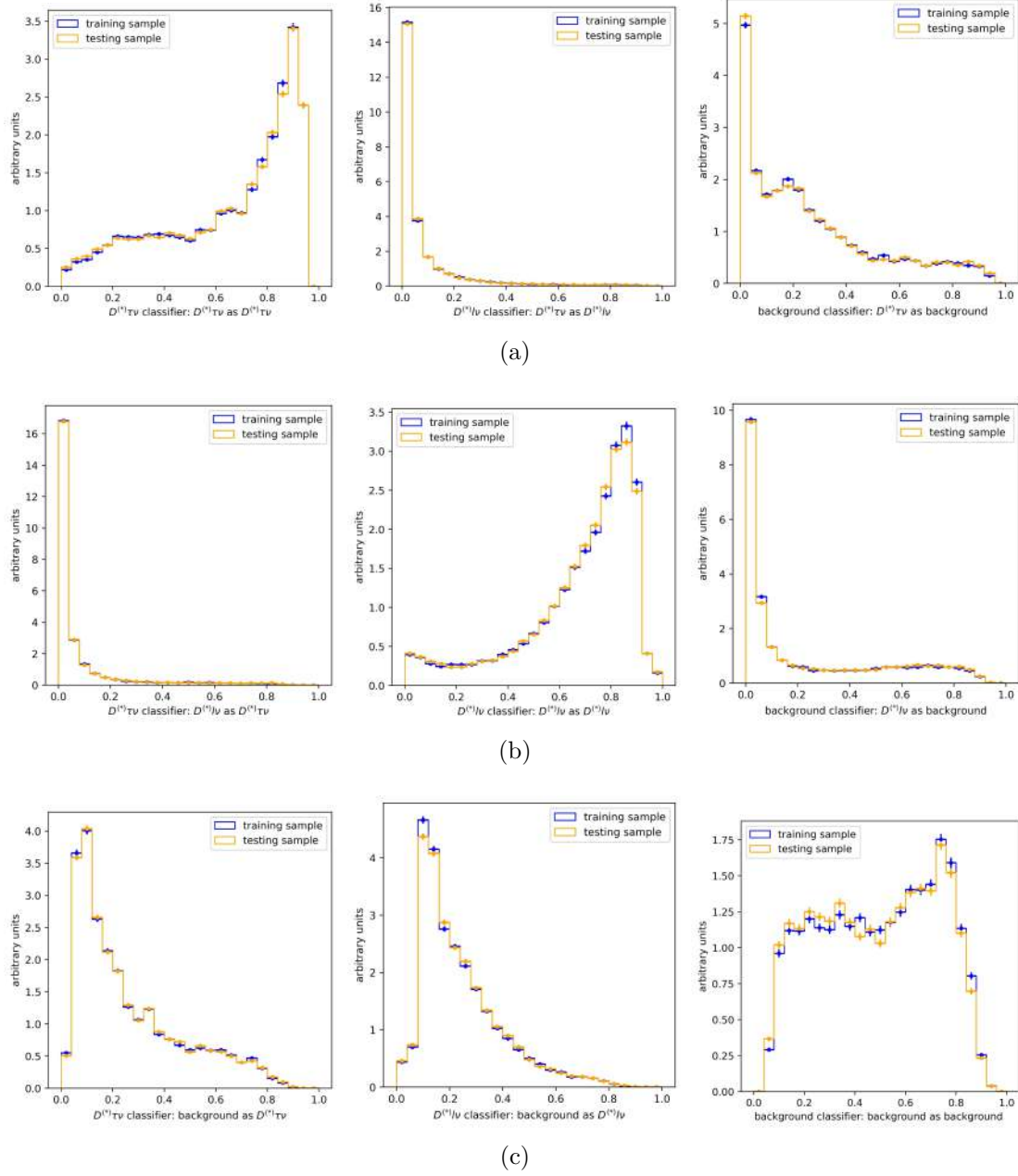


FIG. 11: Classifier output distributions for independent training and testing samples using events from each of the three categories:

(a) $D^{(*)}\tau\nu$

(b) $D^{(*)}\ell\nu$

(c) backgrounds

8.2. Signal extraction variables

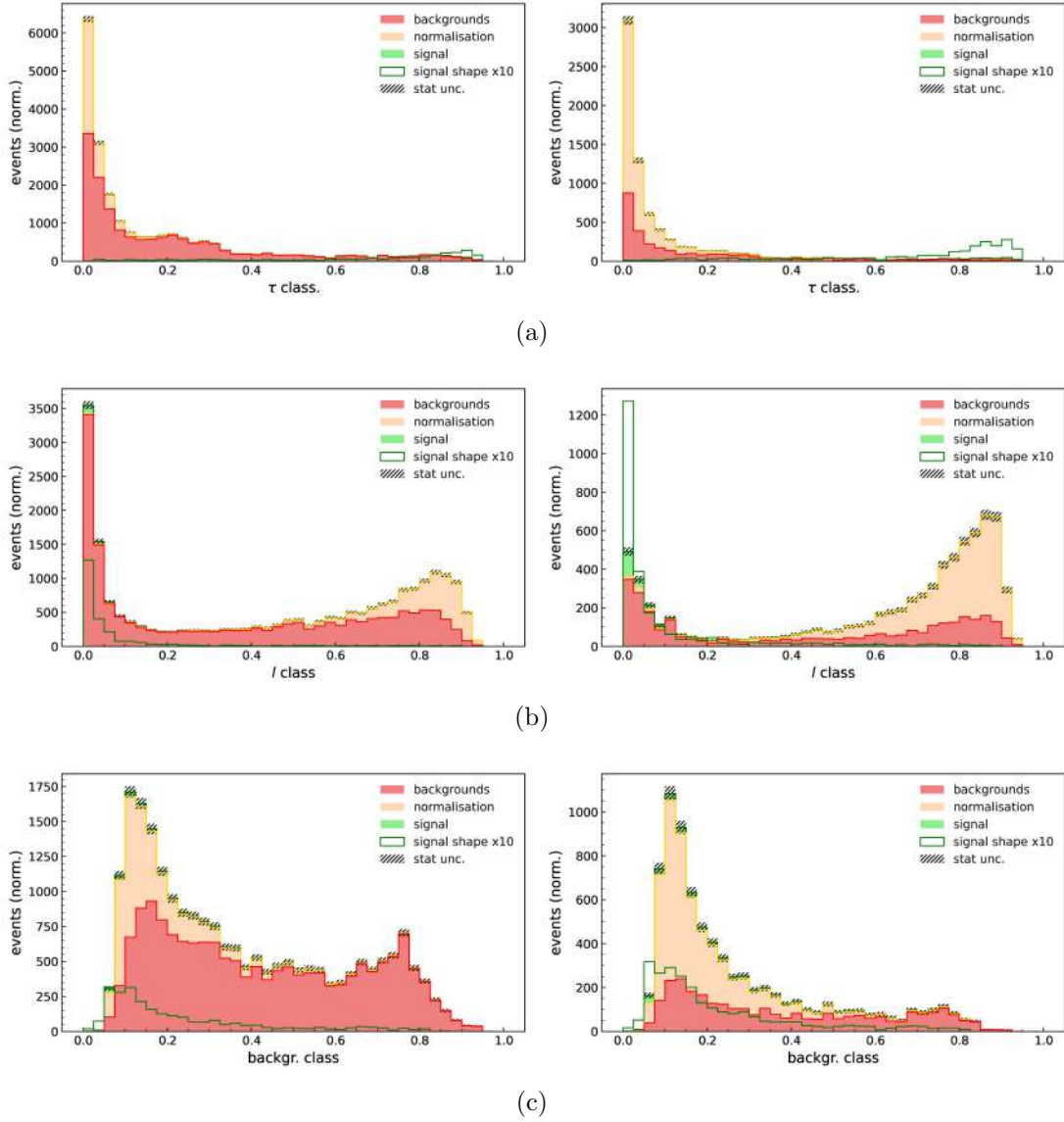


FIG. 12: Distributions of the three event categories in the three classifier outputs:

(a) $p(D^{(*)}\tau\nu)$

(b) $p(D^{(*)}l\nu)$

(c) $p_{\text{backgrounds}}$

The left plot shows $B \rightarrow D(\ell/\tau)\nu_{\ell/\tau}$ events while the right one is for $B \rightarrow D^{*}(\ell/\tau)\nu_{\ell/\tau}$ events.

The fit for the signal extraction makes use of all three classifier outputs. Fig. 12 shows how the three event categories that the fit is aimed to separate are distributed in these variables. The aim is now to achieve the best possible separation between the templates. In order to increase this separation, the classifier outputs may be transformed using a logarithmic function:

$$z_p = \text{logit}(p) = \log\left(\frac{p}{1-p}\right), \quad (4)$$

where p labels the probability of an event belonging to the respective category. This maps the probabilities, which range from 0 to 1, to a larger range between $-\infty$ and ∞ , pulling

apart the distributions of the three templates. Thus, it is easier for the fit to discriminate between the three event categories. Fig. 13 reveals the transformed distributions.

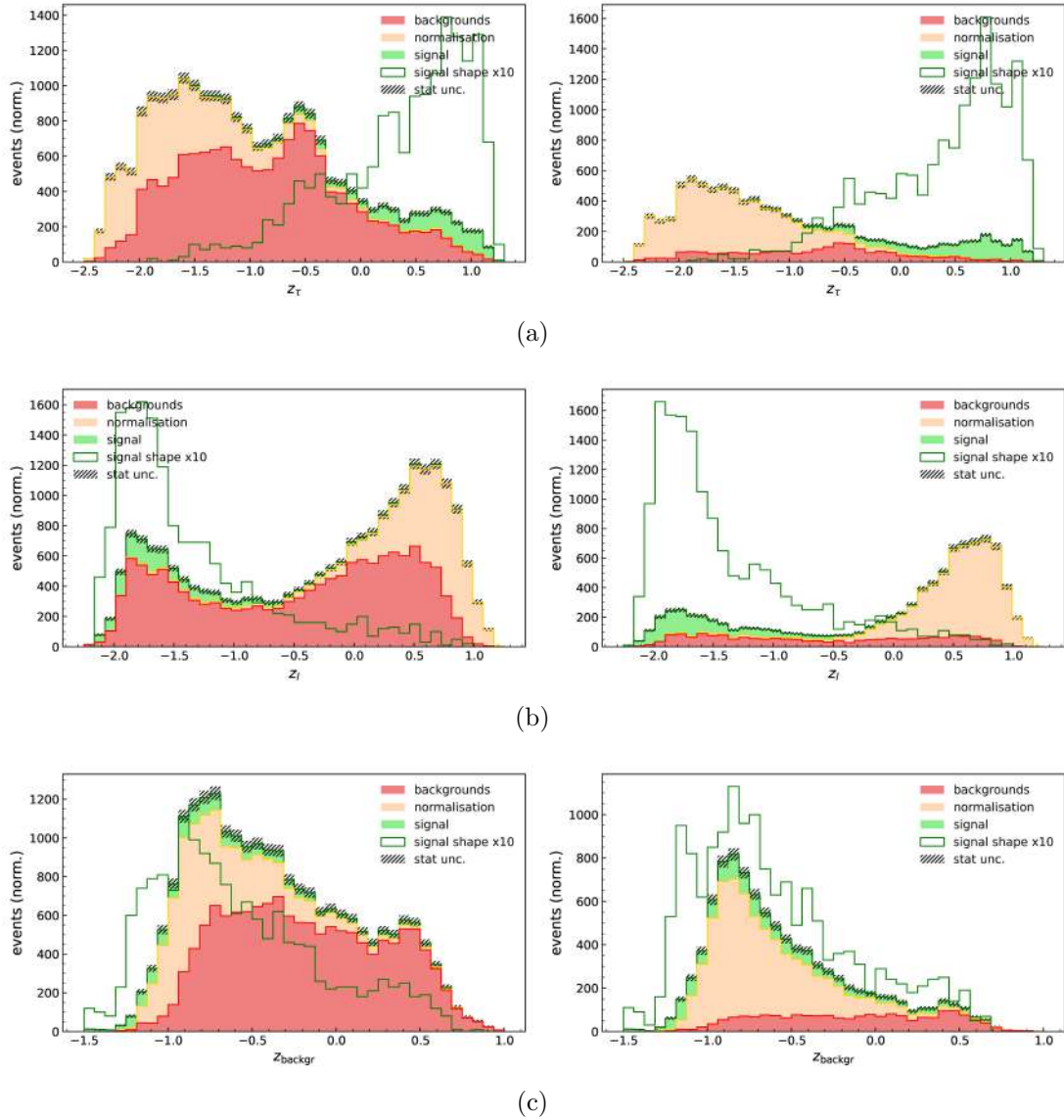


FIG. 13: Transformed classifier outputs using eq.4 for:

(a) $p(D^{(*)}\tau\nu)$

(b) $p(D^{(*)}l\nu)$

(c) $p_{\text{backgrounds}}$

The left plot shows $B \rightarrow D(\ell/\tau)\nu_{\ell/\tau}$ events while the right one is for $B \rightarrow D^*(\ell/\tau)\nu_{\ell/\tau}$ events.

The figure suggests that the signal classifier score already separates the signal template very well from the other templates. What might be problematic is the performance of the background classifier, especially its ability to identify background events against normalisation events. Thus, one may consider a quantity that measures the difference between the probabilities of belonging to either the normalisation or background categories. Compared to just using p_l or p_{backgr} as a fit variable, a difference offers the possibility to at the same time place the signal events rather in the middle of the distribution and thus separate all three categories from one another. Yet, the simple difference $p_l - p_{\text{backgr}}$ still features a small peak of the background values at the same place where normalisation

621 peaks (cf. Fig.14(a)). Thus, again the logarithmic version seems a reasonable choice.
 622 Using just $\log(p_l)/\log(p_{\text{backgr}})$ (cf. Fig.14(b)) would in principle have the same effect as
 623 using the pre-defined score $z = \text{logit}(p)$ (Fig.14(c)). Since the latter is line with the
 624 choice of the first fit variable z_τ , this combination of fit variables seems reasonable.

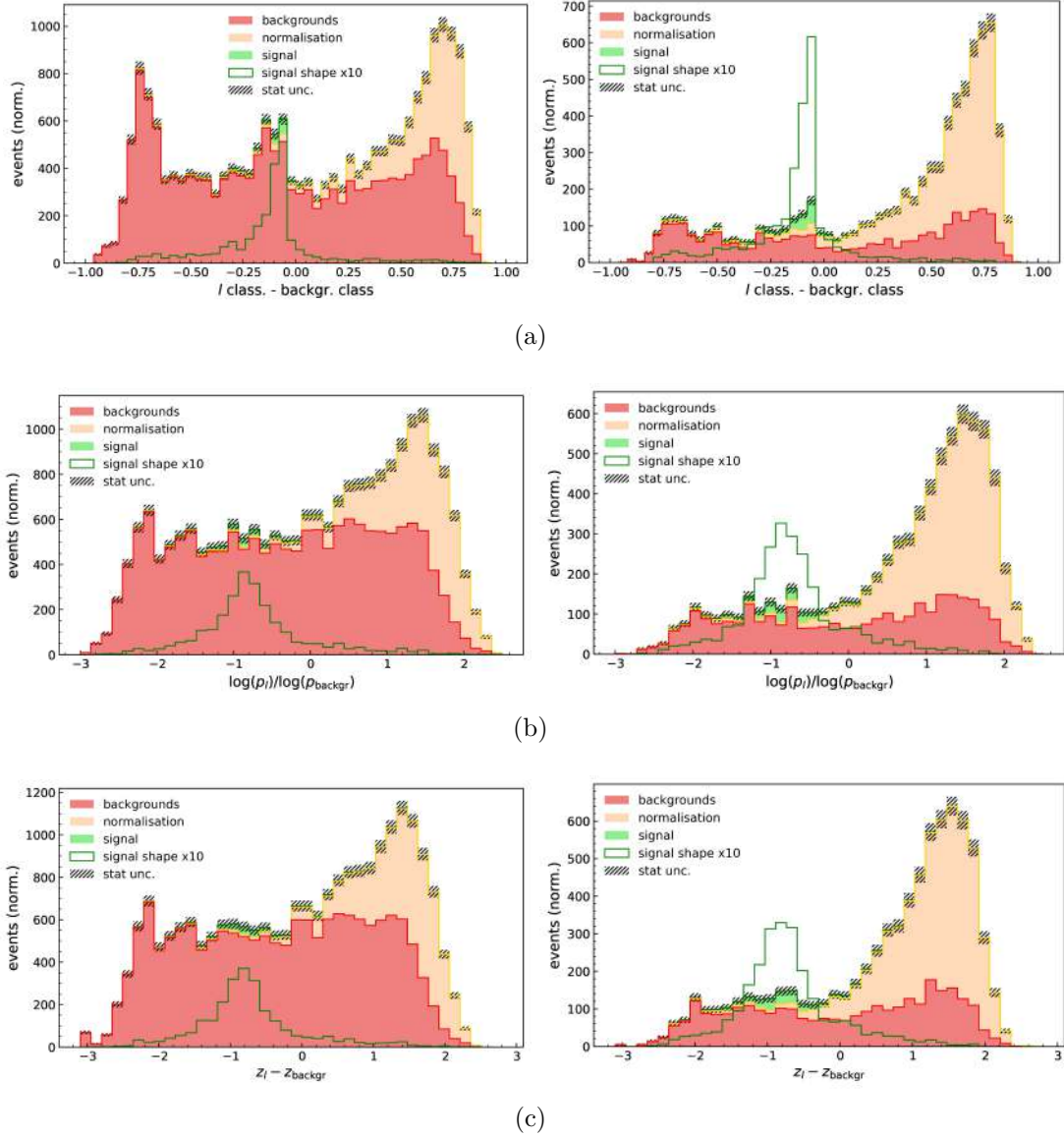


FIG. 14: Distributions of the three event categories in different variables that measure the difference between an event being normalisation- or background-like:

- (a) $p_l - p_{\text{backgr}}$
- (b) $\log(p_l)/\log(p_{\text{backgr}})$
- (c) $z_l - z_{\text{backgr}}$

The left plot shows $B \rightarrow D(\ell/\tau)\nu_{\ell/\tau}$ events while the right one is for $B \rightarrow D^*(\ell/\tau)\nu_{\ell/\tau}$ events.

625 In order to make a choice of the signal extraction variables which is justified also
 626 in a quantitative way, a yield fit is performed as described in Subsec.8.8.3, taking into
 627 account only statistical uncertainties. In order to match the statistical power of the final
 628 analysis dataset, 400 fb^{-1} of generic MC is used during these fits and the employed signal
 629 MC samples ($D^{(*)}\tau\nu$, $D^{(*)}\ell\nu$) are scaled accordingly. The fit is then repeated without

TABLE IX: Parameters considered to find the optimal fit configuration: fit significance and rel. uncertainty on the signal, normalisation and background yields. The values in brackets are those obtained with a signal/normalisation definition via `isSignalAcceptMissingNeutrino`

mode	signal significance	rel. unc. N_{sig} [%]	rel. unc. N_{norm} [%]	rel. unc. $N_{\text{backgr.}}$ [%]
$D(l/\tau)\nu$	5.5 (3.95)	19 (25)	2.0 (2.3)	1.1 (0.8)
$D^*(l/\tau)\nu$	12.5 (8.31)	10 (13)	1.6 (2.1)	3.9 (2.6)

including a signal template. Subsequently, the signal significance is calculated as:

$$-2 \ln \frac{L_p(x|H_1)}{L_p(x|H_0)}, \quad (5)$$

where L_p is the likelihood function, H_1 denotes the hypothesis including the signal, while H_0 is the hypothesis without the signal (cf. Eq.3.14 in [47]).

The so-calculated highest signal significance is obtained using a 2D fit in the $D^{(*)}\tau\nu$ classifier score z_τ and the difference between the $D^{(*)}l\nu$ and background classifier scores, $z_l - z_{\text{backgr.}}$.

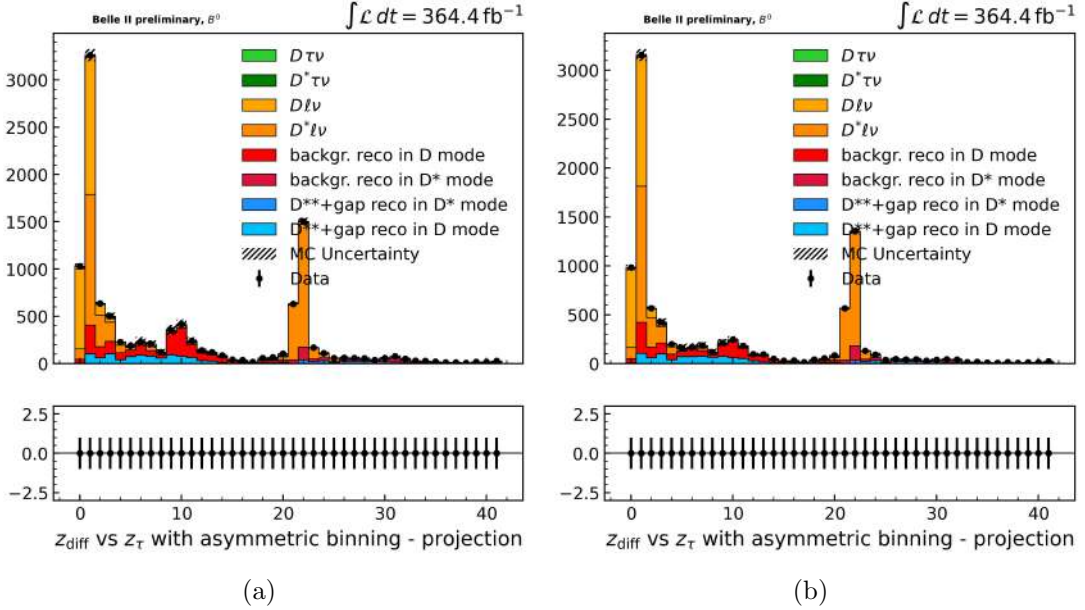


FIG. 15: Post-fit histograms from a full asimov fit showing the number of events within the bins; the number on the x axis corresponds to the bins as indicated in Figs.24 and 25. The binnumbers for the D^* mode (right side of the histogram) are shifted wrt. those of the D mode (left side) by the total number of bins:

(a) electron mode

(b) muon mode

The two-dimensional distributions can be seen in Figs. 16- 19. The latter plots indicate that a two-dimensional fit using these signal extraction variables performs better than a simultaneous one-dimensional one, since the former is capable of taking account of the

separation between the templates in the 2D plane. This was confirmed by comparing the significances for both fit configurations. In order to confirm that the feed-down events for the normalisation and signal categories are located in the same area in 2D-classifier space as the respective correctly reconstructed events, we plot them separately. Thus, we can confirm that these events are indeed assessed in a similar way by the classifier due to their similar kinematics.

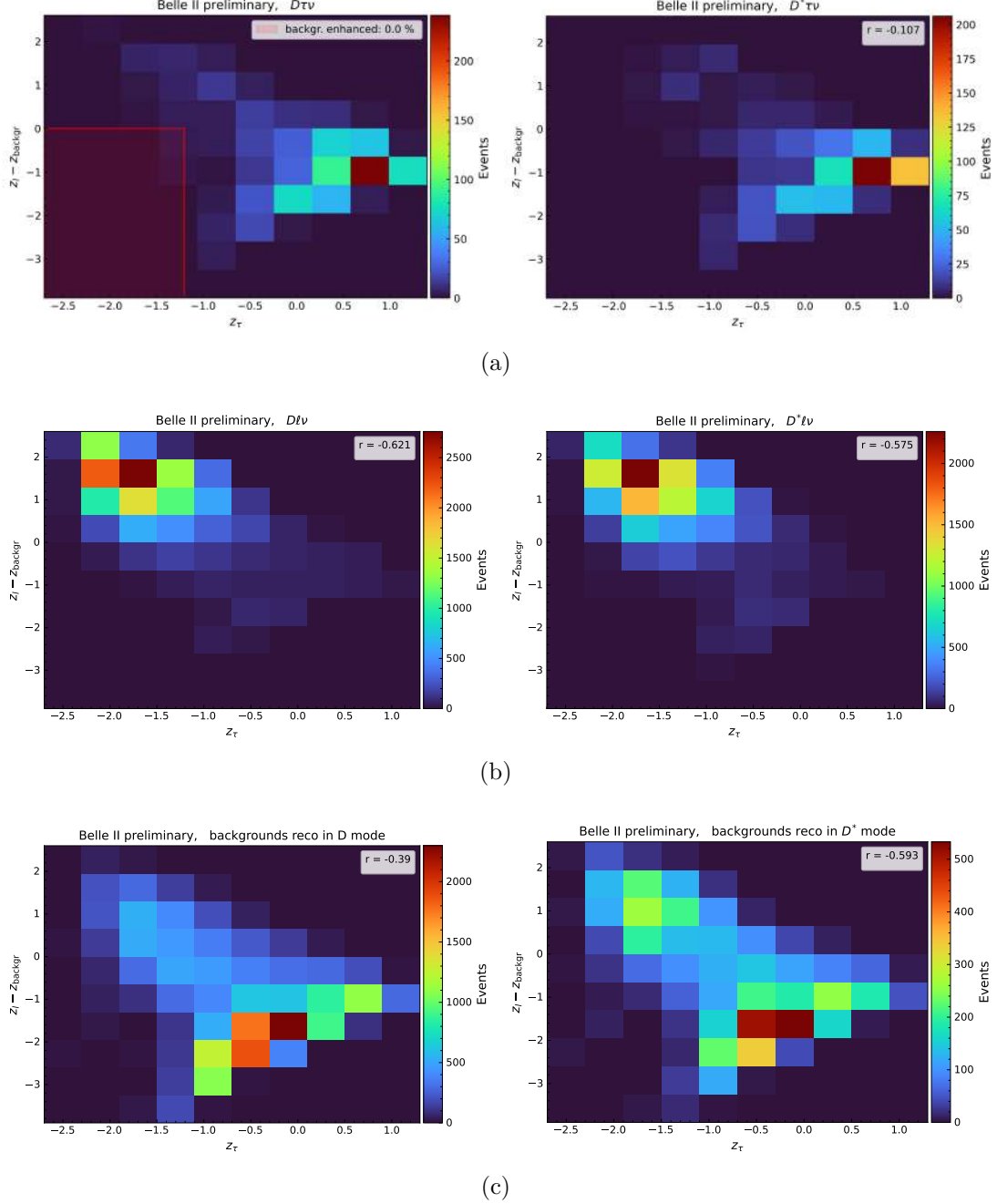
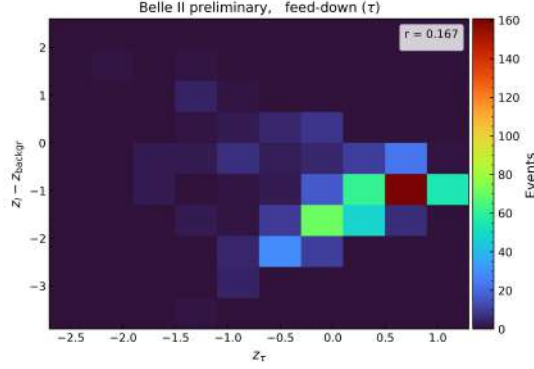


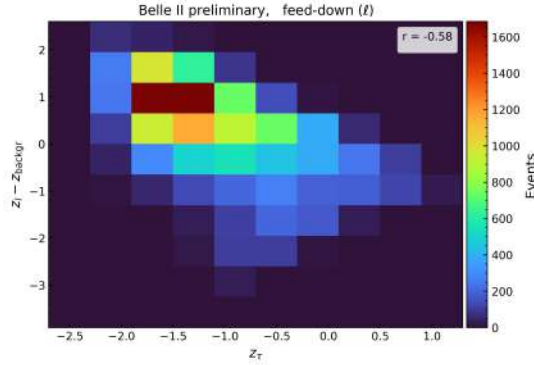
FIG. 16: Two-dimensional distributions of z_τ against $z_l - z_{\text{backgr}}$ for the D mode (left) and D^* mode (right) with the signal lepton being an electron, showing:

- (a) $D^{(*)}\tau\nu$ events, excluding feed-down
- (b) $D^{(*)}e\nu$ events, excluding feed-down
- (c) backgrounds

We also perform a separate validation that large statistical fluctuations do not pose



(a)



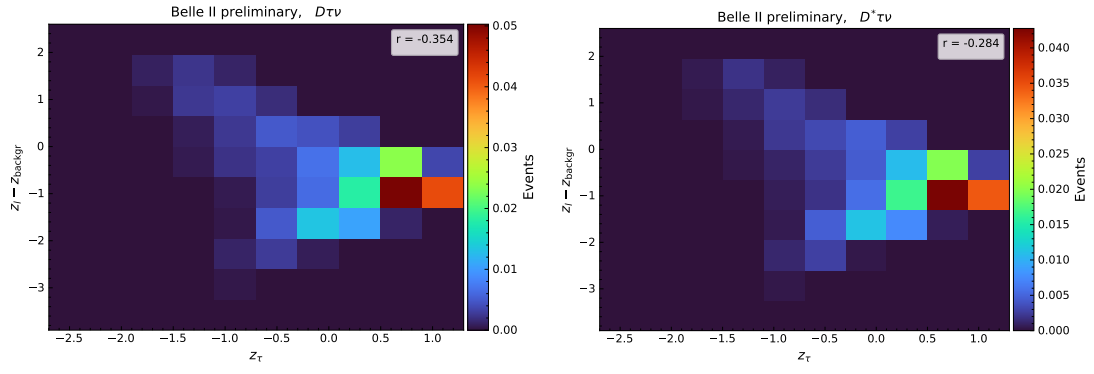
(b)

FIG. 17: Same as fig.16, but showing:

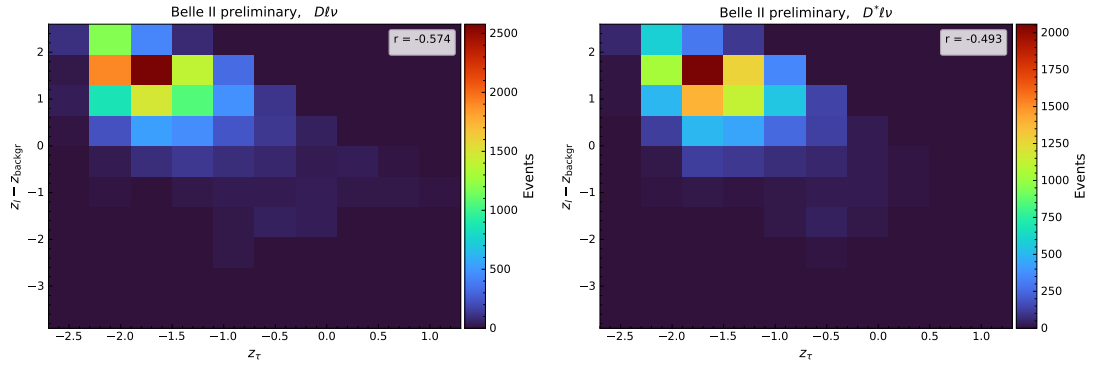
(a) $D^{(*)}\tau\nu$ feed-down events

(b) $D^{(*)}e\nu$ feed-down events

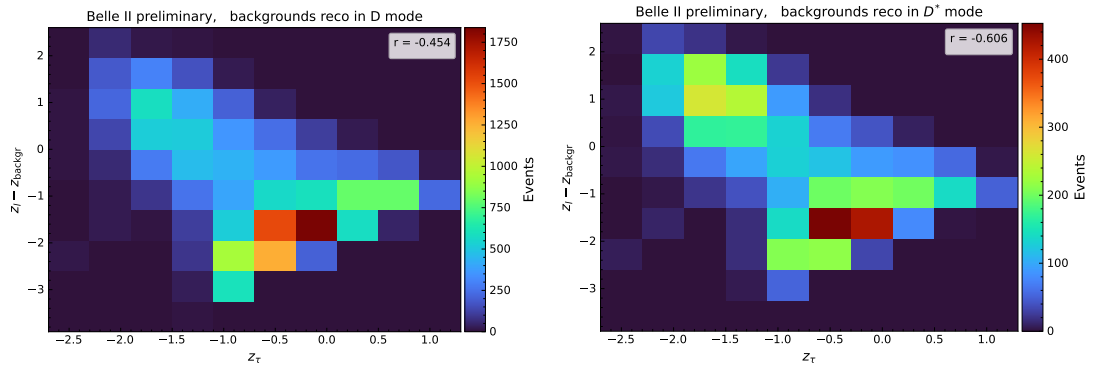
any problem for the signal extraction. the asimov dataset does not include any statistical fluctuation - instead, it perfectly fits to the MC shape. Thus, the same fits are performed using half of the dataset to define the MC templates and the other half to construct the data template as this introduces the aforementioned statistical fluctuations which will be present in data. If our MC simulation models the data correctly, this will give a realistic estimate of the actual fluctuations while the overall shapes of the templates remain roughly the same. With this, we can test whether the fit is able to handle these statistical fluctuations and to determine the yields with the same precision as a full asimov fit. We repeat the fit 200 times. Fig. 20 shows the resulting distribution of fit significances, which is of Gaussian shape. Fitting a Gaussian function to the histograms yields 5.8 ± 1.1 (D case) and 10.4 ± 0.9 (D^* case). This is lower than the significance of the full asimov fit, however, one has to take into account that only half of the statistics is used compared to the case above (200fb^{-1}).



(a)



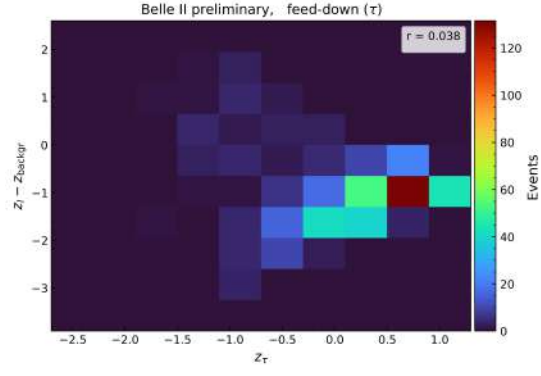
(b)



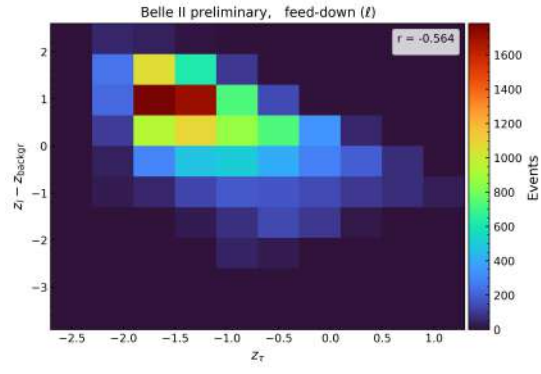
(c)

FIG. 18: Same as fig.16, but with a muon as signal lepton:

- (a) $D^{(*)}\tau\nu$ events, excluding feed-down
- (b) $D^{(*)}\mu\nu$ events, excluding feed-down
- (c) backgrounds reconstructed in $D\mu$ and $D^*\mu$ modes



(a)



(b)

FIG. 19: Same as fig.18, but showing:
 (a) $D^{(*)}\tau\nu$ feed-down events
 (b) $D^{(*)}\mu\nu$ feed-down events

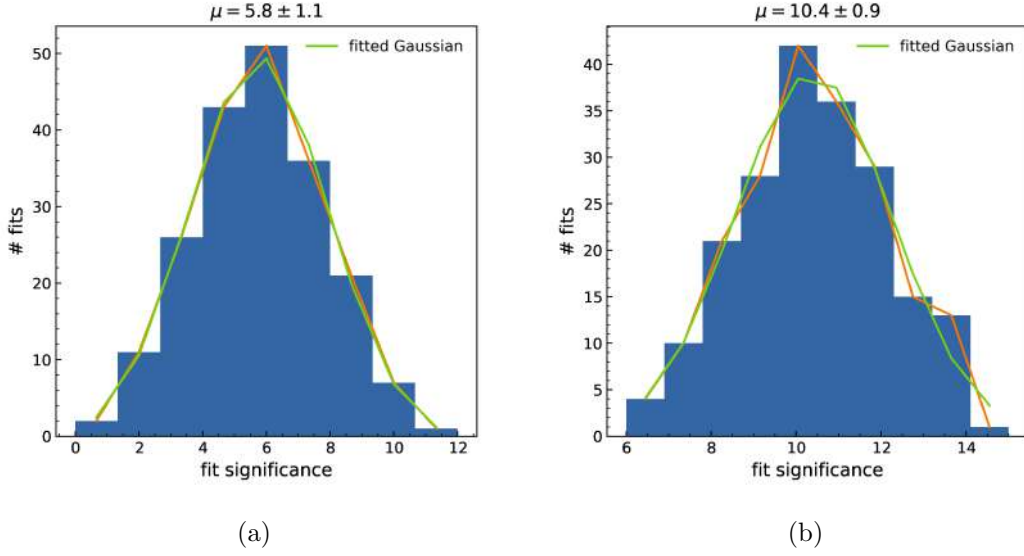


FIG. 20: Distribution of obtained fit significances when fitting one half of the simulated dataset (200 fb^{-1}) to the other half. A Gaussian is fitted to each distribution and the projected column heights using the Gaussian fit function are shown in green. This is done for:

(a) $D(l/\tau)\nu$

(b) $D^*(l/\tau)\nu$

The fitted mean is given in the title of the plot. The fitted Gaussian width can be considered as an uncertainty estimate on the significance.

8.3. Fit strategy using bin-wise NPs (not used for the final results)

The yields of normalisation and signal events in data is to be obtained by performing a binned negative log-likelihood fit using the distributions of the observables z_τ and $z_l - z_{\text{backgr}}$ as described in Subsec.8.8.2.

For this aim, we firstly investigate a fit implementation using `BinFit` [54], a fitting package with python interface based on a minimisation with `iMinuit`. Note that we decided to transition to a different implementation using `pyhf` for obtaining the final results (see Sections 11.11.6-11.11.8). However, the two fitting approaches differ only in their treatment of nuisance parameters for incorporating systematic uncertainties—whether bin-wise or source-wise—while the fitted nominal PDFs of the templates remain the same.

As anticipated in Subsection 8.8.1, 800 fb^{-1} of the simulated events is used for the training, while 2000 fb^{-1} is employed for the generation of a fit template.

First, data and simulated samples undergo the selection criteria described in Section 6. After that, event weights are applied which will be further described in Section 7.

We perform the fit to obtain $\mathcal{R}(D)$ and $\mathcal{R}(D^*)$ simultaneously. This is because there is a large fraction of $B^0 \rightarrow D^*(\ell/\tau)\nu_{\ell/\tau}$ events reconstructed as $B^0 \rightarrow D(\ell/\tau)\nu_{\ell/\tau}$ events, because the slow pion from the D^* decay is missing in the reconstruction or is attributed to the D decay instead. Yet, these are all true $B^0 \rightarrow D^*(\ell/\tau)\nu_{\ell/\tau}$ events, composed of both $B^0 \rightarrow [D^{*+} \rightarrow D^+\pi^0](\ell/\tau)\nu_{\ell/\tau}$ and $B^0 \rightarrow [D^{*+} \rightarrow D^0\pi^+](\ell/\tau)\nu_{\ell/\tau}$ [42] events, and as $\mathcal{R}(D^{*+})$ is the same for all D^{*+} decays, we can use those events as well to determine $\mathcal{R}(D^*)$. Specifically, the normalisation feed-down component present in the D mode shares a yield with the correctly reconstructed $B^0 \rightarrow D^*\ell/\nu_\ell$ events, and similarly, the signal feed-down component in the D mode shares a yield with the correctly reconstructed $B^0 \rightarrow D^*(\tau)\nu_\tau$ events. That way, we can reduce the uncertainties on $\mathcal{R}(D^*)$. For clarification which events fall into the feed-down category we dedicate a small subsection to those events, see Subsec.8.8.4. Thus, we have four signal and normalisation templates which are defined by accessing the generator information of the reconstructed simulated events:

- **signal D:** $D\tau\nu$ events estimated from dedicated signal MC which are required to be reconstructed correctly according to the signal definition presented in Section 5
- **normalisation D:** $D\ell\nu$ events from generic MC passing the truthmatching criteria from Section 5
- **signal D*:** $D^*\tau\nu$ events estimated from dedicated signal MC reconstructed in both the D^* and D modes (feed-down)
- **normalisation D*:** $D^*\ell\nu$ events from generic MC reconstructed in both the D^* and D modes (feed-down)

Concerning the backgrounds, it is useful to perform a further split-up: As we are unaware of the origin of the so-called “gap” modes, their composition as well as BF’s, and also the BF’s of the resonant D^{**} events come with a large BF uncertainty, it is important to assure that the shape of these events does not impact the background yield determination if the fraction of those events relative to the other backgrounds is not estimated correctly in MC. In similar analyses, some problematic issues were seen related to those components, as their postfit yield became negative. In order to be able to directly assess whether this may happen in case of this analysis as well, the non-resonant and resonant D^{**} decays have to be segregated from the rest of the background events.

Thus, there are four distinct background templates contributing to a total of eight fit templates in the simultaneous fit:

- $D^{**} + \text{gap}$ events reconstructed in the D mode:
 - $B\bar{B}$ events from generic MC to model resonant $D^{**}(\ell/\tau)\nu$ decays
 - $B\bar{B}$ events from dedicated signal MC simulating the non-resonant (“gap”) modes that decay into $D^{(*)}(\pi\pi/\eta)(\ell/\tau)\nu$ via the decay into an intermediate broad D^{**} resonance
- backgrounds reconstructed in the D mode:
 - $q\bar{q}$ events estimated from off-resonant data
 - $B\bar{B}$ events from generic MC including events with wrongly reconstructed leptons, events with wrongly reconstructed $D^{(*)}$ decays, $X_u(l/\tau)\nu$ events
- $D^{**} + \text{gap}$ events reconstructed in the D^* mode: see above
- backgrounds reconstructed in the D^* mode: see above

At first, we perform the fit for the electron mode and muon mode separately. If no significant differences are observed, both will ultimately be combined to obtain the final result for $\mathcal{R}(D^{(*)})$.

In order to obtain an overview of the different background sources and where they are situated in the distributions of the fit variables, see Fig. 21 In particular, these kinds of background sources are identified:

- $D^{(*)}\ell/\tau\nu_{\ell/\tau}$ gap events - simulated as decays via an intermediate broad D^{**} resonance (purple - ℓ case; pink - τ case)
- resonant $B \rightarrow D^{**}\ell/\tau\nu_{\ell/\tau}$ decays, not including those events described in the first item (yellow)
- feed-down events: decays involving $D^* \rightarrow D\pi_{\text{slow}}$, where the slow pion has been missed and only the D has been reconstructed (turquoise)
- other events in which a D^* has been reconstructed as a D - e.g. in cases where the D^* originates from the tag side (dark turquoise)
- events in which a D has been reconstructed as a D^* by mistakenly adding a slow pion (dark brown)
- events with a real D but of wrong flavour (brown)
- events with a fake $D^{(*)}$ (green)
- events with a fake lepton (violet)
- $B \rightarrow X_u\ell/\tau\nu_{\ell/\tau}$ events (grey)
- continuum events (blue)

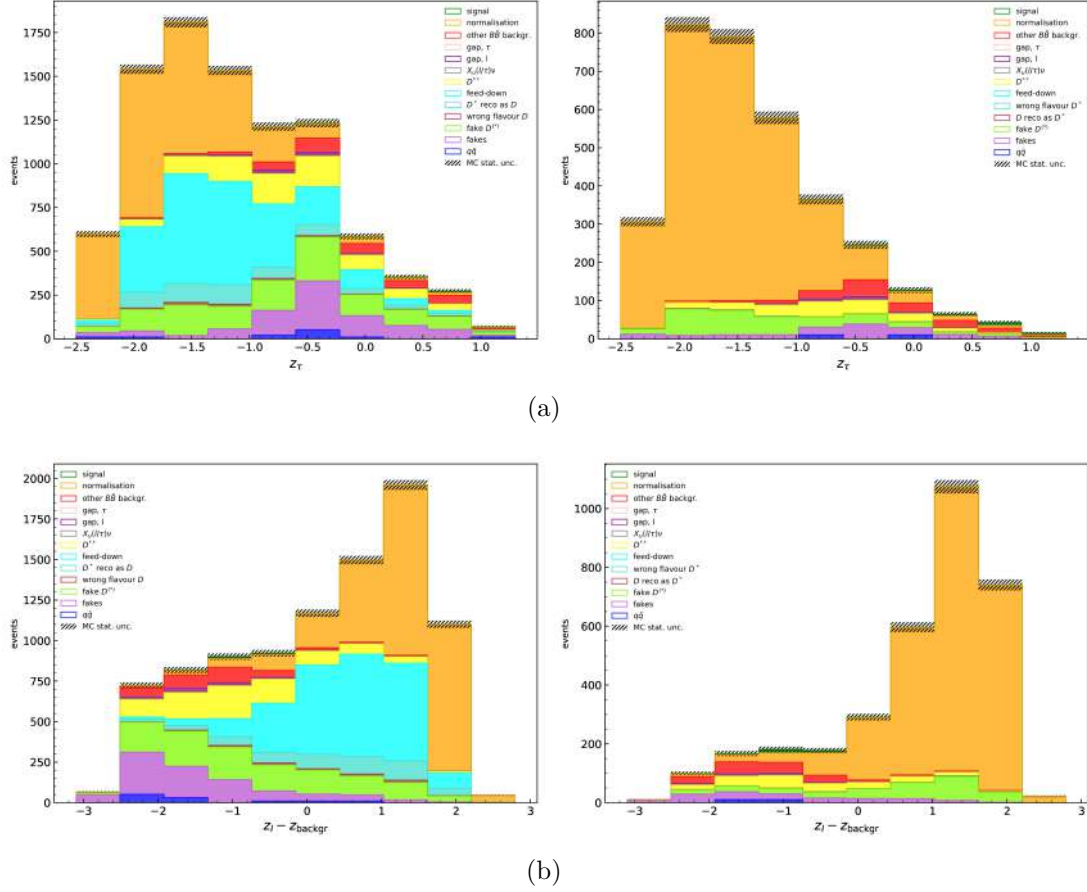


FIG. 21: Distribution of the fit variables, showing signal, normalisation, feed-down (both signal and normalisation) and different types of backgrounds as specified in the itemised list in Subsec.88.3. $B \rightarrow D(\ell/\tau)\nu_{\ell/\tau}$ events can be seen on the left and $B \rightarrow D^*(\ell/\tau)\nu_{\ell/\tau}$ events on the right:

(a) z_τ

(b) $z_l - z_{\text{backgr}}$

Each of the fit templates is related to a yield ($N_{\text{sigD}}, N_{\text{normD}}, N_{D^{**}+\text{gapD}}, N_{\text{backgrD}}, N_{\text{sigD}^*}, N_{\text{normD}^*}, N_{D^{**}+\text{gapD}^*}, N_{\text{backgrD}^*}$) that is to be determined by the fit. The fit is performed maximising the binned extended negative log-likelihood, which is, for a simple yield fit, defined as:

$$-2 \log \mathcal{L} = -2 \log \prod_i \text{Poisson}(N_i^{\text{obs}}, N_i^{\text{exp}}) \quad (6)$$

where the index i stands for a single bin and N_i^{obs} denotes the number of events observed in bin i while N_i^{exp} denotes the expected number of events and is calculated as:

$$N_i^{\text{exp}} = \sum_j N^j \frac{p_i^j}{\sum_k p_k^j}, \quad (7)$$

where j represents the template. N^j is the corresponding yield and p_i^j is the probability that an event of type j appears in bin number i , which is determined from the pdfs obtained from MC simulation. Accordingly, there are eight pdfs belonging to the eight templates: $p^{\text{sigD}}, p^{\text{normD}}, p^{D^{**}+\text{gapD}}, p^{\text{backgrD}}, p^{\text{sigD}^*}, p^{\text{normD}^*}, p^{D^{**}+\text{gapD}^*}, p^{\text{backgrD}^*}$. In case of

Eq.??, there are only eight parameters that are floating (i.e. the scale factors on the yields belonging to these pdfs). The uncertainty obtained from a fit like this is of only statistical nature.

Beyond that, the fit handles systematic uncertainties via bin-wise nuisance parameters (NPs). This allows the fit not only to adjust the yield scale factors for the templates, but also do slight modifications of their shapes within the given uncertainties of each bin. This can be advantageous since the same uncertainties are shared across the different modes, such that the fit might be able to constrain those uncertainties to obtain a more realistic estimate of their size than in a version without systematic nuisance parameters. The systematic uncertainties are added via systematic covariance matrices that either account for flat variations or Gaussian variations of certain weights. The negative log-likelihood can be modified in the following way when adding these covariance matrices:

$$-2 \log \mathcal{L} = -2 \log \prod_i \text{Poisson}(\nu_i^{\text{obs}}, \nu_i^{\text{exp}}) + \theta^T C_\theta^{-1} \theta \quad (8)$$

while ν_i^{exp} describes the expected number of events to be calculated as:

$$\nu_i^{\text{exp}}(\nu^j, \theta_i^j) = \sum_j \nu^j \frac{p_i^j (1 + \theta_i^j)}{\sum_k p_k^j (1 + \theta_k^j)}, \quad (9)$$

where p_i^j is the probability that an event of type j appears in bin number i . The NPs θ allow for variations of each bin by taking into account the systematic uncertainties as well as the limitation in statistics. Their correlation throughout the bins is taken account for by the covariance matrix C . In particular, the following systematic uncertainties are added via flat variations:

- BF of resonant B^0 decays:
Here we use the uncertainty on the BF given in the PDG[5] for the respective up- and down-weights. We vary each BF as listed in tableV separately.
- BF of B^0 decays into gap modes (100% uncertainty):
Since their true composition is uncertain, we vary all BFs at once.
- FF models of BLRXP ($D^{(*)}$), BLR (narrow D^{**} s), BLR (broad D^{**} s):
With respect to the FF systematic uncertainties, the software HAMMER[24] allows for rotating the FF parameters of a certain model into an uncorrelated eigenbasis and, by varying them within their $\pm 1\sigma$ uncertainties, creating up- and down-variations of the FF weights corresponding to each variation. These up- and down-variations can be used for estimating the systematic uncertainties related to a certain FF model. Specifically, for each FF model, we vary each of the parameters using the so-calculated up- and down-variations.
The uncertainties related to the broad D^{**} resonances are estimated using 1.5σ instead of 1σ eigenvariations of their FF parameters, which will yield larger uncertainties. This is due to some mismodelling of their mass distributions when generating these events, leading to D^{**} mass distributions that extend towards the initial B meson mass. This mismodelling is described in detail in Ref.[8], where the proposed procedure is to introduce mass windows of $3 \cdot \Gamma_{D^{*0}}$ and $2.5 \cdot \Gamma_{D_1^*}$ and to subsequently up-scale the remaining events to match their original number. We follow this procedure here as well and try to capture the additional uncertainty introduced by this method with the larger FF variations, i.e. using 1.5σ instead of 1.0σ eigenvariations of the parameters.

- FF models of BLR for the broad D^{**} s in case of the gap modes:
As for the BFs, the gap modes are treated in a special way again due to the lack of knowledge about these modes. Here, each FF model is varied separately for each decay mode. As the gap modes are simulated to decay via intermediate broad D^{**} resonances, there are in total four FF uncertainties related to the gap modes, i. e. one for each of the two broad D^{**} s and one for either τ s or light leptons. As the gap modes are modelled as decays via intermediate broad D^{**} resonances, the case of using 1.5σ eigenvariations applies here as well.

- tracking efficiency:
For the uncertainty related to the tracking efficiency, except for the one related to the charged slow pion, it is recommended to assume an uncertainty of 0.23% per reconstructed track. Thus, for each of the reconstructed D decay modes, the corresponding reconstructed tracks are counted. These sum up together with the track of the reconstructed light lepton to the total number of tracks per event (omitting the tracks on the tag side, which are assumed to be covered by the FEI efficiency). With this, we can calculate an uncertainty related to tracking for each event in the following way:

$$\sigma(\epsilon_{\text{track}}) = 1.003^{n_{\text{(tracks)}}} - 1 \quad (10)$$

Further, we use Gaussian variations of the weights for:

- PID:
To obtain an estimate of the PID systematics, we generate variations of the PID weights using `PIDvar`, once assuming a lepton's systematic uncertainty is fully uncorrelated with leptons from other correction bins.
- slow pion efficiency:
In case of the slow pion efficiency weights, there are different weights depending on the $p_{\pi^\pm}$ range in which we find the momentum of the respective slow pion, each of them coming with a statistical and a systematic uncertainty [18]. With the assumption of full correlations between bins, we generate variations of the slow pion efficiency weights.

Thus, we can get a direct estimate of the full uncertainty within the nominal fit, and we are able to separate only the systematic part by performing the fit twice, once without NPs (Eq.6), and once with NPs (Eq.8).

8.4. Feed-down component

The feed-down component comprises of true $B^0 \rightarrow D^*(\ell/\tau)\nu_{\ell/\tau}$ events which are reconstructed in the wrong channel, i.e. as $B^0 \rightarrow D(\ell/\tau)\nu_{\ell/\tau}$. Specifically, we require for those events:

- no fake/secondary leptons
- neutral B as D^* mother, i.e. rejecting B^+ events
- neither the reconstructed B^0 nor the D^* should stem from the tag side
- the true lepton sister (ℓ case) or true lepton aunt (τ case) was a D^*

TABLE X: Number of $B^0 \rightarrow D^*(\ell/\tau)\nu_{\ell/\tau}$ events reconstructed in the e and μ modes: all of these events together make up the signal/normalisation D^* templates. While the first three (true) decay modes are reconstructed in the D mode, respectively, the fourth one is reconstructed in the D^* mode. The number of $B^0 \rightarrow D(\ell/\tau)\nu_{\ell/\tau}$ events (reconstructed in the D mode) is added for reference.

generated decay mode	reconstructed decay mode	e mode	μ mode
$B^0 \rightarrow [D^{*+} \rightarrow D^+\pi^0]\ell\nu_\ell$	D	1871 ± 17	1741 ± 16
$B^0 \rightarrow [D^{*+} \rightarrow D^+\gamma]\ell\nu_\ell$	D	95 ± 4	97 ± 4
$B^0 \rightarrow [D^{*+} \rightarrow D^0\pi^+]\ell\nu_\ell$	D	419 ± 8	410 ± 8
$B^0 \rightarrow D^{*+}\ell\nu_\ell$	D^*	2202 ± 18	1866 ± 16
$B^0 \rightarrow D^+\ell\nu_\ell$	D	2628 ± 21	2350 ± 19
$B^0 \rightarrow [D^{*+} \rightarrow D^+\pi^0]\tau\nu_\tau$	D	73 ± 3	54 ± 3
$B^0 \rightarrow [D^{*+} \rightarrow D^+\gamma]\tau\nu_\tau$	D	3 ± 1	4 ± 1
$B^0 \rightarrow [D^{*+} \rightarrow D^0\pi^+]\tau\nu_\tau$	D	10 ± 1	12 ± 1
$B^0 \rightarrow D^{*+}\tau\nu_\tau$	D^*	123 ± 4	88 ± 3
$B^0 \rightarrow D^+\tau\nu_\tau$	D	145 ± 5	101 ± 4

- the true first and second daughters of this D^* were either a D^0 and a π^+ or a D^+ and a π^0/γ (or their charge conjugates)

Table X provides a specification of the amount of reconstructed feed-down events in the electron and muon modes. We applied the FEI efficiency scale factor (see Sec. 9.9.2 as well as the luminosity scale factor in order to show the amount of feed-down events that we expect to find in data. While in the $D^{*+} \rightarrow D^+\pi^0$ case, the reconstruction error consists in omitting the slow π^0 and reconstructing the D^+ as lepton sister (or aunt), in the $D^{*+} \rightarrow D^0\pi^+$ case, the D^{*+} is mistaken for the reconstructed D^+ in most of the cases.

8.5. Binning

This subsection deals with the studies performed in order to optimise the binning. E.g. zero entries have to be avoided since these have no uncertainty, which will lead to difficulties for the fit performance. Similarly, we obviate low-statistics bins. Specifically, an optimal binning should satisfy the following conditions:

- no bin has zero total entries in data while there are entries in MC / no bins with < 5 entries in data
- no bin has zero total entries in MC while there are entries in data / no bins with < 5 entries in MC
- no bin contains entries from solely one MC template
- the binning sufficiently fine to allow for differentiation of the fit templates via the shapes of their histograms

We perform asimov fits, firstly without any additional NPs, and evaluate the uncertainties on signal and normalisation yields as well as on the ratio $\mathcal{R}(D^{(*)})$. Fig.23, which shows those uncertainties in % for the four binning versions described above as well as for the binning which is finally chosen for the fit, reveals that the influence of the binning on these uncertainties is comparatively small, as there are only marginal differences in the obtained values. Further, we can conclude that the uncertainty on $\mathcal{R}(D^{(*)})$ is dominated by the uncertainty on the signal yield, while the normalisation yield uncertainty is comparatively small in any case. Considering the 2D distribution, the normalisation events seem well separated from the signal and background ones, such that it is possible to cover the majority of the normalisation events within one or two larger bins only, without losing much sensitivity on $\mathcal{R}(D^{(*)})$. Taking the outcome of this study into consideration, we can specify the aims which the new binning should achieve:

- constraining the signal yield by having a finer binning in the signal region
- constraining the uncertainty on the gap BF, which is implemented as a 100% uncertainty, thus overestimating the real uncertainty associated with it; this requires a finer binning in the transition region between signal and backgrounds, where the gap events are concentrated

To attain this aim, we avail ourselves of the algorithm described above, but instead of using the signal and normalisation events as an input, the input histogram is made up of an equal amount of signal and gap events, such that a binning with more bins in regions where those events are concentrated is constructed. The outcome is adjusted by hand by dividing up certain bins into smaller ones if it helped to reduce the signal yield or gap BF uncertainty. Further, we optimise the binning in an advantageous way concerning the $D^{**}+\text{gap}$ template in the fit: In principle, the normalisation events, which are located in the upper left corner, could have been covered by one large bin. Yet, the latter contains also events from the $D^{**}+\text{gap}$ template. Splitting up this one bin into two bins is reasoned by the fact that it is useful to have one bin in which we find nearly only events from the resonant D^{**} part of this template, since the fit can adjust this template via pulling on the yield, but also by pulling on the binparameters, because of the gap BF uncertainty of 100%. So providing a bin comprised of only the D^{**} part of the distribution, we assure that their yield is correctly adjusted by the fit, not compensating for a too high amount of gap events due to the lack of freedom to pull on the binparameters, since this one bin does not come with a 100% gap BF uncertainty.

The outcome can be seen in Fig.24. Fig. 25 shows the binning overlaid over the 2D distributions of different event categories, highlighting that a large amount of the normalisation events is covered by the bins number 0 and 1 in the upper left corner, while the signal events on the right are distributed across a larger number of bins, which holds also true for the gap category to be located in the transition region towards the highest background concentration in the lower left corner. Further, the plots show the combined $D^{**}+\text{gap}$ template as well as both the resonant and non-resonant (i.e. gap) components, in order to demonstrate that the resonant one extends further towards positive values of z_{diff} . The bin edges are listed in table XI. The numbers shown in the figure correspond to the bin numbers in table XI. In order to show the pre- and postfit agreement using this asymmetric binning, we define another variable which takes on the value of the respective bin number that the event can be categorised in based on its values of z_{τ} and z_{diff} . Counting the events belonging to each bin number in data and simulation leads to the pre- and postfit distributions shown in Fig. 15.

A further study is performed in order to test the chosen binning against a version in which the distributions are rotated which is motivated by the 2D structures seen in Fig.25.

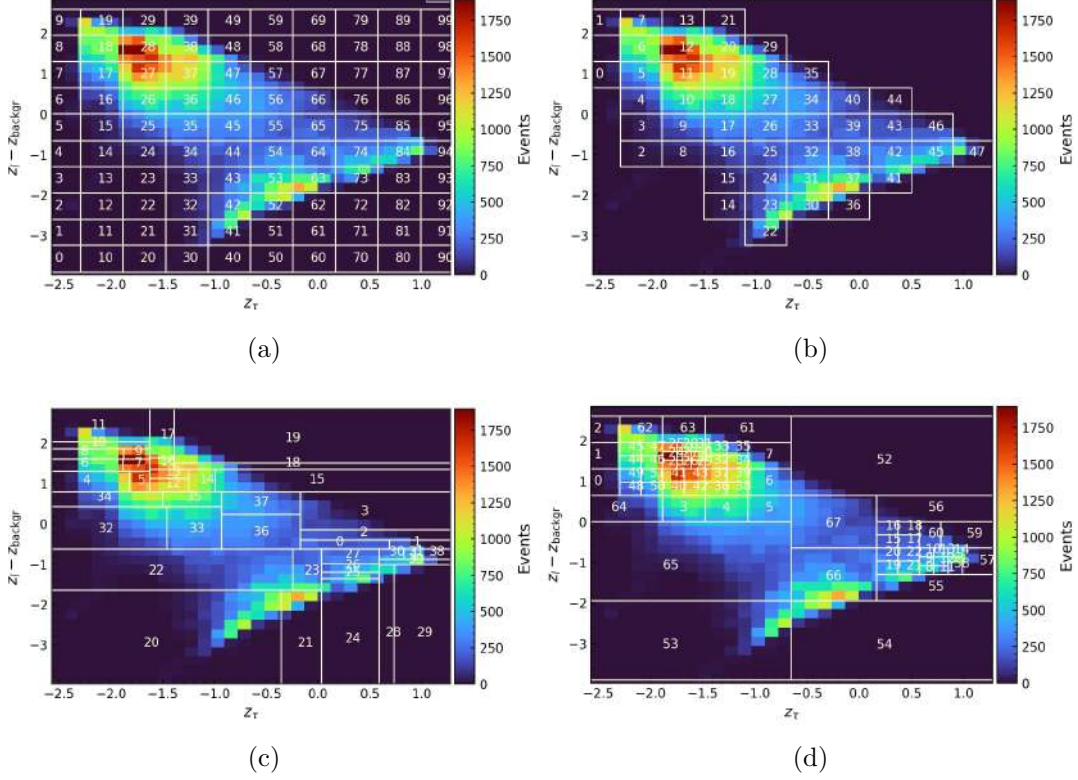


FIG. 22: Different possibilities to construct the binning:

- (a) symmetric 10×10
- (b) same as above, but omitting low-statistics bins
- (c) asymmetric, obtained by algorithm
- (d) asymmetric, by hand

Note that the numbers inside the bins are just an arbitrary numbering of the bins. The 2D distribution shown here contains all events from all categories, including both those reconstructed in the D as well as in the D^* mode.

These may give the impression that a rotation in 2D space could lead to an enhancement of the sensitivity. In order to test this hypothesis, we rotate the distribution by 40 degrees and construct a binning in the same way as described above, using the algorithm with the signal and gap events as an input. The rotated distribution (all events from the four modes combined) together with this binning can be seen in Fig.26. The corresponding pre- and postfit distributions are shown in Fig. 27. We obtain the following asimov fit results:

$$\mathcal{R}(D)_e = 0.299 \pm 0.083(\text{stat}) \pm 0.048(\text{syst}) \quad (11)$$

$$\mathcal{R}(D)_\mu = 0.299 \pm 0.109(\text{stat}) \pm 0.053(\text{syst}) \quad (12)$$

$$\mathcal{R}(D^*)_e = 0.258 \pm 0.044(\text{stat}) \pm 0.019(\text{syst}) \quad (13)$$

$$\mathcal{R}(D^*)_\mu = 0.258 \pm 0.050(\text{stat}) \pm 0.020(\text{syst}) \quad (14)$$

Comparing to the asimov sensitivity achieved with the un-rotated binning (cf. the following Section 8.6), we do not gain a larger sensitivity. Thus, in order to not add a further level of abstraction to the fit variables, we discard this version of the binning and use the above-defined un-rotated version for our nominal fit.

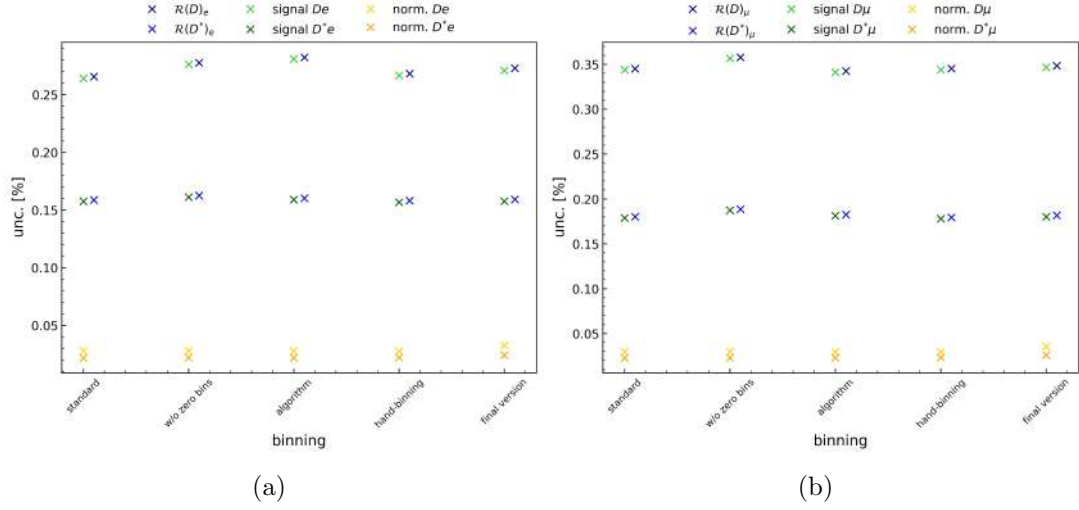


FIG. 23: Relative statistical uncertainties on signal and normalisation yields and on the resulting $\mathcal{R}(D^{(*)})$ using different binnings, the latter of which are indicated on the x axis:

(a) electron mode

(b) muon mode

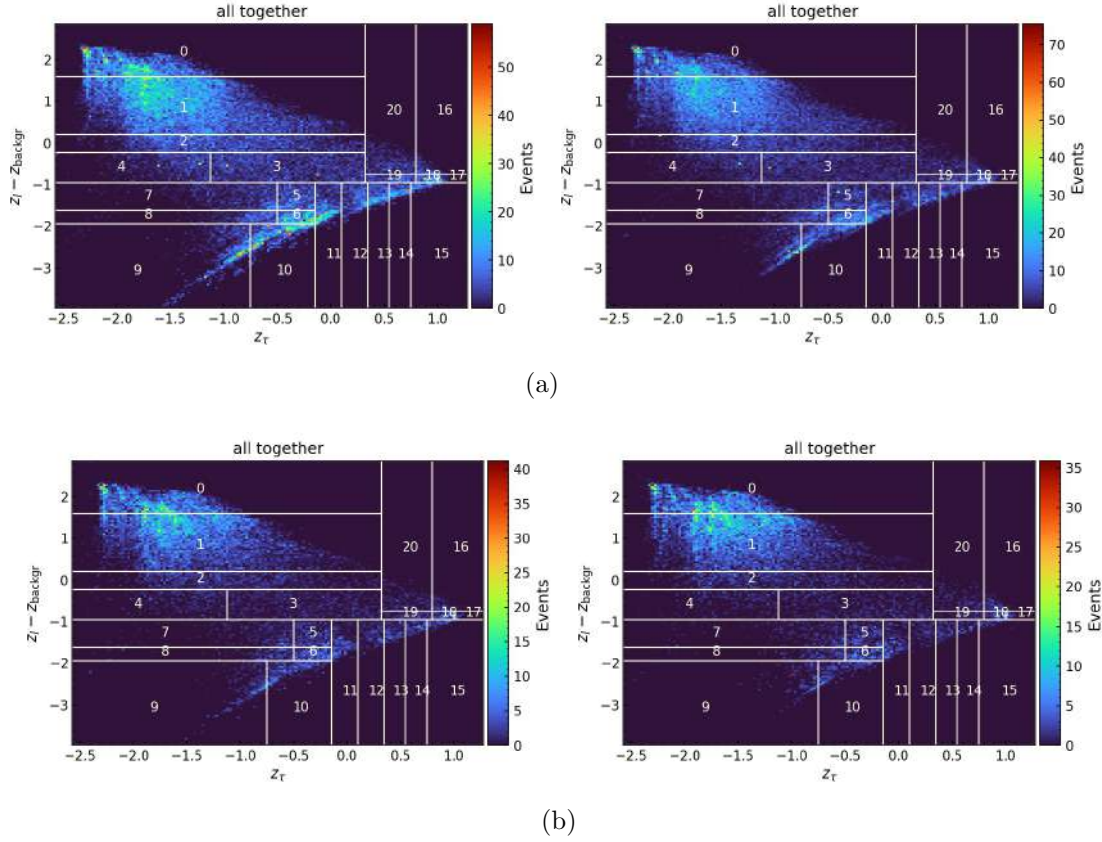


FIG. 24: Same as Fig.22, but for the version of the binning which is finally used, showing the electron mode on the left and the muon mode on the right for D (a) and D^* (b)

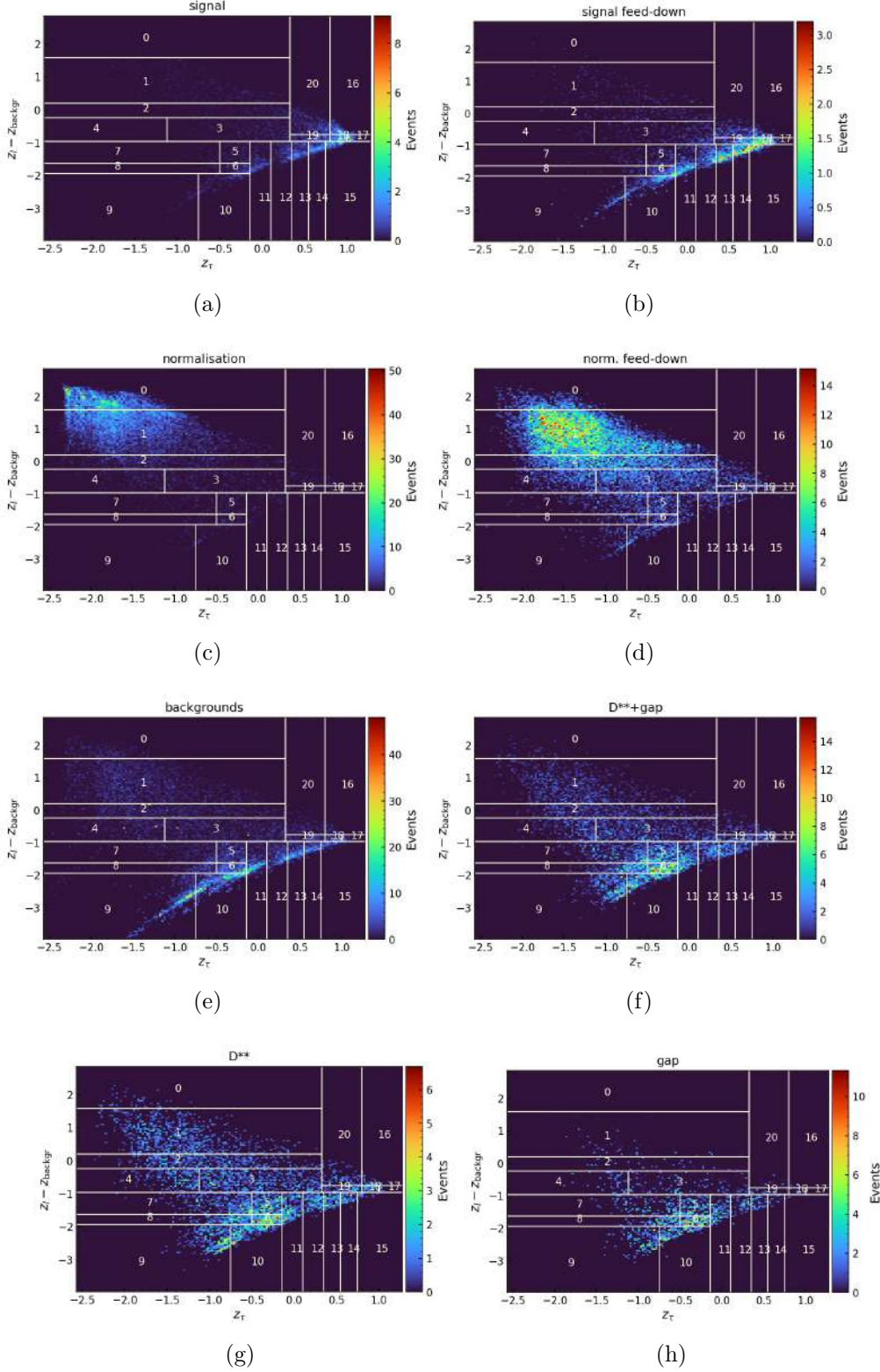


FIG. 25: Same as Fig.24, but only showing the 2D distribution for certain categories of events, exemplary for the De mode: (a) signal (excluding feed-down), (b) signal feed-down, (c) normalisation (excluding feed-down), (d) normalisation feed-down, (e) non- D^{**} backgrounds, (f) $D^{**} + \text{gap}$, (g) D^{**} only, (h) gap only. Please find the remaining modes in the Appendix (Figs. 153-155).

TABLE XI: Bin edges of the final binning in z_τ and z_{diff}

bin number	bin edges z_τ	bin edges z_{diff}
0	$[-2.57, 0.33]$	$[1.583, 2.85]$
1	$[-2.57, 0.33]$	$[0.2, 1.583]$
2	$[-2.57, 0.33]$	$[-0.23, 0.2]$
3	$[-1.12, 0.33]$	$[-0.96, -0.23]$
4	$[-2.57, -1.12]$	$[-0.96, -0.23]$
5	$[-0.5, -0.14]$	$[-1.62, -0.96]$
6	$[-0.5, -0.14]$	$[-1.94, -1.62]$
7	$[-2.57, -0.5]$	$[-1.62, -0.96]$
8	$[-2.57, -0.5]$	$[-1.94, -1.62]$
9	$[-2.57, -0.75]$	$[-3.97, -1.94]$
10	$[-0.75, -0.14]$	$[-3.97, -1.94]$
11	$[-0.14, 0.10]$	$[-3.97, -0.96]$
12	$[0.10, 0.35]$	$[-3.97, -0.96]$
13	$[0.35, 0.55]$	$[-3.97, -0.96]$
14	$[0.55, 0.75]$	$[-3.97, -0.96]$
15	$[0.75, 1.28]$	$[-3.97, -0.96]$
16	$[0.80, 1.28]$	$[-0.75, 2.85]$
17	$[1.0, 1.28]$	$[-0.96, -0.75]$
18	$[0.80, 1.0]$	$[-0.96, -0.75]$
19	$[0.33, 0.80]$	$[-0.96, -0.75]$
20	$[0.33, 0.80]$	$[-0.75, 2.85]$

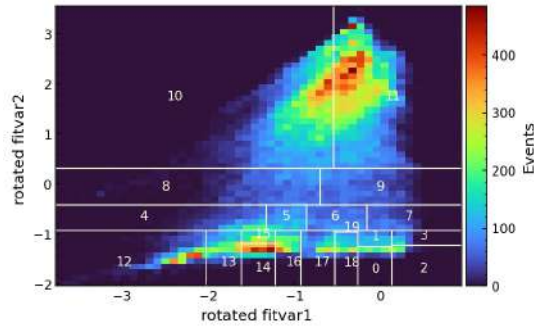


FIG. 26: Applying a rotation of 40 degrees in 2D-classifier space and constructing a binning in a similar way as for the nominal fit leads to this distribution. Events from all categories are shown in all four modes of De , $D\mu$, D^*e and $D^*\mu$ combined.

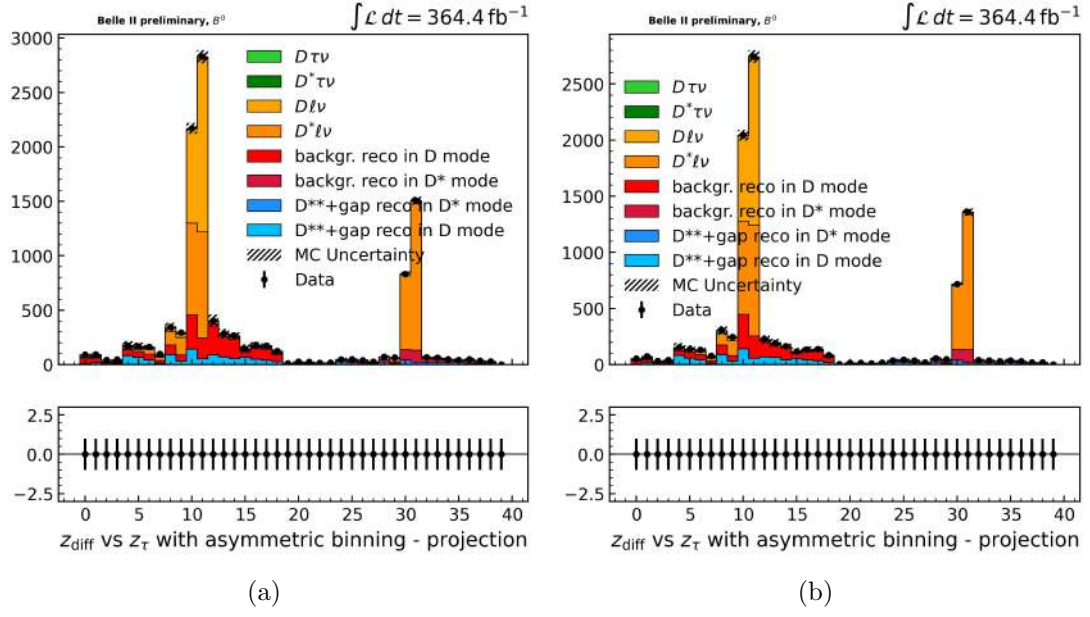


FIG. 27: Post-fit histograms from a full asimov fit using the rotated distributions, similar to Fig.15. The number on the x axis indicates the bin number as illustrated in Fig.26. The y axis measures the number of events within each of these bins.

(a) electron mode

(b) muon mode

8.6. Closure test

In order to validate the signal extraction fit setup, we calculate $\mathcal{R}(D^{(*)})$ from asimov fits. $\mathcal{R}(D^{(*)})$ is calculated in the following way when keeping electron and muon modes separately:

$$\mathcal{R}(D^{(*)}) = \frac{N_{\text{signal}}}{N_{\text{norm}}} \frac{\epsilon_{\text{norm}}}{\epsilon_{\text{signal}}} \frac{1}{\mathcal{B}(\tau \rightarrow (e/\mu)\nu\nu)} \quad (15)$$

When combining electron and muon mode as it is planned for the final result, the equation is modified in the following way:

$$\mathcal{R}(D^{(*)}) = \frac{N_{\text{signal}}}{N_{\text{norm}}/2} \frac{\epsilon_{\text{norm}}}{\epsilon_{\text{signal}}} \frac{1}{\mathcal{B}(\tau \rightarrow \ell\nu\nu)} \quad (16)$$

Here, the normalisation yield is divided by 2, if we fit the electron and muon modes together. Further, $\mathcal{B}(\tau \rightarrow \ell\nu\nu) = 0.178175 + 0.173937$ is the sum of the BF's of τ leptons decaying to electrons and muons. The efficiencies are the sum of the efficiencies for the electron and muon modes, respectively.

The values used for the τ decay BF's are taken from the decay file [2].

For the BF's of $B \rightarrow D^{(*)}(\ell/\tau)\nu$ decays, we apply a BF reweighting to update the generated BF's to the latest PDG values. Thus, we expect the value of $\mathcal{R}(D^{(*)})$ from the asimov fit to be the ratio of those BF's:

$$\mathcal{R}(D) = \frac{6.698 \cdot 10^{-3}}{2.24 \cdot 10^{-2}} = 0.299 \quad (17)$$

$$\mathcal{R}(D^*) = \frac{1.3133 \cdot 10^{-3}}{5.11 \cdot 10^{-2}} = 0.258 \quad (18)$$

The efficiencies of selecting events from the signal category are calculated in the following way:

$$\epsilon_{\text{signal}} = \frac{N_{\text{selected,signal}}}{2 \cdot N_{B^0\bar{B}^0} \cdot \mathcal{B}(B \rightarrow D^{(*)}\tau\nu) \cdot \mathcal{B}(\tau \rightarrow \ell\nu\nu)} \quad (19)$$

The factor of 2 is from the two B^0 mesons in the event. We do not consider the BF's of the selected $D^{(*)}$ modes in the denominator since we do not require the $D^{(*)}$ to be reconstructed correctly in our truthmatching. Correspondingly, those of selecting events from the normalisation category are:

$$\epsilon_{\text{norm}} = \frac{N_{\text{selected,norm}}}{2 \cdot N_{B^0\bar{B}^0} \cdot \mathcal{B}(B \rightarrow D^{(*)}\ell\nu)} \quad (20)$$

Here, $\mathcal{B}(B \rightarrow D^{(*)}\ell/\tau\nu)$ correspond to the corrected BF's as used in eqs.17-18. All other BF's are not updated in the MC samples and are thus taken from the decay file [2]. The number of produced $B^0\bar{B}^0$ pairs corresponds to $51 \cdot 10^6$ for 100fb^{-1} , and thus we find in 2000fb^{-1} (i.e. the amount of MC used in the fit) $N_{B^0\bar{B}^0} = 1.02 \cdot 10^9$. N_{selected} is determined from the sum of all weights of either the signal or normalisation category in each mode. The so-calculated signal and normalisation efficiencies are listed in table XII.

From the asimov fits assuming an integrated luminosity of 364fb^{-1} (as used in the fit to data), we obtain the following yields for the electron (muon) mode:

$$N_{\text{signalD}} = 143_{-48}^{+46} \quad (100_{-42}^{+41}) \quad (21)$$

TABLE XII: Signal and normalisation efficiencies obtained in the D and D^* modes, separating between electrons and muons

mode	ϵ_{signal}	ϵ_{norm}
De	$3.3 \cdot 10^{-4}$	$3.2 \cdot 10^{-4}$
$D\mu$	$2.5 \cdot 10^{-4}$	$2.8 \cdot 10^{-4}$
D^*e	$2.6 \cdot 10^{-4}$	$2.4 \cdot 10^{-4}$
$D^*\mu$	$2.1 \cdot 10^{-4}$	$2.1 \cdot 10^{-4}$

$$N_{\text{signal}D^*} = 207_{-40}^{+38} (155_{-35}^{+34}) \quad (22)$$

$$N_{\text{normalisation}D} = 2625_{-85}^{+84} (2347_{-100}^{+100}) \quad (23)$$

$$N_{\text{normalisation}D^*} = 4535_{-139}^{+143} (4062_{-157}^{+161}) \quad (24)$$

Inserting these yields as well as the efficiencies from tableXII into eq.15, we retrieve the values for $\mathcal{R}(D^{(*)})$ given in eqs.17-18:

$$\mathcal{R}(D)_e = 0.299 \pm 0.097 \quad (\pm 0.083(\text{stat}) \pm 0.050(\text{syst})) \quad (25)$$

$$\mathcal{R}(D)_\mu = 0.299 \pm 0.122 \quad (\pm 0.110(\text{stat}) \pm 0.053(\text{syst})) \quad (26)$$

$$\mathcal{R}(D^*)_e = 0.258 \pm 0.046 \quad (\pm 0.043(\text{stat}) \pm 0.016(\text{syst})) \quad (27)$$

$$\mathcal{R}(D^*)_\mu = 0.258 \pm 0.054 \quad (\pm 0.050(\text{stat}) \pm 0.020(\text{syst})) \quad (28)$$

Due to the simultaneous fit, the values for $\mathcal{R}(D)$ and $\mathcal{R}(D^*)$ are correlated. We obtain the following correlations between both values when we include all sources of systematic uncertainty (only the statistical uncertainty) in the fit:

$$\rho(\mathcal{R}(D/D^*)_e) = -0.1803(-0.2497) \quad (29)$$

$$\rho(\mathcal{R}(D/D^*)_\mu) = -0.2217(-0.2589) \quad (30)$$

These are visualised as ellipses in Figs.28, together with previous results of $\mathcal{R}(D/D^*)$ as shown in Fig.1[5]. Note that these are separated into electron and muon modes for this analysis, while both modes are combined in all other analyses.

As further check for consistency, we can set one or more of the yield parameters fix in the fit. When only fixing the signal yield, we obtain:

$$\mathcal{R}(D)_e = 0.299 \pm 0.0096 \quad (31)$$

$$\mathcal{R}(D^*)_e = 0.258 \pm 0.0075 \quad (32)$$

$$\mathcal{R}(D)_\mu = 0.299 \pm 0.0107 \quad (33)$$

$$\mathcal{R}(D^*)_\mu = 0.258 \pm 0.0082 \quad (34)$$

These are exactly the same values for $\mathcal{R}(D^{(*)})$ as in the case with all yields floating. The much smaller uncertainty here reflects the limited amount of signal statistics; the contribution of the normalisation uncertainty to the total one is much smaller.

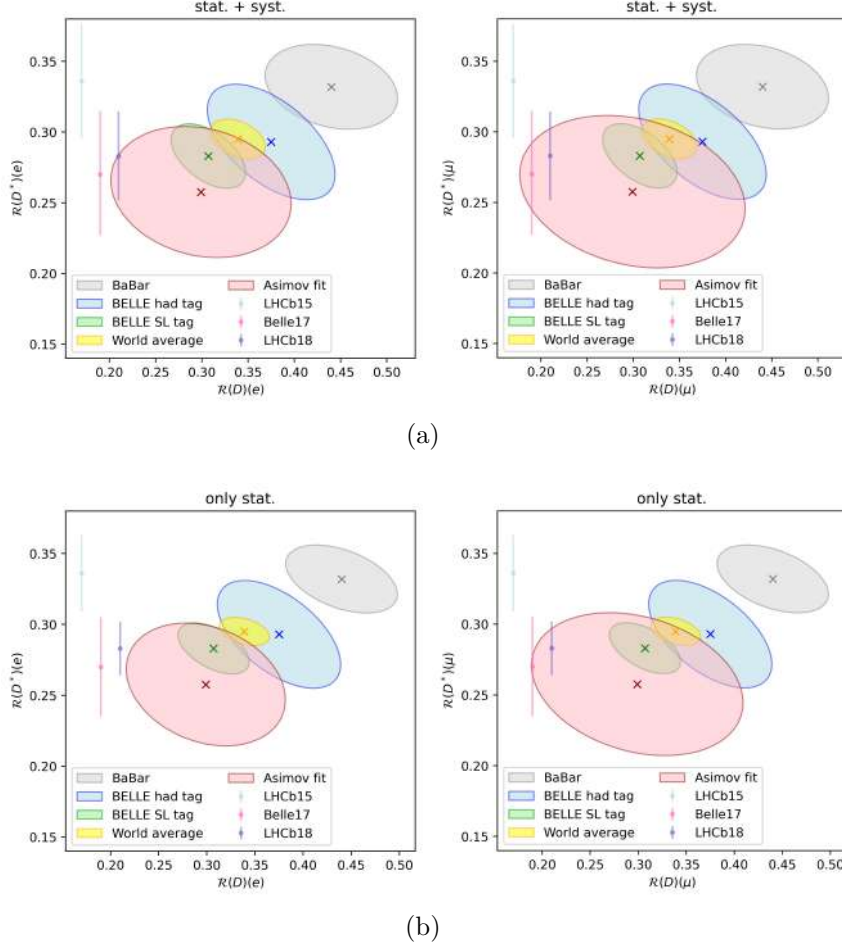


FIG. 28: Visualisation of correlations between $\mathcal{R}(D)$ and $\mathcal{R}(D^*)$ as ellipses, together with previous results[5]:

(a) including all sources of uncertainty

(b) statistical uncertainties only.

The electron mode is shown on the left and the muon mode on the right.

Fixing both signal and normalisation yields is also possible:

$$\mathcal{R}(D)_e = 0.299 \pm 0.0012 \quad (35)$$

$$\mathcal{R}(D^*)_e = 0.258 \pm 0.0009 \quad (36)$$

$$\mathcal{R}(D)_\mu = 0.299 \pm 0.0017 \quad (37)$$

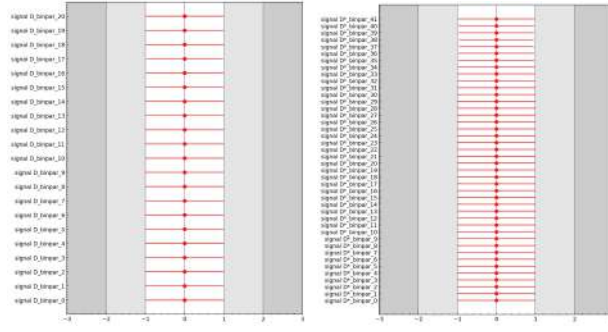
$$\mathcal{R}(D^*)_\mu = 0.258 \pm 0.0011 \quad (38)$$

Again, the same values for $R(D^{(*)})$ are obtained. The remaining uncertainties are due to the correlations between signal and normalisation yields.

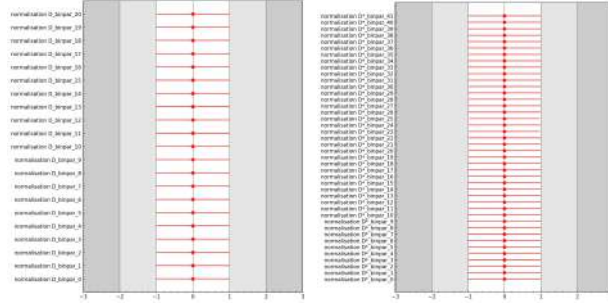
Further, we can look at the nuisance parameter pulls for all categories. Figs. 29-30 show the pulls for the bin-parameters for each of the fit templates. Note that the signal/normalisation D^* templates have twice as much bin-parameters as the other templates, as they have events in both the D and D^* modes due to the feed-down component. We can confirm that all of the NP pulls are concentrated at zero. In case of the signal and normalisation categories, the bin-parameters are not further constrained in the asimov fit. The D^{**} +gap components, however, do feature constrained NPs. This behaviour is

1009 expected and can be explained by the BF uncertainty attributed to the “gap” events: As
1010 we have no further knowledge about these events, their BFs and composition, we treat
1011 this with a 100% BF uncertainty, which, however, overestimates the real uncertainty. The
1012 fit with bin-wise NPs can further constrain this uncertainty, thus lowering the errorbars of
1013 the respective pulls. The truth of the latter statement can be confirmed by repeating the
1014 same fit, but without adding in the 100% BF uncertainty for the “gap” events, leading
1015 to the same 1σ pulls as for the signal and normalisation categories.

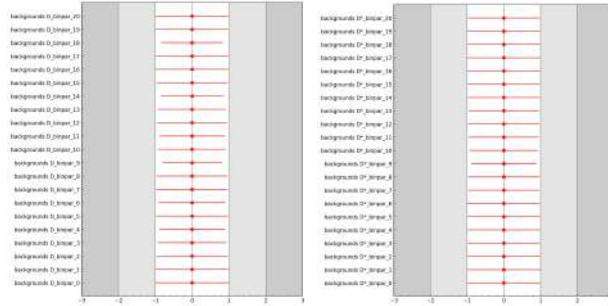
1016 In case of the backgrounds, only a few of the bin-parameters are further constrained.
1017 The origin of these constraints can be traced back to a small amount of offresonant
1018 events with large weights, which are part of these templates. For further details, see
1019 Section 11.5.



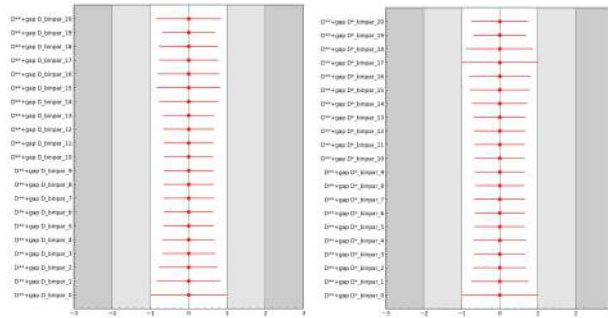
(a)



(b)



(c)



(d)

FIG. 29: NP pull distributions when performing an asimov fit for:

(a) signal

(b) normalisation

(c) backgrounds

(d) $D^{**} + \text{gap}$.

Each plot shows the pulls for the 21 bin-wise NPs for the respective category. The left plot corresponds to the D case; the one on the right to the D^* case. The pull distributions in this figure are for the electron mode.

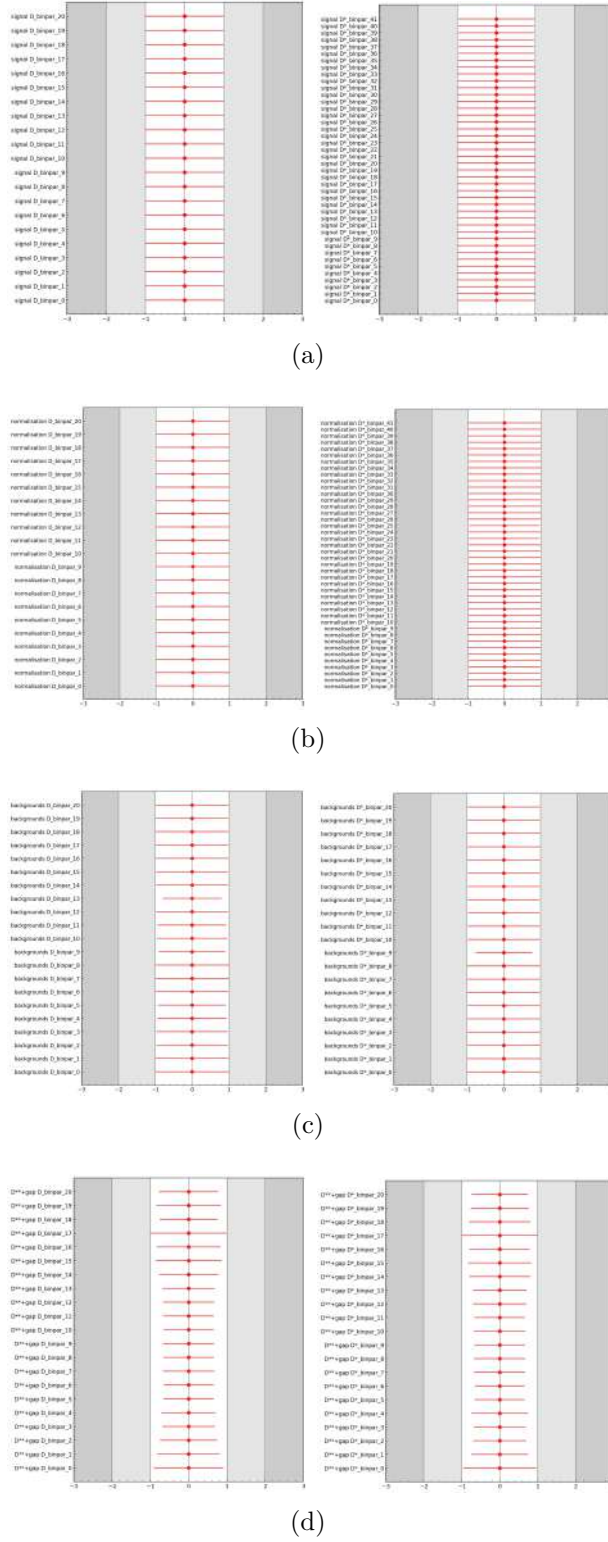


FIG. 30: Same as Fig.29, but for the muon mode

1020 8.7. Pull tests

1021 For the purpose of validating the fit model and our uncertainty estimate, we perform
 1022 pull tests, meaning that we generate toy datasets including statistical fluctuations with

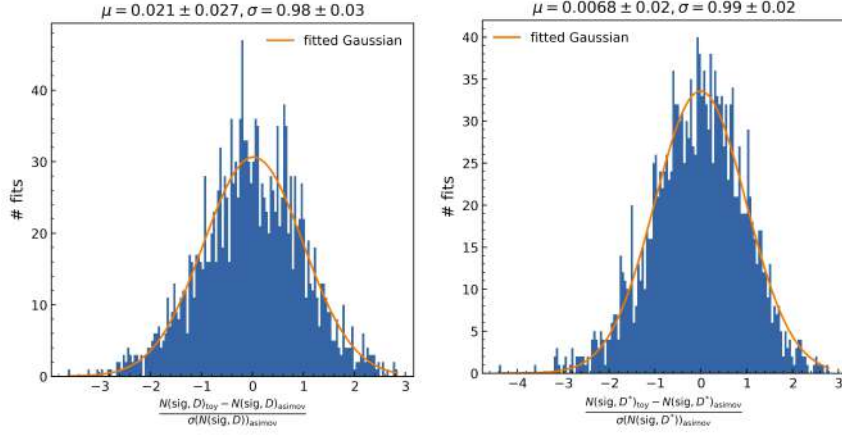
1023 which we repeat the fit, allowing us to investigate the resulting pull distributions:

$$\text{pull} = \frac{N_{\text{toy}} - N_{\text{asimov}}}{\sigma_{\text{toy}}} \quad (39)$$

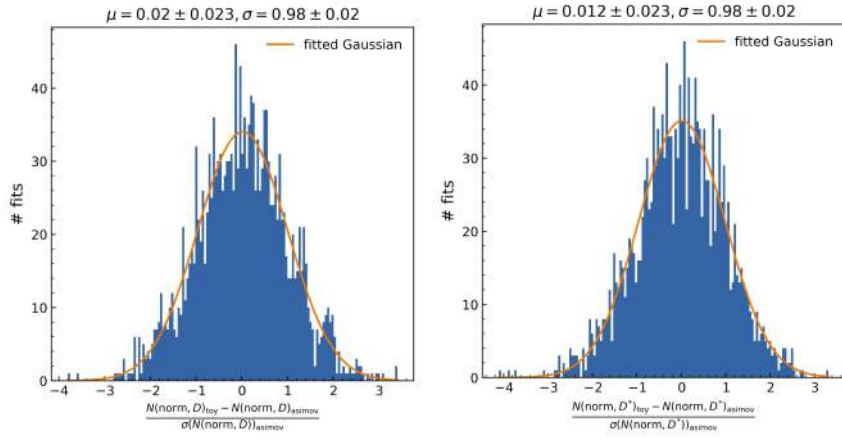
1024 We perform this test using 2000toys derived from the asimov distribution by adding
 1025 Poisson fluctuations to the bin entries. We measure the signal and normalisation yields
 1026 and plot their pull distributions. By running `minos` in the fit for each toy we can obtain
 1027 asymmetric uncertainties for the yields. While the deviation from the nominal uncertainty
 1028 is only about 0.3% on average in case of the normalisation yields, we find an average
 1029 asymmetry of about 3% in case of the signal yields. Thus, as the signal yield uncertainty
 1030 will dominate the final uncertainty on $\mathcal{R}(D^{(*)})$, we consider this asymmetry in the pull
 1031 distribution in the following way: If the toy yield is above the asimov expectation, the
 1032 negative yield uncertainty is taken to calculate the pull. If, on the other hand, the toy
 1033 yield is smaller than the asimov one, the positive uncertainty is chosen. This corrects for
 1034 the slight asymmetry in the pull distribution.

1035 In addition, we also calculate $\mathcal{R}(D^*)$ from the refitted yields to evaluate also the pull
 1036 distribution of the ratio we are ultimately measuring with the fit. In order to take the
 1037 asymmetry of the yields into account, we choose from the asymmetric uncertainties of
 1038 the signal and normalisation yields as outlined above in order to find the uncertainty on
 1039 $\mathcal{R}(D^*)$.

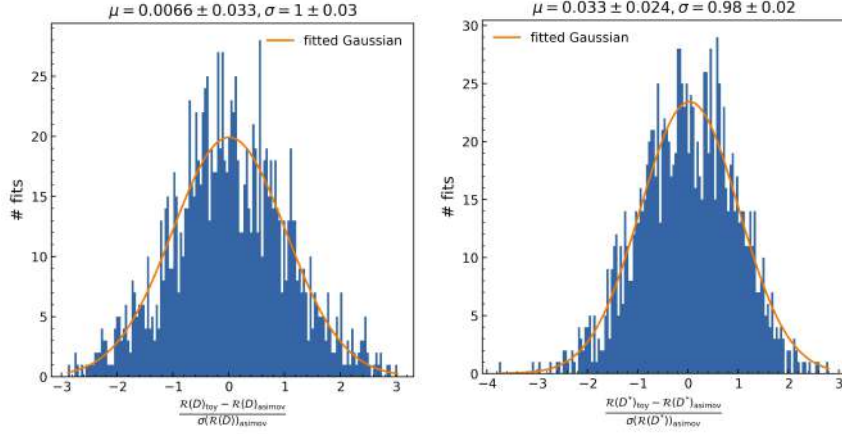
1040 Subsequently, we fit a Gaussian distribution to the latter to obtain its mean and
 1041 standard deviation. This can be seen in Figs.31-32, where the μ and σ obtained from
 1042 the fit, including their uncertainties, are displayed in the title of each plot. Doing so, we
 1043 can confirm that the mean is consistent with 0 and the width is consistent with 1 within
 1044 their respective uncertainties. This suggests that the uncertainty estimate from the fit is
 1045 consistent with the statistical fluctuations.



(a)



(b)

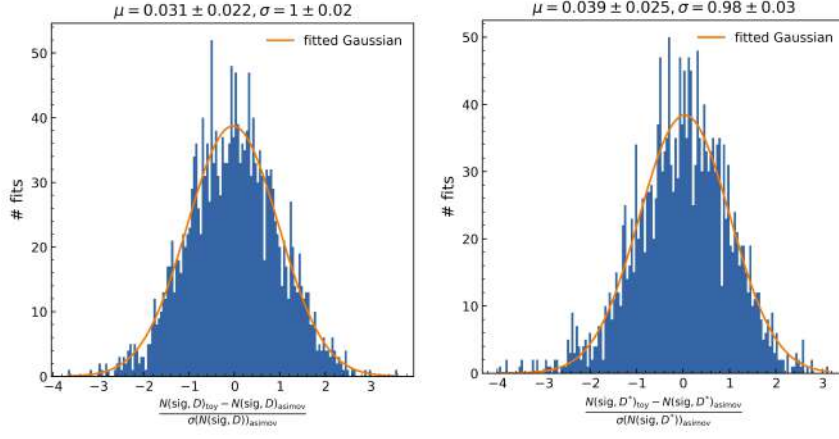


(c)

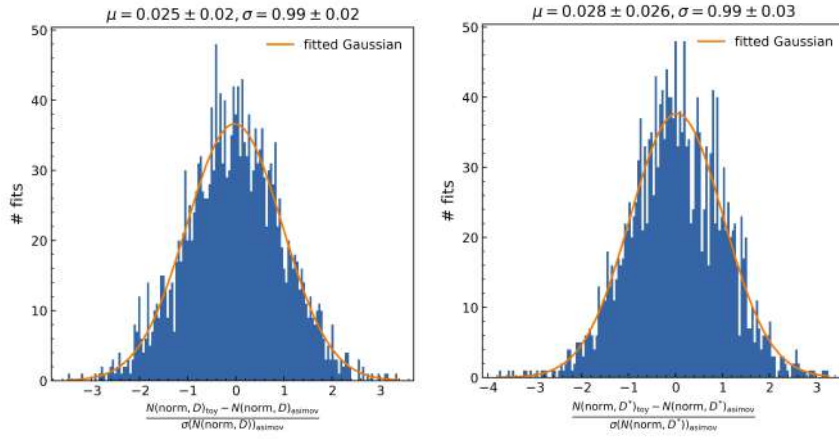
FIG. 31: Pull distributions for:

- (a) N_{signal}
- (b) $N_{\text{normalisation}}$
- (c) $\mathcal{R}(D^*)$.

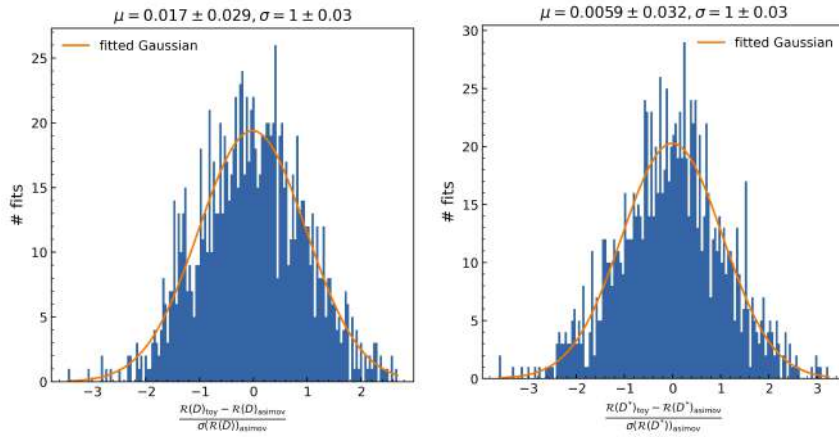
The plot on the left corresponds to the D mode while the D^* mode is displayed on the right. Each plot includes a Gaussian distribution that is fitted to the histogram. On the top of each plot, the fitted mean and width of this Gaussian is given. The distributions are for the electron mode



(a)



(b)



(c)

FIG. 32: Same as Fig.31, but for the muon mode

1046 9. SIDEBAND STUDIES

1047 In this Section, we define several sidebands that are aimed to check for the agreement
 1048 in specific categories of events. TableXIII provides an overview of all sidebands used in
 1049 this analysis.

They will be used in the following for several cross-checks.

Furthermore, in this Section, we derive scale factors wrt. the FEI efficiency (Section Sec9.9.2). Note that these scale factors are already applied in all of the plots in this Section. The fits to obtain the scale factors are performed in sidebands which have partial overlap with the entire fit region. Thus, when doing a χ^2 test on a variable distribution, checking the agreement of the histograms in data vs. MC, we have to subtract degrees of freedom in order to calculate the corresponding p value. Due to a potential overlap, the number of degrees of freedom to subtract is ≤ 2 . The most “pessimistic” p value would be the case of subtracting 2 degrees of freedom, which is what we apply in each of the plots in this Section.

9.1. Sidebands enhanced in the three event categories

Using the classifier outputs, it is possible to define several sidebands which allow to test the data-MC agreement. Specifically, we can examine certain sidebands which are enhanced in a specific category of events. Thus, this Section deals with the definition of such sidebands in order to use them in the following subsections for data-MC studies.

The sidebands are defined via the two fit variables, using their two-dimensional distribution (in the following referred to as “2D-classifier space”).

Firstly, we define a signal-enhanced region just via the signal classifier score, using $z_\tau \geq -1.2$. This region will be kept blinded for the sideband studies. Only for the investigations regarding the tag-side decay mode ID distributions (Section 9.9.2.2.2, we will determine the data-MC ratio in this region per tag-side decay mode ID. The orthogonal region, i.e. $z_\tau < -1.2$, will be referred to as “ z_τ sideband” in the following. The signal-enhanced region comprises 97%-98% of all signal events.

To check the agreement for the normalisation category independently from the one of the background category, we define the normalisation-enriched sideband with an additional cut on the difference classifier score between normalisation and backgrounds, $z_{\text{diff}} > 0$, such that more than 70% of all normalisation events fall into this sideband.

The remaining region will be used for the aim of testing the agreement in the background category, i.e. using the cuts $z_\tau < -1.2$ and $z_{\text{diff}} \leq 0$. Although the fraction of background events wrt. the total amount of background events is not very high, as the backgrounds are spread out across the whole 2D classifier space, the region is dominated by the backgrounds, i.e. the background fraction in this region is $> 50\%$ in both the electron and muon modes.

In the normalisation- and background-enhanced regions, the signal events only contribute a fraction of $\leq 3\%$.

The definition of all sidebands is visualised in Figs. 143-148 in the Appendix, also showing the fraction of included events in each sideband for each of the three event categories.

TABLE XIII: Sideband definitions

sideband	cuts	usage
q^2	$q^2 < 3\text{GeV}^2$	verification of BDT input variable shapes (2D classifier space sidebands not yet available)
signal-enhanced	$z_\tau > -1.2$	verification of cancellation of FEI efficiency by looking at the data/MC ratio per DMID; inverse region used for deriving FEI eff. scale factors;
normalisation-enhanced	$z_\tau \leq -1.2$ and $z_{\text{diff}} > 0$	verification of BDT output variable shapes; verification of cancellation of FEI efficiency
background-enhanced	$z_\tau \leq -1.2$ and $z_{\text{diff}} \leq 0$	verification of BDT output variable shapes
signal region	$0.3 < z_\tau$ and $-1.95 < z_{\text{diff}} \leq 0.0$	investigating background composition in simulation
D^{**} enhanced	$-1.1 < z_\tau \leq -0.1$ and $-2.6 < z_{\text{diff}} \leq -0.65$	checks wrt. to D^{**} modelling
around the signal region	$0 < z_\tau$ and $-2.6 < z_{\text{diff}} \leq -0.65$ and NOT signal region (definition see above)	verification of BDT output variable shapes

9.2. FEI efficiency studies

First of all, we need to address the FEI efficiency. There have been several efforts for calibrating the SL FEI but no official SL FEI calibration exists. Several of the complications related to the FEI are summarised in ref. [48]. Ideally, such a calibration is not necessary for this analysis as the measured BF's for $\mathcal{R}(D^{(*)})$ are corrected by the efficiencies ϵ_{sig} , ϵ_{norm} of both modes. Thus, the difference in signal and normalisation tag efficiencies is included in ϵ_{sig} and ϵ_{norm} as long as those are well-modelled in simulation. An additional systematic may be introduced if the FEI efficiency was different for the signal and normalisation modes, impeding the desired cancellation in the $\mathcal{R}(D^{(*)})$ ratio. The following subsections are dedicated to several studies in that respect.

9.2.1. Bad tag ratios and resulting FEI efficiency scale factors

We assume that there is a uniform FEI efficiency scale factor that equally applies to signal as well as normalisation modes. Several studies suggest that this scale factor

is connected to the ratio of bad tags, i.e. wrongly reconstructed tag-side B mesons. Theoretically, there should be no signal side dependence of the FEI efficiency. However, the FEI efficiency for events with good tags, i.e. correctly reconstructed B_{tag} mesons, is different from those with bad tags. This can effectively lead to different FEI efficiency scale factors for certain components, if their bad tag ratios do not coincide. To put it the other way around, we expect events with similar bad tag ratios in MC to scale the same way in when compared to data. Thus, we investigate the bad tag ratios for different signal side categories. These turn out to be:

- signal events (excluding feed-down): 23%
- normalisation events (excluding feed-down): 21%
- feed-down events: 23%
- background events: 46%

As ‘bad tags’, we consider all events in which the variable `isSignalAcceptMissingNeutrino` is not true for the tag-side B^0 meson. As the background events have a much larger fraction of bad tags than the signal and normalisation categories, we expect to find a different FEI efficiency scale factor for those events. In order to test this hypothesis and determine these scale factors for the normalisation and background categories, we perform a fit to data in two different sidebands: the signal classifier sideband (labelled as z_τ in the plots) as well as the $q^2 < 3 \text{ GeV}$ sideband. Via a two-dimensional negative log-likelihood fit set up in the same way as described in Sec.8 (but without adding a separate signal template [43]), we extract the scale factors for normalisation, feed-down (only in case of the D mode), and backgrounds for both the D and D^* modes. Note that this is not a simultaneous fit of both modes, because the purpose is to check whether we find a consistent scale factor for the feed-down events independently from the one for those D^* events reconstructed in the correct mode. The obtained scale factors are visualised in figure 33. The uncertainties quoted in this table only reflect the statistical uncertainty obtained from the fit, since we do not add any nuisance parameters for the systematic uncertainties. We then compute the variance weighted mean of those scale factors, which is also indicated in the plots as a grey line. Subsequently, we can compute a χ^2 and corresponding p value if the scale factors are all compatible with the variance-weighted mean. This p value is shown in the title of the plot. It also takes into account the correlations due to the partial overlap of the sidebands. The p value suggests that there is indeed a consistent scale factor for normalisation and feed-down events in the D and D^* modes. Thus, we can use the variance-weighted mean to scale the normalisation events in all prefit plots. Although we find a lower p value for the background scale factors, the scale factor appears to be consistent in this case as well. Similarly to the normalisation case, it is used to scale the background events in the prefit plots unless otherwise stated[44]. We further check whether the bad tag ratios are the same for signal and normalisation across all tag-side decay mode IDs. For this, we compute the ratio of signal and normalisation bad-tag ratios (in the respective enhanced regions) and test their compatibility with one across all tag-side decay mode IDs (see Fig.34). These appear to be consistent. The checks performed wrt. the signal and normalisation bad-tag ratios and tag-side DMID distributions (see Sec. 9.9.2.2) suggest that signal events should scale in the same way as normalisation events. Thus, the scale factor derived for the latter is also applied to the signal (including signal feed-down) events. The fact that we find reasonable prefit agreement in the unboxed prefit distributions (see Section 11) is an indication that the difference between normalisation and background scale factors

can indeed be traced back to their different bad tag ratios and is another assurance that signal and normalisation FEI efficiency scale factors cancel in the ratio of $\mathcal{R}(D^{(*)})$ due to similar bad tag ratios.

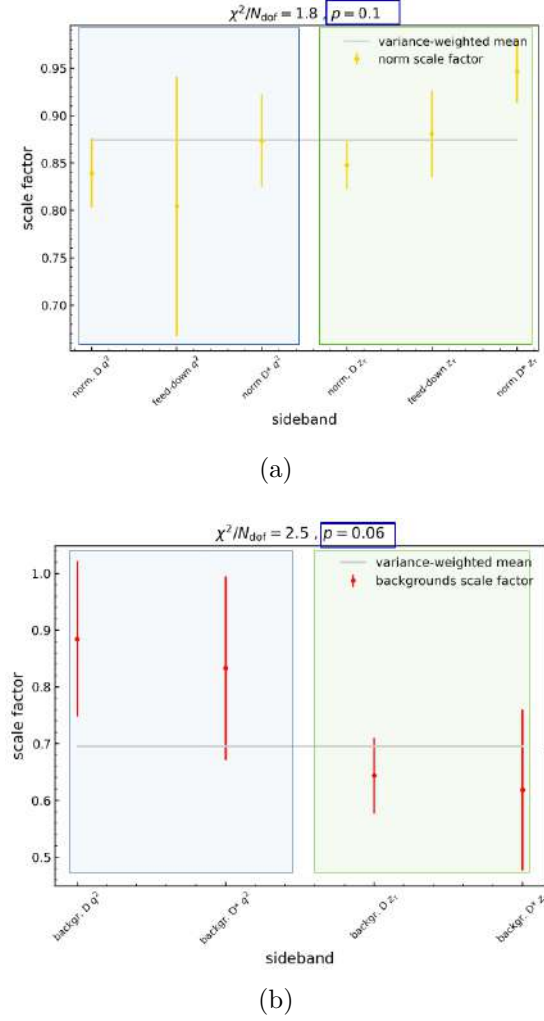


FIG. 33: Scale factors derived in two different sidebands for the D and D^* modes with errorbars indicating the statistical uncertainty as obtained from the fit:

- (a) $D\ell\nu$, $D^*\ell\nu$ and feed-down events (normalisation mode)
- (b) background events

Above each plot, the reduced χ^2 and p value reflecting the compatibility of the scale factors with the variance-weighted mean (grey line) within the respective uncertainties is displayed.

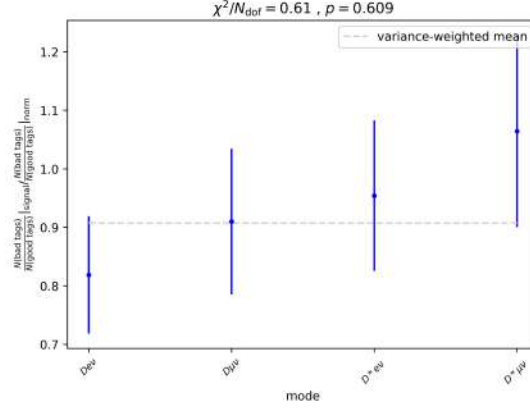
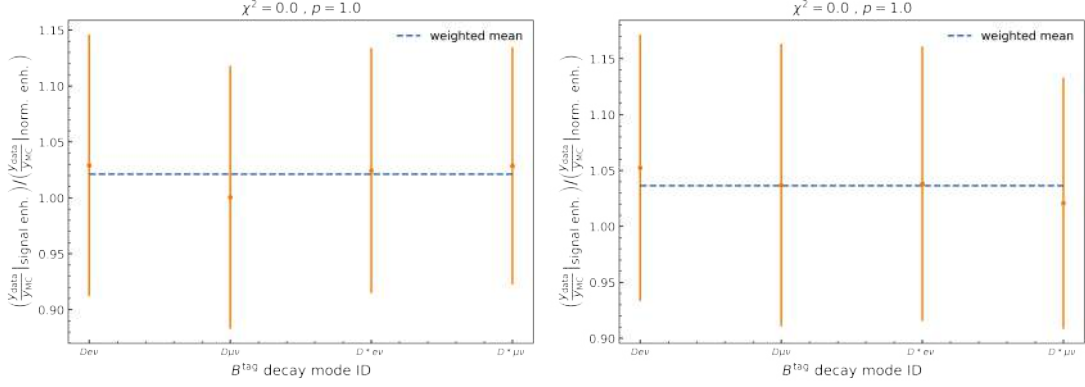


FIG. 34: Ratio of bad tag ratios in the signal and normalisation enhanced regions across different tag-side decay mode IDs.

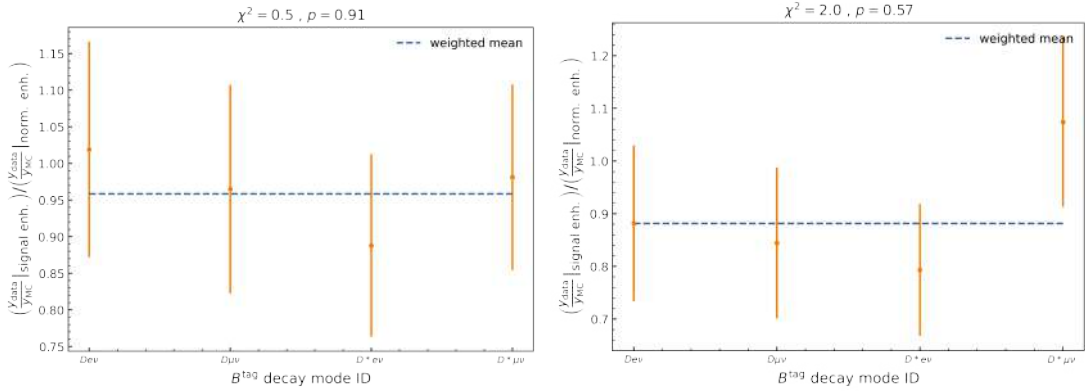
9.2.2. Tag-side decay mode ID

Further checks have to be performed regarding the distributions of FEI-reconstructed decay modes (“DMID”), which might be different for signal and normalisation modes. To test their modelling we avail ourselves of the normalisation- and signal-enhanced regions. For each DMID, we determine the ratio of data to MC in the normalisation-enhanced and signal-enhanced regions. For this we already apply the scale factors derived in Section 9.9.2. In the signal-enhanced region, we are using signal MC to model the $D^{(*)}\tau\nu$ events. In Fig.35, we show the ratio of the signal-enhanced data-MC ratio and the normalisation-enhanced data-MC ratio. In order to verify that this ratio is consistent over all DMIDs, we need to bear in mind that both the normalisation- as well as the signal-enhanced regions are not solely composed of those event categories, but also contain backgrounds, so we cannot directly conclude towards $\mathcal{R}(D^{(*)})$ from those ratios. If the ratio was equal to one for all DMIDs, this would mean that the data behave as expected by the SM, which might not be the case. Thus, to test the consistency, we apply a χ^2 test of their deviation of the plotted ratios from the variance-weighted mean. The resulting χ^2 and corresponding p values are given in the title of each plot.

Given the large p values, we can conclude that the ratios of data to MC of the normalisation and signal categories are indeed compatible across tag-side decay modes.



(a)



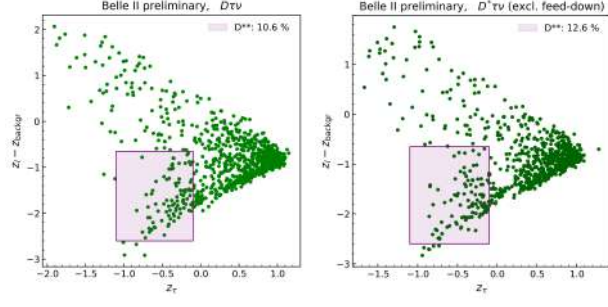
(b)

FIG. 35: Ratio of the signal-enhanced data-MC ratio with respect to the normalisation-enhanced data-MC ratio for $D(\ell/\tau)\nu$ (a) and $D^*(\ell\tau)\nu$ (b) in the electron mode (left) and muon mode (right). Above each plot, a χ^2 and p value indicate the compatibility of these ratios with one over all tag-side decay modes.

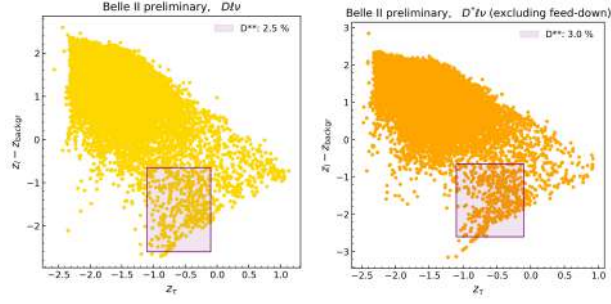
9.3. Studies of D^{**} modelling

In order to test the modelling of the D^{**} resonances, we can isolate those events in 2D-classifier space and check the agreement in this D^{**} -enriched sideband. Two types of events should be distinguished here: Firstly, the resonant D^{**} decays as simulated in generic MC, and secondly, non-resonant decays into $D^{(*)}\pi\pi / D^{(*)}\eta$ final states via an intermediate, broad D^{**} resonance, simulated by dedicated signal MC in order to replace the “gap” events existing in generic MC, which have been modelled via phase space considerations. We identify an area in 2D classifier space which contains approximately 50% of those events while there are only about 10% signal events, so that it is possible to look at the data-MC agreement in this region. See Figs.36-37 for the definition of this region in 2D-classifier space.

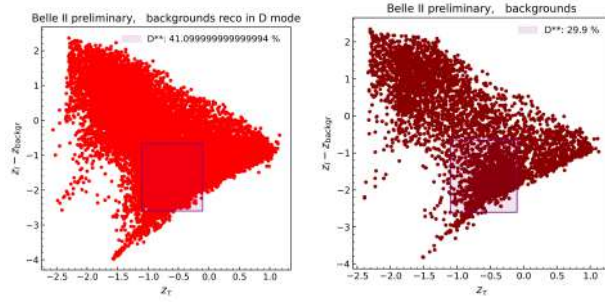
We look at the additional energy in the calorimeter to check the modelling in this sideband, cf. Fig.38, while applying the derived scale factors for signal and normalisation, and find reasonable agreement.



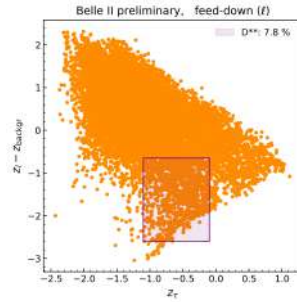
(a)



(b)

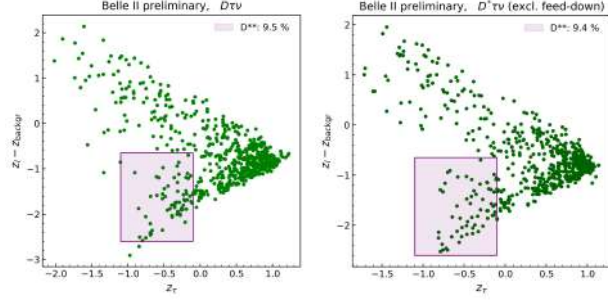


(c)

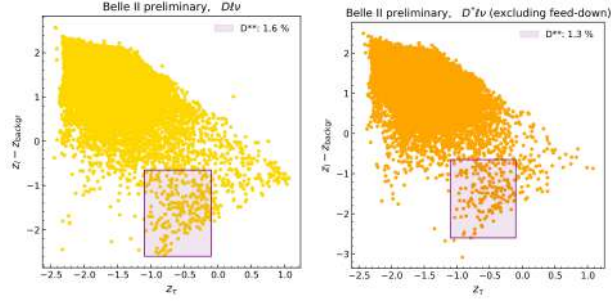


(d)

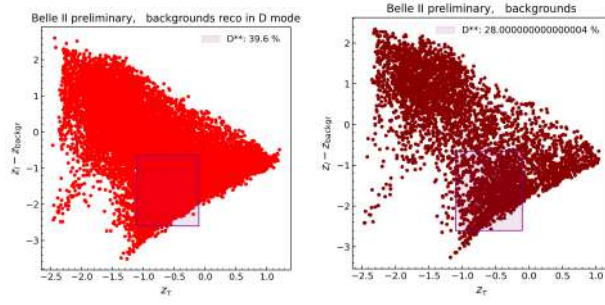
FIG. 36: Definition of the D^{**} enhanced region in 2D-classifier space (showing only the electron mode)



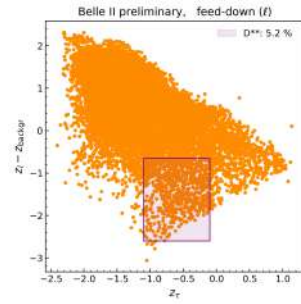
(a)



(b)



(c)



(d)

FIG. 37: Same as Fig.36, but for the muon mode

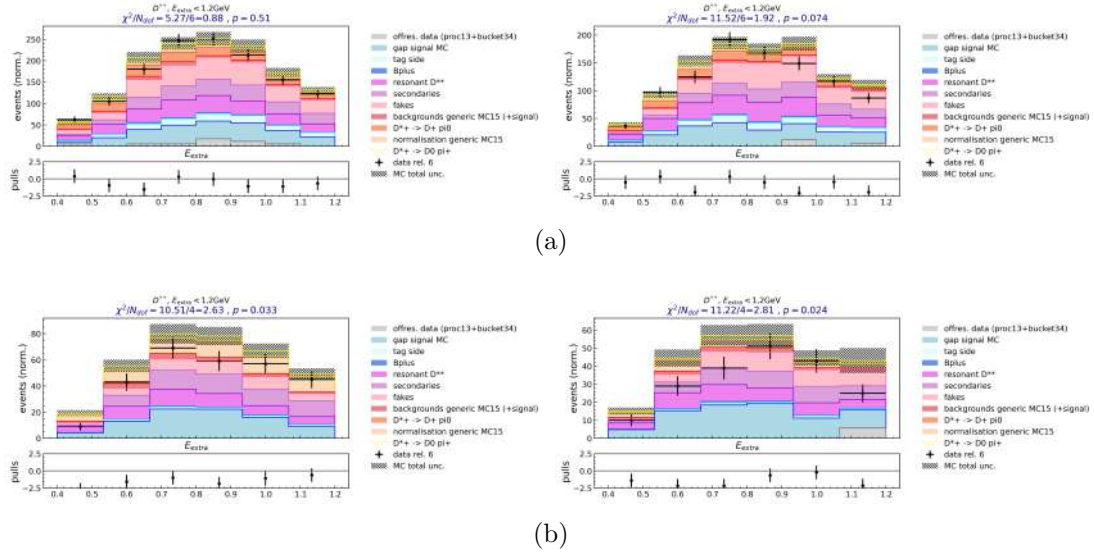


FIG. 38: E_{extra} in the D^{**} enhanced sideband for the electron mode (left) and muon mode (right):

(a) D mode

(b) D^* mode

Note that the number of degrees of freedom considered here is reduced by 2 due to the two scale factors applied (see also the beginning of this Section for further explanation)

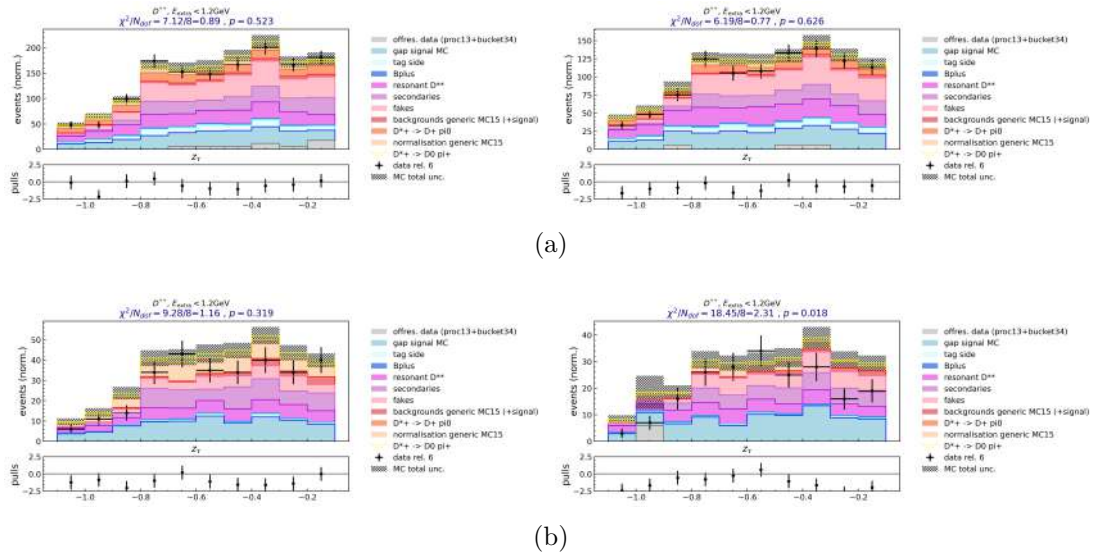


FIG. 39: Same as Fig. 38, but for z_τ

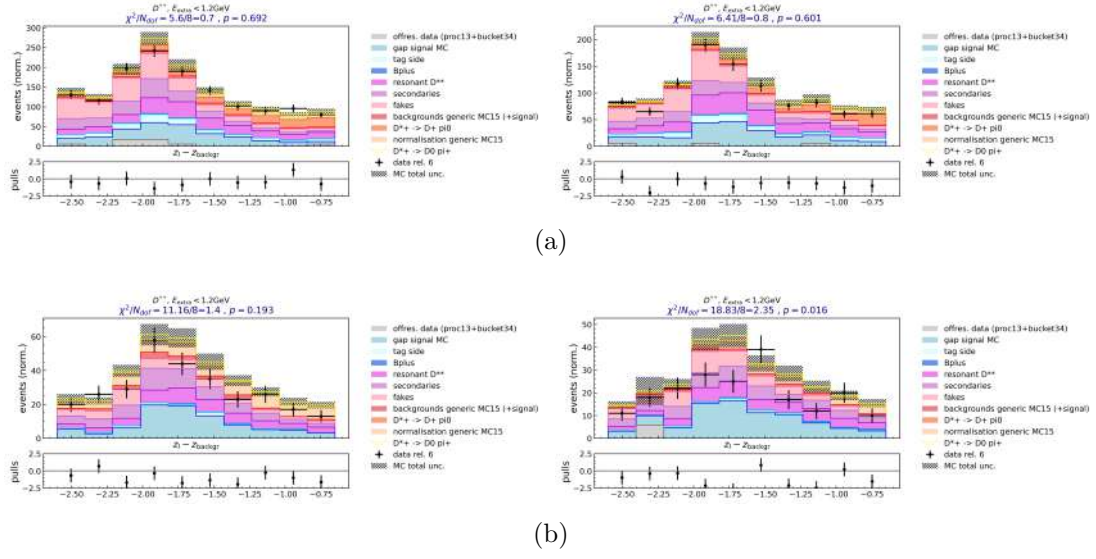


FIG. 40: Same as Fig. 38, but for z_{diff}

9.4. Studies of background composition in signal region

In order to investigate better the background composition in the region where the signal events are located, we define a region in 2D classifier space around the signal events (see tableXIII for the exact cuts). We plot solely the simulated events and identify different background categories and compute their fractions for the D (D^*) modes [45]:

- fake leptons: 11% (15%)
- secondary leptons: 16% (13%)
- resonant D^{**} decays (from generic MC): 5% (5%)
- non-resonant D^{**} decays (from signal MC): 5% (4%)
- B^+ events: 1% (1%)
- tag-side combinatorics: 1% (2%)

The remaining events can be attributed to the normalisation and feed-down categories.

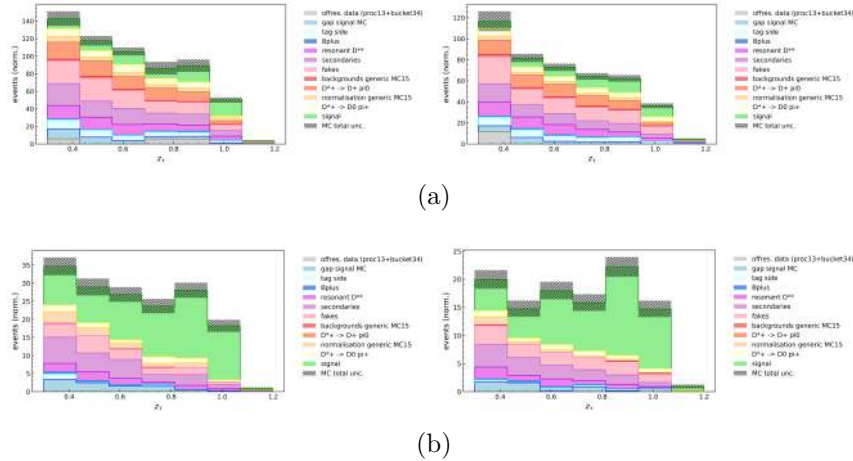
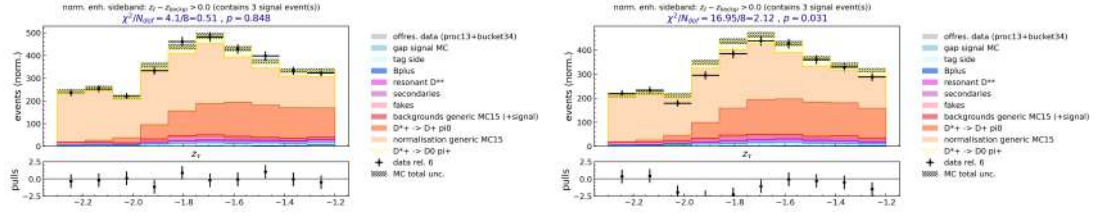


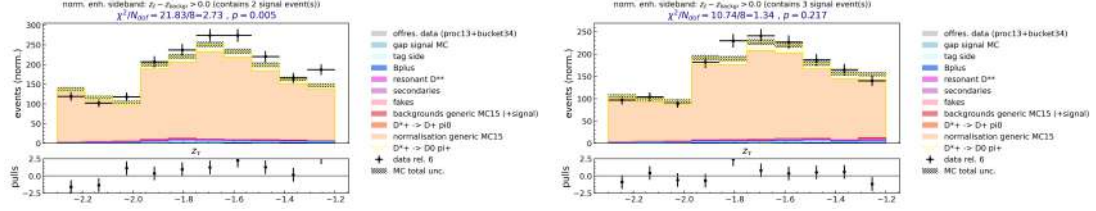
FIG. 41: z_τ in the signal region for the electron mode (left) and muon mode (right), decomposing the distribution into different components:

(a) D mode

(b) D^* mode



(a)

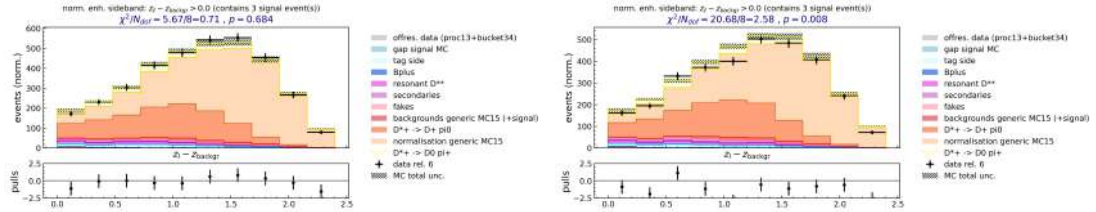


(b)

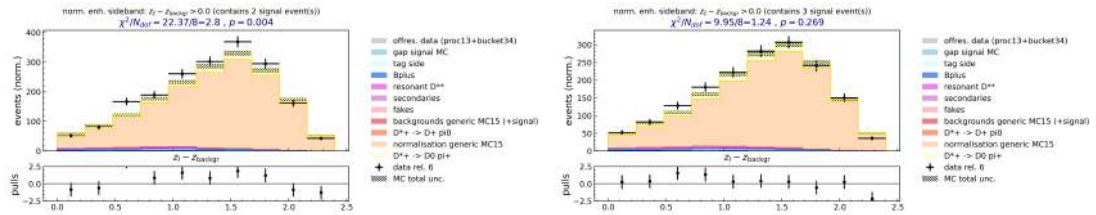
FIG. 43: z_l in the normalisation enhanced sideband for the electron mode (left) and muon mode (right):

(a) D mode

(b) D^* mode



(a)



(b)

FIG. 44: Same as Fig.43, but for $z_l - z_{\text{backgr}}$

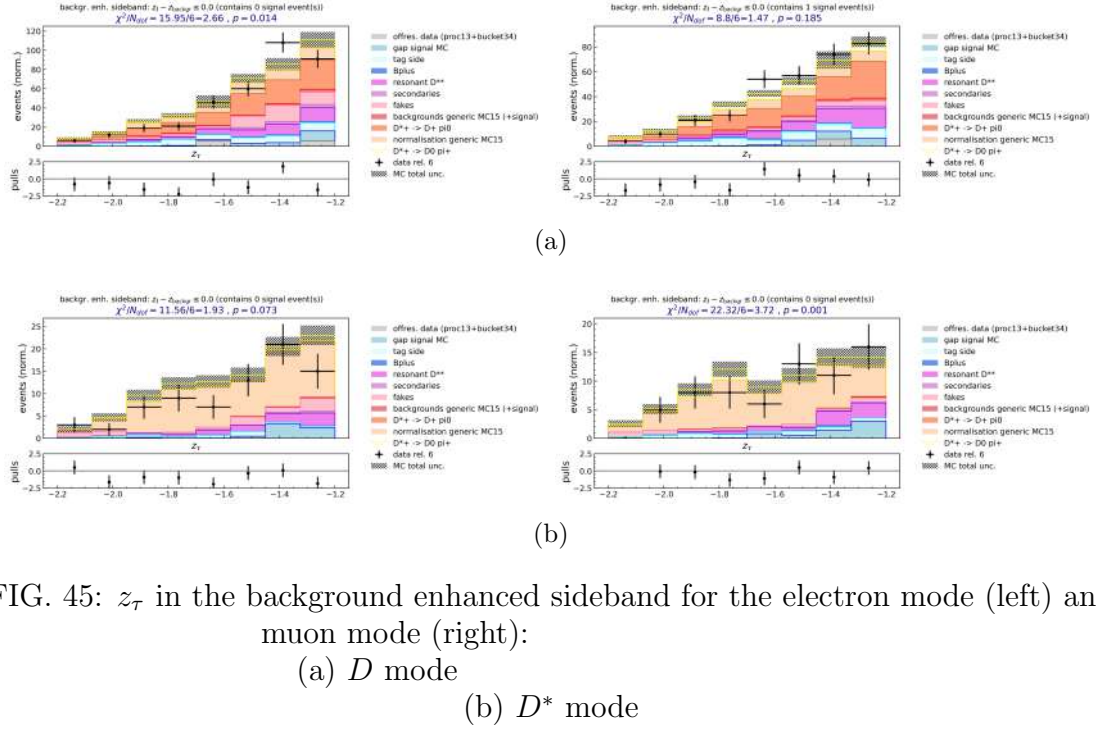


FIG. 45: z_τ in the background enhanced sideband for the electron mode (left) and muon mode (right):

(a) D mode

(b) D^* mode

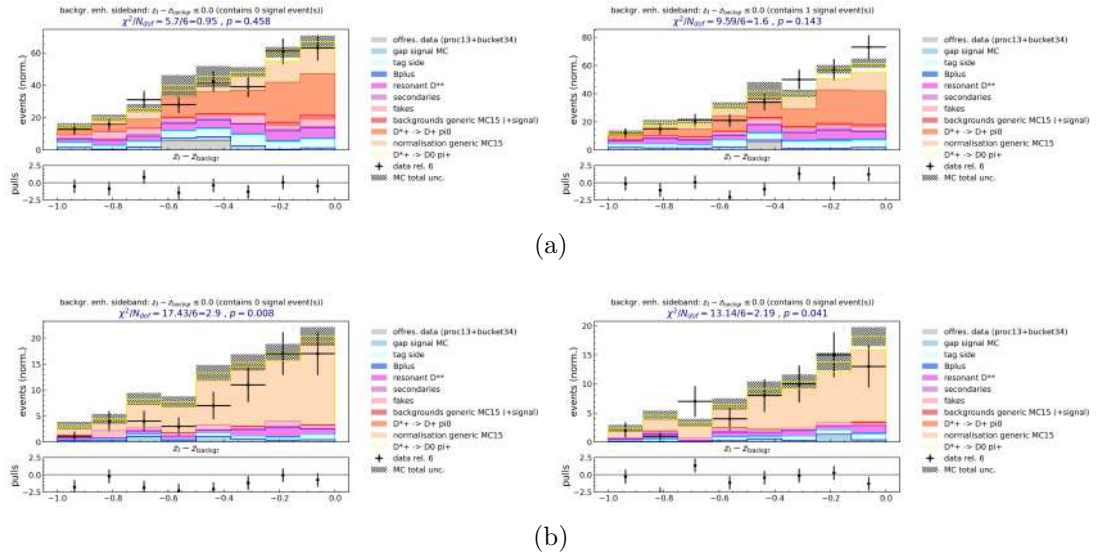


FIG. 46: Same as Fig.45, but for $z_l - z_{\text{backgr}}$

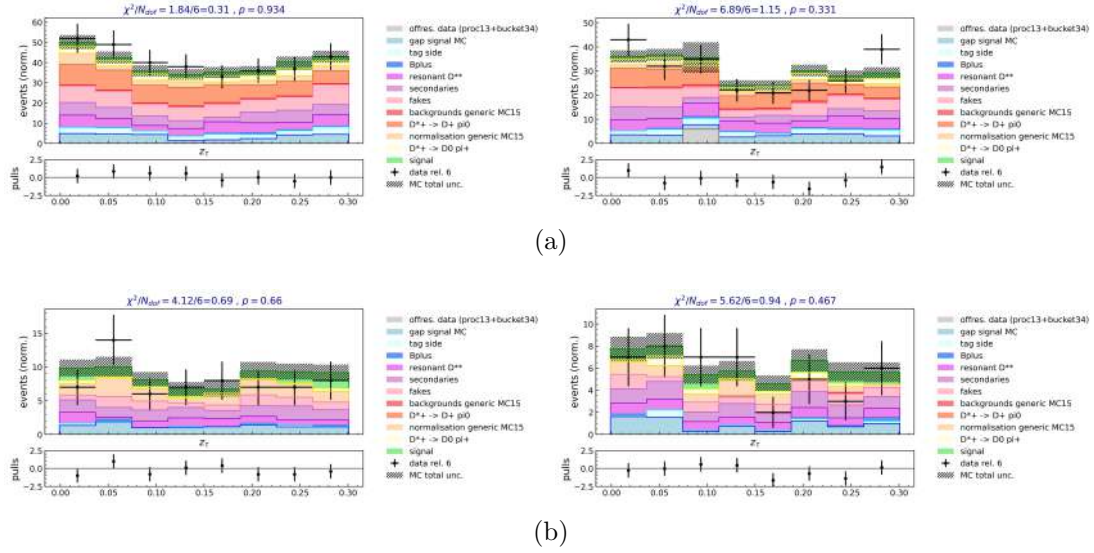


FIG. 47: z_τ in the sideband around the signal region for the electron mode (left) and muon mode (right):

(a) D mode

(b) D^* mode

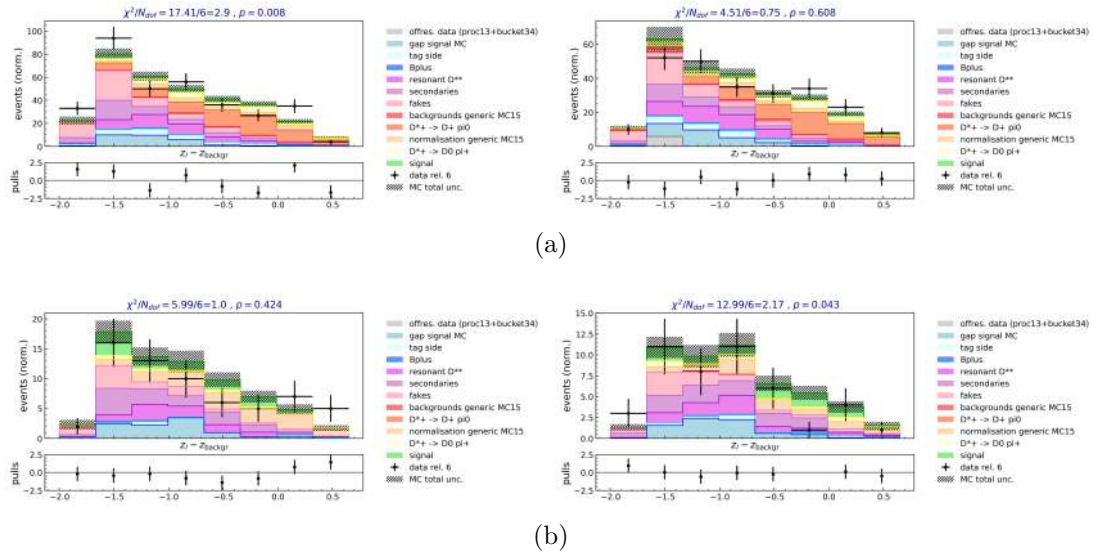


FIG. 48: Same as Fig.47, but for $z_l - z_{\text{backgr}}$

1226 10. SYSTEMATIC UNCERTAINTIES

1227 In this section, we investigate possible systematic effects on the signal extraction.

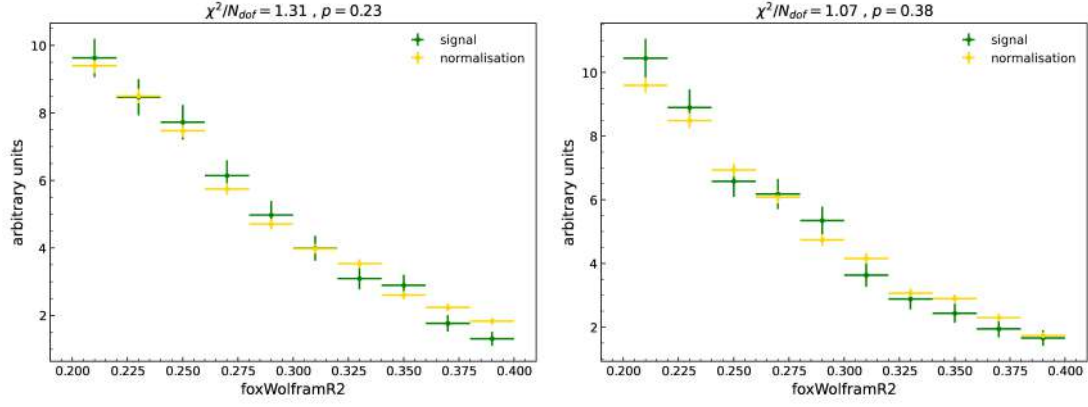
1228 Firstly, those could be caused by any cut on one of the selection variables, if they
1229 are modelled poorly, such that cutting on those variables leads to different portions of
1230 signal and normalisation events surviving the selection. These will be described in the
1231 first subsections.

1232 Secondly, in the following part of this Section, we aim to get an estimate of the size
1233 of several other sources of systematic uncertainty that may be of relevance for the signal
1234 extraction. We do this via repeating the nominal fit, but adding only a single source of
1235 systematic uncertainty.

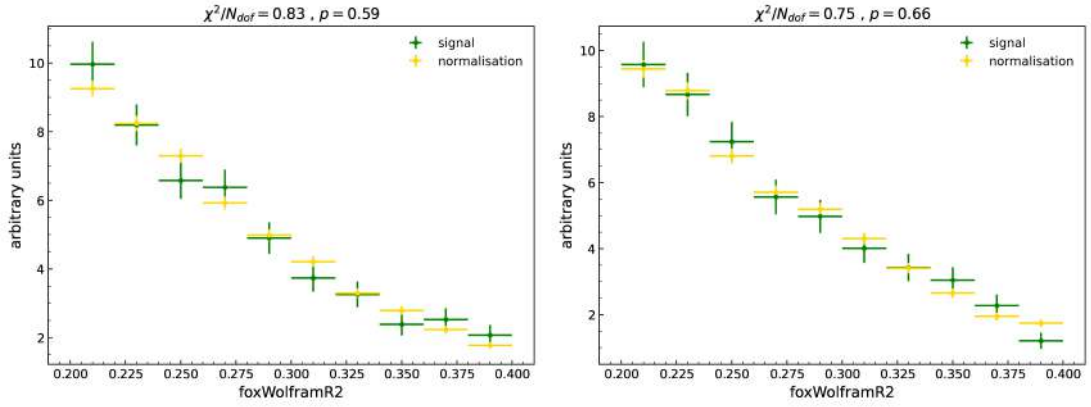
1236 10.1. Studies related to potential systematics introduced by the selection

1237 10.1.1. $R2$

1238 The first variable we investigate here is the second-order Fox-Wolfram moment, which
1239 is known to be modelled poorly, in the sense that other analyses observe a shift in data
1240 vs MC. If the modelling error affects signal and normalisation modes in a different way, a
1241 data-MC discrepancy would introduce an additional systematic in the analysis. To test
1242 this, we plot the normalised shapes of signal and normalisation distributions in MC of
1243 this variable together with the corresponding statistical uncertainty as an errorbar (cf.
1244 Fig.49). For the modelling of the normalisation events, we use generic MC, while the
1245 signal mode is represented by signal MC. To check the compatibility of their shapes, we
1246 compute a χ^2 and corresponding p value, which are displayed above each plot. Here
1247 we are only looking at the high- $R2$ region (from 0.2 to 0.4), since different signal and
1248 normalisation shapes in this region might mean that the cut at 0.4 affects both samples
1249 in a different way. The resulting p values are acceptably high, so we take this as evidence
1250 that the cut on the second-order Fox-Wolfram moment does not introduce a significant
1251 bias in the analysis. If the $R2$ distribution is shifted in data, the resulting efficiency
1252 differences between data and MC will divide out in the ratio.



(a)



(b)

FIG. 49: Normalised distributions of truthmatched signal and normalisation components of the second-order Fox-Wolfram moment for $D(\ell/\tau)\nu$ (left) and $D^*(\ell/\tau)\nu$ (right), with χ^2 and p value quantifying the consistency of the two distributions:

(a) electron mode

(b) muon mode

1253 10.1.2. $\cos \theta_{BY}^{\text{tag}}$

1254 Secondly, the tightened selection of $\cos \theta_{BY}^{\text{tag}}$ on the tag side may have a different
 1255 effect on signal and normalisation events. Thus we repeat the study for this variable, also
 1256 leading to reasonably high p values (see Fig.50).

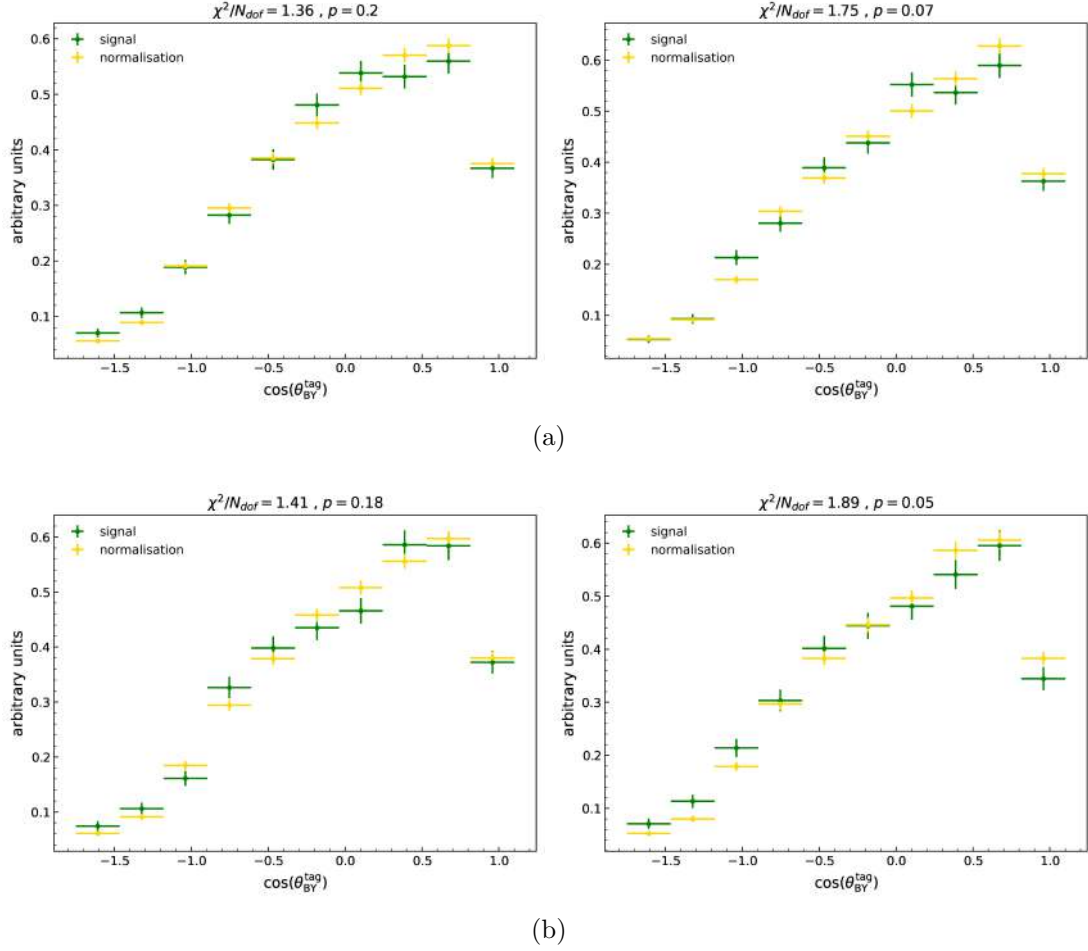
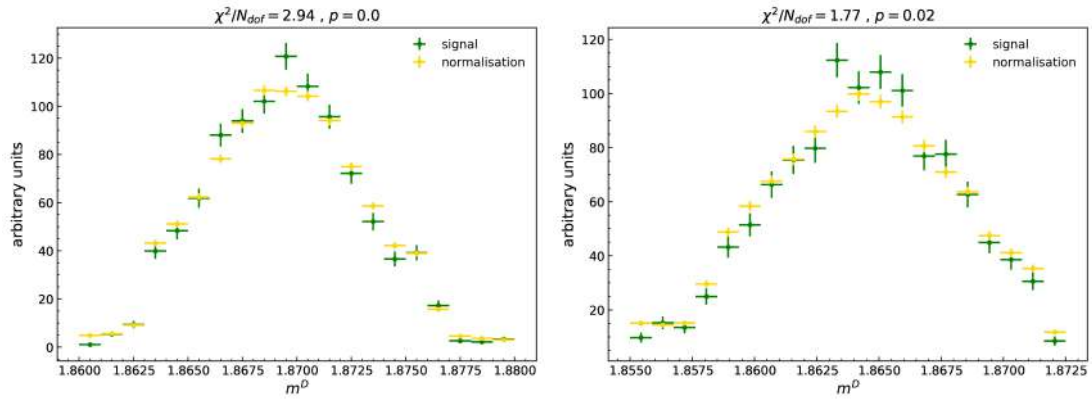


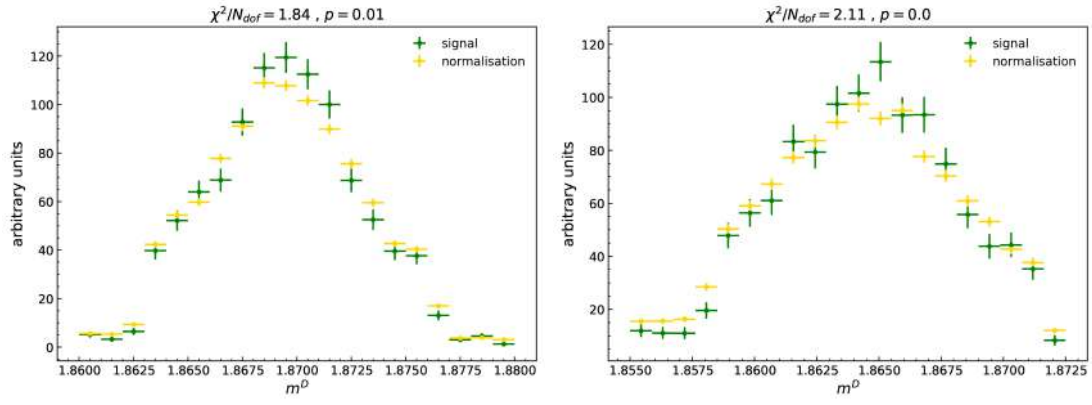
FIG. 50: Same as Fig.49, but for $\cos \theta_{BY}^{\text{tag}}$

10.1.3. $\Delta m(D^* - D)$

Thirdly, the selection of mass windows for the aim of reconstructing a pure sample of $D^{(*)}$ mesons might introduce a systematic, since certain mass peaks might be shifted in data. We repeat the same investigations as before also for the D mass, m_D , and the mass difference between the D^* and D mesons, $\Delta m(D^* - D)$ (see Figs.51-52). Also here, the distributions and p values suggest that the corresponding selections do not prefer any of the signal or normalisation categories as there seems to be no obvious shift.

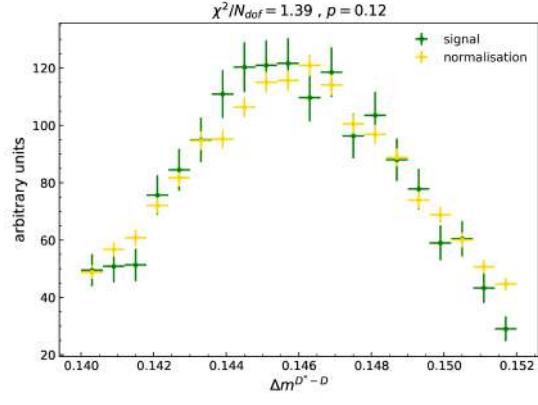


(a)

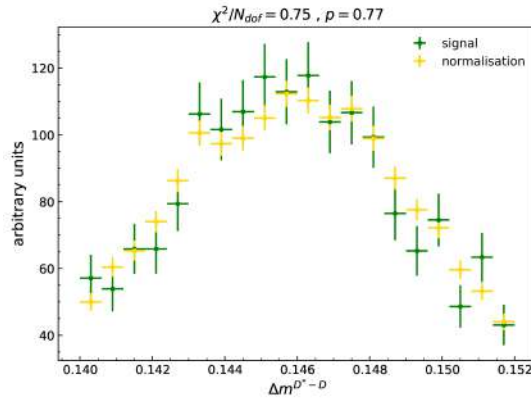


(b)

FIG. 51: Same as Fig.49, but for m_D



(a)



(b)

FIG. 52: Same as Fig.49, but for $\Delta m(D^* - D)$ (only showing the D^* mode, because the selection on this variable is only used for reconstructing D^* mesons)

1265 10.1.4. χ^2 from D vertex fit

1266 Further, a systematic might be introduced by the best candidate selection using the
 1267 χ^2 probability from **TreeFit**. Fig.53 demonstrates that also the χ^2 distributions are
 1268 compatible.

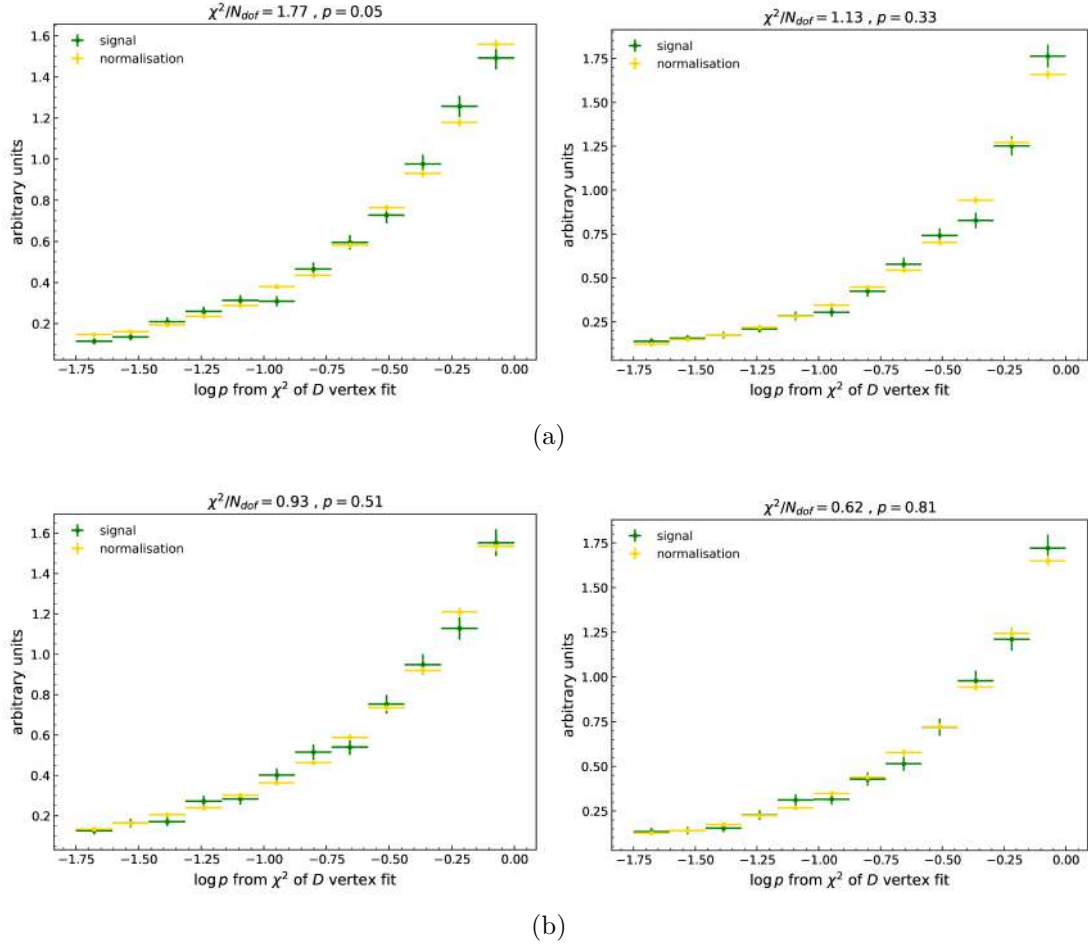


FIG. 53: Same as Fig.49, but for $\log(p)$ from χ^2 of **TreeFit**.

1269 10.1.5. E_{ECL}

1270 Finally, the selection on the extra energy in the calorimeter of $< 1.2 \text{ GeV}$ is investi-
 1271 gated. Here, it is especially important to check the modelling in the high- E_{extra} region
 1272 near the cut. Fig.54 shows a consistency of the two distributions also in this case.

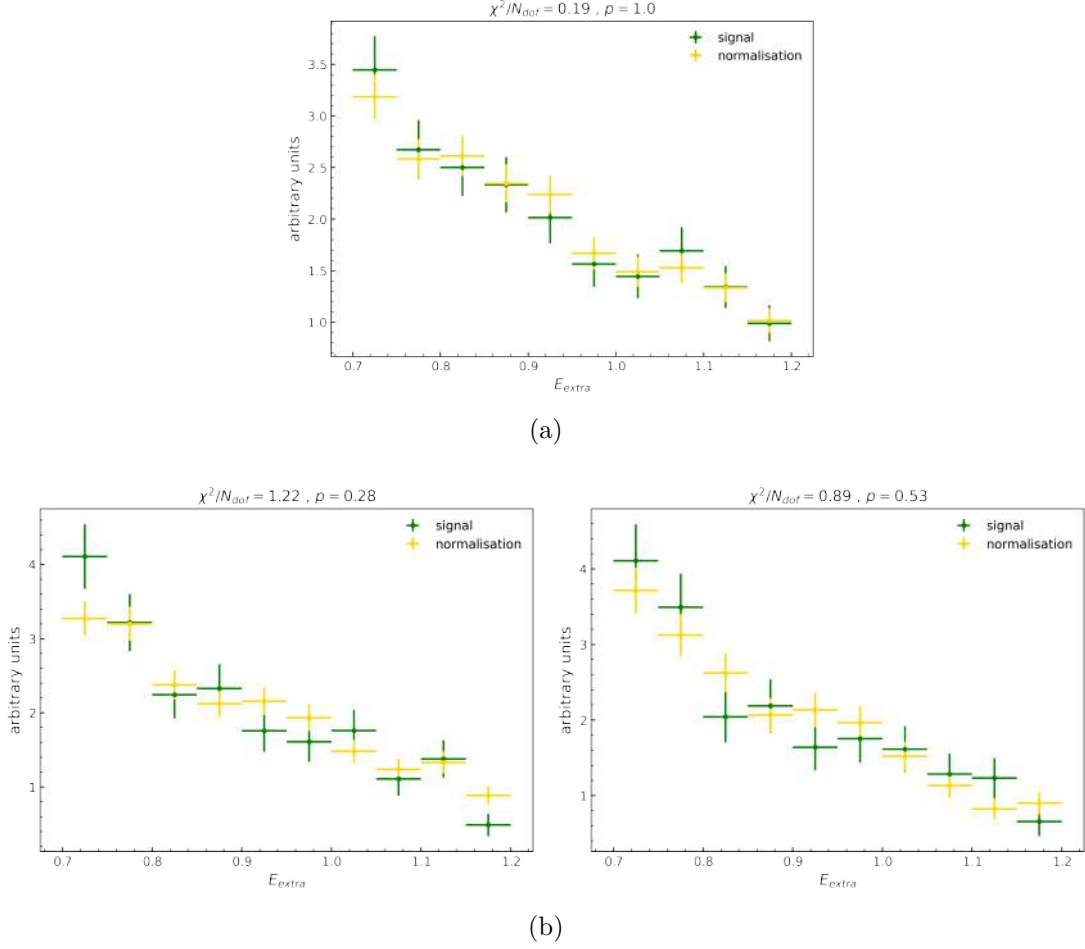


FIG. 54: Same as Fig.49, but for E_{ECL} .

10.1.6. Bad D^* fraction

Fig. 55 reveals that the fraction of “bad D^* ” events, i.e. events in which the reconstructed slow pion does not stem from a D^* (cf. the definition in [23]), is considerably small, such that a fit to the Δm in order to determine the uncertainty on this fraction, ultimately allowing for the estimation of a corresponding additional systematic uncertainty, is not feasible. Yet, the slow pion efficiency uncertainty may already cover the uncertainty related to the bad D^* fraction.

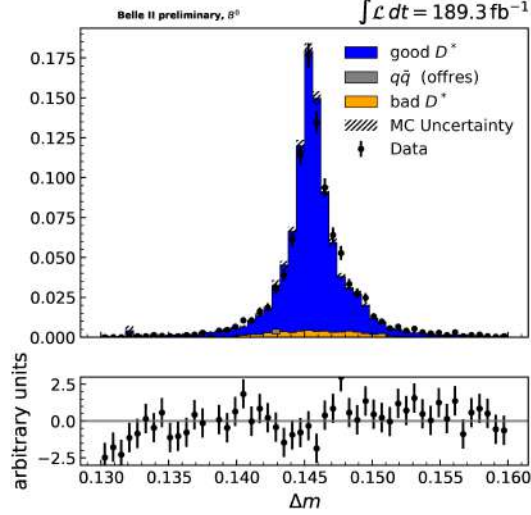


FIG. 55: Normalised distribution of $\Delta m(D^* - D)$ for data and simulation, separating the latter histogram into “good” and “bad” D^* contributions, using the definition in [23]

10.2. Estimation of the size of systematic uncertainties using bin-wise NPs

We can estimate the contribution of a certain source of systematic uncertainty to the total systematic uncertainty by adding only a single source of systematic uncertainty in the fit and considering how much the result differs from the nominal fit result. Table XIV provides an overview of the size of these uncertainties.

From the table, we can conclude that the largest uncertainty for this measurement arises from the modelling of the gap modes, for which we assume the large BF uncertainty. For the electron mode, also the PID systematics play a major role. Yet, all systematic effects are smaller than the uncertainty attributed to data statistics, such that this analysis is rather statistically than systematically limited.

TABLE XIV: Additive uncertainties in the fit with NPs

source of syst.	unc. $\mathcal{R}(D)_e$	unc. $\mathcal{R}(D^*)_e$	unc. $\mathcal{R}(D)_\mu$	unc. $\mathcal{R}(D^*)_\mu$
slow pion	0.0014 (0.5 %)	0.0006 (0.2 %)	0.0015 (0.5 %)	0.0007 (0.3 %)
tracking	0.0005 (0.2 %)	0.0002 (0.1 %)	<0.0001 (<0.1 %)	<0.001 (<0.1 %)
PID	0.0204 (6.8 %)	0.0055 (2.2 %)	0.0027 (0.9 %)	0.0006 (0.2 %)
FFs	0.0028 (0.9 %)	0.0006 (0.2 %)	0.003 (1.0 %)	0.0006 (0.2 %)
FFs (gap)	0.0022 (0.7 %)	0.0008 (0.3 %)	0.0029 (1.0 %)	0.0027 (1.1 %)
BFs	0.0032 (1.1 %)	0.0009 (0.4 %)	0.0013 (0.4 %)	0.0012 (0.5 %)
BFs (gap)	0.0222 (7.4 %)	0.0018 (0.7 %)	0.0302 (10.1 %)	0.0053 (2.0 %)

11. UNBOXING PROGRESS

In this section, we present the results of the consecutive unboxing steps as described in Sec.11. In the following plots, we include not only the statistical MC uncertainty, but also the systematic uncertainties related to BF's, FF's, slow pion efficiency, general tracking efficiency and PID, in order to be able to assess the agreement in a more quantitative way when looking at p values and pulls. Note that for the computation of the p values, we subtract 2 from the number of degrees of freedom due to the two different scale factors applied for the signal/normalisation and background categories. Although this is not fully accurate, since the scale factors are derived using only a smaller part of the dataset, it represents a rather pessimistic version of the p value, since subtracting degrees of freedom will reduce the latter.

11.1. First step: prefit agreement in selection and BDT input variables

As a first step in the unboxing procedure, we look at the data-MC agreement in our selection variables, in order to make sure we do not introduce a bias by cutting on any of those variables. The following plots present the event-level, tag-side and signal-side selection variables as well as BDT input variables separately for the case of reconstructing De , $D\mu$, D^*e and $D^*\mu$ events.

11.1.1. Event-level preselection

The event-level preselection variables include the second-order Fox-Wolfram moment, which appears to be modelled sufficiently well. If there is still a slight shift in data vs. MC, this will affect signal and normalisation in the same way as checked in Subsec.10.10.1. Further, it comprises some selection variables defined in the skimming procedure. A perfect modelling of those is not essential since the distributions mainly start beyond where the cuts are placed.

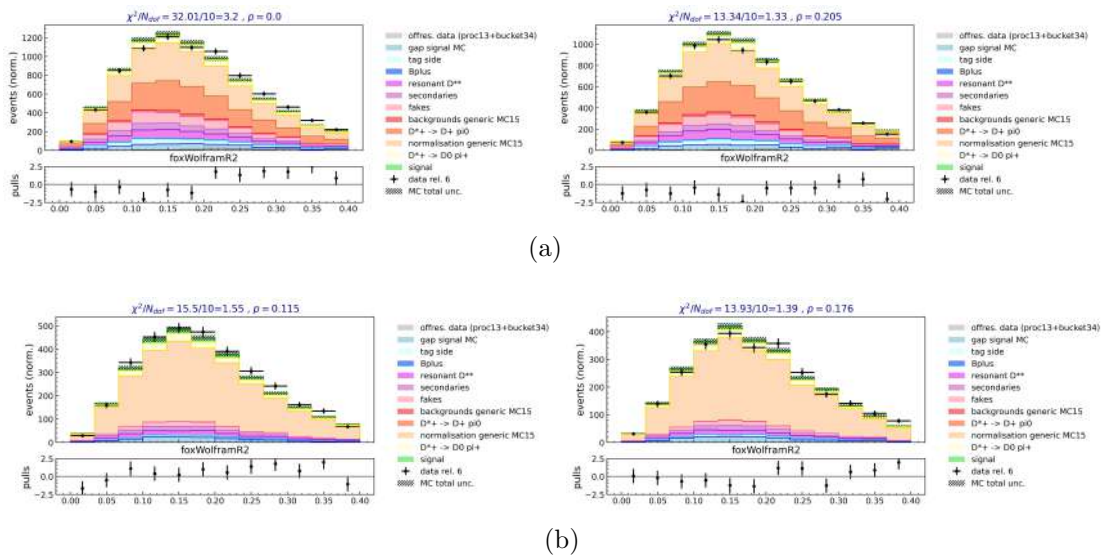


FIG. 56: Distribution of the second-order Fox-Wolfram moment in data vs simulation, separately for signal-side electrons (left) and muons (right), for the D mode (a) and D^* mode (b). The selection is $j0.4$.

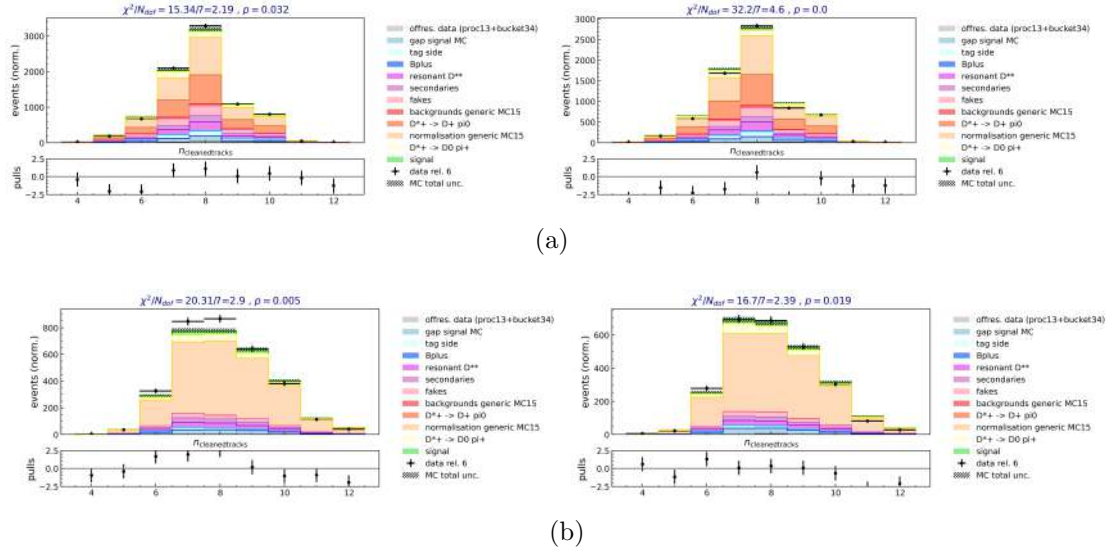


FIG. 57: Same as Fig.56, but for $n_{\text{cleantracks}}$. Although the cut is at > 2 here, the distribution is shown for $n_{\text{cleantracks}} > 3$ since there are no events with 3 tracks (completeness constraint)

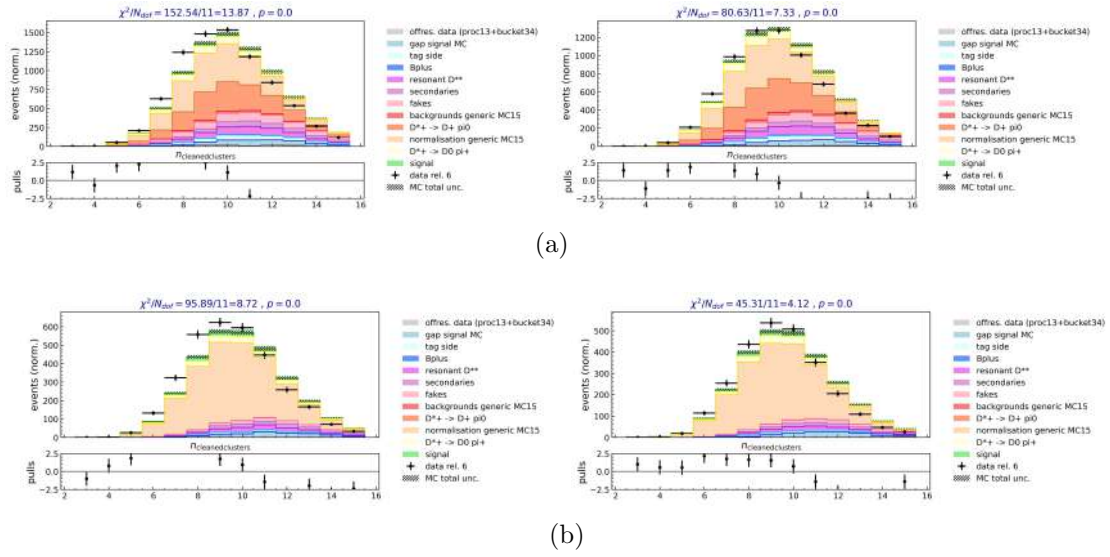


FIG. 58: Same as Fig.56, but for $n_{\text{cleanECLclusters}}$. The selection is > 2 .

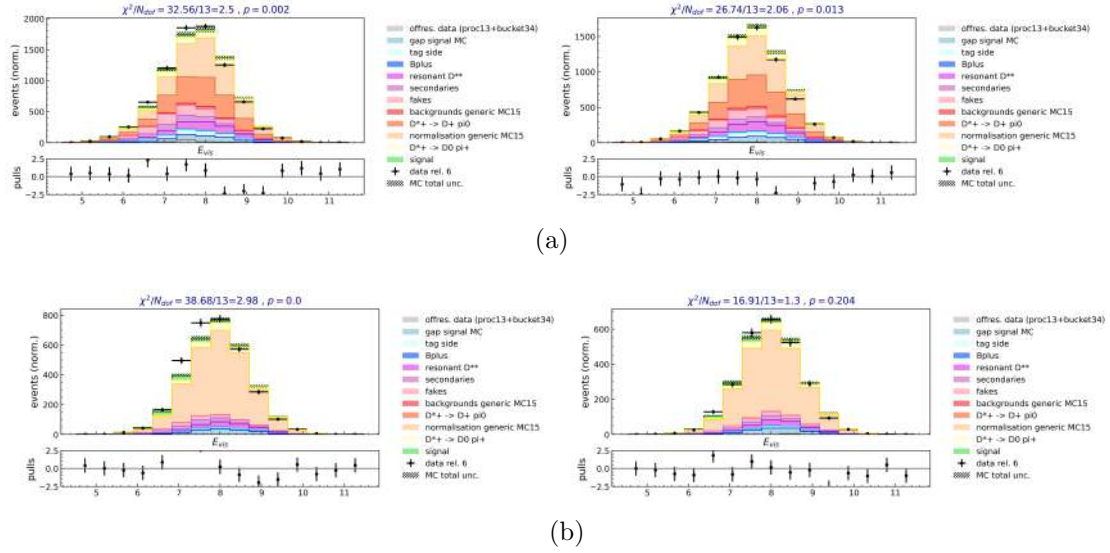


FIG. 59: Same as Fig.56, but for E_{vis} . The selection in the FEI skims is at $\sqrt{s} = 4$ GeV and is not tightened any further.

11.1.2. Tag-side selection

On the tag side, there are several cuts already defined in the skimming procedure, most of which are tightened in this analysis. While tightening the cut on the signal probability to $\log_{10} \mathcal{P} > -1$ leads to the fact that the B_{tag}^0 decay modes of $D^* \pi e \nu$ and $D^* \pi \mu \nu$ do not contribute anymore, considering the p_l^* spectrum of the modes $D \pi e \nu$ and $D \pi \mu \nu$ (cf. Fig.61) motivates our decision to exclude those modes as well. In addition to the fact that those modes predominantly consist of bad tags, they barely contribute any signal events, thus justifying this decision. The cut on the tag-side DMID is therefore tightened to < 4 (which is already applied in the other plots here). As can be seen in the DMID distributions, this does not have a large impact on the analysis, since the total number of events in these modes is very small. Further, those decay modes with IDs ≥ 4 are reconstructing a charged pion together with a D or D^* meson, which might lead to the reconstruction of the phase-space modelled generic “gap” modes on the tag side. As those modes are replaced on the signal side, removing them on the tag side as well leads to more consistency here.

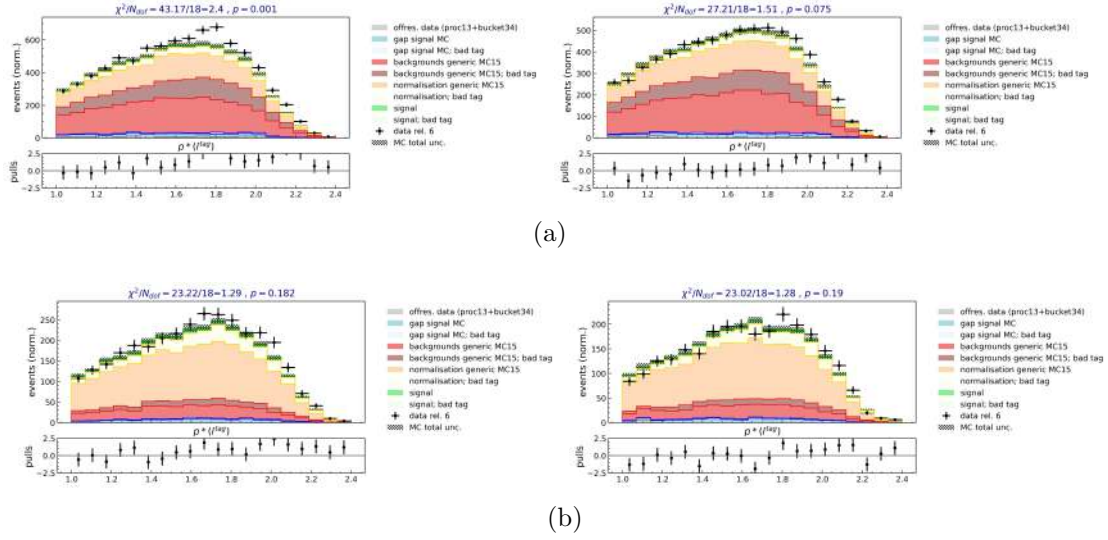


FIG. 60: Same as Fig.56, but for $p_{l^*}^*$ in case of $DMID < 4$. The cut in the FEI skins is at $p_{l^*}^* > 1$ and is not tightened any further.

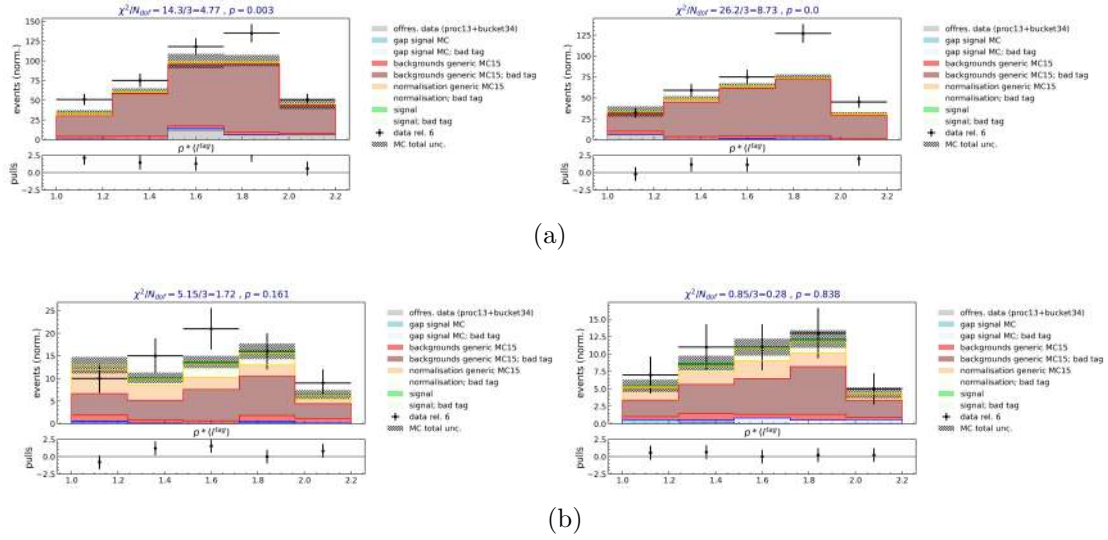


FIG. 61: Same as Fig.56, but for $p_{l^*}^*$ in case of $DMID \geq 4$. These modes are not used in the fit anymore.

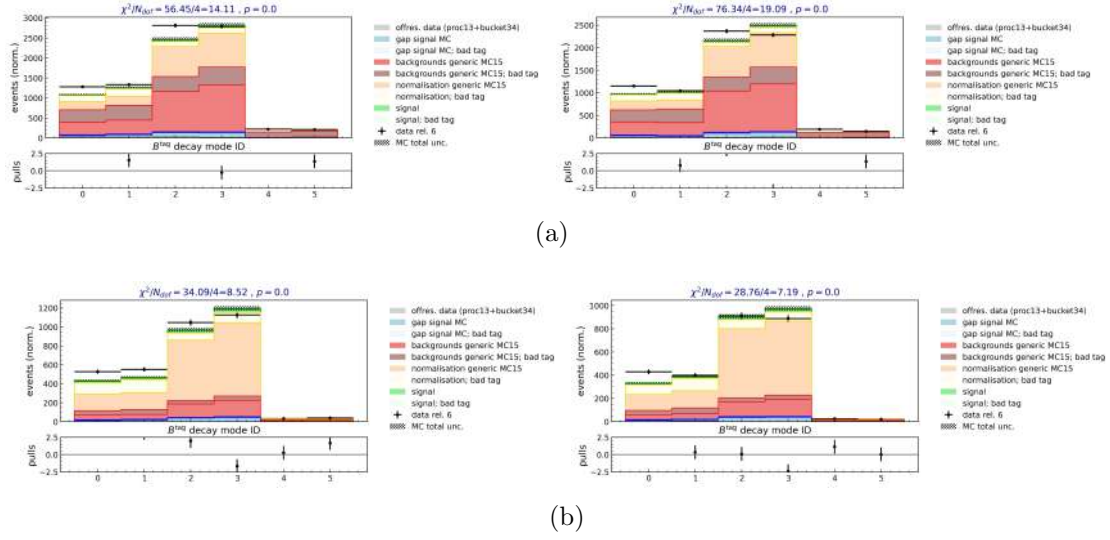


FIG. 62: Same as Fig.56, but for DMID. Note that only DMIDs 0-4 are used in the analysis.

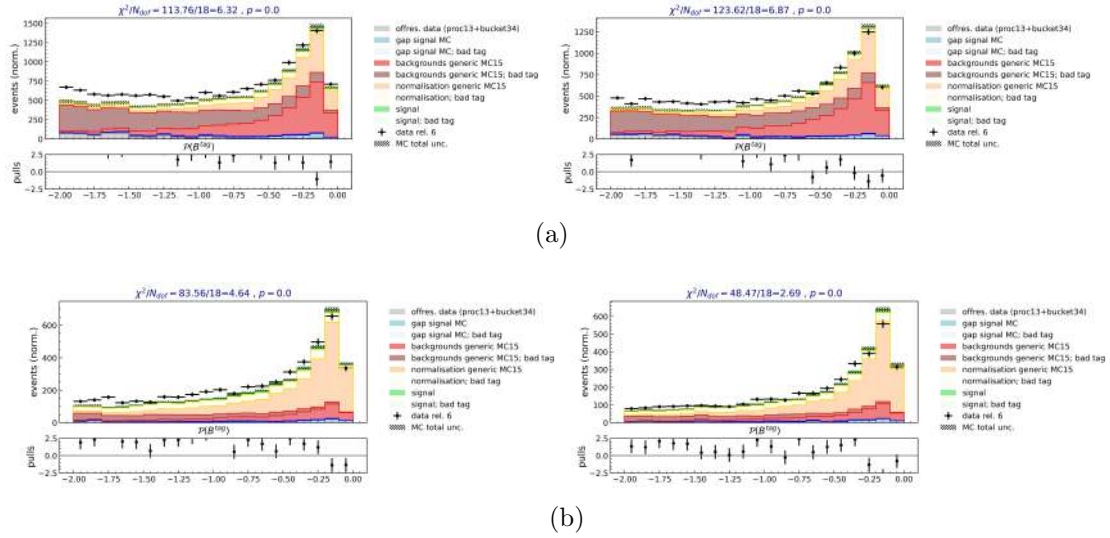


FIG. 63: Same as Fig.56, but for $\log_{10} \mathcal{P}_{\text{tag}}$. Although the cut is at $\log_{10} \mathcal{P}_{\text{tag}} > -1$ here, the entire distribution is shown as the cut on signal probability is tightened in the online selection.

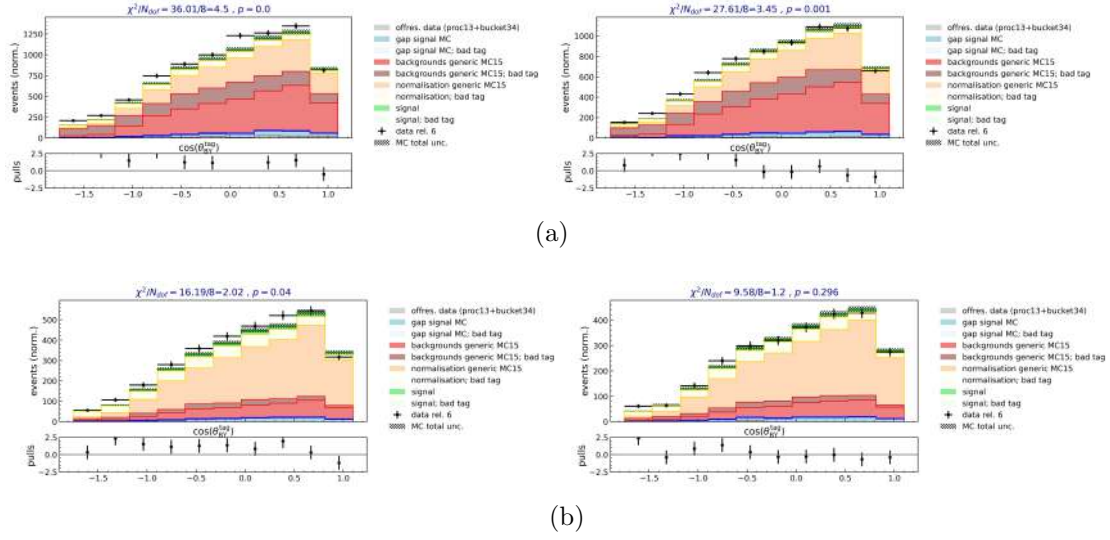


FIG. 64: Same as Fig.56, but for $\cos\theta_{B^{\text{tag}}}$. The cut window is from -1.75 to +1.1

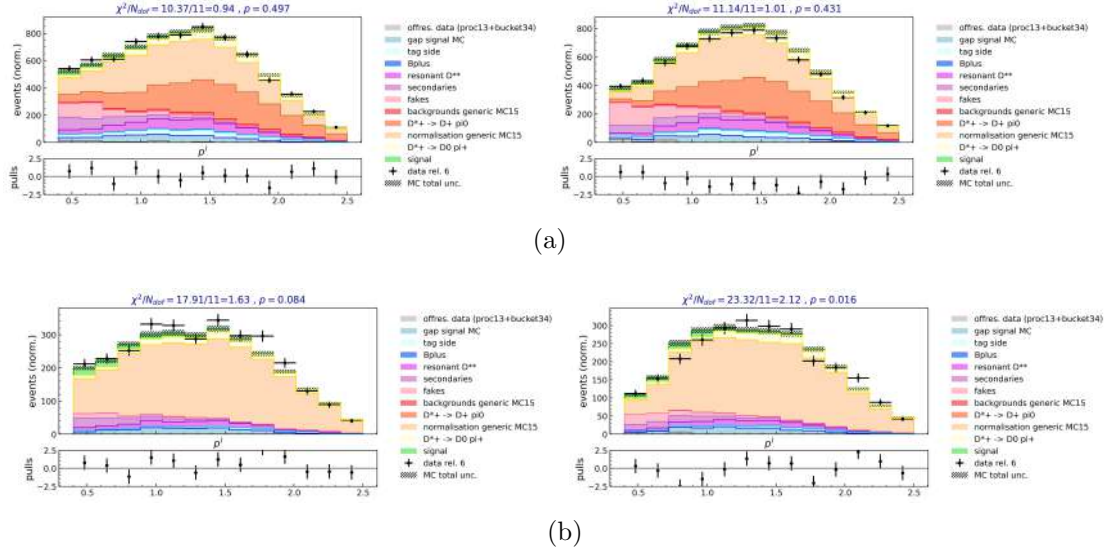


FIG. 65: Same as Fig.56, but for p_l . The selection is $p_e > 0.2 \text{ GeV}$ or $p_\mu > 0.4 \text{ GeV}$, respectively

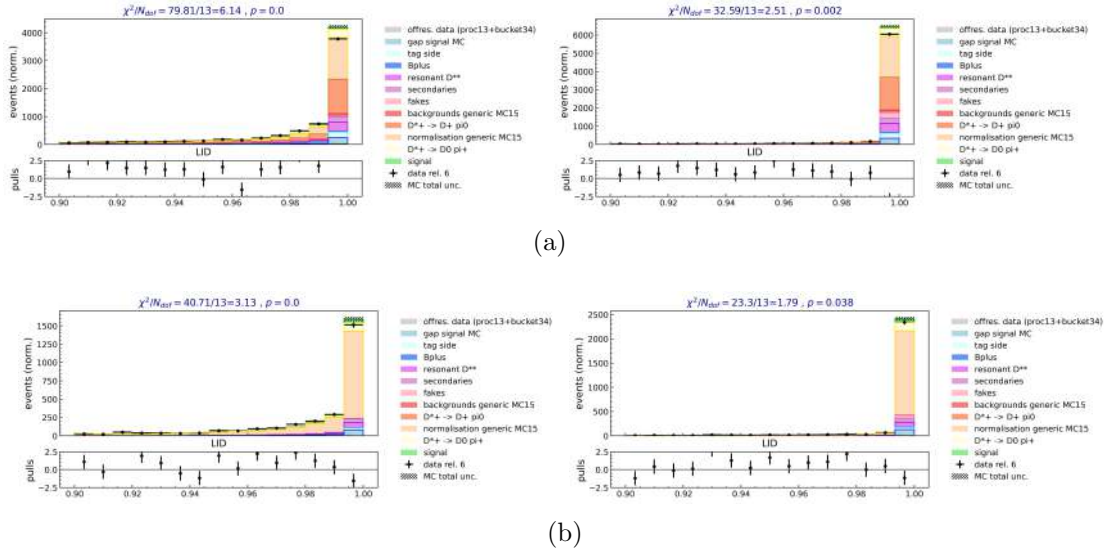
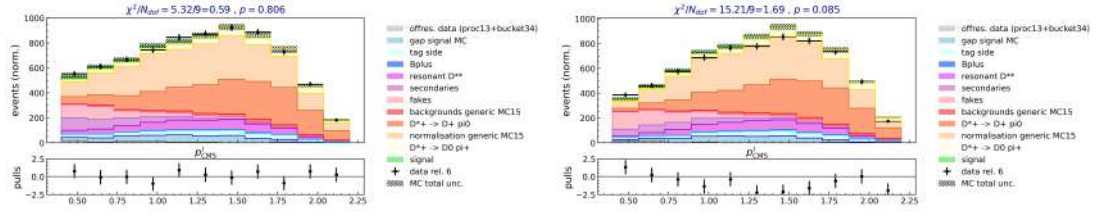
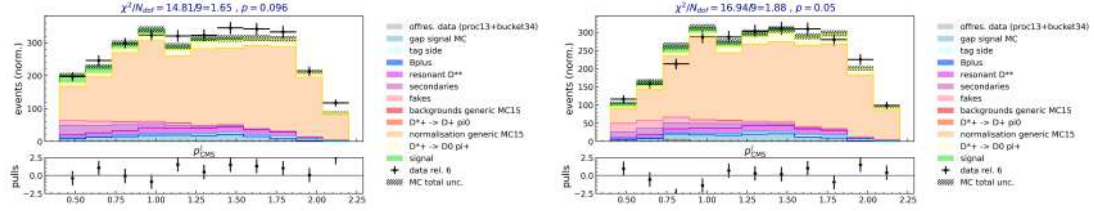


FIG. 66: Same as Fig.56, but for lepton ID. The selection is > 0.5 for electrons and > 0.9 for muons

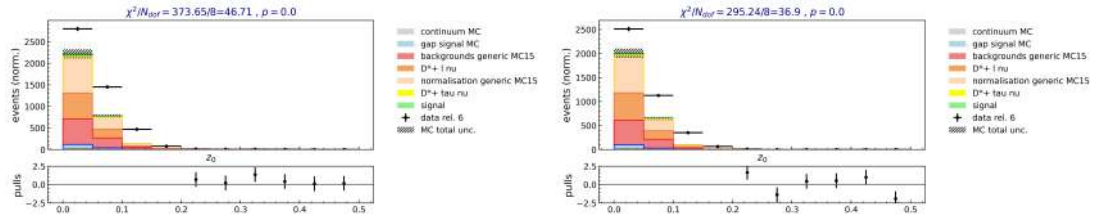


(a)

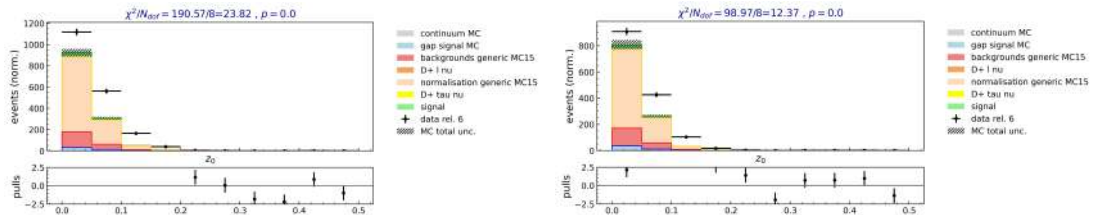


(b)

FIG. 67: Same as Fig.56, but for p_l^* . This is one of the input variables for the BDT.



(a)



(b)

FIG. 68: Same as Fig.56, but for z_0^l

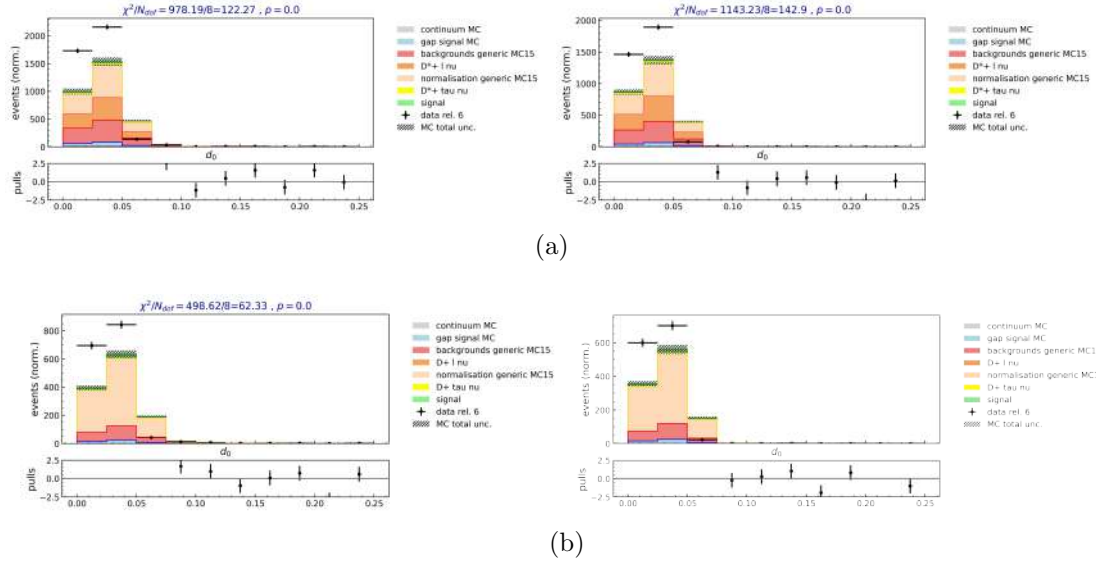


FIG. 69: Same as Fig.56, but for d_0^l

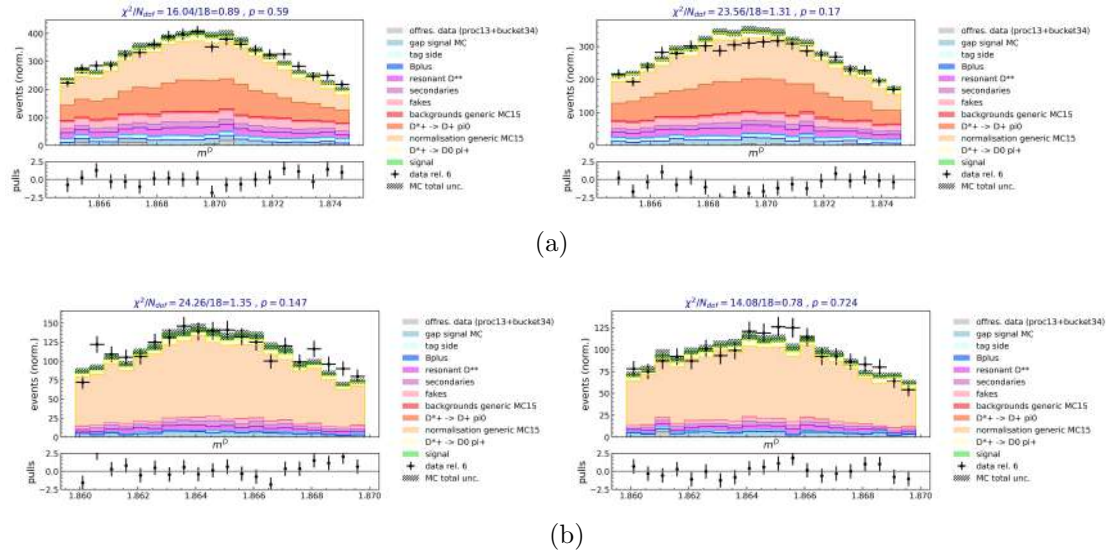


FIG. 70: Same as Fig.56, but for m_D . The plot range is identical to the cut window.

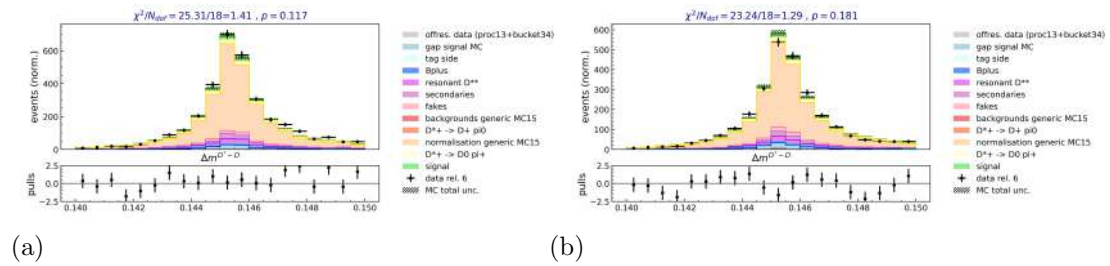


FIG. 71: Same as Fig.56, but for Δm_{D^*-D} . The plot range is identical to the cut window.

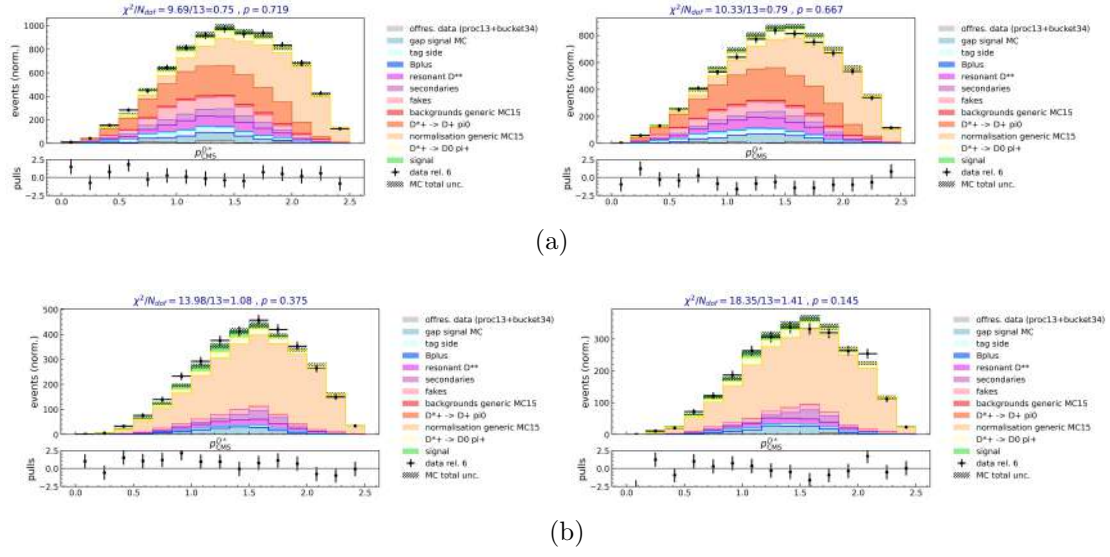


FIG. 72: Same as Fig.56, but for $p_{D^{*+}}^{sig}$. This is one of the input variables for the BDT.

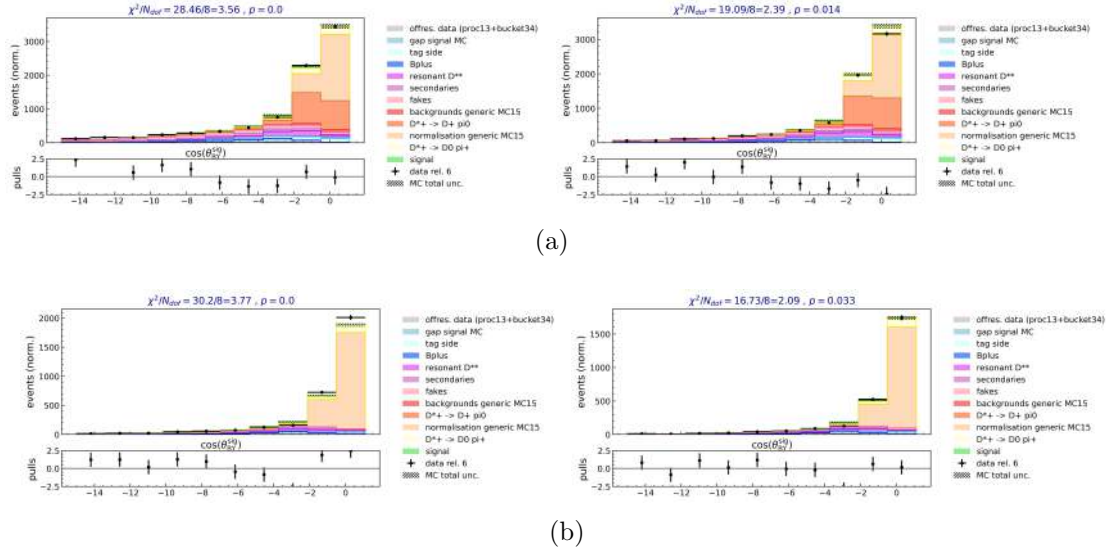


FIG. 73: Same as Fig.56, but for $\cos \theta_{BY}^{sig}$. The selection window is from -15 to +1.1. It is also one of the input variables for the BDT.

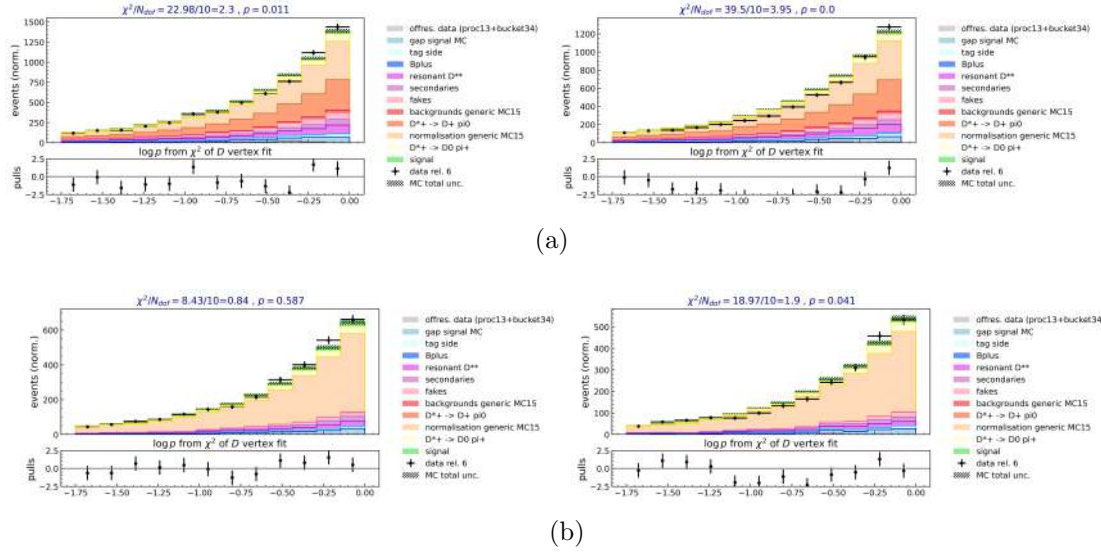


FIG. 74: Same as Fig.56, but for $\mathcal{P}(\chi_{\text{DTreeFit}}^2)$. This variable is used for the Best Candidate Selection on the signal side, i.e. the signal B^0 candidate with the highest value of $\mathcal{P}(\chi_{\text{DTreeFit}}^2)$ is selected. Thus, this BCS has not been applied for these plots. The x axis is logarithmic to better investigate those events with large values of $\mathcal{P}(\chi_{\text{DTreeFit}}^2)$.

11.1.4. Whole event after selecting the signal side

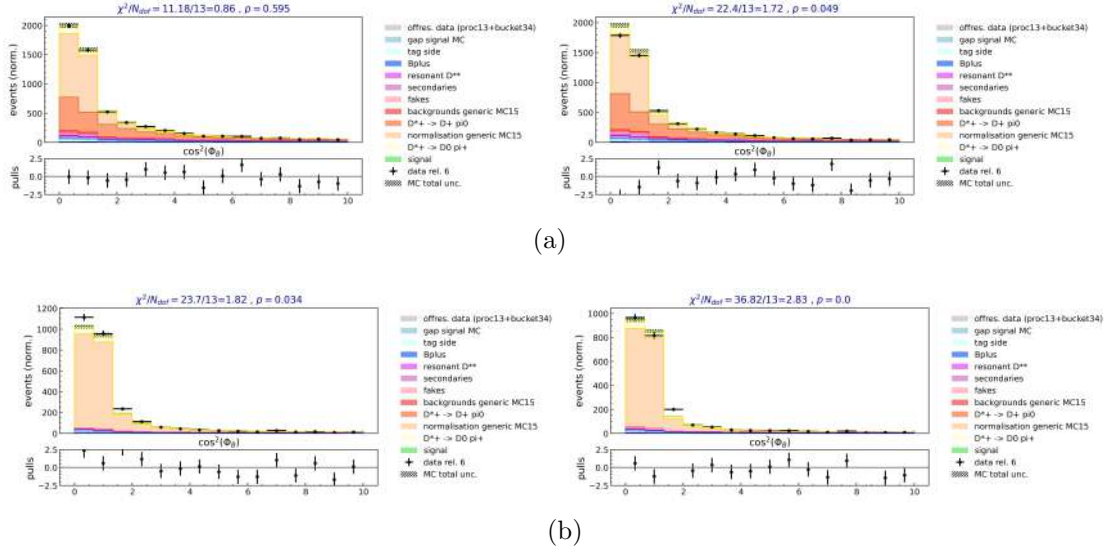


FIG. 75: Same as Fig.56, but for $\cos^2 \Phi_B$. This is one of the input variables for the BDT.

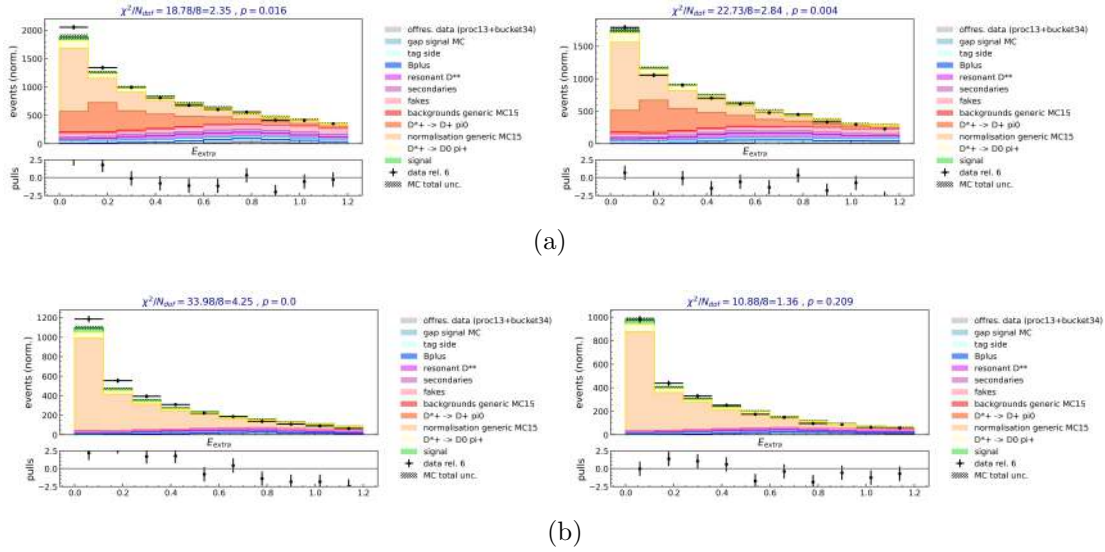
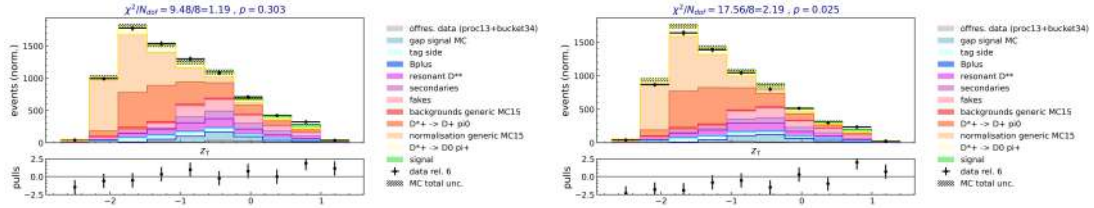


FIG. 76: Same as Fig.56, but for E_{extra} . The cut is at 1.2 GeV. This is also one of the input variables for the BDT.

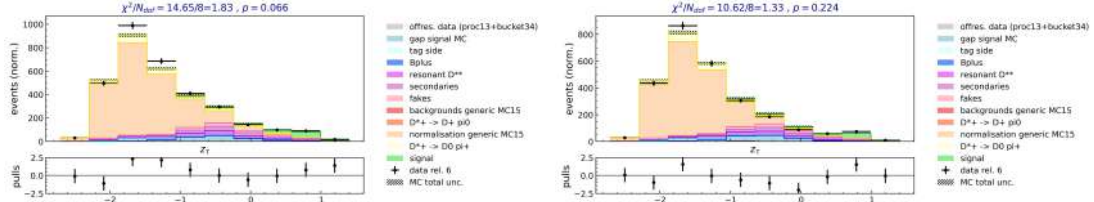
11.2. Second step: prefit agreement in BDT output variables

See <https://indico.belle2.org/event/9806/contributions/63535/attachments/>

23433/34681/SLRDstar_unboxed_BDToutput_distributions.pdf

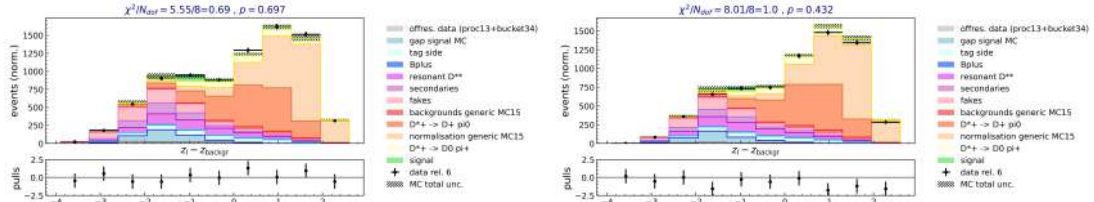


(a)

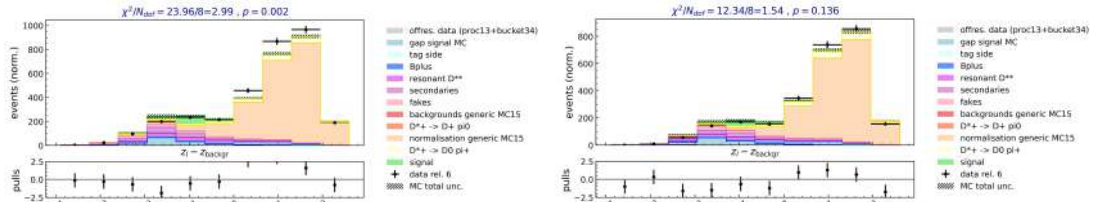


(b)

FIG. 77: Same as Fig. 56, but for z_τ



(a)

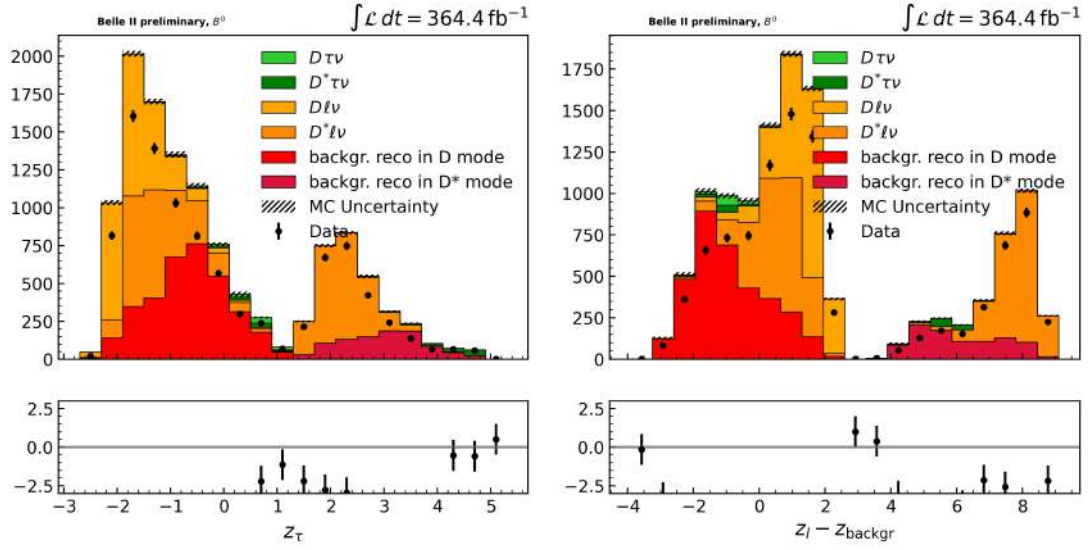


(b)

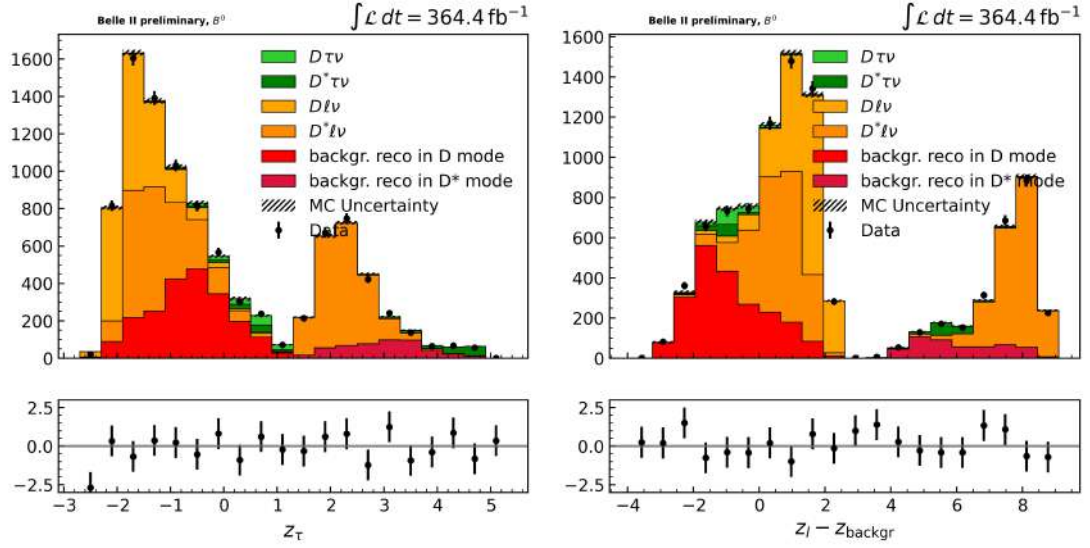
FIG. 78: Same as Fig. 56, but for z_{diff}

11.3. Third step: fit in the muon mode with $\mathcal{R}(D^{(*)})$ results blinded

See https://indico.belle2.org/event/9833/contributions/63687/attachments/23580/34839/SLRDstar_blinded_fit_muon_mode.pdf



(a)



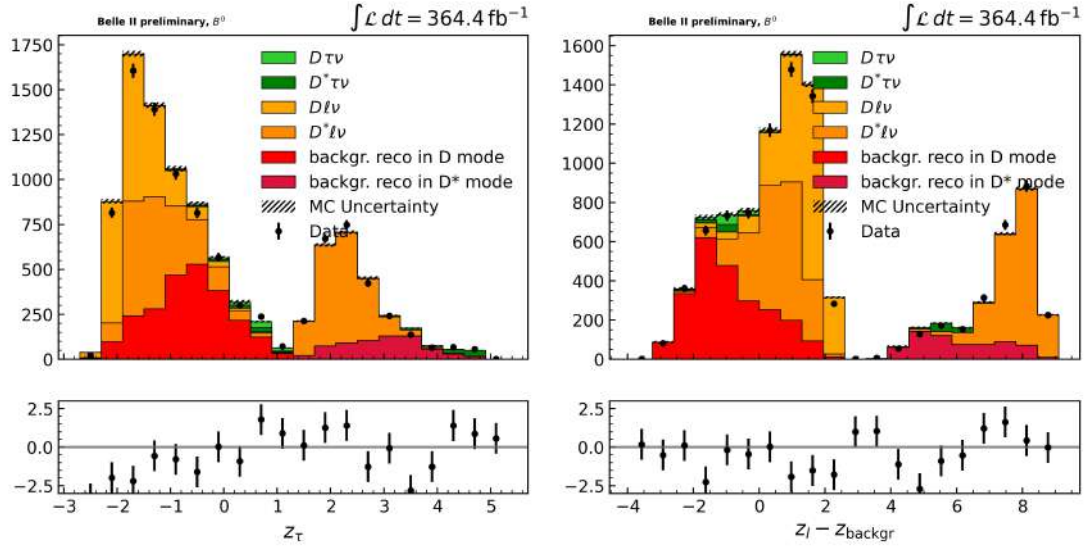
(b)

FIG. 79: One-dimensional projections onto the axes of the two fitvariables:

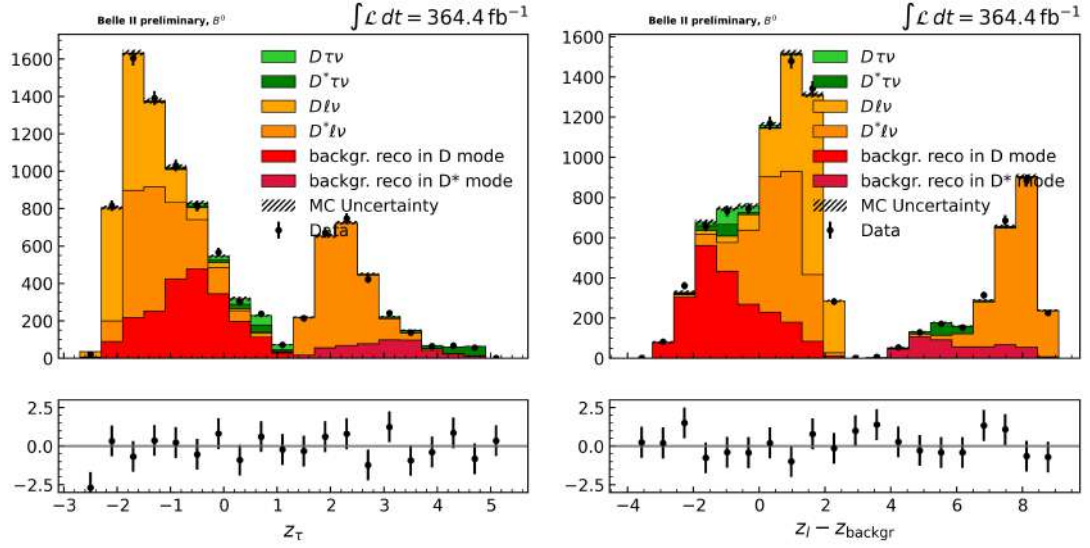
(a) prefit

(b) postfit

The prefit plots are without applying the derived FEI efficiency scale factors.



(a)



(b)

FIG. 80: Same as Fig. 79, but the prefit plots are **with** application of the derived FEI efficiency scale factors.

11.4. Fourth step: fit in the electron mode with $\mathcal{R}(D^{(*)})$ results blinded

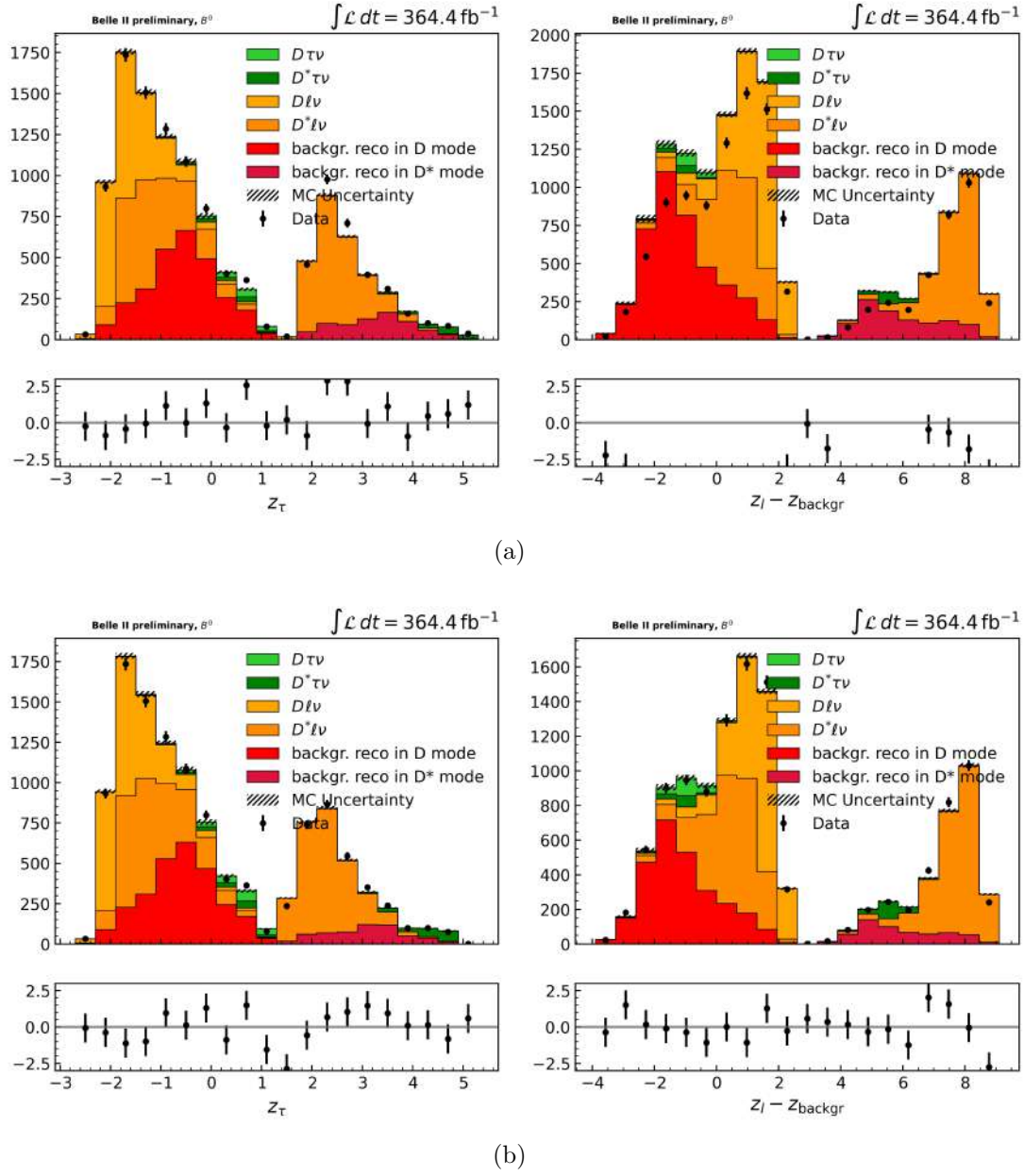
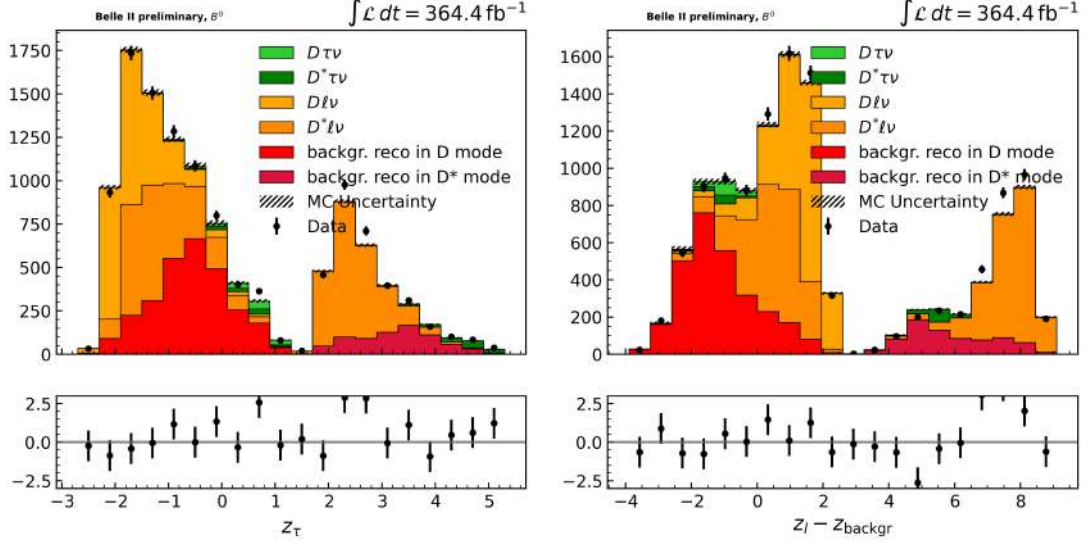


FIG. 81: One-dimensional projections onto the axes of the two fitvariables:

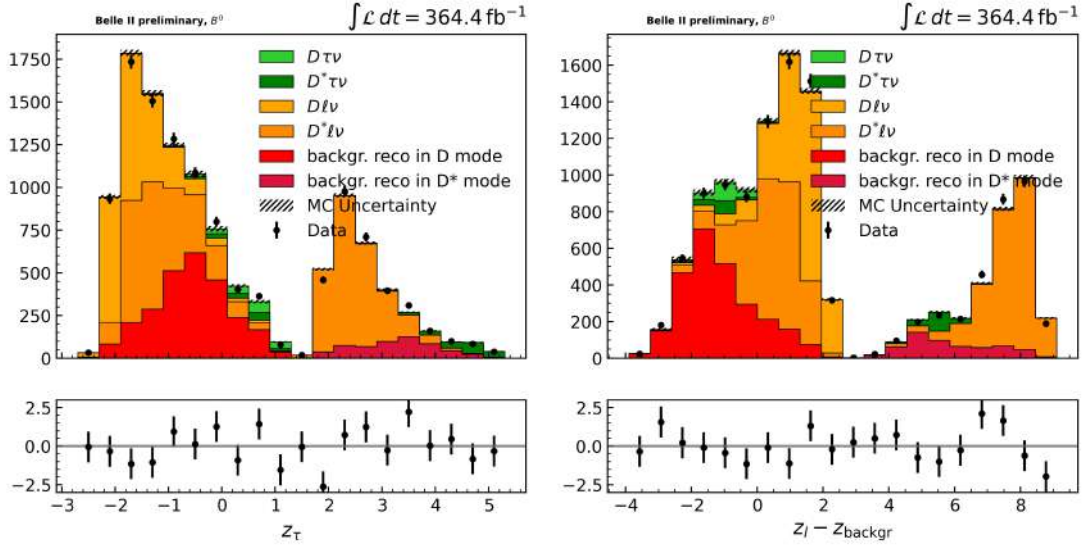
(a) prefit

(b) postfit

The prefit plots are **with** application of the derived FEI efficiency scale factors.



(a)



(b)

FIG. 82: One-dimensional projections onto the axes of the two fitvariables:

(a) prefit

(b) postfit

The prefit plots are **with** application of the derived FEI efficiency scale factors.

11.5. Repetition of step 3 and 4 with new binning as well as bin-wise NPs

Using the asymmetric binning as well as the bin-wise NPs, we can evaluate the pre- and postfit plots when fitting data (cf. Figs.83-84) as well as the pre- and post-fit χ^2/N_{dof} values obtained when taking into account the NPs:

$$\chi^2/N_{\text{dof}}(e; \text{post}) = 0.744 \quad \rightarrow \quad p = 0.860 \quad (40)$$

$$\chi^2/N_{\text{dof}}(\mu; \text{post}) = 0.603 \quad \rightarrow \quad p = 0.967 \quad (41)$$

The post-fit agreement seems reasonable and also the post-fit χ^2/N_{dof} is reasonably small, leading to sufficiently high p values.

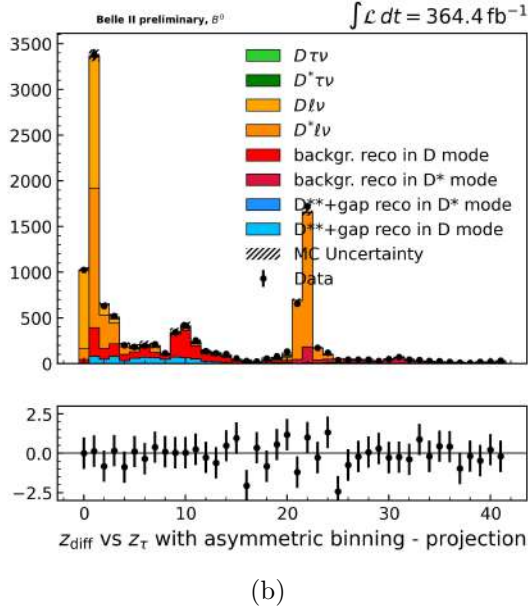
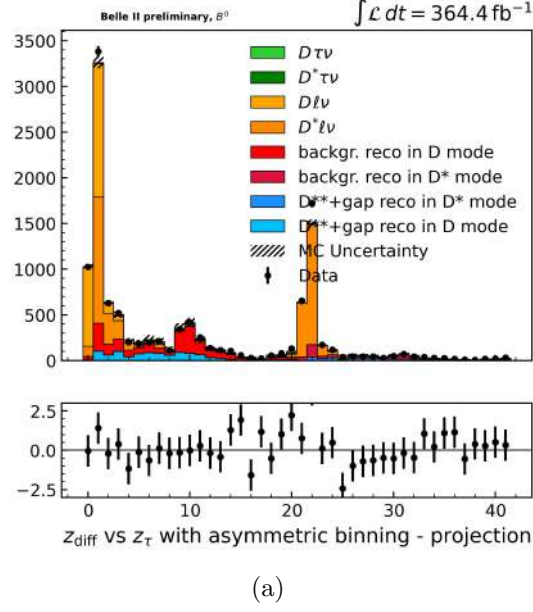


FIG. 83: Histograms showing the number of events in each of the bins defined in table XI For the D mode (left side of the histogram), the number on the x axis corresponds to the bin number as written in the table; for the D^* mode (right side of the histogram), these numbers are shifted by the total bin number. The pre- (a) and postfit (b) histograms are for the electron mode.

While the results for the signal and normalisation yields are still blinded here, it is possible to already evaluate the post-fit background and $D^{**}+\text{gap}$ yields. The following results are obtained for the electron (muon) mode wrt. the non- D^{**} backgrounds:

$$N_{\text{back}}^{(D;\text{pre})} = 1936 \pm 76 \quad (1509 \pm 38) \quad \rightarrow \quad N_{\text{back}}^{(D;\text{post})} = 1979 \pm 179 \quad (1875 \pm 179) \quad (42)$$

$$N_{\text{back}}^{(D^*;\text{pre})} = 460 \pm 14 \quad (409 \pm 12) \quad \rightarrow \quad N_{\text{back}}^{(D^*;\text{post})} = 581 \pm 127 \quad (632 \pm 140) \quad (43)$$

The post-fit yields deviate from the pre-fit ones by 1σ (2σ) of the respective fit result.

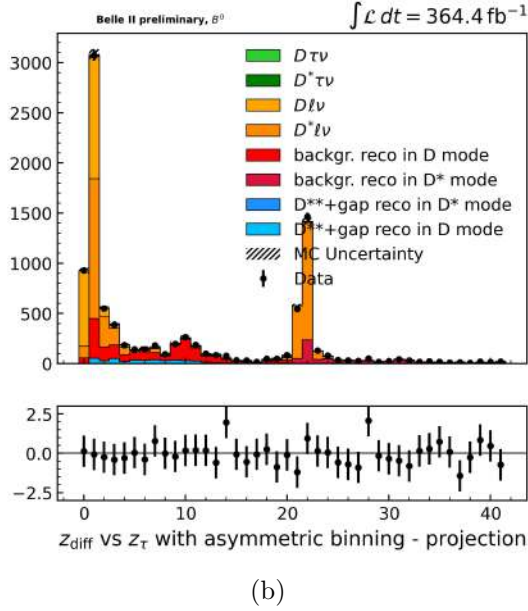
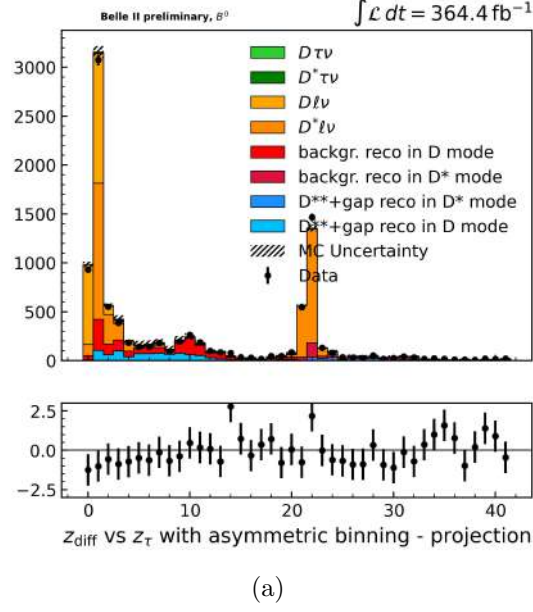


FIG. 84: Same as Fig.83, but for the muon mode.

For the $D^{**} + \text{gap}$ template, the fit adjusts the yields in the following way:

$$N_{\text{back}}^{(D;\text{pre})} = 970 \pm 402 \quad (874 \pm 355) \quad \rightarrow \quad N_{\text{back}}^{(D;\text{post})} = 751 \pm 151 \quad (433 \pm 136) \quad (44)$$

$$N_{\text{back}}^{(D^{*};\text{pre})} = 282 \pm 152 \quad (242 \pm 136) \quad \rightarrow \quad N_{\text{back}}^{(D^{*};\text{post})} = 83 \pm 74 \quad (25 \pm 72) \quad (45)$$

Since we attribute a 100% uncertainty to the BF of the gap events, the fit may as well use the NPs in order to adjust the gap yield. Thus, we study further the influence of the assumed gap BF uncertainty on the fitted yields. The plots in Fig.85 illustrate the post-fit $D^{**} + \text{gap}$ yields together with their uncertainties as errorbars when provide these events with different prefit gap BF uncertainties. Specifically, we test the nominal case of an uncertainty of 100% against the cases of no uncertainty at all, an uncertainty of 50% as well as an uncertainty of 150%. The mean of those yields is shown as a dashed line.

Further, we perform a fit in which the D^{**} +gap template contains double the gap events as well as a fit in which this template is made up of only the D^{**} events while omitting all gap events. The difference between the yields which we obtain in both cases is shown as a blue band around the mean. Apart from that, a the results of a fit is shown in which all the D^{**} +gap bin-parameters are fixed to 0 (grey). This case is slightly different from the one with zero gap BF uncertainty as also none of the remaining systematic uncertainties (e.g. D^{**} BF, FFs) may contribute. Further, we repeat this fit but with fixing the NPs to their best-fit value (at 100% gap BF uncertainty) instead of to their pre-fit value (dark grey). All data points can be seen to agree with each other and are situated inside the blue band, indicating an independency of the fit results for the D^{**} +gap yields of the input gap BF uncertainty. In addition, the pre-fit value for the respective yields is shown in grey and a different shading indicates the pre-fit yield uncertainties in terms of standard deviations. This illustrates that those pre-fit yields agree with the post-fit ones within 1σ (De), 1.5σ (D^*e and $D\mu$), as well as 2σ ($D^*\mu$). The down-scaling of this component is in line with what had already been observed in other $\mathcal{R}(D^{(*)})$ analyses. The yields in case of the D^* modes agree with the corresponding fitted yield of the recent hadronically tagged $\mathcal{R}(D^*)$ analysis[35] of 33.8 ± 19.2 within their, however very large, uncertainties.

Figs.86-87 show the NP pulls for the different event categories. For the muon mode, we observe the same constrained background NPs in bins 13 and 9 of the D and D^* modes, respectively, that had already been found in the asimov fit (cf. Section8.8.6), which, beyond that, are strongly pulled into one direction. The origin of this constraint is the following: The background template consists of several different sources of $B\bar{B}$ backgrounds from generic MC (2000fb^{-1}) as well as of continuum events estimated from offresonant data (42fb^{-1}). In order to scale the continuum events accordingly to the data luminosity of 364fb^{-1} , a large weight has to be applied. Due to the selections applied, not many of the offresonant events remain. Fig.88 shows the 2D distributions only for those offresonant events with the binning overlaid, similar to Fig.25, revealing that there is only one offresonant event in the $D^*\mu$ mode, situated in bin number 9. In the $D\mu$ mode, bin number 13 contains two offresonant events. The constraints are caused by the large weight that those events have. Further, the pulls can be explained by the fit pulling on the NPs in order to adjust the continuum yield relative to the one of the remaining backgrounds.

In order to confirm this hypothesis, another fit has been performed in which the continuum template was separated from the remaining backgrounds. A constraint was added on its yield in order to ensure convergence of the fit. See Fig.89 for the resulting NP pull distributions for the backgrounds as well as the separate continuum template. Indeed, these do not show the same constraints in bin 9 and 13. The D^* binparameters for the continuum template are not constrained or pulled at all since this is only one event for which the yield can be adjusted. The D binparameters still show a pull in bin 13 - there may be too much continuum in this bin relative to the other ones, such that the fit has to adjust this via the NPs here. Knowing the origin of the background NP behaviour, we aim to continue using the previously described one as the nominal fit, since a separate continuum template with so few events might make the fit unstable.

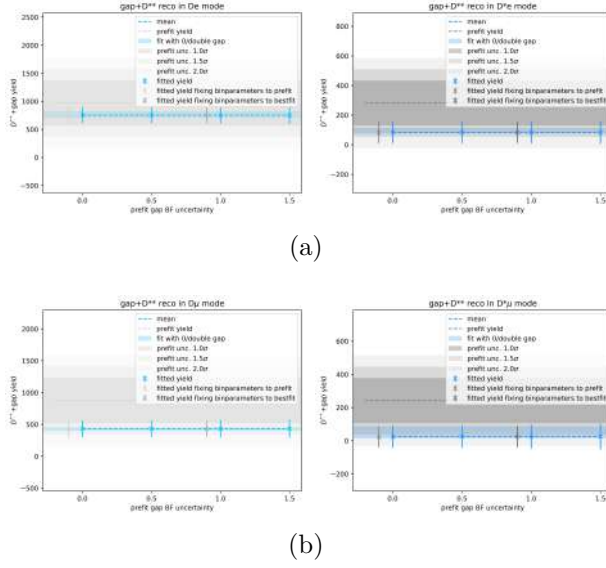
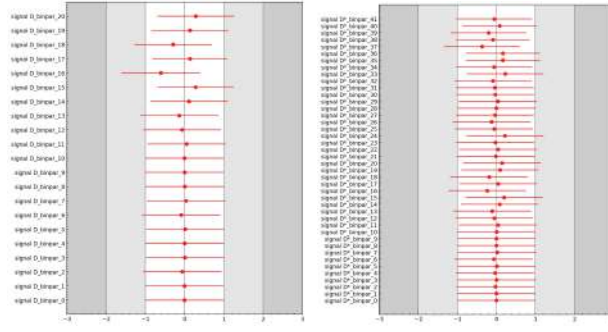


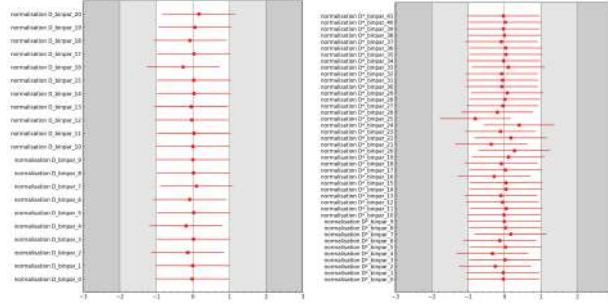
FIG. 85: Fitted yields for the $D^{**}+\text{gap}$ component when assuming different pre-fit gap BF uncertainties (blue crosses). The uncertainties on the respective yields obtained from the fit are displayed as an errorbar. The mean of these yields is indicated as a dashed line. Additionally, the same yields are shown when fixing the $D^{**}+\text{gap}$ bin-parameters to their pre-fit values (i.e. 0; light grey) and to their best-fit values (i.e. assuming a gap BF uncertainty of 100%; dark grey). The fits with double the gap events as well as without all events of the latter category lead to the blue band around the dashed line by subtracting the corresponding fit results for the $D^{**}+\text{gap}$ template. In the latter case, no gap BF uncertainty is presumed. Finally, the grey dashed line shows their pre-fit yields and the uncertainty from MC expectation in terms of standard deviations using different shadings. The D mode is shown on the left and the D^* mode on the right:

(a) electron mode

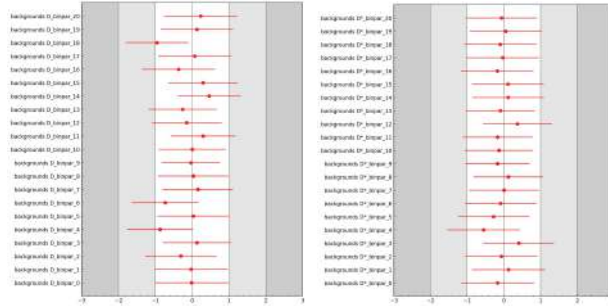
(b) muon mode



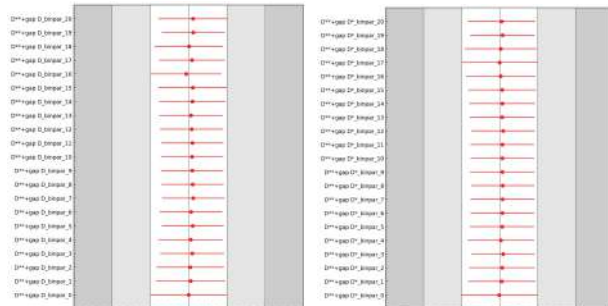
(a)



(b)



(c)

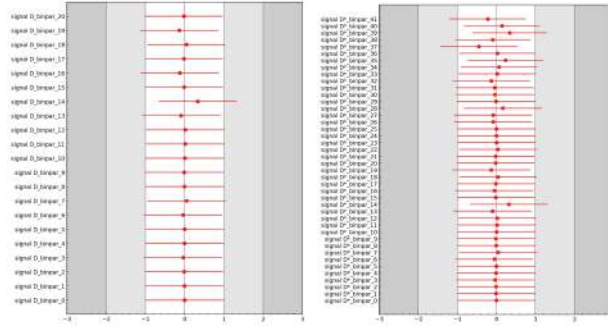


(d)

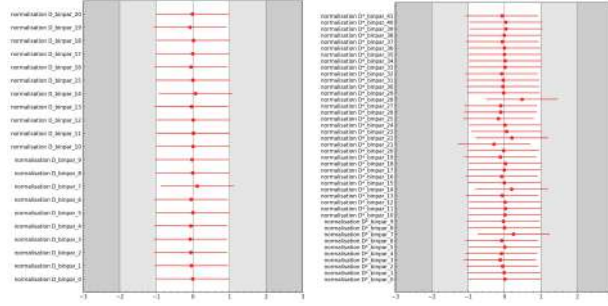
FIG. 86: NP pull distributions when performing a fit to data for:

- (a) signal
- (b) normalisation
- (c) backgrounds
- (d) $D^{**} + \text{gap}$.

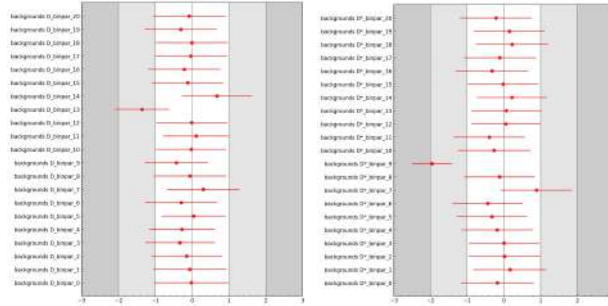
This corresponds to Fig.29, where the pull distributions for asimov data are shown. The pull distributions in this figure are for the electron mode.



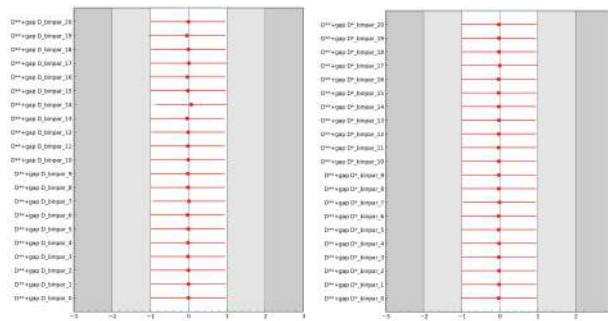
(a)



(b)

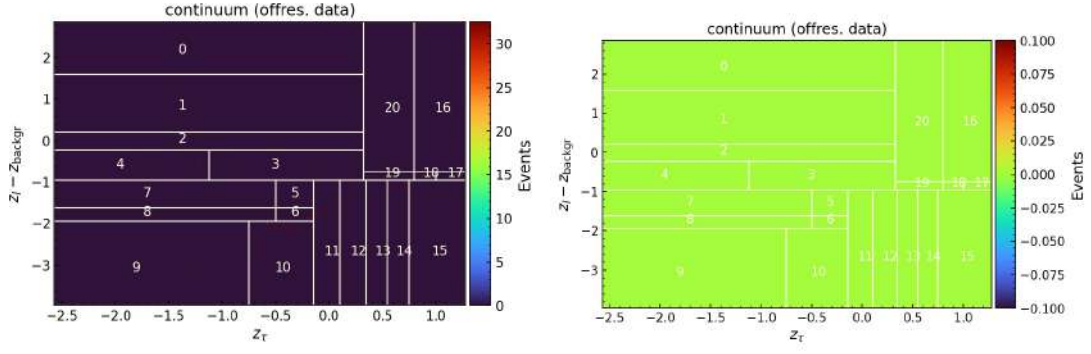


(c)

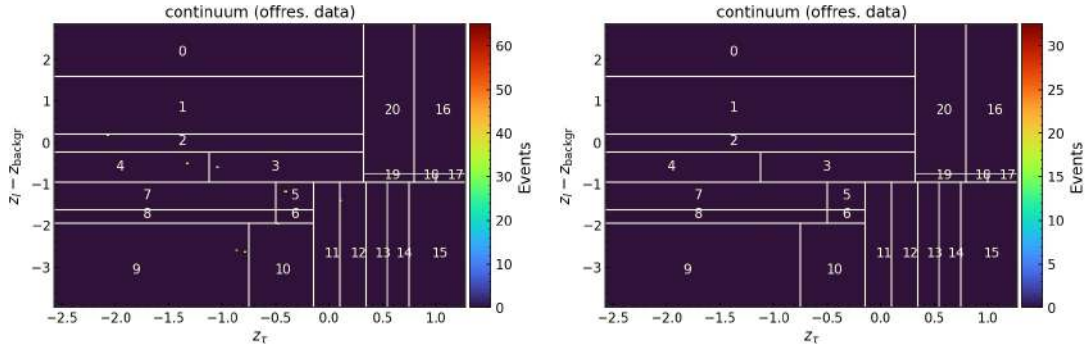


(d)

FIG. 87: Same as Fig.86, but for the muon mode (corresponding to Fig.30)



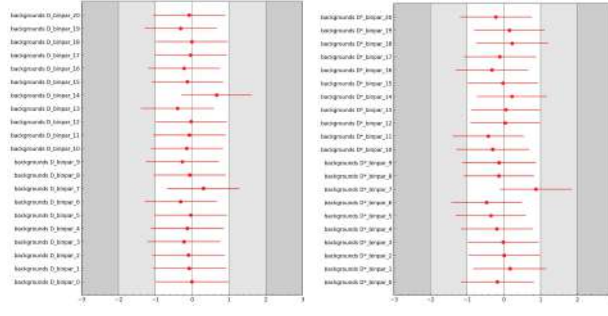
(a)



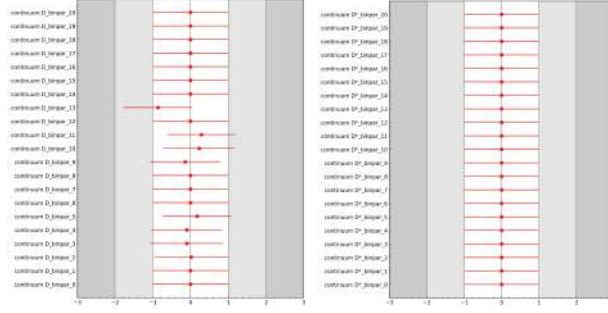
(b)

FIG. 88: Offresonant events used for continuum modelling in 2D-classifier space with the asymmetric binning overlaid, showing the D mode on the left and the D^* mode on the right:

- (a) electron mode (in this case, the D^* mode does not provide any offresonant events)
- (b) muon mode.



(a)



(b)

FIG. 89: NP pull distributions when performing a fit with a separate continuum template:

(a) remaining backgrounds

(b) continuum

These pull distributions are for the muon mode, in which NP pulls as well as constraints have been observed in bin 13 for the D mode and bin 9 for the D^* mode.

11.6. Fit with source-wise NPs using pyhf

We decided to transition from a bin-wise to a source-wise implementation of nuisance parameters due to several significant advantages.

- It facilitates a more precise estimation of the impact from each individual source of systematic uncertainty. By defining the NPs source-wise, it is easier to construct models where these parameters are (nearly) orthogonal, meaning their effects on the final results are not as correlated as in the case of bin-wise NPs. This method enhances our ability to accurately quantify and manage the contributions from different sources.
- We observe asymmetric likelihood curves for the signal and normalisation yields when performing the fit with the bin-wise NP implementation, see Fig. 90. These yields are used to calculate the ratios $\mathcal{R}(D^{(*)})$ after the fit. Yet, it is not trivial how to propagate asymmetric uncertainties on the yields to the ratios. The `pyhf` implementation, however, enables to fit the ratios directly. What is more, it allows for performing a 2D contour scan over the ratio parameters, while re-minimising wrt. all other parameters, in order to obtain a more accurate confidence ellipse. See Fig.91 for the 2D contour scan obtained in the aforementioned manner in comparison to other measurements.

The source-wise fit is performed using the `pyhf` package, which is based on `HistFactory` and `ROOT`, with a detailed description presented in [32] and [19]. The minimisation is carried out using the `iMinuit` package [40]. The notation used here is adapted to the documentation of the `pyhf` package.

The fit likelihood can be expressed as:

$$f(n, a \mid \eta, \chi) = \prod_{c \in \text{channels}} \prod_{b \in \text{bins}_c} \text{Pois}(n_{cb} \mid \nu_{cb}(\eta, \chi)) \quad (46)$$

$$\times \prod_{\chi \in \mathcal{X}} c_{\chi}(a_{\chi} \mid \chi), \quad (47)$$

Here, n_{cb} corresponds to the number of events observed in data. Further, c denotes the four channels in which the measurement is performed, i.e. the reconstructed decays categorised into De , D^*e , $D\mu$ and $D^*\mu$ final states. The parameter b represents the bins of the histograms in the two-dimensional confidence scores, using the asymmetric binning with 21 bins within each channel described in Section 8.5. The parameter s in the likelihood definition indicates the sample type, i.e. signal $B^0 \rightarrow D^{(*)}\tau\bar{\nu}_{\tau}$, normalisation $B^0 \rightarrow D^{(*)}\ell\bar{\nu}_{\ell}$ with $\ell = e, \mu$, the background from resonant and non-resonant (“gap”) D^{**} decays, and other backgrounds. The expected value of a given bin and category is computed as the sum of the expected values ν_{scb} for the individual samples, which can be modified by either constrained (χ) or unconstrained parameters (η):

$$\nu_{cb}(\phi) = \sum_{s \in \text{samples}} \nu_{scb}(\eta, \chi) = \sum_{s \in \text{samples}} \left(\prod_{\kappa \in \kappa} \kappa_{scb}(\eta, \chi) \right)_{\text{multiplicative modifiers}} \left(\nu_{scb}^0(\eta, \chi) + \sum_{\Delta \in \Delta} \Delta_{scb}(\eta, \chi) \right)_{\text{additive modifiers}} \quad (48)$$

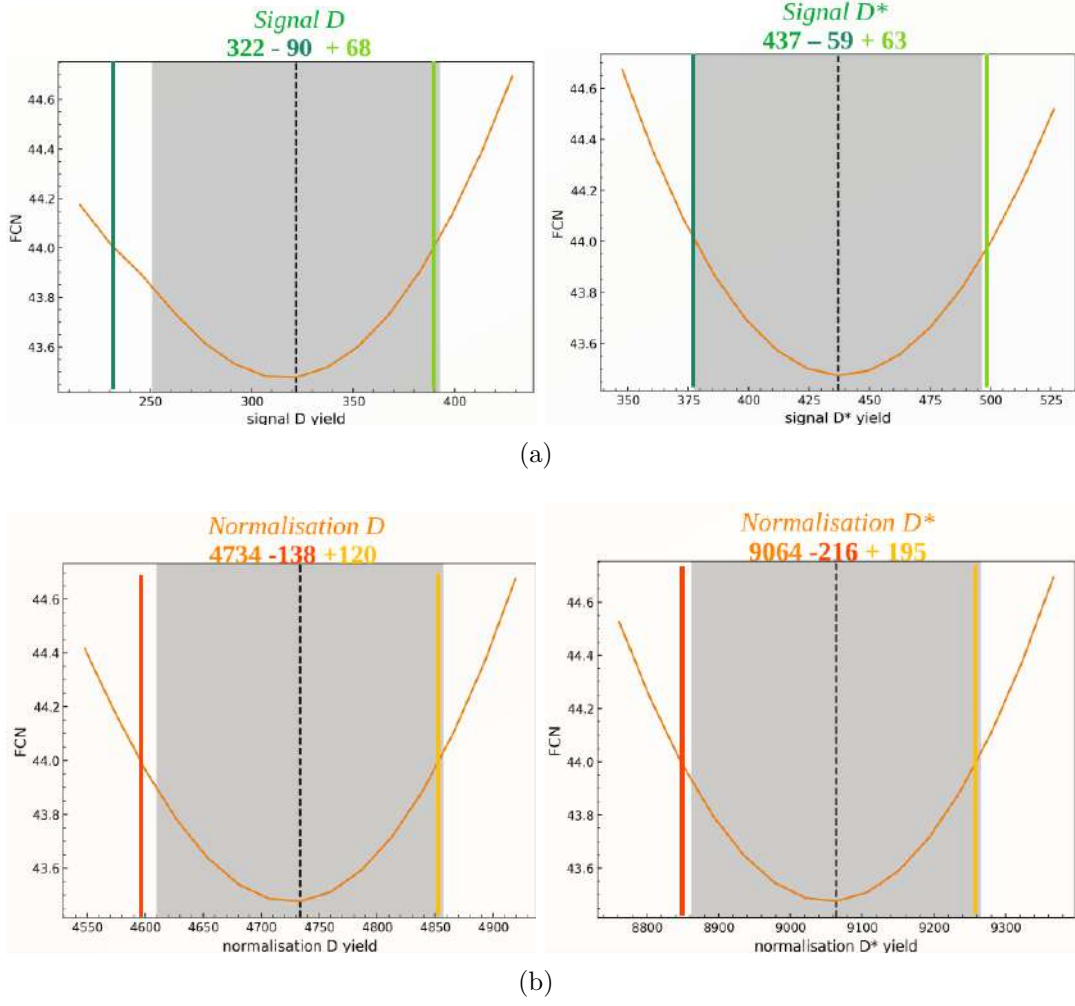


FIG. 90: Likelihood curves when using the bin-wise NP implementation, obtained from a scan over the scale-factors for (a) signal $D^{(*)}$ (b) normalisation $D^{(*)}$.

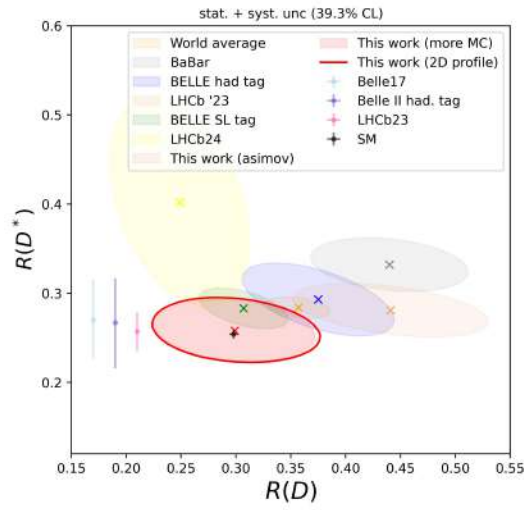


FIG. 91: Confidence ellipse in the $\mathcal{R}(D)$ - $\mathcal{R}(D^*)$ plane (asimov results; darker red: if 7000 fb^{-1} instead of 2000 fb^{-1} was included in the fit) with previous measurements

1439

1440 These modifiers allow for a change of the scaling (“multiplicative modifiers”, κ) as well
 1441 as a correlated shape modification of the histograms (“additive modifiers”, Δ). The term
 1442 “correlated” here refers to the fact that a single modifier is able to alter all the expected
 1443 sample rates in the bins of a certain channel, unlike “uncorrelated” modifiers, which would
 1444 mean having one modifier per bin, with individual constraint terms[46].

1445 They depend upon model parameters $\alpha = \eta, \chi$, with η denoting an unconstrained
 1446 model parameter and χ labelling a constrained one. The unconstrained parameters in
 1447 our fit are related to the normalisations of the four sample types, allowing to change
 1448 their scaling. Specifically, we directly fit the ratios $\mathcal{R}(D)$ and $\mathcal{R}(D^*)$ in addition to scale
 1449 factors for the normalisation ($D^{(*)}\ell\nu$) component in the D and D^* modes, respectively.
 1450 Furthermore, the scale factors for the yields of the background components (D^{**} and
 1451 remaining backgrounds) are completely unconstrained within each channel. The con-
 1452 strained parameters are used to include systematic uncertainties via the terms c_χ and
 1453 are constrained by auxiliary data a_χ . In general, these parameters can affect both the
 1454 shape and the scaling of the histograms. These terms c_χ can be expressed as Gaussian
 1455 constraints:

$$\text{Gauss}(a = 0 \mid \alpha, \sigma = 1), \text{ where :} \quad (49)$$

- 1456 • α denotes the model parameter (NP)
- 1457 • the mean a (auxiliary data) is at zero, which is the value the corresponding NP
 1458 assumes in case of an asimov fit
- 1459 • the width σ denotes the relative uncertainty in the constrained parameter, i.e. the
 1460 relative uncertainty of the event rate.

1461 The overall likelihood function used in the fit combines the Poisson likelihood for the
 1462 observed event counts with these Gaussian constraints for the NPs.

1463 TablesXV-XVI provide an overview of all modifiers used in the fit, their types as well
 1464 as the samples and regions they are applied to.

1465 In addition, tablesXIX-XXIV lists all fitted values for these modifiers in an asimov fit.

1466 The sample normalisations listed in tableXV are multiplicative modifiers of type κ .
 1467 For these, we do not input any auxiliary data but instead let them float freely.

1468 The majority of systematic uncertainties in this analysis are implemented in the form
 1469 of **Histosys** modifiers, which can be classified as “additive” modifiers of type Δ . These
 1470 apply absolute adjustments to the sample data based on provided upward and downward
 1471 variations.

1472 An exception is the uncertainty related to the reconstruction efficiency. It is repre-
 1473 sented by a **Normsys** modifier, a type κ modifier. This modifier scales the sample rate
 1474 multiplicatively according to the upward and downward uncertainties specified, adjusting
 1475 the rate by interpolating between the given values.

1476 MC statistical uncertainty in each bin is also handled with type κ modifiers. However,
 1477 each of these MC statistical modifiers scales only the yield in a single bin by a factor.

1478 The uncertainty resulting from the limitation in MC statistics is implemented using
 1479 the lite Beeston-Barlow method further described in Ref. [19]. The general idea coincides
 1480 with the implementation of the remaining NPs, to introduce a Gaussian-distributed NP
 1481 per bin in the histogram. These NPs are then used to encode the variation in the expected
 1482 event count due to statistical fluctuations in the MC simulation. As a difference to the
 1483 aforementioned Gaussian constraints, the constraint term for each bin is a Gaussian which
 1484 is centred at $a_{\gamma_{cb}} = 1$, since these modifiers are implemented as multiplicative modifiers

TABLE XV: Parameters corresponding to sample scale factors

Modifier	Type	Affected Samples	Affected regions	Parameters	η/χ
$\mathcal{R}(D)$	κ	signal D	De / D μ	1	η
$\mathcal{R}(D^*)$	κ	signal D	De / D μ / D *e / D $^*\mu$	1	η
Scale factor normalisation D	κ	norm D / signal D	De / D μ	1	η
Scale factor normalisation D *	κ	norm D * / signal D *	De / D μ / D *e / D $^*\mu$	1	η
Scale factor Ddst+gap D	κ	Ddst+gap D	De / D μ	2 for e / μ	η
Scale factor Ddst+gap D *	κ	Ddst+gap D *	D *e / D $^*\mu$	2 for e / μ	η
Scale factor backgrounds D	κ	backgrounds D	De / D μ	2 for e / μ	η
Scale factor backgrounds D *	κ	backgrounds D *	D *e / D $^*\mu$	2 for e / μ	η

(modifier type κ). For each sample s within the bin b in channel c , there is an individual yield uncertainty δ_{csb} , derived from the statistical uncertainties of the simulated events. The total uncertainty for bin b in channel c , denoted as σ_{cb} , is computed by summing the individual uncertainties δ_{csb} in quadrature. This means:

$$\sigma_{cb} = \sqrt{\sum_s \delta_{csb}^2} \quad (50)$$

The width of the Gaussian in the corresponding constraint term, δ_b , is normalised to the total event rate in the bin. It is calculated as:

$$\delta_{cb} = \frac{\sigma_{cb}}{\sum_s v_{csb}^0}, \quad (51)$$

where v_{csb}^0 represents the nominal (expected) yield in bin b for sample s .

Thus, the pdf of this Gaussian constraint can be written as:

$$\text{Gaus}(a_{\gamma_{cb}} = 1 \mid \gamma_{cb}, \delta_{cb}), \text{ where:} \quad (52)$$

- γ_{cb} is the model parameter (NP) for bin b and channel c
- $a_{\gamma_{cb}}$ is the auxiliary data for bin b and channel c
- δ_{cb} is the total yield uncertainty in bin b and channel c , derived from the individual uncertainties of all samples as described above.

TABLE XVI: Parameters encoding systematic uncertainties

Modifier	Type	Affected Samples	Affected regions	Parameters	η/χ
MC stat	κ	all	De / $D\mu$ / D^*e / $D^*\mu$	4 * 21 for each bin in each region	χ
slow pion	Δ	all	D^*e / $D^*\mu$	6 orthogonal variations	χ
tracking	Δ	all	De / $D\mu$ / D^*e / $D^*\mu$	1 variation within the given unc of 0.23%	χ
PID	Δ	all	De / $D\mu$ / D^*e / $D^*\mu$	6 orthogonal variations	χ
FFs D^*	Δ	signal D/D^* / norm D/D^*	De / $D\mu$ / D^*e / $D^*\mu$	9 up/down variations of FF parameters + 1 accounting for difference between BGL & BLRXP	χ
FFs broad	Δ	Ddst+gap D/D^*	De / $D\mu$ / D^*e / $D^*\mu$	3 up/down variations of FF parameters	χ
FFs narrow	Δ	Ddst+gap D/D^*	De / $D\mu$ / D^*e / $D^*\mu$	4 up/down variations of FF parameters	χ
FF (gap)	Δ	Ddst+gap D/D^*	De / $D\mu$ / D^*e / $D^*\mu$	2 * 2 * 3 up/down variations of FF parameters for each broad resonance for ℓ & τ	χ
BF	Δ	Ddst+gap D/D^*	De / $D\mu$ / D^*e / $D^*\mu$	2 * 4 variations within the BF unc. For each D^{**} resonance for ℓ & τ	χ
BF (gap)	Δ	signal D/D^* / norm D / D^*	De / $D\mu$ / D^*e / $D^*\mu$	1 variation assuming 100% BF unc	χ
reco eff	κ	signal D/D^* / norm D / D^*	De / $D\mu$ / D^*e / $D^*\mu$	2*2 for signal/norm in D & D^* mode + 2 for the feed-down component	χ

1497 The constraint term of the MC statistical uncertainty for all bins is obtained as the
 1498 product of these Gaussian constraints:

$$\prod_b \text{Gaus}(a_{\gamma_{cb}} = 1 \mid \gamma_{cb}, \delta_{cb}) \quad (53)$$

1499 The use of the lite Beeston-Barlow method is a difference to the previously used bin-
 1500 wise NP implementation, where one NP per bin as well as per template is assigned, which
 1501 includes the MC statistical uncertainty. The reason for deviating from this approach can
 1502 be understood considering the following:

- 1503 • The number of NP increases linearly with the number of bins and templates, lead-
 1504 ing to a large number of parameters, which can make the model computationally
 1505 intensive and harder to fit.
- 1506 • The conservative approach might lead to over-parameterisation, especially if some
 1507 bins have very low statistics, resulting in poorly constrained NPs.

The reduced number of NPs associated with the limitation in MC statistics compared to the “binfit” implementation implies that the associated uncertainty might differ slightly in both fit results.

What is more, also for the remaining systematic uncertainties, the sizes of uncertainty from the same source can change between the fit using bin-wise NPs and the one using source-wise NPs, due to differences in how these approaches model and propagate uncertainties through the likelihood function. I.e., the bin-wise approach does not account for any correlations between the uncertainties in different bins arising from the same source, while the source-wise approach, by using one NP per source that affects all bins, inherently assumes that the uncertainty is correlated across bins.

This implies that not only the sizes of uncertainty differ between both fit versions, but also the fitted central values might be different.

We performed several tests that to be confident that both fit implementations are valid approaches without any bias. Firstly, we perform both fits without any NPs added, such that the only uncertainty is of statistical nature. We obtain the same results as with the bin-wise fit. Secondly, we perform asimov fits with systematics to compare their sizes. As central values, we obtain with binfit:

$$R(D) = 0.299 \pm 0.076 \quad (\pm 0.067 \text{ stat} \pm 0.039 \text{ syst}) \quad (54)$$

$$R(D^*) = 0.258 \pm 0.035 \quad (\pm 0.034 \text{ stat} \pm 0.012 \text{ syst}) \quad (55)$$

With pyhf, the results are:

$$R(D) = 0.299 \pm 0.0765 \quad (\pm 0.0667 \text{ stat} \pm 0.0375 \text{ syst}) \quad (56)$$

$$R(D^*) = 0.258 \pm 0.0368 \quad (\pm 0.0337 \text{ stat} \pm 0.0148 \text{ syst}) \quad (57)$$

Table XVII reveals that the systematic uncertainties are not identical of similar size, as explained above. As expected, the data statistical uncertainty coincides between both fits.

TABLE XVII: Uncertainty comparison between asimov fits in binfit and pyhf

	Unc. R(D) binfit	Unc. R(D*) binfit	Unc. R(D) pyhf	Unc. R(D*) pyhf
MC stat	0.0298 (9.97 %)	0.0110 (4.28 %)	0.0281 (9.38 %)	0.0139 (5.41 %)
Slow pion	0.0003 (0.09 %)	0.0006 (0.24 %)	<0.0001 (0.00 %)	0.0006 (0.24 %)
Tracking	0.0001 (0.02 %)	<0.0001 (0.00 %)	<0.0001 (0.00 %)	0.0001 (0.03 %)
PID	0.0139 (4.64 %)	0.0032 (1.25 %)	0.0141 (4.7 %)	0.003 (1.18 %)
FFs	0.0014 (0.48 %)	0.0003 (0.12 %)	0.0006 (0.21 %)	0.0028 (1.07 %)
FF (gap)	0.0039 (1.3 %)	0.002 (0.77 %)	0.0038 (1.26 %)	0.0019 (0.75 %)
BF	0.0047 (1.57 %)	0.0006 (0.23 %)	0.0047 (1.57 %)	0.0006 (0.23 %)
BF (gap)	0.0208 (6.94 %)	0.0029 (1.13 %)	0.0189 (6.33 %)	0.0028 (1.09 %)
Sum	0.03934 (13.17 %)	0.01201 (4.675 %)	0.0372 (12.45 %)	0.0149 (5.81 %)

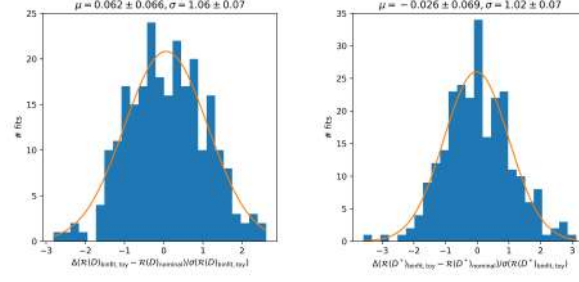
To investigate the aforementioned potential changes in central value, we can make use of toy fits. Specifically, we generate statistical toys based on the asimov histograms by randomly drawing from a Poisson distribution, utilising the function `random.poisson` from python’s package NumPy [31]. We fit the same toys with pyhf and binfit. Firstly, we can check for closure when performing the fits with data statistical uncertainties only.

1535 To align both fitting algorithms in a better way, we can also fit the yields $N_{\text{sig},D}$ and
1536 $N_{\text{norm},D}$ (N_{sig,D^*} and N_{norm,D^*}) instead of their ratio $\mathcal{R}(D)$ ($\mathcal{R}(D^*)$) in the **pyhf** version.
1537 We calculate $\mathcal{R}(D)$ and $\mathcal{R}(D^*)$ afterwards in the same way as it is done in the **binfit**
1538 version. Figs. 92 verifies that for each algorithm, the mean is consistent with zero within
1539 the uncertainties, and the width of the corresponding Gaussian is consistent with one.
1540 Beyond that, the pull plots of all three versions are compatible.

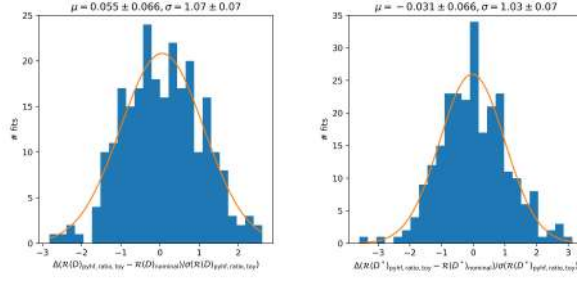
1541 In addition, see Fig.93 for the resulting difference between the $\mathcal{R}(D^{(*)})$ ratios in **binfit**
1542 compared to **pyhf** when fitting the ratios directly in **pyhf** (a) and when using the signal
1543 and normalisation yields as fit parameters in **pyhf** (b). We utilise 250 equal toys. These
1544 distributions have the same mean and rms values. The small difference seen here can
1545 be due to numerical precision, especially since the values for $\mathcal{R}(D)$ and $\mathcal{R}(D^*)$ are only
1546 calculated after the fit with inputs given up to a certain precision in **binfit**, while being
1547 the direct fit outputs in the **pyhf** case. Moreover, different convergence criteria may also
1548 affect the final result, i.e. the minor differences in criteria applied for the fit algorithm
1549 to stop. Also starting from slightly different initial conditions, also given that yields are
1550 fitted in the **binfit** case and ratios in the **pyhf** case, might lead to convergence at slightly
1551 different local minima.

1552 Further, Fig.93 includes the difference between both **pyhf** fit versions with different
1553 fit parameters. These differences are of the order 10^{-6} with means compatible with 0.
1554 Again, these may be attributed to numerical precision.

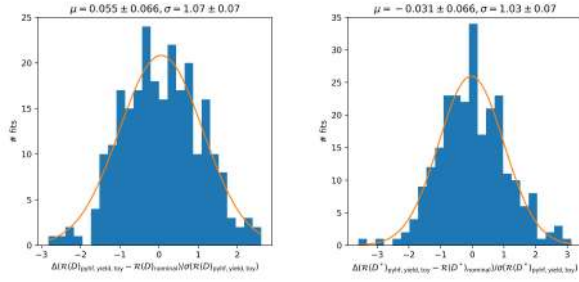
1555 A second step involves verifying that the fits remain unbiased even when systematic
1556 NPs are included. Fig. 94 shows that also in this case, the behaviour is as expected: The
1557 mean is still consistent with zero and the σ corresponds to the statistical part of the total
1558 uncertainty as given in Eqs.55-57.



(a)

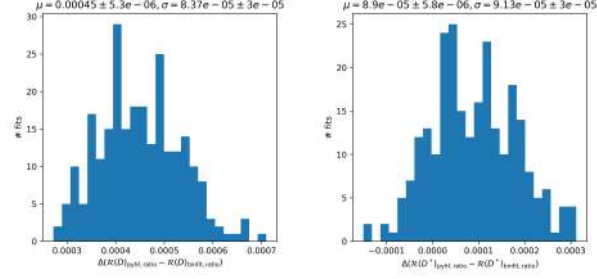


(b)

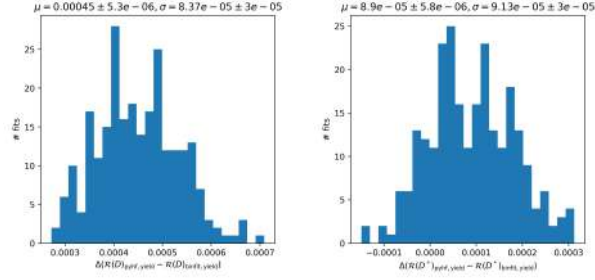


(c)

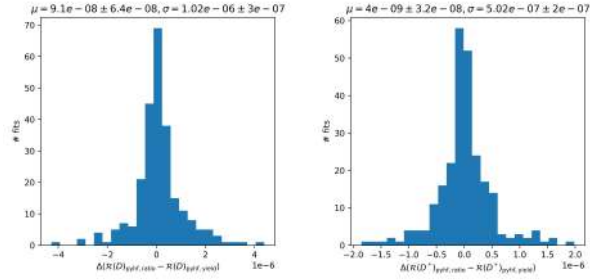
FIG. 92: Fits to 250 toy data without systematic NPs:
(a) **binfit** case, showing $\mathcal{R}(D)$ on the left and $\mathcal{R}(D^*)$ on the right;
(b) **pyhf** case, fit parameters are $\mathcal{R}(D)$ and $\mathcal{R}(D^*)$;
(c) **pyhf** case, fit parameters are $N_{\text{sig},D/D^*}$ and $N_{\text{norm},D/D^*}$



(a)



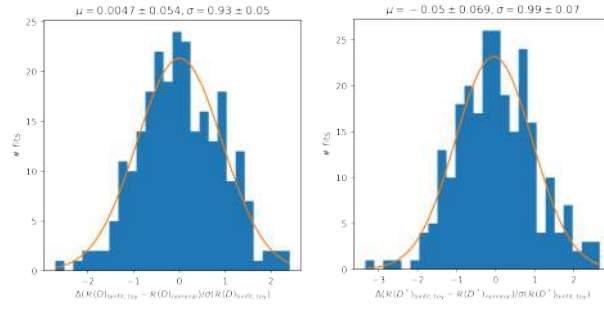
(b)



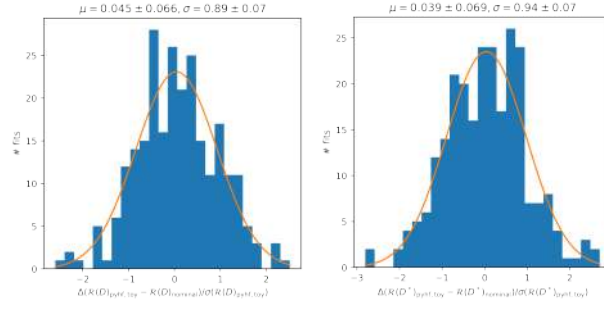
(c)

FIG. 93: Differences between the fitted values fitting the same 250 toys with different fit versions:

- (a) difference between **binfit**'s and **pyhf**'s $\mathcal{R}(D)$ and $\mathcal{R}(D^*)$ values, fitting the ratios in **pyhf**;
- (b) difference between **binfit**'s and **pyhf**'s $\mathcal{R}(D)$ and $\mathcal{R}(D^*)$ values when fitting signal and normalisation yields instead of $\mathcal{R}(D^{(*)})$ ratios in both fitters;
- (c) difference between **pyhf**'s $\mathcal{R}(D)$ and $\mathcal{R}(D^*)$ values when the $\mathcal{R}(D^{(*)})$ ratios compared to fitting signal and normalisation yields



(a)



(b)

FIG. 94: Fits to toy data using systematic NPs:
 (a) binfit case;
 (b) pyhf case

TABLE XVIII: Statistical and systematic uncertainties in an asimov fit using pyhf

source of unc.	$\Delta\mathcal{R}(D)$	$\Delta\mathcal{R}(D^*)$
data statistics	0.0659 (22.04 %)	0.0328 (12.7 %)
Monte Carlo statistics	0.0281 (9.4 %)	0.014 (5.42 %)
slow pion eff.	0.0002 (0.05 %)	0.0003 (0.12 %)
tracking	0.0001 (0.03 %)	0.0001 (0.03 %)
PID	0.0142 (4.73 %)	0.0031 (1.2 %)
FFs ($D^{(*)}$)	0.0006 (0.19 %)	0.0024 (0.94 %)
FFs (broad D^{**})	0.0005 (0.16 %)	0.0004 (0.14 %)
FFs (gap)	0.0038 (1.27 %)	0.002 (0.77 %)
BFs (D^{**})	0.0047 (1.58 %)	0.0006 (0.24 %)
BFs (gap)	0.0189 (6.34 %)	0.0029 (1.14 %)
reco. eff.	0.0092 (3.08 %)	0.0054 (2.08 %)

η	value
BDstnorm	1.0000 ± 0.0292
RDstar	0.2580 ± 0.0363
DststnormDste	1.0000 ± 0.3887
backgroundnormDste	1.0000 ± 0.2424
DststnormDstmu	1.0000 ± 0.3836
backgroundnormDstmu	1.0000 ± 0.2466
BDnorm	1.0000 ± 0.0277
RD	0.2990 ± 0.0764
DststnormDe	1.0000 ± 0.2816
backgroundnormDe	1.000 ± 0.1002
DstnormDmu	1.000 ± 0.1006
backgroundnormDmu	1.000 ± 0.1066

TABLE XIX: Fit results with uncertainties for the asimov fit with pyhf (part 1, showing only the unconstrained η parameters).

11.7. Systematic uncertainties in an asimov fit using the pyhf fit version

In the source-wise fit, the contribution of each source of systematic uncertainty can be estimated by repeating the fit including only a single one of the systematic NPs and subtracting the data statistical uncertainty in quadrature. Table XVIII shows the size of each source of systematic uncertainty in an asimov fit.

NP	value
FFmodeldiffnorm	0.000 ± 0.9566
PIDvar0	0.000 ± 0.8093
PIDvar1	0.000 ± 0.9756
PIDvar2	0.000 ± 0.9933
PIDvar3	0.000 ± 0.9933
PIDvar4	0.000 ± 0.9933
PIDvar5	0.000 ± 0.9932
FFBtoDorDstvar0	0.000 ± 0.9923
FFBtoDorDstvar1	0.000 ± 0.9932
FFBtoDorDstvar2	0.000 ± 0.9932
FFBtoDorDstvar3	0.000 ± 0.9933
FFBtoDorDstvar4	0.000 ± 0.9932
FFBtoDorDstvar5	0.000 ± 0.9932
FFBtoDorDstvar6	0.000 ± 0.9932
FFBtoDorDstvar7	0.000 ± 0.9932
FFBtoDorDstvar8	0.000 ± 0.9932
trackEfficiencyUncert	0.000 ± 0.9932
slowpivar0	0.000 ± 0.9932
slowpivar1	0.000 ± 0.9932
slowpivar2	0.000 ± 0.9932
slowpivar3	0.000 ± 0.9932
slowpivar4	0.000 ± 0.9932
slowpivar5	0.000 ± 0.9932
BFDdst12411Tau	0.000 ± 0.9933
BFDdst12413Tau	0.000 ± 0.9933

TABLE XX: Fit results with uncertainties for the asimov fit with `pyhf` (part 2, showing only the constrained χ parameters).

NP	value
BFDdst22411Tau	0.000 ± 0.9932
BFDdst22413Tau	0.000 ± 0.9932
BFDdst3415	0.000 ± 0.9932
BFDdst4115	0.000 ± 0.9933
BFBtoDdstBroadvar0	0.000 ± 0.9798
FFBtoDststBroadvar1	0.0000 ± 0.9933
FFBtoDststBroadvar2	0.0000 ± 0.9926
FFBtoDststNarrowvar0	0.0000 ± 0.9884
FFBtoDststNarrowvar1	0.0000 ± 0.9919
FFBtoDststNarrowvar2	0.0000 ± 0.9933
FFBtoDststNarrowvar3	0.0000 ± 0.9932
gapFFBtoDstst0stEllvar0	0.0000 ± 0.9781
gapFFBtoDstst0stEllvar1	0.0000 ± 0.9933
gapFFBtoDstst0stEllvar2	0.0000 ± 0.9908
gapFFBtoDstst0stTauvar0	0.0000 ± 0.9933
gapFFBtoDstst0stTauvar1	0.0000 ± 0.9933
gapFFBtoDstst0stTauvar2	0.0000 ± 0.9933
gapFFBtoDstst1stEllvar0	0.0000 ± 0.9551
gapFFBtoDstst1stEllvar1	0.0000 ± 0.9933
gapFFBtoDstst1stEllvar2	0.0000 ± 0.9918
gapFFBtoDstst1stTauvar0	0.0000 ± 0.9933
gapFFBtoDstst1stTauvar1	0.0000 ± 0.9933
gapFFBtoDstst1stTauvar2	0.0000 ± 0.9933
gapBF	0.0000 ± 0.4195
recoeffnormDstar	0.000 ± 0.9902
recoeffsignalDstar	0.000 ± 0.9930
recoeffnormD	0.000 ± 0.9900
recoeffsignalD	0.000 ± 0.9930
recoeffnormDstarfeeddown	0.000 ± 0.9902
recoeffsignalDstarfeeddown	0.000 ± 0.9930
staterrorDstare[0]	1.000 ± 0.0159
staterrorDstare[1]	1.000 ± 0.0105
staterrorDstare[2]	1.000 ± 0.0361
staterrorDstare[3]	1.000 ± 0.0383
staterrorDstare[4]	1.000 ± 0.0511
staterrorDstare[5]	1.000 ± 0.0511
staterrorDstare[6]	1.000 ± 0.0523
staterrorDstare[7]	1.000 ± 0.0537

TABLE XXI: Fit results with uncertainties for the asimov fit with pyhf (part 3, showing only the constrained χ parameters).

NP	value
staterrorDstare[8]	1.000 ± 0.0649
staterrorDstare[9]	1.000 ± 0.0522
staterrorDstare[10]	1.000 ± 0.0505
staterrorDstare[11]	1.000 ± 0.0552
staterrorDstare[12]	1.000 ± 0.0712
staterrorDstare[13]	1.000 ± 0.0753
staterrorDstare[14]	1.000 ± 0.0667
staterrorDstare[15]	1.000 ± 0.1067
staterrorDstare[16]	1.000 ± 0.1057
staterrorDstare[17]	1.000 ± 0.1422
staterrorDstare[18]	1.0000 ± 0.0945
staterrorDstare[19]	1.0000 ± 0.0947
staterrorDstare[20]	1.0000 ± 0.0734
staterrorDstarmu[0]	1.0000 ± 0.0168
staterrorDstarmu[1]	1.0000 ± 0.0111
staterrorDstarmu[2]	1.0000 ± 0.0346
staterrorDstarmu[3]	1.0000 ± 0.0419
staterrorDstarmu[4]	1.0000 ± 0.0618
staterrorDstarmu[5]	1.0000 ± 0.0583
staterrorDstarmu[6]	1.0000 ± 0.0622
staterrorDstarmu[7]	1.0000 ± 0.0595
staterrorDstarmu[8]	1.0000 ± 0.0742
staterrorDstarmu[9]	1.0000 ± 0.0627
staterrorDstarmu[10]	1.0000 ± 0.0602
staterrorDstarmu[11]	1.0000 ± 0.0617

TABLE XXII: Fit results with uncertainties for the asimov fit with pyhf (part 4, showing only the constrained χ parameters).

NP	value
staterrorDstarmu[12]	1.0000 ± 0.0982
staterrorDstarmu[13]	1.0000 ± 0.0965
staterrorDstarmu[14]	1.0000 ± 0.1234
staterrorDstarmu[15]	1.0000 ± 0.1446
staterrorDstarmu[16]	1.0000 ± 0.1037
staterrorDstarmu[17]	1.0000 ± 0.1555
staterrorDstarmu[18]	1.0000 ± 0.1113
staterrorDstarmu[19]	1.0000 ± 0.1171
staterrorDstarmu[20]	1.0000 ± 0.0837
staterrorDe[0]	1.0000 ± 0.0126
staterrorDe[1]	1.0000 ± 0.0071
staterrorDe[2]	1.0000 ± 0.0157
staterrorDe[3]	1.0000 ± 0.0178
staterrorDe[4]	1.0000 ± 0.0261
staterrorDe[5]	1.0000 ± 0.0291
staterrorDe[6]	1.0000 ± 0.0263
staterrorDe[7]	1.0000 ± 0.0278
staterrorDe[8]	1.0000 ± 0.0365
staterrorDe[9]	1.0000 ± 0.0216
staterrorDe[10]	1.0000 ± 0.0200
staterrorDe[11]	1.0000 ± 0.0254
staterrorDe[12]	1.0000 ± 0.0331
staterrorDe[13]	1.0000 ± 0.0361
staterrorDe[14]	1.0000 ± 0.0421
staterrorDe[15]	1.0000 ± 0.0602

TABLE XXIII: Fit results with uncertainties for the asimov fit with pyhf (part 5, showing only the constrained χ parameters).

NP	value
staterrorDe[16]	1.0000 ± 0.0638
staterrorDe[17]	1.0000 ± 0.0920
staterrorDe[18]	1.0000 ± 0.0512
staterrorDe[19]	1.0000 ± 0.0480
staterrorDe[20]	1.0000 ± 0.0387
staterrorDmu[0]	1.0000 ± 0.0129
staterrorDmu[1]	1.0000 ± 0.0072
staterrorDmu[2]	1.0000 ± 0.0166
staterrorDmu[3]	1.0000 ± 0.0193
staterrorDmu[4]	1.0000 ± 0.0280
staterrorDmu[5]	1.0000 ± 0.0314
staterrorDmu[6]	1.0000 ± 0.0307
staterrorDmu[7]	1.0000 ± 0.0294
staterrorDmu[8]	1.0000 ± 0.0372
staterrorDmu[9]	1.0000 ± 0.0273
staterrorDmu[10]	1.0000 ± 0.0259
staterrorDmu[11]	1.0000 ± 0.0296
staterrorDmu[12]	1.0000 ± 0.0399
staterrorDmu[13]	1.000 ± 0.0406
staterrorDmu[14]	1.000 ± 0.0553
staterrorDmu[15]	1.000 ± 0.0728
staterrorDmu[16]	1.000 ± 0.0688
staterrorDmu[17]	1.000 ± 0.1005
staterrorDmu[18]	1.000 ± 0.0607
staterrorDmu[19]	1.000 ± 0.0537
staterrorDmu[20]	1.000 ± 0.0429

TABLE XXIV: Fit results with uncertainties for the asimov fit with pyhf (part 6, showing only the constrained χ parameters).

11.8. Unboxed results using the pyhf fit version

We perform the fit to 365 fb^{-1} of data using the fit with source-wise NPs as described in Sect. 11.11.6. Our results are

$$\mathcal{R}(D^+) = 0.418 \pm 0.074 \text{ (stat)} \pm 0.051 \text{ (syst)}, \quad (58)$$

$$\mathcal{R}(D^{*+}) = 0.306 \pm 0.034 \text{ (stat)} \pm 0.018 \text{ (syst)}, \quad (59)$$

with experimental and statistical correlations of $\rho_{\text{tot}} = -0.256$ and $\rho_{\text{stat}} = -0.272$. Fig. 95 shows 39.3% confidence region of $\mathcal{R}(D^{(*)})$ and compares it to other measurements of these ratios. Our measurement is compatible with the current world average within 0.86 standard deviations. It agrees with the SM expectation within 1.66 standard deviations. Fig. 96 allows for a comparison of the pre-fit and post-fit agreement between the observed

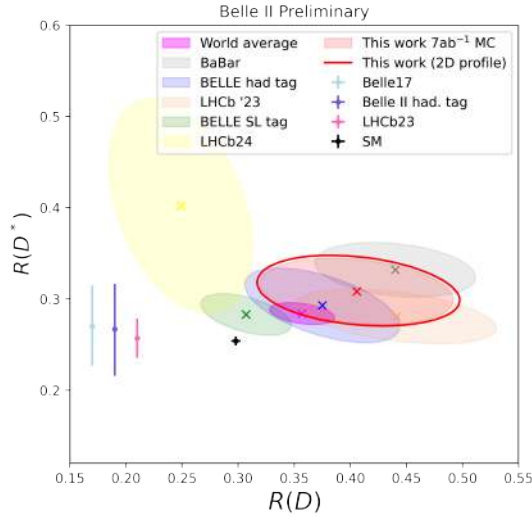


FIG. 95: The 39.3% confidence region in the $\mathcal{R}(D)$ - $\mathcal{R}(D^*)$ plane is shown (red curve) and compared to the world average and other measurements (red filled ellipse: if 7 ab^{-1} instead of 2 ab^{-1} was included in the fit).

data and the fitted histograms in the two-dimensional bins defined in Fig. 24. Fig. 97 shows the latter in a zoomed-in version, providing a better visibility of those bins that are not enhanced in events from the normalisation category.

Table XXXV presents the sizes of statistical and systematic uncertainties of the measurement. The largest uncertainties are due to the size of the recorded and simulated event samples, the second-largest significant uncertainty on $\mathcal{R}(D^+)$ is due to the modelling of the “gap” modes and the third-largest uncertainty is associated with lepton identification.

See tables XXVI-XXVII for the best-fit values of all fit parameters.

NP	value
FFmodeldiffnorm	-0.5171 ± 0.9584
PIDvar0	-0.4920 ± 0.7758
PIDvar1	0.2541 ± 0.9940
PIDvar2	0.4890 ± 1.0182
PIDvar3	0.1077 ± 0.9959

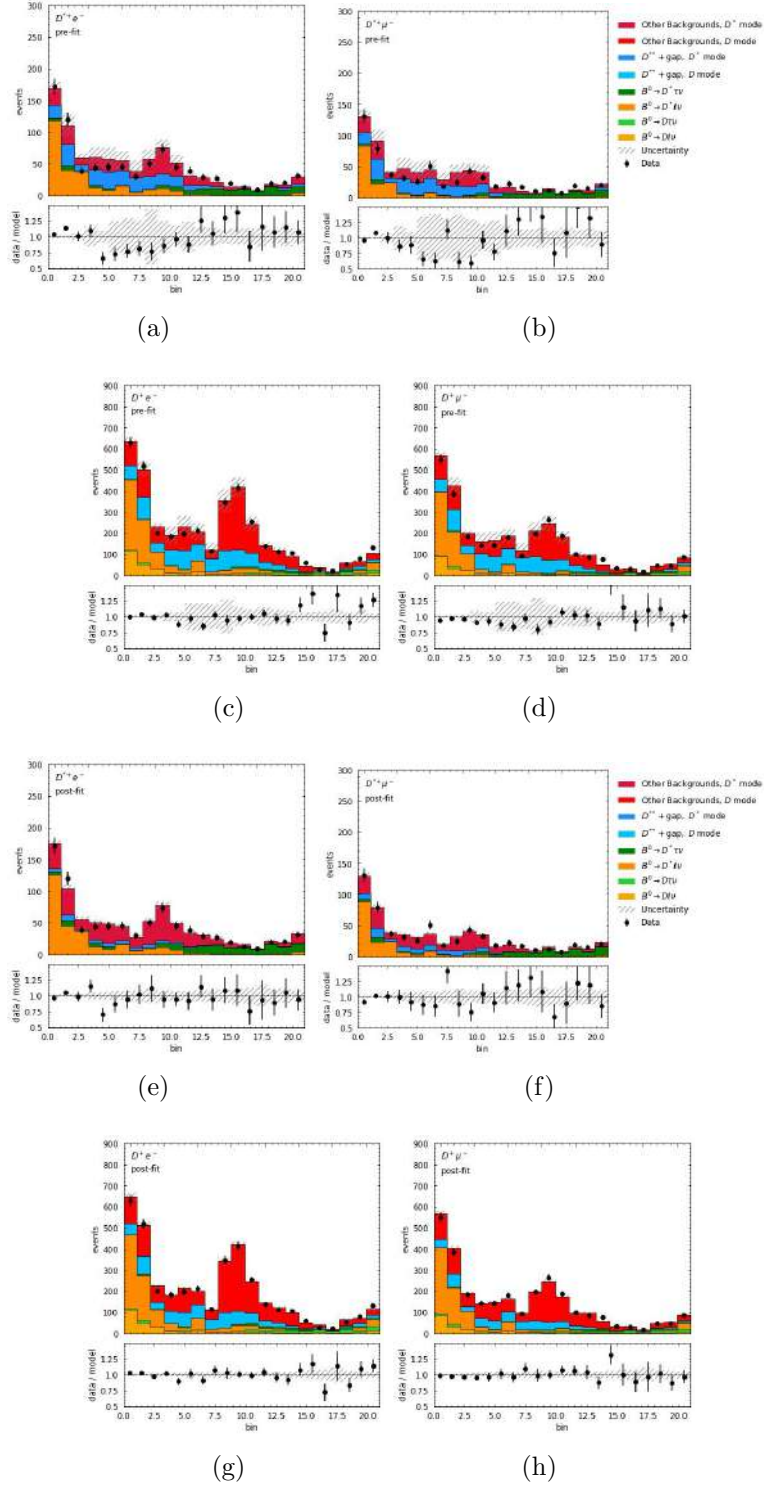


FIG. 97: Zoomed-in version of Fig.96

FFBtoDorDstvar1	-0.0451±0.9670
FFBtoDorDstvar2	-0.1045±1.0111
FFBtoDorDstvar3	0.1107±0.9871
FFBtoDorDstvar4	0.0274±1.0025
FFBtoDorDstvar5	0.0153±0.9977

TABLE XXV: Statistical and Systematic Uncertainties

Source of Unc.	$\Delta\mathcal{R}(D^+)$	$\Delta\mathcal{R}(D^{*+})$
Statistical	0.0737 (18.2 %)	0.034 (11.0 %)
MC sample sizes	0.0335 (8.3 %)	0.0161 (5.2 %)
Slow pion eff.	< 0.0001 (nil)	0.0002 (0.1 %)
Tracking	< 0.0001 (nil)	< 0.0001 (nil)
PID	0.0163 (4.0 %)	0.0043 (1.4 %)
FFs ($D^{(*)}$)	0.0012 (0.3 %)	0.0032 (1.0 %)
FFs (D^{**})	< 0.0001 (nil)	0.0001 (0.1 %)
FFs (“gap”)	0.0048 (1.2 %)	0.0008 (0.3 %)
Branching fractions (D^{**})	0.0038 (0.9 %)	0.0001 (0.1 %)
Branching fractions (“gap”)	0.0303 (7.5 %)	0.0005 (0.2 %)
Reco. eff.	0.0125 (3.1 %)	0.0065 (2.1 %)
$\cos\theta_{BY}$ modelling	0.015 (3.7 %)	0.005 (1.6 %)

TABLE XXVI: Fit results with uncertainties for the data fit with `pyhf`, showing only the unconstrained η parameters.

η	value
BDstnorm	1.0525±0.0282
RDstar	0.3083±0.0387
DststnormDste	0.2849±0.3560
backgroundnormDste	1.3907±0.2565
DststnormDstmu	0.4891±0.4308
backgroundnormDstmu	1.1171±0.2359
BDnorm	0.9509±0.0277
RD	0.4057±0.0905
DststnormDe	0.8717±0.3917
backgroundnormDe	1.0954±0.1065
DststnormDmu	0.6506±0.2446
backgroundnormDmu	1.1092±0.115

FFBtoDorDstvar6	-0.0036±0.9986
FFBtoDorDstvar7	-0.1579±0.9954
FFBtoDorDstvar8	0.0573±1.0002
trackEfficiencyUncert	0.0124±0.9998
slowpivar0	-0.2788±0.9914
slowpivar1	-0.0330±1.0010
slowpivar2	-0.0472±0.9993
slowpivar3	0.0017±0.9998

slowpivar4	-0.0227±0.9999
slowpivar5	0.0197±0.9998
BFDdst1z411	-0.0692±1.0030
BFDdst1z411tau	0.0152±1.0000
BFDdst1z413	0.0063±0.9982
BFDdst1z413tau	0.0088±1.0000
BFDdst2z413	0.0999±0.9938
BFDdst2z413tau	0.0012±1.0000
BFDdst415	-0.0355±1.0005
BFDdst415tau	0.0006±1.0000
FFBtoDststBroadvar0	0.0905±1.0477
FFBtoDststBroadvar1	0.0003±1.0000
FFBtoDststBroadvar2	-0.0036±1.0016
FFBtoDststNarrowvar0	-0.0601±1.0012
FFBtoDststNarrowvar1	-0.0445±1.0033
FFBtoDststNarrowvar2	0.0011±1.0000
FFBtoDststNarrowvar3	0.0133±0.9996
gapFFBtoDstst0stEllvar0	0.2169±1.1021
gapFFBtoDstst0stEllvar1	0.0032±1.0000
gapFFBtoDstst0stEllvar2	0.0627±1.0094
gapFFBtoDstst0stTauvar0	0.0012±1.0004
gapFFBtoDstst0stTauvar1	0.0000±1.0000
gapFFBtoDstst0stTauvar2	0.0002±1.0001
gapFFBtoDstst1stEllvar0	0.1011±1.0881
gapFFBtoDstst1stEllvar1	-0.0018±1.0000
gapFFBtoDstst1stEllvar2	-0.0531±1.0061
gapFFBtoDstst1stTauvar0	-0.0007±0.9998
gapFFBtoDstst1stTauvar1	0.0000±1.0001
gapFFBtoDstst1stTauvar2	0.0001±0.9993
gapBF	-0.2418±0.5839
recoeffnormDstar	0.1207 ± 0.9973
recoeffsignalDstar	-0.0284 ± 1.0004
recoeffnormD	-0.0000 ± 0.9984
recoeffsignalD	0.0000 ± 0.9999
recoeffnormDstarfeeddown	-0.1209 ± 0.9982
recoeffsignalDstarfeeddown	0.0283 ± 0.9992
staterrorD*e[0]	0.9935 ± 0.0159
staterrorD*e[1]	1.0104 ± 0.0104
staterrorD*e[2]	0.9971 ± 0.0302
staterrorD*e[3]	1.0255 ± 0.0379
staterrorD*e[4]	0.9464 ± 0.0522
staterrorD*e[5]	0.9788 ± 0.0520
staterrorD*e[6]	0.9913 ± 0.0529
staterrorD*e[7]	1.0029 ± 0.0544

staterrorD*e[8]	1.0141 ± 0.0653
staterrorD*e[9]	0.9906 ± 0.0527
staterrorD*e[10]	0.9891 ± 0.0467
staterrorD*e[11]	0.9857 ± 0.0556
staterrorD*e[12]	1.0277 ± 0.0702
staterrorD*e[13]	0.9881 ± 0.0749
staterrorD*e[14]	1.0175 ± 0.0852
staterrorD*e[15]	1.0196 ± 0.1042
staterrorD*e[16]	0.9491 ± 0.1059
staterrorD*e[17]	0.9854 ± 0.1404
staterrorD*e[18]	0.9747 ± 0.0939
staterrorD*e[19]	1.0091 ± 0.0939
staterrorD*e[20]	0.9873 ± 0.0729
staterrorD*mu[0]	0.9845 ± 0.0168
staterrorD*mu[1]	1.0042 ± 0.0110
staterrorD*mu[2]	1.0016 ± 0.0345
staterrorD*mu[3]	1.0001 ± 0.0423
staterrorD*mu[4]	0.9867 ± 0.0622
staterrorD*mu[5]	0.9821 ± 0.0594
staterrorD*mu[6]	0.9793 ± 0.0635
staterrorD*mu[7]	1.0571 ± 0.0593
staterrorD*mu[8]	0.9861 ± 0.0761
staterrorD*mu[9]	0.9639 ± 0.0645
staterrorD*mu[10]	1.0087 ± 0.0605
staterrorD*mu[11]	0.9861 ± 0.0626
staterrorD*mu[12]	1.0261 ± 0.0977
staterrorD*mu[13]	1.0362 ± 0.0949
staterrorD*mu[14]	1.0675 ± 0.1194
staterrorD*mu[15]	1.0192 ± 0.1416
staterrorD*mu[16]	0.9320 ± 0.1050
staterrorD*mu[17]	0.9781 ± 0.1542
staterrorD*mu[18]	1.0477 ± 0.1085
staterrorD*mu[19]	1.0377 ± 0.1151
staterrorD*mu[20]	0.9711 ± 0.0842
staterrorDe[0]	1.0048 ± 0.0126
staterrorDe[1]	1.0045 ± 0.0070
staterrorDe[2]	0.9947 ± 0.0159
staterrorDe[3]	1.0027 ± 0.0180
staterrorDe[4]	0.9807 ± 0.0264
staterrorDe[5]	1.0033 ± 0.0291
staterrorDe[6]	0.9838 ± 0.0266
staterrorDe[7]	1.0120 ± 0.0278
staterrorDe[8]	1.0041 ± 0.0366
staterrorDe[9]	1.0008 ± 0.0217

staterrorDe[10]	0.9983 ± 0.0201
staterrorDe[11]	1.0071 ± 0.0254
staterrorDe[12]	0.9923 ± 0.0332
staterrorDe[13]	0.9853 ± 0.0362
staterrorDe[14]	1.0147 ± 0.0417
staterrorDe[15]	1.0349 ± 0.0590
staterrorDe[16]	0.9938 ± 0.0531
staterrorDe[17]	1.0226 ± 0.0635
staterrorDe[18]	0.9931 ± 0.0541
staterrorDe[19]	1.0225 ± 0.0620
staterrorDe[20]	0.9937 ± 0.0467
staterrorDmu[0]	1.0040 ± 0.0144
staterrorDmu[1]	1.0020 ± 0.0097
staterrorDmu[2]	0.9896 ± 0.0234
staterrorDmu[3]	0.9912 ± 0.0274
staterrorDmu[4]	1.0000 ± 0.0348
staterrorDmu[5]	1.0137 ± 0.0374
staterrorDmu[6]	0.9882 ± 0.0390
staterrorDmu[7]	0.9970 ± 0.0365
staterrorDmu[8]	1.0025 ± 0.0503
staterrorDmu[9]	1.0018 ± 0.0395
staterrorDmu[10]	0.9858 ± 0.0317
staterrorDmu[11]	0.9988 ± 0.0387
staterrorDmu[12]	1.0022 ± 0.0552
staterrorDmu[13]	0.9858 ± 0.0514
staterrorDmu[14]	1.0132 ± 0.0692
staterrorDmu[15]	0.9716 ± 0.0720
staterrorDmu[16]	1.0271 ± 0.0731
staterrorDmu[17]	1.0262 ± 0.0790
staterrorDmu[18]	1.0165 ± 0.0762
staterrorDmu[19]	1.0146 ± 0.0835
staterrorDmu[20]	1.0150 ± 0.0651
staterrorlD*e[0]	0.9847 ± 0.0142
staterrorlD*e[1]	1.0048 ± 0.0090
staterrorlD*e[2]	0.9916 ± 0.0230
staterrorlD*e[3]	0.9998 ± 0.0300
staterrorlD*e[4]	0.9900 ± 0.0396
staterrorlD*e[5]	1.0154 ± 0.0410
staterrorlD*e[6]	0.9810 ± 0.0460
staterrorlD*e[7]	0.9885 ± 0.0417
staterrorlD*e[8]	1.0050 ± 0.0555
staterrorlD*e[9]	0.9761 ± 0.0461
staterrorlD*e[10]	1.0151 ± 0.0468
staterrorlD*e[11]	0.9966 ± 0.0498

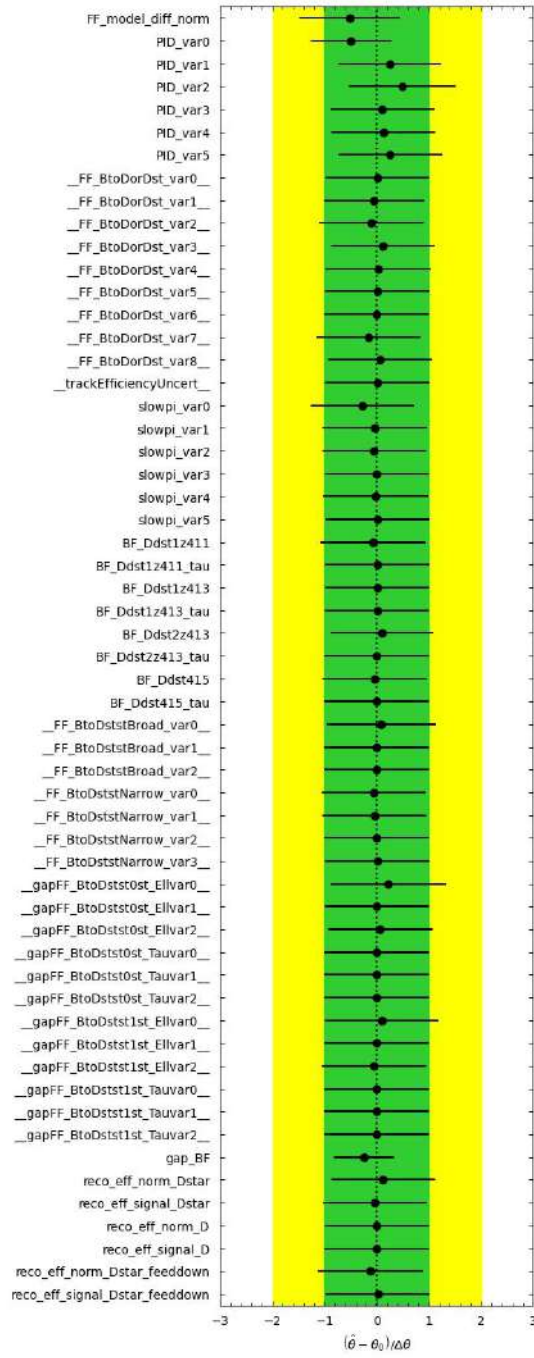
staterrorlD*e[12]	0.9897 ± 0.0602
staterrorlD*e[13]	0.9838 ± 0.0602
staterrorD*e[13]	0.9881 ± 0.0749
staterrorD*e[14]	1.0175 ± 0.0852
staterrorD*e[15]	1.0196 ± 0.1042
staterrorD*e[16]	0.9491 ± 0.1059
staterrorD*e[17]	0.9854 ± 0.1404
staterrorD*e[18]	0.9747 ± 0.0939
staterrorD*e[19]	1.0091 ± 0.0939
staterrorD*e[20]	0.9873 ± 0.0729
staterrorD*mu[0]	0.9845 ± 0.0168
staterrorD*mu[1]	1.0042 ± 0.0110
staterrorD*mu[2]	1.0016 ± 0.0345
staterrorD*mu[3]	1.0001 ± 0.0423
staterrorD*mu[4]	0.9867 ± 0.0622
staterrorD*mu[5]	0.9821 ± 0.0594
staterrorD*mu[6]	0.9793 ± 0.0635
staterrorD*mu[7]	1.0571 ± 0.0593
staterrorD*mu[8]	0.9861 ± 0.0761
staterrorD*mu[9]	0.9639 ± 0.0645
staterrorD*mu[10]	1.0087 ± 0.0605
staterrorD*mu[11]	0.9861 ± 0.0626
staterrorD*mu[12]	1.0261 ± 0.0977
staterrorD*mu[13]	1.0362 ± 0.0949
staterrorD*mu[14]	1.0675 ± 0.1194
staterrorD*mu[15]	1.0192 ± 0.1416
staterrorD*mu[16]	0.9320 ± 0.1050
staterrorD*mu[17]	0.9781 ± 0.1542
staterrorD*mu[18]	1.0477 ± 0.1085
staterrorD*mu[19]	1.0377 ± 0.1151
staterrorD*mu[20]	0.9711 ± 0.0842
staterrorDe[0]	1.0048 ± 0.0126
staterrorDe[1]	1.0045 ± 0.0070
staterrorDe[2]	0.9947 ± 0.0159
staterrorDe[3]	1.0027 ± 0.0180
staterrorDe[4]	0.9807 ± 0.0264
staterrorDe[5]	1.0033 ± 0.0291
staterrorDe[6]	0.9838 ± 0.0266
staterrorDe[7]	1.0120 ± 0.0278
staterrorDe[8]	1.0041 ± 0.0366
staterrorDe[9]	1.0008 ± 0.0217
staterrorDe[10]	0.9983 ± 0.0201
staterrorDe[11]	1.0071 ± 0.0254
staterrorDe[12]	0.9923 ± 0.0332

staterrorDe[13]	0.9853 ± 0.0362
staterrorDe[14]	1.0147 ± 0.0417
staterrorDe[15]	1.0349 ± 0.0590
staterrorDe[16]	0.9447 ± 0.0648
staterrorDe[17]	1.0293 ± 0.0905
staterrorDe[18]	0.9669 ± 0.0514
staterrorDe[19]	1.0175 ± 0.0475
staterrorDe[20]	1.0278 ± 0.0382
staterrorDmu[0]	0.9968 ± 0.0129
staterrorDmu[1]	0.9951 ± 0.0072
staterrorDmu[2]	0.9941 ± 0.0167
staterrorDmu[3]	0.9918 ± 0.0194
staterrorDmu[4]	0.9928 ± 0.0282
staterrorDmu[5]	1.0020 ± 0.0318
staterrorDmu[6]	0.9935 ± 0.0311
staterrorDmu[7]	1.0151 ± 0.0294
staterrorDmu[8]	0.9985 ± 0.0378
staterrorDmu[9]	0.9997 ± 0.0274
staterrorDmu[10]	1.0127 ± 0.0257
staterrorDmu[11]	1.0113 ± 0.0295
staterrorDmu[12]	1.0072 ± 0.0399
staterrorDmu[13]	0.9758 ± 0.0409
staterrorDmu[14]	1.0618 ± 0.0540
staterrorDmu[15]	1.0213 ± 0.0614
staterrorDmu[16]	1.0067 ± 0.0669
staterrorDmu[17]	1.0208 ± 0.1012
staterrorDmu[18]	1.0033 ± 0.0590
staterrorDmu[19]	0.9991 ± 0.0652
staterrorDmu[20]	1.0093 ± 0.0420

TABLE XXVII: Fit results with uncertainties for the data fit with `pyhf`, showing only the constrained χ parameters).

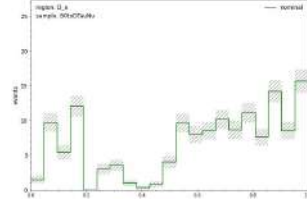
1586 Fig. 98 shows the pulls of the nuisance parameters.

1587 In order to better illustrate the shapes of the individual template histograms, those
1588 are shown for all four sample types in each fit region in Figs.99-104. All pre-fit corrections
1589 are included in these plots.

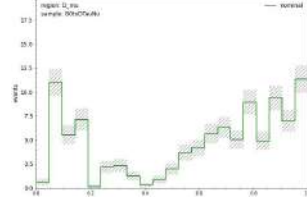


(a)

FIG. 98: Nuisance parameter pulls in the fit to data using pyhf

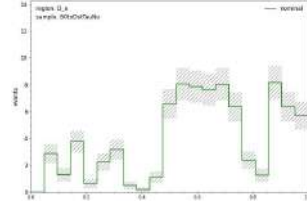


(a)

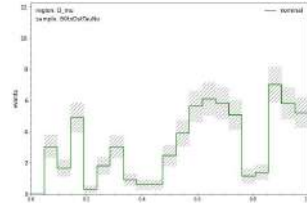


(b)

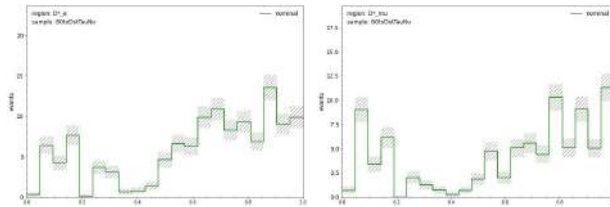
FIG. 99: Pre-fit shapes of the histograms of the “signal D ” category in the De (a), $D\mu$ (b) regions, including all corrections



(a)



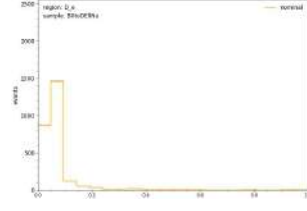
(b)



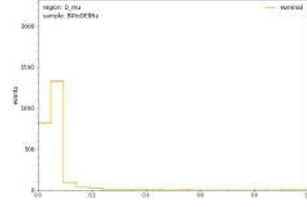
(c)

(d)

FIG. 100: Pre-fit shapes of the histograms of the “signal D^* ” category in the De (a), $D\mu$ (b), D^*e (c), $D^*\mu$ (d) regions, including all corrections

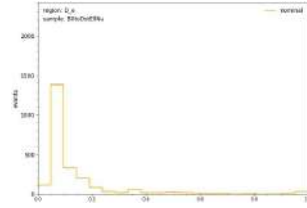


(a)

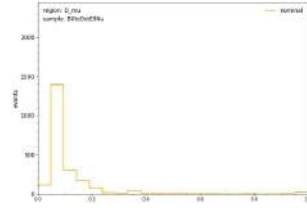


(b)

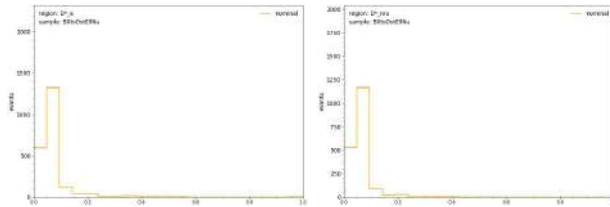
FIG. 101: Pre-fit shapes of the histograms of the “normalisation D ” category in the De (a), $D\mu$ (b) regions, including all corrections



(a)



(b)



(c)

(d)

FIG. 102: Pre-fit shapes of the histograms of the “normalisation D^* ” category in the De (a), $D\mu$ (b), D^*e (c), $D^*\mu$ (d) regions, including all corrections

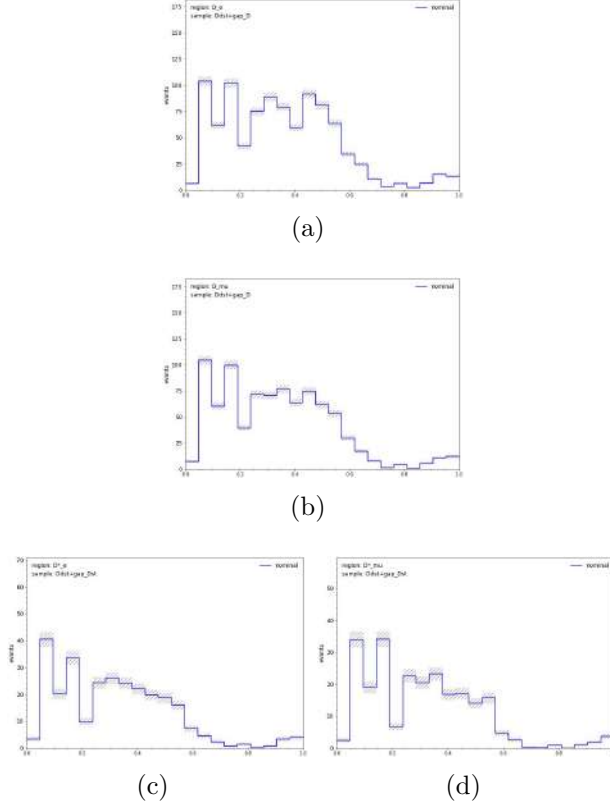


FIG. 103: Pre-fit shapes of the histograms of the “ $D^{**}+\text{gap}$ ” category in the De (a), $D\mu$ (b), D^*e (c), $D^*\mu$ (d) regions, including all corrections

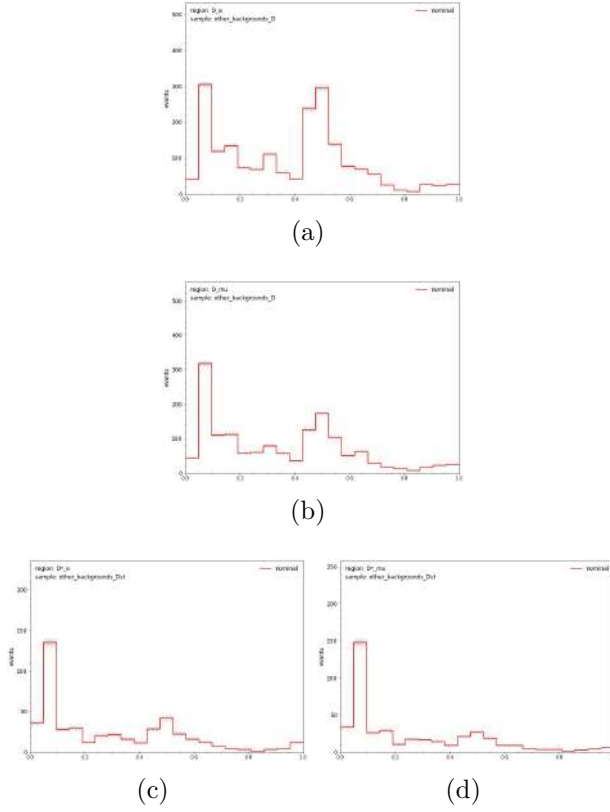
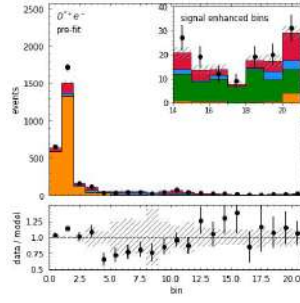
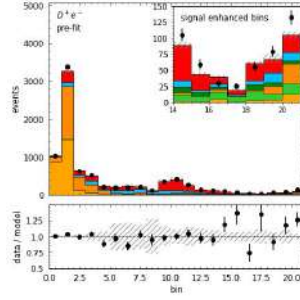


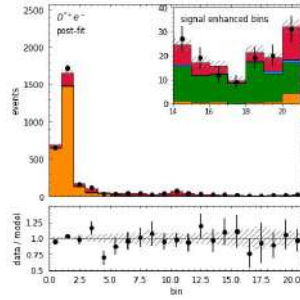
FIG. 104: Pre-fit shapes of the histograms of the “other backgrounds” category in the De (a), $D\mu$ (b), D^*e (c), $D^*\mu$ (d) regions, including all corrections



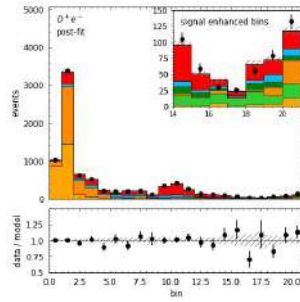
(a)



(b)

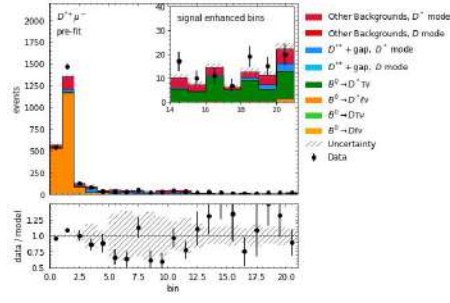


(c)

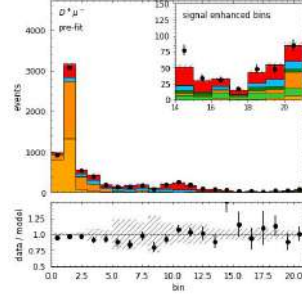


(d)

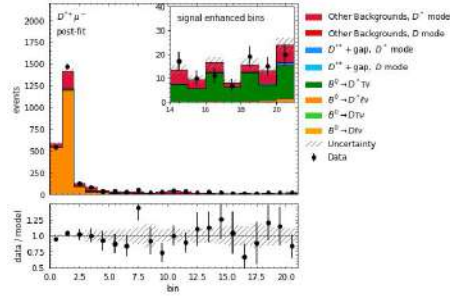
FIG. 105: Pre-fit (a-b) and post-fit (c-d) distributions in the two relevant fit regions De (a/c) and D^*e (b/d) when performing the electron-only fit



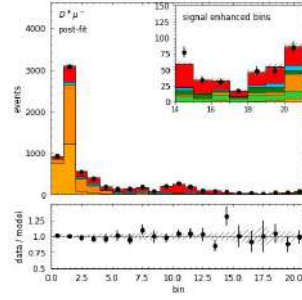
(a)



(b)



(c)



(d)

FIG. 106: Pre-fit (a-b) and post-fit (c-d) distributions in the two relevant fit regions $D\mu$ (a/c) and $D^*\mu$ (b/d) when performing the muon-only fit

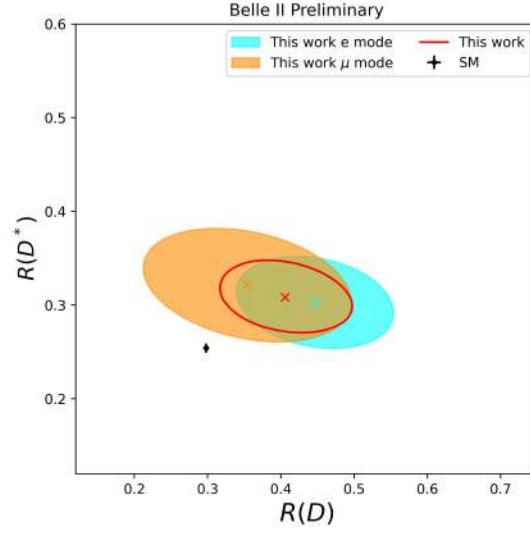


FIG. 107: The 39.3% confidence region in the $\mathcal{R}(D)$ - $\mathcal{R}(D^*)$ plane for the nominal fit (red curve) in comparison to those of the electron (cyan) and muon (orange) fits.

11.9. Split sample fits

To assess the stability of the fit results, several dataset splits are conducted: The dataset is divided into multiple approximately equal halves by applying various cuts and their corresponding inverse cuts. This approach allows for a thorough evaluation of the consistency and robustness of the fit results across different subsets of the data.

For each split, two separate values for $\mathcal{R}(D)$ and $\mathcal{R}(D^*)$ are fitted. In these fits, all normalisations are unlinked between the respective categories, across the categories remain unlinked, while most systematic uncertainties are maintained as linked, except for those that are not formally independent.

For instance, in the $e - \mu$ split, all systematics are linked except for the PID and fake rate systematics. The following itemised lists presents an overview of the splits performed:

- e/μ : Fitting $\mathcal{R}(D^{(*)})$ separately in the electron and muon channels.
- D/D^* : Splitting the dataset by unlinking the D and D^* modes. In this scenario, $\mathcal{R}(D)$ and $\mathcal{R}(D^*)$ can, in principle, be fitted in the D mode alone using the feed-down events, while only $\mathcal{R}(D^*)$ can be fitted in the D^* mode. However, due to the low number of signal feed-down events and their histogram overlap with the signal D events, it becomes challenging to separate $\mathcal{R}(D)$ and $\mathcal{R}(D^*)$ in the fit, resulting in a high anti-correlation between them. On the other hand, the $\mathcal{R}(D^*)$ result remains stable, even without utilising the feed-down events, and the difference in central values remains small (0.002).
- $\pm$: Splitting according to the charge of the signal lepton.
- $\cos \theta_\ell < 0.22$: Splitting based on the polar angle of the signal lepton. The cut value here is chosen such that it divides the dataset in approximately half the size.
- $N_{\text{tracks}} < 4$: Splitting according to the number of reconstructed tracks on the signal side. In this split, the tracks are defined excluding the one of the signal lepton. Due to the choice of D^* and D^0 modes, there are between 2 and 5 tracks present. It is important to note that this split is not exactly equal in size; for instance, there are more D^+ modes with a lower number of tracks than with a larger number when applying the above criterion. Nonetheless, this represents the most effective approach when imposing a cut based on this integer variable.
- dataset: Splitting according to the data taking period, comparing **proc13** versus the **buckets**. Both contain approximately half of the nominal statistics. The MC templates fitted to the datasets remain as before. The results of this split reveal a discrepancy between these datasets, which can be attributed to variations in lepton identification performance, as confirmed by the performance group[1]. The lepton identification performance is observed to be poorer in the **buckets** compared to **proc13**. This difference is further accentuated in this analysis due to the presence of a lepton on the tag side where lepton identification is also utilised. Additional plots in the Appendix demonstrate that, across various variables, fewer true leptons are reconstructed in the **buckets** compared to **proc13**. Beyond that, all corrections applied to the MC templates are provided for the combined datasets, which complicates the comparison between data and MC in this case.

The first test involves fitting $\mathcal{R}(D)$ and $\mathcal{R}(D^*)$ separately in the electron and muon channels. Systematics that are consistent across both channels remain linked, while those specific to a particular channel, such as lepton ID uncertainty, are treated independently.

The electron-only fit yields:

$$\mathcal{R}(D^+) = 0.447 \pm 0.091_{\text{stat}} \pm 0.057_{\text{syst}} \quad (60)$$

$$\mathcal{R}(D^{*+}) = 0.303 \pm 0.043_{\text{stat}} \pm 0.023_{\text{syst}} \quad (61)$$

The correlation between $\mathcal{R}(D^+)$ and $\mathcal{R}(D^{*+})$ amounts to:

$$\rho = -0.230 \text{ (stat. part: } \rho = -0.268\text{)}. \quad (62)$$

The results for the muon-only fit are:

$$\mathcal{R}(D^+) = 0.354 \pm 0.125_{\text{stat}} \pm 0.067_{\text{syst}} \quad (63)$$

$$\mathcal{R}(D^{*+}) = 0.321 \pm 0.054_{\text{stat}} \pm 0.027_{\text{syst}} \quad (64)$$

Here, the correlation coefficient reads:

$$\rho = -0.315 \text{ (stat. part: } \rho = -0.286\text{)}. \quad (65)$$

Fig. 105 shows the pre- and postfit plots for the electron mode, while those for the muon mode are presented in Fig.106. Fig. 107 shows the confidence ellipse of the nominal fit compared to those of the individual electron and muon fits.

For this specific split, we can perform further cross-checks with respect to the SM expectation: They can be utilised to compute the light lepton ratios $\mathcal{R}(D)_{e/\mu}$ and $\mathcal{R}(D^*)_{e/\mu}$, which are expected to be equal to one according to the SM: The fit results of the yields for the normalisation channel are corrected for the reconstruction efficiencies in the e and μ channels, obtaining the results visualised in Fig.108.

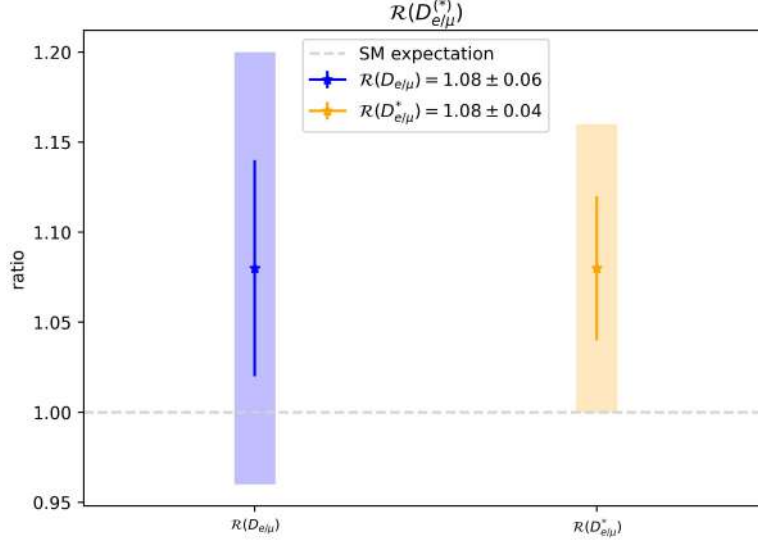
The one-sigma error bars and two larger bands corresponding to the two-sigma uncertainties are plotted to demonstrate that both ratios are compatible with one within two sigma.

Another verification of the SM expectation involves computing the ratio of branching ratios for $B \rightarrow D\ell\nu$ over $B \rightarrow D^*\ell\nu$. The SM expectation for this ratio, however, differs between B^0 and B^+ . An average can therefore be computed, represented by the grey line in Fig.109.

Alternatively, a comparison can be made with the value obtained in Ref[10], from which the plot in Fig.109(b) is taken. In that study, a new form factor parametrisation is derived, resulting in an expected ratio represented by the pink line in Fig109(a). The results for the e and μ split sample fits as well as the nominal fit result is consistent with the latter.

TABLE XXVIII: Summary of $\mathcal{R}(D)$ split sample fits

cut criterion	$\mathcal{R}(D)_{\text{cut}}$	$\mathcal{R}(D)_{\text{anti-cut}}$	$\Delta(\mathcal{R}(D))$	χ^2	p_1	p_2
e/μ	$0.447 \pm 0.091_{\text{stat}} \pm 0.057_{\text{syst}}$	$0.354 \pm 0.125_{\text{stat}} \pm 0.067_{\text{syst}}$	0.09 ± 0.17	0.289	0.591	0.856
D/D^*	$0.284 \pm 0.119_{\text{stat}} \pm 0.062_{\text{syst}}$	-	-	-	-	0.361
$\pm$	$0.443 \pm 0.097_{\text{stat}} \pm 0.057_{\text{syst}}$	$0.389 \pm 0.109_{\text{stat}} \pm 0.063_{\text{syst}}$	0.05 ± 0.16	0.110	0.740	0.488
$\cos \theta_\ell < 0.22$	$0.334 \pm 0.110_{\text{stat}} \pm 0.056_{\text{syst}}$	$0.597 \pm 0.094_{\text{stat}} \pm 0.051_{\text{syst}}$	0.10 ± 0.16	0.402	0.526	0.738
$N_{\text{tracks}} < 4$	$0.399 \pm 0.080_{\text{stat}} \pm 0.046_{\text{syst}}$	$0.355 \pm 0.151_{\text{stat}} \pm 0.069_{\text{syst}}$	0.01 ± 0.19	0.002	0.966	0.715
decaymodes	$0.368 \pm 0.149_{\text{stat}} \pm 0.067_{\text{syst}}$	$0.390 \pm 0.086_{\text{stat}} \pm 0.053_{\text{syst}}$	0.02 ± 0.19	0.013	0.908	0.399
proc13/buckets	$0.258 \pm 0.099_{\text{stat}} \pm 0.049_{\text{syst}}$	$0.597 \pm 0.110_{\text{stat}} \pm 0.054_{\text{syst}}$	0.34 ± 0.16	4.611	0.032	0.098



(a)

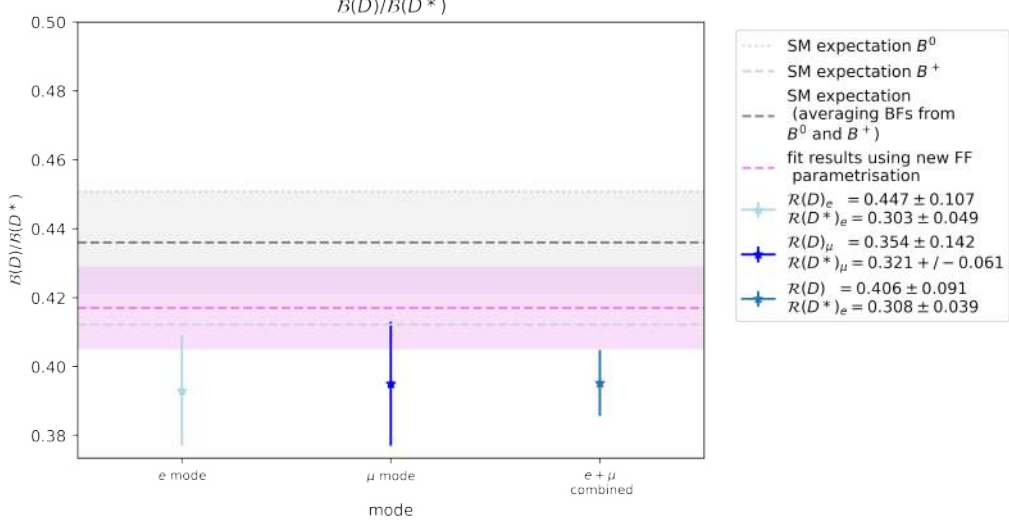
FIG. 108: Light lepton ratios with 1 and 2 σ error bars compared to SM expectation

TABLE XXIX: Summary of $\mathcal{R}(D^*)$ split sample fits

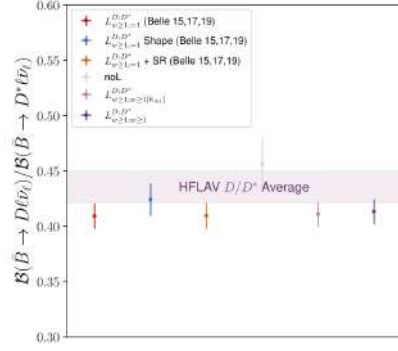
cut criterion	$\mathcal{R}^*(D)_{\text{cut}}$	$\mathcal{R}(D^*)_{\text{anti-cut}}$	$\Delta(\mathcal{R}(D))$	χ^2	p_1	p_2
e/μ	$0.303 \pm 0.043_{\text{stat}} \pm 0.023_{\text{syst}}$	$0.321 \pm 0.054_{\text{stat}} \pm 0.027_{\text{syst}}$	0.02 ± 0.08	0.57	0.811	0.856
D/D^*	$0.477 \pm 0.193_{\text{stat}} \pm 0.094_{\text{syst}}$	$0.306 \pm 0.034_{\text{stat}} \pm 0.018_{\text{syst}}$	0.17 ± 0.22	0.620	0.431	0.361
$\pm$	$0.363 \pm 0.050_{\text{stat}} \pm 0.027_{\text{syst}}$	$.298 \pm 0.046_{\text{stat}} \pm 0.023_{\text{syst}}$	0.07 ± 0.08	0.748	0.387	0.488
$\cos \theta_\ell < 0.22$	$0.338 \pm 0.051_{\text{stat}} \pm 0.027_{\text{syst}}$	$0.291 \pm 0.044_{\text{stat}} \pm 0.024_{\text{syst}}$	0.02 ± 0.08	0.099	0.753	0.738
$N_{\text{tracks}} < 4$	$0.287 \pm 0.054_{\text{stat}} \pm 0.025_{\text{syst}}$	$0.339 \pm 0.041_{\text{stat}} \pm 0.022_{\text{syst}}$	0.05 ± 0.08	0.517	0.472	0.715
decaymodes	$0.323 \pm 0.049_{\text{stat}} \pm 0.023_{\text{syst}}$	$0.340 \pm 0.042_{\text{stat}} \pm 0.023_{\text{syst}}$	0.01 ± 0.07	0.037	0.848	0.399
proc13/buckets	$0.321 \pm 0.046_{\text{stat}} \pm 0.019_{\text{syst}}$	$0.291 \pm 0.049_{\text{stat}} \pm 0.017_{\text{syst}}$	0.03 ± 0.07	0.165	0.685	0.098

In the following, we discuss the results of the remaining sample splits: The fitted ratios for each split are presented in tables XXVIII and XXIX, and by utilising their uncertainties as well as the correlation matrix from the fit, the compatibility of these two values is assessed. The uncertainty in their difference is given in tables XXVIII-XXIX as well. Beyond that, it is possible to attribute a p value to this difference quantifying the compatibility, by computing a χ^2 value from the inverted covariance matrix. These p values are included in the tables as well. To assess the consistency of the entire fit with the nominal one, a different test can be implemented: Each split-sample fit produces two measurements, each of which corresponds to one of the nominal fit results, resulting in a total of four measurements. For each split-sample fit, we obtain four measurements, with two expected to be identical. To assess the consistency of these results, a chi-squared function is defined as:

$$\chi^2 = \begin{pmatrix} R(D)_1 - R(D) \\ R(D)_2 - R(D) \\ R(D^*)_1 - R(D^*) \\ R(D^*)_2 - R(D^*) \end{pmatrix}^T \text{Cov}^{-1} \begin{pmatrix} R(D)_1 - R(D) \\ R(D)_2 - R(D) \\ R(D^*)_1 - R(D^*) \\ R(D^*)_2 - R(D^*) \end{pmatrix} \quad (66)$$



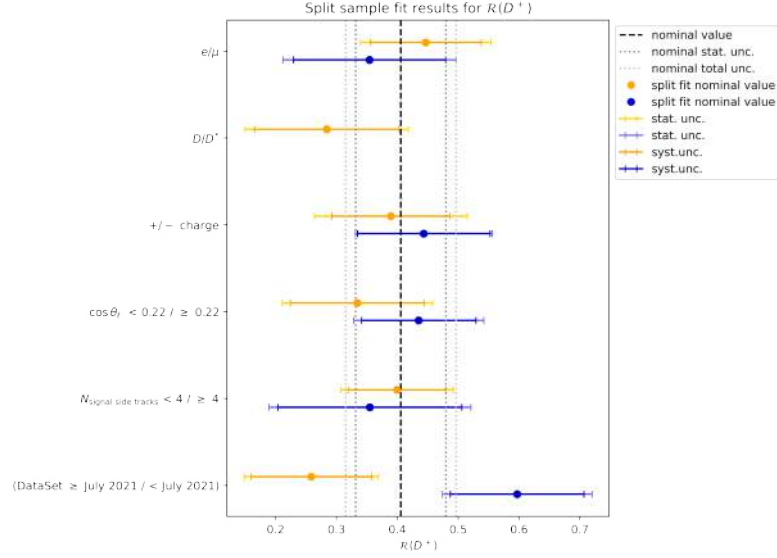
(a)



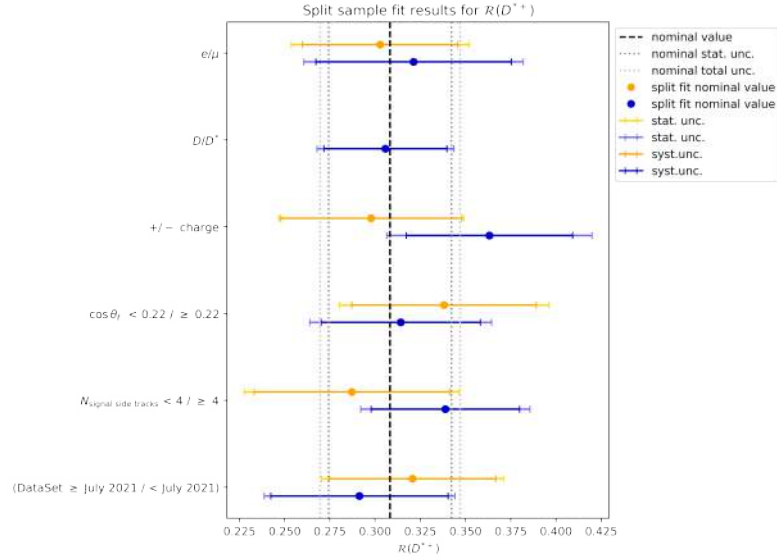
(b)

FIG. 109: (a) ratio of branching ratios for $B \rightarrow D\ell\nu$ over $B \rightarrow D^*\ell\nu$ for e , μ and combined fits as compared to expectations (b) from Ref.[10]

The covariance matrix (Cov) used here is derived from the fit. A corresponding p -value is then calculated using two degrees of freedom. The results are included in the table. These indicate that the values of all performed splits are compatible, except for the split according to the data taking period and the fit version where the feed-down events only were used to calculate $R(D^*)$ in order to compare with the value in the D^* channels. The latter origin lies in the performance of `leptonID`, which is degraded in the `bucket` datasets as compared to `proc13` [1]. The former is reasoned by the low separation power of the signal D and signal feed-down events. Fig.110 provides a summary of all split sample fits, visualising the central values of each fit along with the sizes of the statistical and systematic uncertainties and comparing to the nominal fit results including its associated uncertainties. With the exception of the dataset split in the D case, all fitted values are consistent with the nominal fit.



(a)



(b)

FIG. 110: Summary of the various dataset splits, presenting the central values of each fit along with the sizes of the statistical and systematic uncertainties. These results are compared to the nominal fit and the associated uncertainties:

(a) $\mathcal{R}(D)$

(b) $\mathcal{R}(D^*)$

11.10. Stability against signal probability

As an additional test, we perform the fit with different cuts on the FEI signal probability.

The nominal cut on this variable is $\log \mathcal{P} > -1$, while in this study, we choose looser as well as tighter cuts.

The $(\log) \mathcal{P}$ distribution for all selected tags across the full range of these cuts performed (with the nominal cut at $\log \mathcal{P} > -1$) is provided in Figs. 111- 112 using the best-fit scale factors, thus enabling an evaluation of the agreement in the absence of an

FEI calibration. As the latter is assumed to depend on the signal probability working point and the nominal fit to extract the best-fit factors is performed for $\log \mathcal{P} > -1$, larger deviations are seen in the $\log \mathcal{P} \leq -1$ region of these distributions.

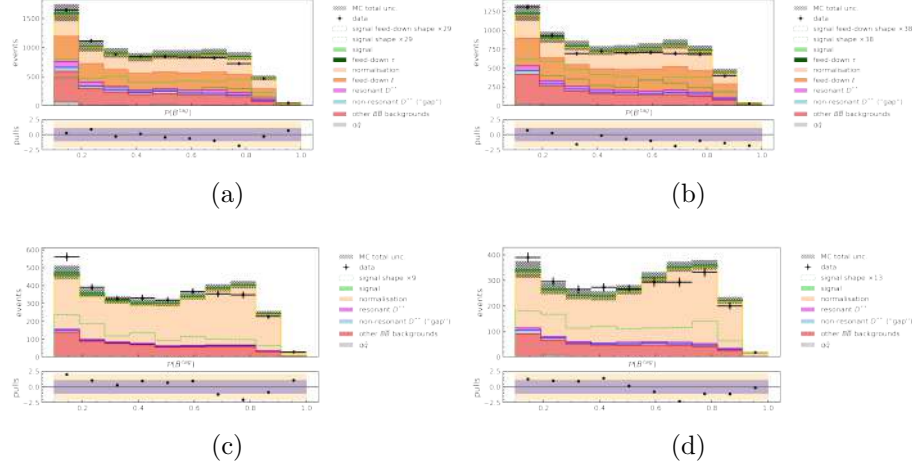


FIG. 111: Distribution of FEI signal probability across the range of applied cuts for the stability test; the templates are scaled according to the best-fit scale factors; systematic uncertainties are included

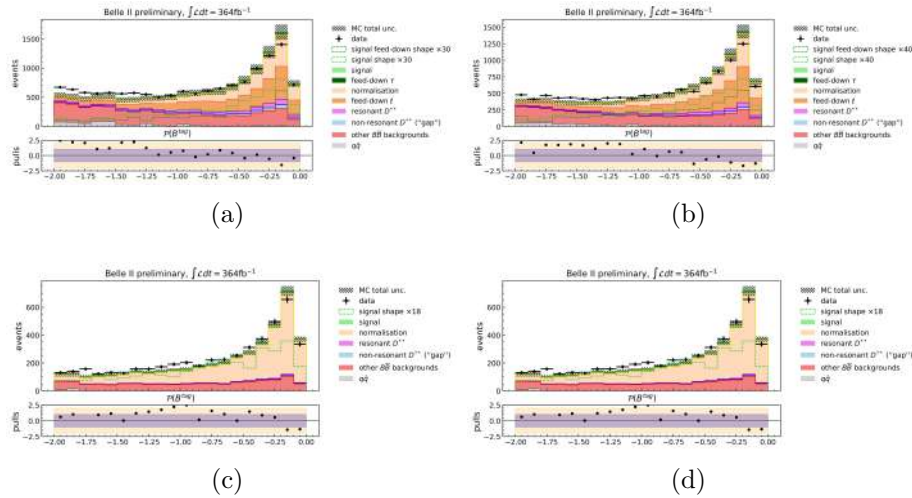


FIG. 112: Same as Fig. 111, but on a logarithmic scale

In this section, we discuss a looser cut at $\mathcal{P} > 0.01$, an intermediate cut at $\mathcal{P} > 0.05$ and a tighter cut at $\mathcal{P} > 0.2$ in detail and provide a summary of different cuts on $\mathcal{P}$ in Fig. ??.

The first of those fits discussed here uses a loose cut on $\log \mathcal{P}$ of greater than > -2 . This fit allows us to examine indirectly how candidates with a lower signal probability affect the fit results, and to assess the influence of the tag selection on the results (we select the best tag according to this variable). Signal overlaps of 36% and 32% are observed in the D and D^* modes, respectively. The results obtained are:

$$\mathcal{R}(D^+) = 0.400 \pm 0.0711_{\text{stat}} \pm 0.068_{\text{syst}} \quad (67)$$

$$\mathcal{R}(D^{*+}) = 0.301 \pm 0.029_{\text{stat}} \pm 0.016_{\text{syst}} \quad (68)$$

and are consistent with the nominal fit results, thereby confirming the stability of the fit outcomes with respect to the choice of the best tag selection variable.

For the intermediate cut at $\mathcal{P} > 0.05$, the fit results remain largely consistent with the original values as well:

$$\mathcal{R}(D^+) = 0.400 \pm 0.072_{\text{stat}} \pm 0.070_{\text{syst}} \quad (69)$$

$$\mathcal{R}(D^{*+}) = 0.289 \pm 0.032_{\text{stat}} \pm 0.017_{\text{syst}} \quad (70)$$

When applying a tighter cut at 0.2, the central values shift more notably (but remain within the uncertainties):

$$\mathcal{R}(D^+) = 0.472 \pm 0.081_{\text{stat}} \pm 0.051_{\text{syst}} \quad (71)$$

$$\mathcal{R}(D^{*+}) = 0.280 \pm 0.037_{\text{stat}} \pm 0.019_{\text{syst}} \quad (72)$$

To further validate these results taking into account systematic correlations, a split fit similar to the split sample fits discussed in Section 11.1.9 was performed with the cut set at 0.2, yielding two separate values for $\mathcal{R}(D^{(*)})$. Both results are found to be compatible considering the uncertainty in their difference when taking into account the correlation matrix obtained from the fit:

$$\mathcal{R}(D^+)_{\text{low}} = 0.425 \pm 0.123_{\text{stat}} \pm 0.068_{\text{syst}} \quad (73)$$

$$\mathcal{R}(D^+)_{\text{high}} = 0.468 \pm 0.081_{\text{stat}} \pm 0.047_{\text{syst}} \quad (74)$$

$$\mathcal{R}(D^{*+})_{\text{low}} = 0.307 \pm 0.044_{\text{stat}} \pm 0.022_{\text{syst}} \quad (75)$$

$$\mathcal{R}(D^{*+})_{\text{high}} = 0.286 \pm 0.038_{\text{stat}} \pm 0.017_{\text{syst}} \quad (76)$$

$$\Delta\mathcal{R}(D^+) = 0.04 \pm 0.16 \rightarrow \chi^2 = 0.071 \rightarrow p = 0.79 \quad (77)$$

$$\Delta\mathcal{R}(D^{*+}) = 0.02 \pm 0.06 \rightarrow \chi^2 = 0.108 \rightarrow p = 0.74 \quad (78)$$

Calculating a combined χ^2 as above yields:

$$\chi^2/\text{dof} = 0.296 \rightarrow p = 0.74 \quad (79)$$

11.11. Candidate multiplicities

This Section deals with the number of B candidates per event before applying best candidate selections. The number of tag candidates before applying the BCS on the

Number of cand. per event after all selections	D mode	D^* mode
data	1.039 ± 0.012	1.068 ± 0.020
MC total	1.043 ± 0.006	1.078 ± 0.008
MC signal	1.02 ± 0.04	1.07 ± 0.04
MC normalisation	1.022 ± 0.008	1.062 ± 0.009

tag side, which is determined according to the FEI signal probability, is not directly accessible as the FEI skims are used, where each skim already ranks the best 20 candidates

according to the signal probability. The BCS is applied online before reconstructing the signal side, ensuring that only one tag candidate per event is considered during the signal side reconstruction. The cross-check using a lower signal probability cut is an indication that the signal probability used as BCS does not introduce a bias (see the previous Section 11.9. In turn, the candidate multiplicities on the signal side, where the BCS is applied offline, can still be examined to determine whether any difference exists between signal and normalisation samples, which could indicate potential bias. The presented numbers in table ?? demonstrate compatibility within the associated uncertainties.

11.12. BDT input variable modelling post-fit

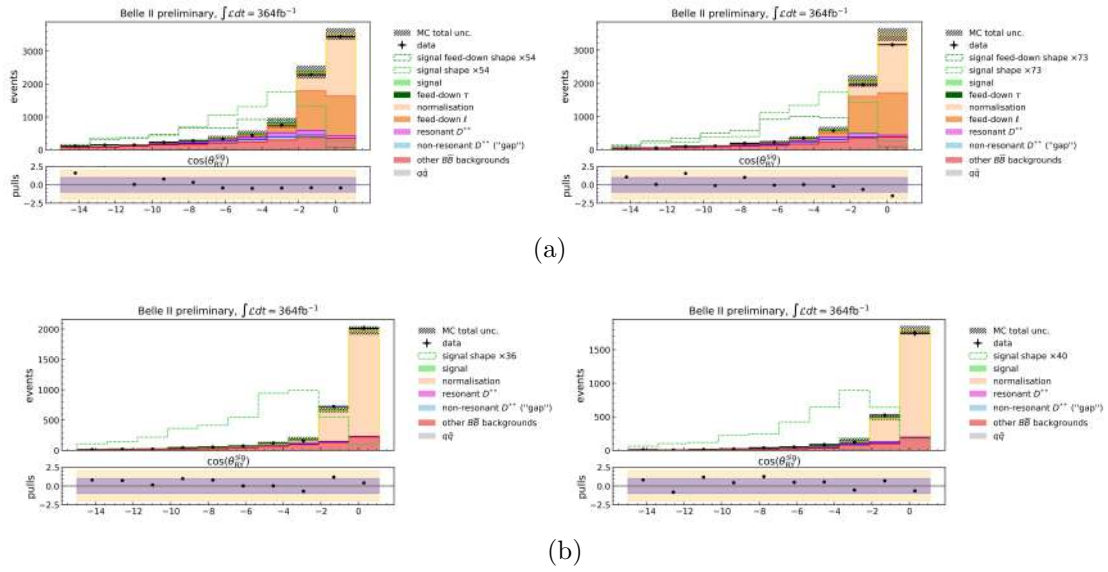


FIG. 113: Distribution of $\cos \theta_{BY}$ with post-fit scaling, including all systematic uncertainties for:

(a) De (b) $D\mu$ (c) D^*e (d) $D^*\mu$

Evaluating the pre-fit agreement in the BDT input variables presents challenges, as there is no FEI calibration available (only scale factors derived from a previous fit, see Sect. 9.9.2). Therefore, the most meaningful way to assess the agreement is by examining the post-fit distributions. These distributions are shown in Figs. 135-138. In all distributions, systematic uncertainties and post-fit scale factors are applied. Upon analysing the pulls of these distributions, a pull exceeding 2σ is observed in the lower $\cos \theta_{BY}$ region of the De mode. As this is the most important BDT input variable, this observation leads us to the development of a strategy to assign a systematic uncertainty to this issue, which will be discussed in this Section. We also provide an overview of the background composition in this region in table XXX.

The proposed strategy involves evaluating the impact of the mis-modelling in the specified region by reweighting the MC samples according to the data distribution and subsequently refitting an asimov distribution. The reweighting is performed using event-based weights. To determine these weights, a spline is fitted to the data-to-MC ratio to assign a weight based on the corresponding $\cos \theta_{BY}$ value. As observed in the plot of the data-to-MC ratio in Fig. ??, there is a discernible linear trend. Consequently, a linear spline is fitted. Since in the high $\cos \theta_{BY}$ region, the agreement is already reasonable and

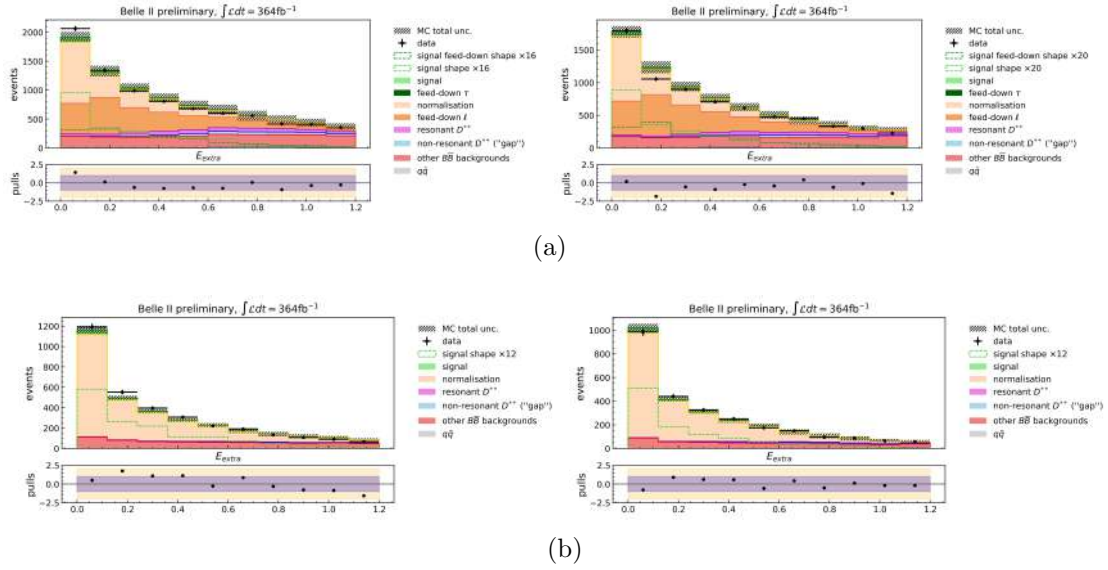


FIG. 114: Distribution of E_{extra} with post-fit scaling, including all systematic uncertainties for:
 (a) De (b) $D\mu$ (c) D^*e (d) $D^*\mu$

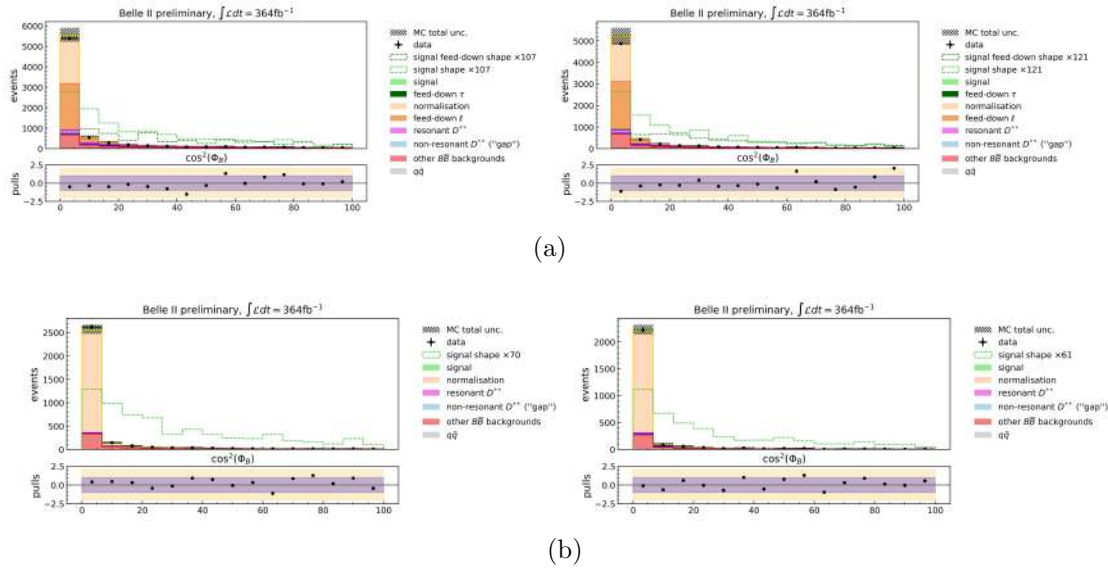


FIG. 115: Distribution of $\cos^2 \Phi_B$ with post-fit scaling, including all systematic uncertainties for:
 (a) De (b) $D\mu$ (c) D^*e (d) $D^*\mu$

1761 should not be addressed by this systematic, once a value of 1 is reached, the weight is
 1762 capped at 1. This approach leads to the fact that the normalisation events are minimally
 1763 affected by the reweighting process.

1765 The reweighting is performed independently within each fitted channel, resulting in
 1766 four distinct spline fits. The reweighted $\cos \theta_{BY}$ distributions obtained by applying event-
 1767 based weights according to the corresponding $\cos \theta_{BY}$ values can be seen in Fig.119. The
 1768 pulls now indicate good agreement across all bins. The so-derived event weights are
 1769 subsequently utilised in an asimov fit. After applying the $\cos \theta_{BY}$ reweighting, each
 1770 template is normalised to match the number of events in the original templates, ensuring

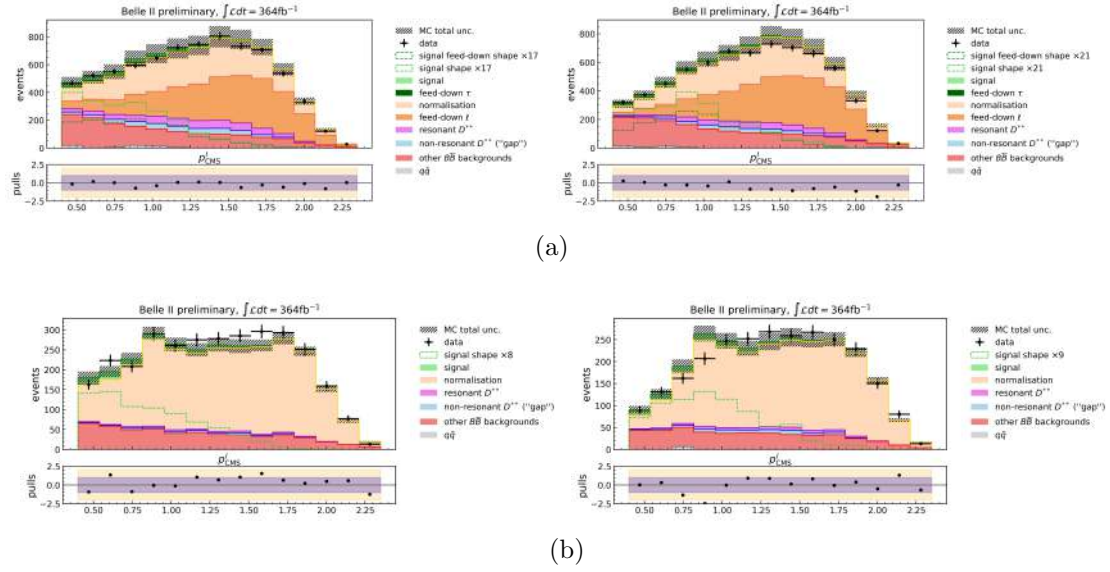


FIG. 116: Distribution of p_i^* with post-fit scaling, including all systematic uncertainties for:

(a) De (b) $D\mu$ (c) D^*e (d) $D^*\mu$

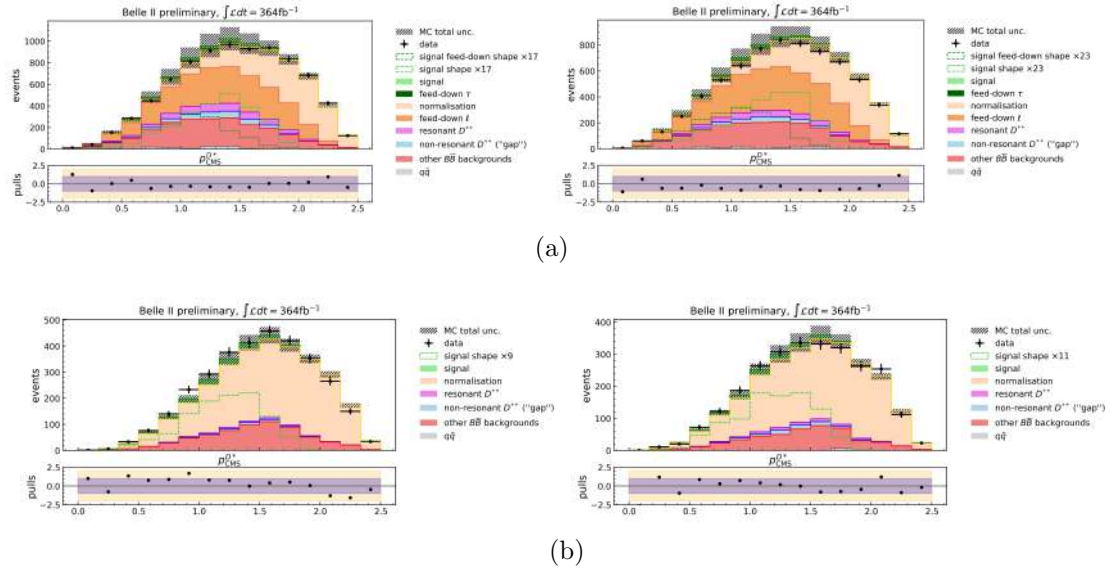


FIG. 117: Distribution of $p_{D(*)}^*$ with post-fit scaling, including all systematic uncertainties for:

(a) De (b) $D\mu$ (c) D^*e (d) $D^*\mu$

that only the impact of the shape difference is examined. Two possibilities exist here:

- The first possibility involves reweighting the asimov data and performing the fit using the nominal fit templates. This approach results in a slight change in the nominal $\mathcal{R}(D)$ and $\mathcal{R}(D^*)$ values by 0.01 and 0.005, respectively.
- The second possibility involves the opposite approach: reweighting the fit templates and performing the fit to the nominal asimov data distribution. In this case, the central values shift by 0.015 and 0.005.

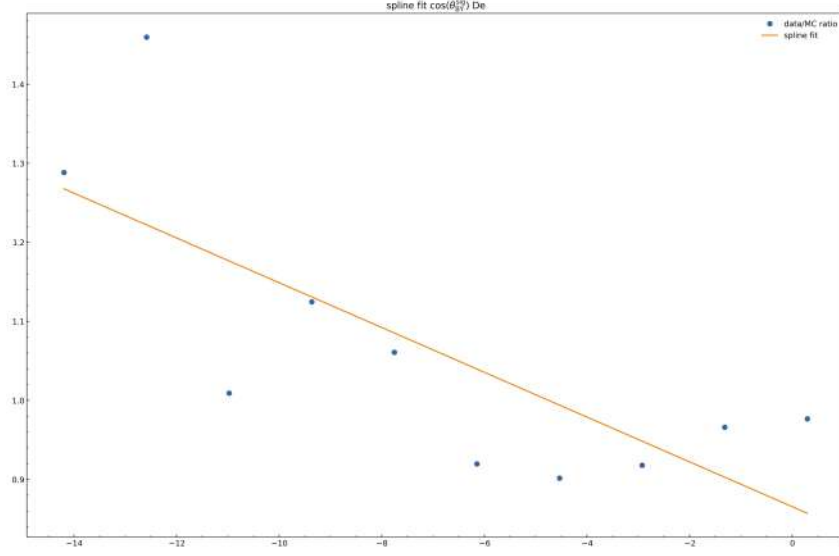


FIG. 118: Data-MC ratio and spline fit

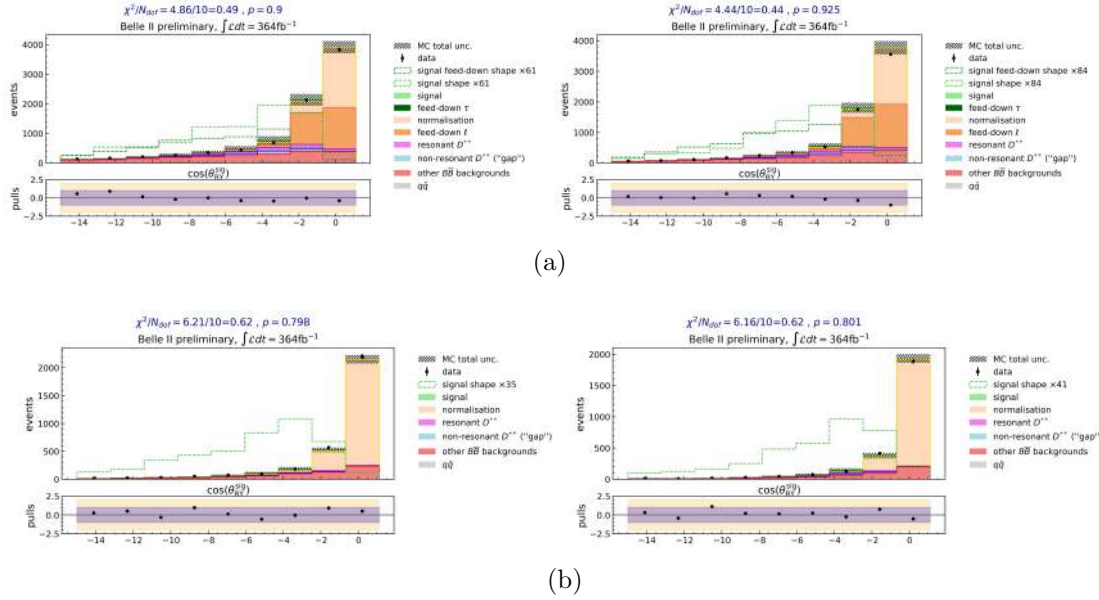


FIG. 119: Distribution of $\cos \theta_{BY}$ applying the reweighting derived from its data-MC ratio for:

(a) De (b) $D\mu$ (c) D^*e (d) $D^*\mu$

Since the difference in the positive direction is not identical to that in the negative direction for $\mathcal{R}(D)$, the larger observed difference is used, and the uncertainty is symmetrised accordingly. The observed differences are then added in quadrature to the systematic uncertainty on $\mathcal{R}(D)$ and $\mathcal{R}(D^*)$.

The final results with uncertainties, incorporating the new systematic uncertainty, are:

$$\mathcal{R}(D^+) = 0.406 \pm 0.074_{\text{stat}} \pm 0.055_{\text{syst}} \quad (80)$$

and

$$\mathcal{R}(D^{*+}) = 0.308 \pm 0.034_{\text{stat}} \pm 0.019_{\text{syst}}. \quad (81)$$

This results in being the fourth largest systematic uncertainty for $\mathcal{R}(D)$ and the third largest one for $\mathcal{R}(D^*)$ (cf. table XXXV). However, as seen in the confidence ellipse plot in

TABLE XXX: Backgrounds in the region $\cos \theta_{BY} < -8$ of the De mode

%	D ⁺	D ⁰	D _s ⁺	τ	B ⁺		
42% true electrons from..							
fraction wrt total backgrounds	16%	9%	5%	3%	3%		
44% pions faking electrons from..	D ⁰	B ⁰	K ^{∗+}	D ⁺	K _S ⁰	ρ ⁺	ω
fraction wrt total backgrounds	5%	5%	4%	4%	3%	3%	3%
11% Kaons faking electrons from..	D ⁰	K ^{∗0}	B ⁰	D ⁺	K ^{∗+}		
fraction wrt total backgrounds	3%	2%	2%	1%	1%		
6% true e _s from..	B ⁰	D _s ⁺	B ⁺	D ⁺			
fraction wrt total backgrounds	3%	2%	1%	0.03%			

Fig.120, this additional uncertainty does not significantly alter the confidence ellipse, as it remains dominated by the limitation in data statistics.

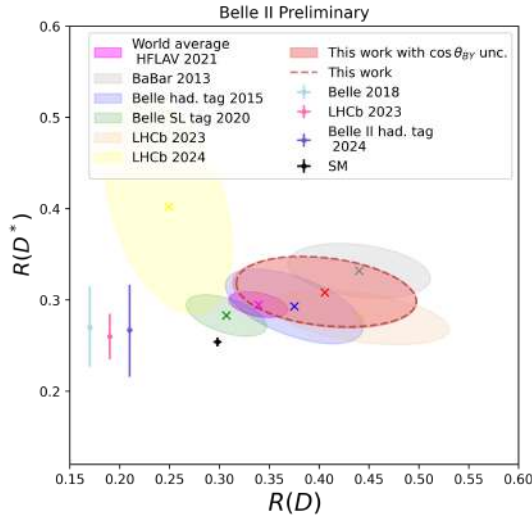


FIG. 120: The 39.3% confidence region in the $\mathcal{R}(D)$ - $\mathcal{R}(D^*)$ plane with updated uncertainties compared to the one without $\cos \theta_{BY}$ modelling uncertainty.

11.13. Fractions of certain other events with true τ leptons

This section investigates the impact of events involving true τ decays within the reconstructed data, as an observed increase in $\mathcal{R}(D^{(*)})$ imply a proportionally higher yield of these events compared to SM predictions.

Firstly, we analyse the fraction of events from $B^+ \rightarrow D^{(*)}\tau\nu$ and $B^+ \rightarrow D^0\ell\nu$ decays within the dataset. Given that isospin symmetry is assumed for $\mathcal{R}(D^{(*)})$, an increased value of $\mathcal{R}(D^{(*)})$ value would suggest a correspondingly higher yield of $B^+ \rightarrow D^{(*)}\tau\nu$ events compared to $B^+ \rightarrow D^0\ell\nu$ events than theoretically expected from the SM. Table XXXI presents the fractions of signal events that originate from a B^+ decay relative to the nominal signal events within each bin, enabling an assessment of their contributions to signal-sensitive regions (bins 14-21). These fractions are provided separately for each fit channel De , $D\mu$, D^*e , $D^*\mu$. Similarly, table XXXII shows the fractions of $B^+ \rightarrow D^0\ell\nu$ decays relative to the nominal normalisation events per bin and per channel. The

1803 identification of both the charged-B signal events and charged-B normalisation events
 1804 follows the same methodology as for neutral-B signal and normalisation events (see Sec.5),
 1805 i.e. also including the feed-down events in which the D from the D^* decay has been
 1806 reconstructed in the D channel; the only distinction is that the lepton (grand-)mother is
 1807 expected to be a B^+ with the corresponding allowed $D^{(*)}$ decay products. These events
 1808 are further illustrated through 2D z_τ - z_{diff} distributions in Figs.121-122. Annotated with
 1809 raw event counts, these distributions show that a greater proportion of events aligns with
 1810 the normalisation category, particularly in the D mode, compared to the signal category.

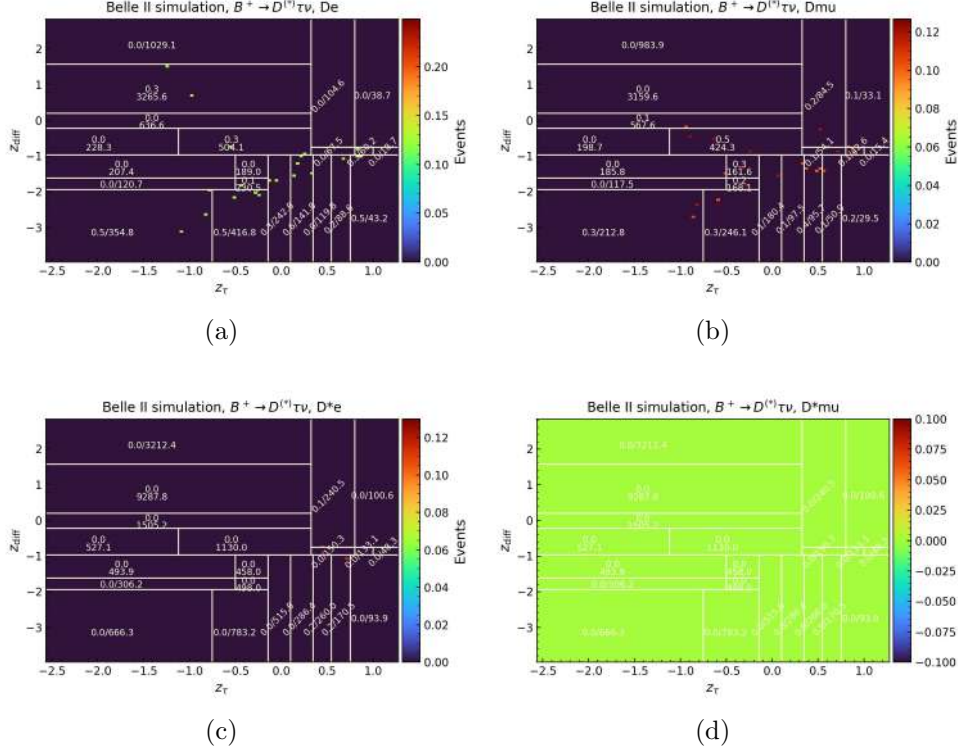


FIG. 121: 2D distributions for charged-B signal events with raw event numbers for $B^+ \rightarrow D^{(*)}\tau\nu$ and $B^0 \rightarrow D^{(*)}\tau\nu$

In the nominal fit, these decays contribute to the “Other Background” components, , with their yields scaled according to the BF’s used for MC production. In order to assess the systematic uncertainties associated with the BF’s of the $B^+ \rightarrow D^*\tau\nu$ decays used in the simulation, as well as the anticipated yields from $B^+ \rightarrow D^0\ell\nu$ decays, we conduct a fit where charged-B events were assigned to signal and normalisation categories rather than to the backgrounds, resulting in a slight decrease in $\mathcal{R}(D)$ but no significant effect on $\mathcal{R}(D^*)$:

$$\mathcal{R}(D) = 0.401 \pm 0.075_{\text{stat}} \pm 0.050_{\text{syst}}$$

$$\mathcal{R}(D^*) = 0.308 \pm 0.034_{\text{stat}} \pm 0.018_{\text{syst}}$$

For reference, the nominal fit results are:

$$\mathcal{R}(D) = 0.406 \pm 0.074_{\text{stat}} \pm 0.053_{\text{syst}}$$

$$\mathcal{R}(D^*) = 0.308 \pm 0.034_{\text{stat}} \pm 0.018_{\text{syst}}$$

1811 The fact that this minor variation is only observed in the D mode is consistent with the
 1812 observation that the fractions of charged-B events are generally lower for the D^* mode

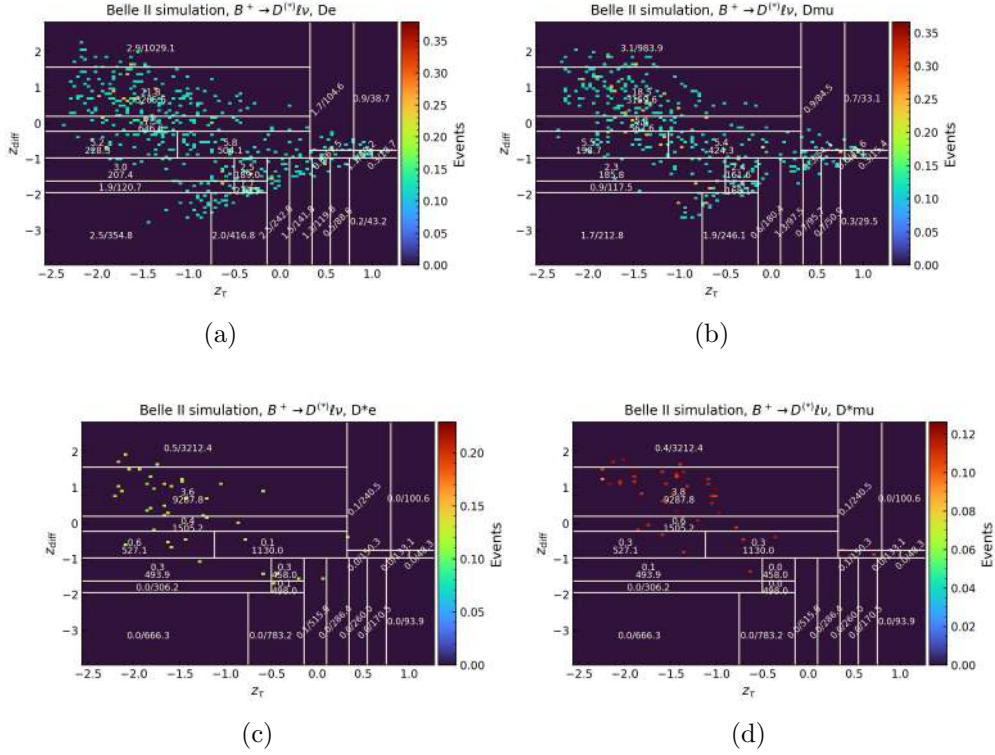


FIG. 122: Same as Fig.121, but for $B^+ \rightarrow D^{(*)} \ell \nu$.

than for the D mode. Furthermore, given the larger fraction of these events categorised within the normalisation category, the lowering of the fitted $\mathcal{R}(D)$ value is reasonable well. Considering that the central value shift is approximately 1%, relative to the total uncertainty of 22%, treating these events as background appears justified. Figs. 123-124 display the pre- and post-fit distributions for this fit.

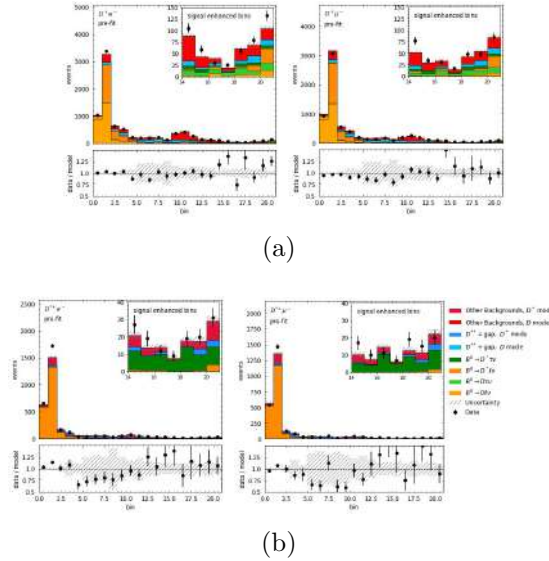


FIG. 123: Pre-fit (a-d) distributions within the four fit channels for the fit that includes charged-B signal and normalisation decays.

TABLE XXXI: Fractions of $B^+ \rightarrow D^{(*)}\tau\nu$ within the four fit channels in %
wrt. nominal signal events
using pre-fit scaling

Bin Number	De	$D\mu$	D^*e	$D^*\mu$
0	0.00	0.00	0.00	0.00
1	0.01	0.00	0.00	0.00
2	0.00	0.02	0.00	0.00
3	0.06	0.11	0.00	0.00
4	0.00	0.00	0.00	0.00
5	0.00	0.18	0.00	0.00
6	0.06	0.09	0.00	0.00
7	0.00	0.00	0.00	0.00
8	0.00	0.00	0.00	0.00
9	0.14	0.13	0.00	0.00
10	0.11	0.12	0.00	0.00
11	0.13	0.08	0.00	0.00
12	0.43	0.13	0.00	0.00
13	0.00	0.40	0.06	0.00
14	0.18	0.26	0.09	0.00
15	1.06	0.54	0.00	0.00
16	0.00	0.39	0.00	0.00
17	0.00	0.00	0.00	0.00
18	0.53	0.31	0.00	0.00
19	0.00	0.28	0.00	0.00
20	0.00	0.18	0.06	0.00
total	0.04	0.04	0.00	0.0

To address the potential cross-feed contributions from non-leptonic τ decays, we examine the ratio of events with hadronic τ decays compared to those with leptonic τ decays following a similar approach as with the charged-B events. Table XXXIV displays these ratios per bin for each fit channel and Fig.125 illustrates their 2D distribution alongside the raw event counts.

In the nominal fit, events from hadronic τ decays are categorised within the "Other backgrounds" components, consistent with the treatment of charged-B leptonic τ decays. To assess the impact of this categorisation, we perform a test reassigning these events to the signal. Similarly to the test above, we also include feed-down events with hadronic τ decays here. The resulting fit outcome shows only marginal differences from the nominal fit results:

$$\mathcal{R}(D) = 0.411 \pm 0.074_{stat} \pm 0.048_{syst}$$

$$\mathcal{R}(D^*) = 0.307 \pm 0.033_{stat} \pm 0.017_{syst}$$

Considering the ratios in table XXXIV, these deviations appear reasonable. For a comprehensive integration of hadronic τ decays into the fit of $\mathcal{R}(D^{(*)})$, however, it would be necessary to account for individual branching fractions of these decays and adjust the signal reconstruction efficiencies accordingly. Nonetheless, the fit outcome indicates that treating these events as part of the background is reasonable, as the observed change

TABLE XXXII: Fractions of $B^+ \rightarrow D^{(*)}\ell\nu$ within the four fit channels in %
wrt. nominal normalisation events
using pre-fit scaling

Bin Number	De	$D\mu$	D^*e	$D^*\mu$
0	0.28	0.32	0.01	0.01
1	0.67	0.58	0.04	0.04
2	1.18	1.01	0.03	0.04
3	1.16	1.27	0.01	0.03
4	2.28	2.79	0.11	0.05
5	1.32	1.47	0.06	0.00
6	0.75	0.81	0.03	0.00
7	1.43	1.23	0.06	0.03
8	1.55	0.77	0.00	0.00
9	0.71	0.78	0.00	0.00
10	0.48	0.76	0.00	0.00
11	1.03	0.33	0.03	0.00
12	1.08	1.36	0.00	0.00
13	1.05	0.77	0.00	0.00
14	0.53	1.34	0.00	0.00
15	0.37	0.95	0.00	0.00
16	2.40	2.24	0.00	0.00
17	1.66	1.81	0.00	0.00
18	1.81	1.39	0.00	0.00
19	1.17	2.48	0.00	0.09
20	1.63	1.05	0.06	0.06
total	0.83	0.79	0.03	0.0

Channel	$q\bar{q}$ fraction
De	5.5%
$D\mu$	3.9%
D^*e	0.0%
$D^*\mu$	1.4%

TABLE XXXIII: Fraction of $q\bar{q}$ events estimated from off-resonance data within each fit channel relative to the total number of the fit category "other backgrounds".

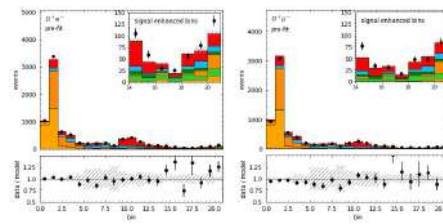
in central values remains small relative to statistical uncertainties. Pre- and post-fit distributions can be seen in Figs. 126-127.

11.14. Impact of `nCDCHits > 0` cut

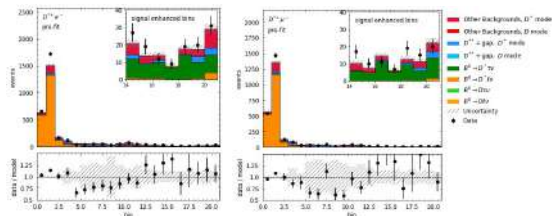
This Section deals with the impact of the application of the `nCDCHits > 0` cut, since the PID corrections were determined without incorporating this criterion. To evaluate the impact of this cut, a subset of $D^{(*)}\tau\nu$ signal MC events and a corresponding subset

TABLE XXXIV: Ratios for $\tau \rightarrow \text{had.}$ (%)
wrt. nominal signal events

Bin Number	De	$D\mu$	D^*e	$D^*\mu$
0	10.7	37.5	0.0	32.9
1	8.3	17.4	4.6	13.3
2	3.7	18.6	8.0	8.2
3	5.2	30.9	4.9	14.3
4	0.0	0.0	0.0	0.0
5	9.6	15.2	0.0	16.6
6	33.9	62.0	4.8	14.3
7	21.4	43.5	0.0	9.3
8	124.2	284.4	0.0	16.1
9	245.1	176.1	53.5	43.0
10	63.7	120.2	8.1	34.8
11	28.1	75.5	2.3	19.6
12	7.0	49.6	4.0	15.5
13	9.7	35.8	4.3	8.3
14	7.3	15.2	0.6	4.1
15	3.0	14.3	0.0	4.3
16	0.0	6.8	0.9	5.4
17	2.2	4.7	2.2	4.8
18	0.8	7.1	0.0	1.6
19	0.0	6.2	1.6	4.4
20	2.0	7.2	3.1	4.1
total	9.9	23.9	3.3	9.3



(a)



(b)

FIG. 126: Pre-fit (a-d) distributions within the four fit channels for the fit that includes charged-B signal and normalisation decays.

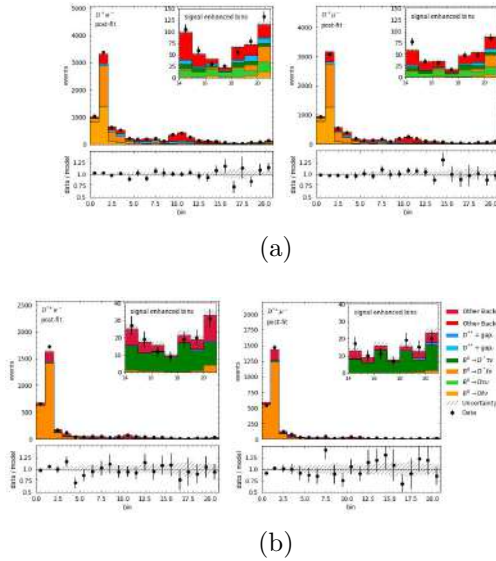


FIG. 127: Post-fit distributions within the four fit channels for the fit that includes charged-B signal and normalisation decays.

- difference in truth-matched normalisation events: 0.47%

Moreover, as these differences are approximately consistent across both signal and normalisation categories, they are likely to effectively cancel in the computation of the $\mathcal{R}(D^{(*)})$ ratio.

Additionally, a related investigation conducted by Alessandro Gaz[1], investigating the impact of an `nCDCHits` > 20 cut, found only a permille-level difference. This further supports the conclusion that the results are robust against leaving out this cut.

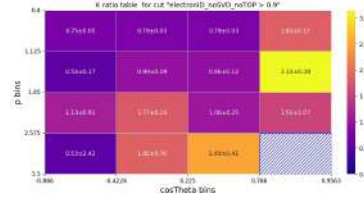
11.15. Stability against PID cut

This section examines the impact of tightening the PID cut for the signal lepton. This involves modifying the PID correction factors and adjusting weight variations to account for systematic uncertainties. Official PID tables provide working points at thresholds of 0.5, 0.9, 0.95, and 0.99. Our nominal selections are `electronID_noSVD_noTOP` > 0.5 and `muonID_noSVD` > 0.9. In this study, tighter thresholds of `electronID_noSVD_noTOP` > 0.9 and `muonID_noSVD` > 0.95 are used. For this purpose, it is necessary to exchange the lepton ID weights for true leptons as well as fakes. However, due to unphysically high fake rate corrections (> 100) within the official fake rate tables, an alternative set of fake rate tables was generated, also allowing for the application of the `nCDCHits` > 0 cut which has not been used for the generation of the official tables. This adjustment, however, could not be extended to efficiency tables because the generation of efficiency weights is only available for $p > 1.5$ GeV. Consequently, the official efficiency tables were retained, as the issue with high weights only pertains to the fake rate corrections. See Fig.128 for the self-generated fake rate correction tables in bins of p and $\cos\theta$.

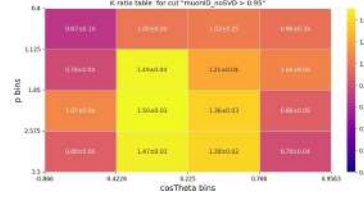
The fit results with the tighter PID threshold are:

$$\mathcal{R}(D) = 0.418 \pm 0.078_{\text{stat}} \pm 0.048_{\text{svst}}$$

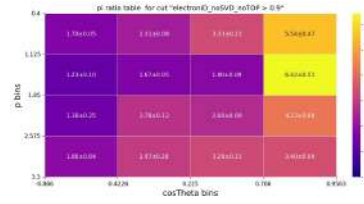
Figs.129-130 shows the pre- and postfit plots for this fit.



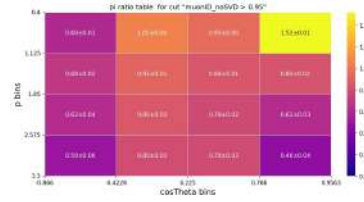
(a)



(b)



(c)



(d)

FIG. 128: Self-generated fake rate tables for:

- (a) Kaon-electron fakes
- (b) Kaon-muon fakes
- (c) pion-electron fakes
- (d) pion-muon fakes

To assess the compatibility of the results with the nominal fit, a bootstrap study, analogous to that used for evaluating signal probability cut dependence (see Section 11.10), is conducted. A bootstrap dataset is generated from the nominal data and fit with and without the tighter PID cut, incorporating the new PID weights and variations in the case of the tighter PID cut. The resulting 2D distribution of $\Delta\mathcal{R}(D) = (\mathcal{R}(D) | \text{loose PID}) - (\mathcal{R}(D) | \text{tight PID})$ and $\Delta\mathcal{R}(D^*) = (\mathcal{R}(D^*) | \text{loose PID}) - (\mathcal{R}(D^*) | \text{tight PID})$ allows for the derivation of a covariance matrix. Using this matrix, a chi-square statistic and associated p -value for the observed differences relative to $(0,0)$ can be calculated, yielding a p -value of 0.22. The 2D bootstrap distribution can be seen in Fig.131.

The observed difference lies within this distribution, suggesting that the difference is likely attributable to statistical variation.

To fully verify the compatibility of the results, it would be necessary to consider correlations between subsets defined as $(\text{loose PID cut} < \text{PID} < \text{tight PID cut})$ and $(\text{PID} > \text{tight$

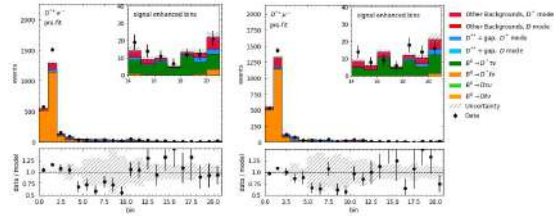
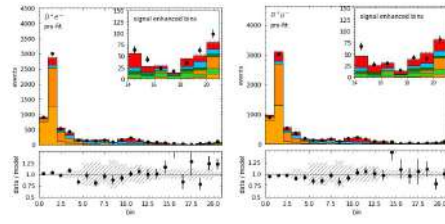


FIG. 129: Pre-fit (a-d) distributions within the four fit channels for the fit with tighter PID threshold.

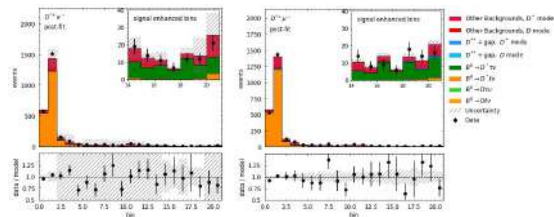
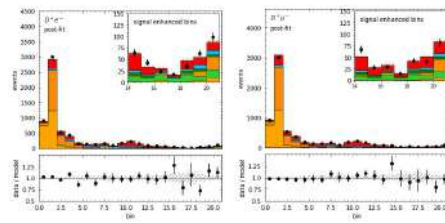


FIG. 130: Post-fit distributions within the four fit channels for the fit with tighter PID threshold.

¹⁸⁷⁶ PID cut). However, this correlation analysis is challenging because no PID corrections
¹⁸⁷⁷ exist for the intermediate subset of (loose PID cut<PID<tight PID cut).

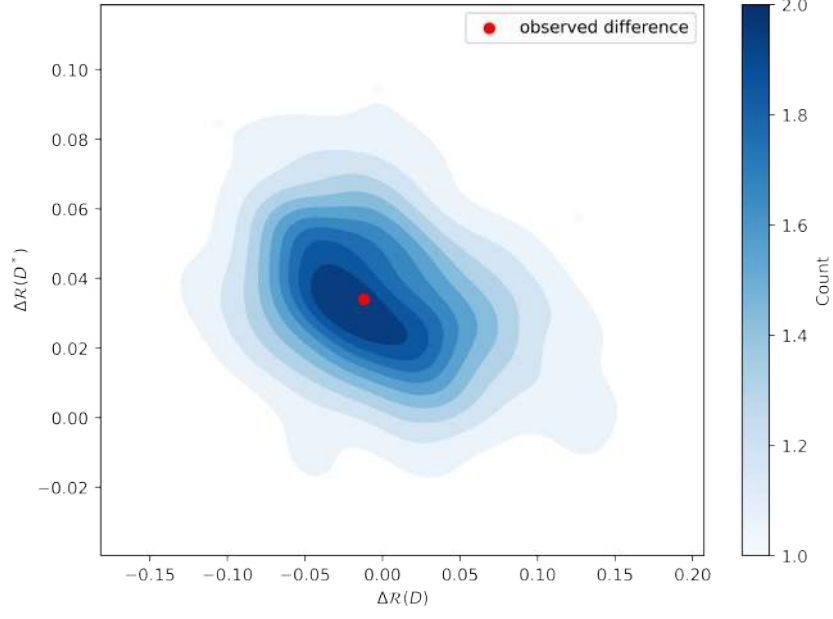


FIG. 131: 2D distribution of $\Delta\mathcal{R}(D) = (\mathcal{R}(D) \mid \text{loose PID}) - (\mathcal{R}(D) \mid \text{tight PID})$ and $\Delta\mathcal{R}(D^*) = (\mathcal{R}(D^*) \mid \text{loose PID}) - (\mathcal{R}(D^*) \mid \text{tight PID})$

12. CHANGES DURING CWR1 AND FINAL RESULTS

During the CWR1 stage we decided to slightly change the fit setup by linking the normalisation of $\bar{B} \rightarrow D^{**} \ell \bar{\nu}_\ell / \bar{B} \rightarrow D_{\text{gap}}^{**} \ell \bar{\nu}_\ell$ events between electron and muon channels. This slightly changed the fit results.

Further, we have introduced a new systematic uncertainty related to the fraction of $q\bar{q}$ events within the background template by varying the latter within $\pm 10\%$ to account for potential shape changes.

This 10% uncertainty is motivated by the observed difference between the processed off-resonance data and continuum MC. With a slightly looser selection on E_{extra} and signal probability to enhance statistical precision, we obtain a relative difference of:

$$\frac{\text{Data} - \text{MC}}{\text{Data}} = (8.5 \pm 3.5)\%$$

When applying identical selection criteria, the difference becomes (19 ± 16) , which is statistically too imprecise to be conclusive.

The final results can be found in this section.

12.1. $\mathcal{R}(D^+)$ and $\mathcal{R}(D^{*+})$

Figure 132 shows the fitted classifiers bins for $D^{(*)+}e^-$ and $D^{(*)+}\mu^-$ categories. We measure

$$\mathcal{R}(D^+) = 0.418 \pm 0.074 \text{ (stat)} \pm 0.051 \text{ (syst)}, \quad (82)$$

$$\mathcal{R}(D^{*+}) = 0.306 \pm 0.034 \text{ (stat)} \pm 0.018 \text{ (syst)}, \quad (83)$$

with a correlation of $\rho = -0.24$. These values are compatible with the SM predictions of $\mathcal{R}(D) = 0.296 \pm 0.005$ and $\mathcal{R}(D^*) = 0.254 \pm 0.004$ [11, 13, 14, 16, 26, 34, 39] within 1.7 standard deviations. The post-fit p-value is 8.3% and evaluated using the saturated likelihood method [6]. Figures 133-134 show the distributions of E_{extra} and p_i^* for the signal enriched bins of the two-dimensional classifier with the post-fit scaling applied, while Figs. 135-139 visualise the post-fit agreement in the entire fit region for the five BDT input variables. Table XXXVI summarises the yields determined using the best-fit scale factors.

Figure 140 shows the leading 20 systematic uncertainties on $\mathcal{R}(D^+)$ and $\mathcal{R}(D^{*+})$, excluding the dominant systematic error due to the limited MC sample size. If stated the number (#) indicates the eigendirection of the uncertainty source. The uncertainty 'Choice of FF model' parametrizes the impact of changing the signal and normalization form factor parameterization, cf. Section ???. Within the FF uncertainties for the D^{**} resonances, we distinguish between D_{broad}^* resonances ($D_0^*(2400)$ and $D_1(2430)$) and D_{narrow}^* resonances ($D_1(2420)$ and $D_2^*(2460)$) as they are described by different parametrizations due to their varying decay widths.

The pull is defined as the difference of the post-fit value ($\hat{\theta}$) with respect to the nominal input value of the nuisance parameter (θ_0), and normalized with the post-fit uncertainty ($\Delta\theta$).

Also shown are the impacts on the relative uncertainties $\Delta\mathcal{R}(D^+)/\mathcal{R}(D^+)$ and $\Delta\mathcal{R}(D^{*+})/\mathcal{R}(D^{*+})$ on the determined ratios. We estimate the pre-fit impact by analyzing how variations in the NP affect the POI, independent of the other fitted parameters: In order to isolate the specific contribution of the NP, we construct an Asimov dataset

TABLE XXXV: Systematic uncertainties on $\mathcal{R}(D^+)$ and $\mathcal{R}(D^{*+})$ ranked by magnitude on the uncertainty on $\mathcal{R}(D^+)$. The percentage values in brackets indicate the relative uncertainty.

Systematic Uncertainty	$\Delta\mathcal{R}(D^+)$	$\Delta\mathcal{R}(D^{*+})$
Additive		
MC sample size	0.033 (8.0%)	0.014 (4.7%)
Gap $\mathcal{B}$	0.027 (6.4%)	0.001 (0.1%)
LID efficiency (μ)	0.022 (5.1%)	0.001 (0.1%)
Fake rates (e)	0.012 (2.9%)	0.003 (0.9%)
Continuum fraction	0.002 (0.6%)	0.001 (0.2%)
Gap FFs	0.002 (0.5%)	0.001 (0.2%)
$\bar{B} \rightarrow D^{(*)}\ell\bar{\nu}_\ell / \tau\bar{\nu}_\tau$ FFs	0.002 (0.5%)	0.002 (0.7%)
$\mathcal{B}(\bar{B} \rightarrow D^{**}\ell\bar{\nu}_\ell)$	0.002 (0.5%)	0.001 (0.1%)
$\bar{B} \rightarrow D^{**}\ell\bar{\nu}_\ell$ FFs	0.001 (0.3%)	0.001 (0.2%)
BDT modeling	0.001 (0.3%)	0.001 (0.2%)
LID efficiency (e)	0.001 (0.1%)	0.001 (0.2%)
Fake rates (μ)	0.001 (0.1%)	0.001 (0.1%)
$\pi^\pm$ from $D^* \rightarrow D\pi$	0.003 (0.7%)	0.001 (0.1%)
Total Additive Uncertainty	0.050 (12%)	0.015 (4.8%)
Multiplicative		
$\bar{B} \rightarrow D^{(*)}\ell\bar{\nu}_\ell / \tau\bar{\nu}_\tau$ FFs	0.009 (2.1%)	0.011 (3.5%)
MC sample size	0.007 (1.7%)	0.004 (1.2%)
LID efficiency (e)	0.001 (0.2%)	0.001 (0.2%)
$\mathcal{B}(\tau^- \rightarrow \ell^-\bar{\nu}_\ell\nu_\tau)$	0.001 (0.2%)	0.001 (0.2%)
LID efficiency (μ)	0.001 (0.1%)	0.001 (0.1%)
Tracking efficiency	0.001 (0.1%)	0.001 (0.1%)
$\pi^\pm$ from $D^* \rightarrow D\pi$	– (–)	0.001 (0.2%)
Total Multiplicative Uncertainty	0.012 (2.8%)	0.011 (3.7%)
Total Syst. Uncertainty	0.051 (12%)	0.018 (6.2%)
Total Stat. Uncertainty	0.074 (18%)	0.034 (11%)
Total Uncertainty	0.090 (22%)	0.039 (13%)

¹⁹¹³ in which the NP under investigation is maintained at its initial value, while all other
¹⁹¹⁴ parameters are set to their best-fit values obtained from the nominal fit. This dataset
¹⁹¹⁵ is then refitted, keeping all parameters except the respective POI ($\Delta\mathcal{R}(\mathcal{D}^+)/\mathcal{R}(\mathcal{D}^+)$ or
¹⁹¹⁶ $\Delta\mathcal{R}(\mathcal{D}^{*+})/\mathcal{R}(\mathcal{D}^{*+})$) fixed to their best-fit values, while shifting the NP to $\theta_0 \pm \Delta\theta_0$, where
¹⁹¹⁷ $\Delta\theta_0$ denotes its pre-fit uncertainty. The pre-fit impact is then quantified as the relative

TABLE XXXVI: Determined yields and uncertainties for all four categories.

Sample	D^+e	$D^+\mu$	$D^{*+}e$	$D^{*+}\mu$
$\overline{B}^0 \rightarrow D^+ \ell \bar{\nu}_\ell$	2519 ± 68	2233 ± 61		
$\overline{B}^0 \rightarrow D^{*+} \ell \bar{\nu}_\ell$	2486 ± 63	2323 ± 58	2344 ± 51	1961 ± 44
$\overline{B}^0 \rightarrow D^+ \tau \bar{\nu}_\tau$	191 ± 41	155 ± 65		
$\overline{B}^0 \rightarrow D^{*+} \tau \nu$	106 ± 14	84 ± 11	155 ± 19	111 ± 14
$\overline{B} \rightarrow D^{**} \ell \bar{\nu}_\ell / \overline{B} \rightarrow D_{\text{gap}}^{**} \ell \bar{\nu}_\ell$	653 ± 112	586 ± 102	87 ± 55	75 ± 46
$B\overline{B}$ and Continuum Bkg.	2177 ± 145	1582 ± 149	611 ± 95	497 ± 83
Data	8219	6854	3241	2621

change in the POI compared to a reference POI value, determined in the same way with the NP set at its initial value.

The post-fit impact is defined as the relative deviation of the POI from its nominal value, considering only post-fit systematic variations. To evaluate this, the best-fit value of the NP from the nominal fit, $\hat{\theta}$, is used and varied within its estimated uncertainties, $\hat{\theta} \pm \Delta_{\pm} \hat{\theta}$, where $\Delta_{\pm} \hat{\theta}$ represents the asymmetric uncertainty obtained from the likelihood scan. In this case, only the NP under consideration is fixed, while all other parameters are allowed to vary freely during the fit to real data.

12.2. LFU tests of electrons versus muons

For the ratio of the semileptonic signal branching fractions of electrons to muons we find,

$$\mathcal{R}(D_{e/\mu}^+) = 1.07 \pm 0.05(\text{stat}) \pm 0.02(\text{syst}), \quad (84)$$

$$\mathcal{R}(D_{e/\mu}^{*+}) = 1.08 \pm 0.04(\text{stat}) \pm 0.02(\text{syst}), \quad (85)$$

consistent with the expectation of LFU within 1.2 and 1.6 standard deviations, respectively.

12.3. Stability Checks

To assess the stability of the result, we redetermine $\mathcal{R}(D^{(*)+})$ using different sample splits, fitting them simultaneously to account for correlations in common systematic uncertainties. The tests are summarized in Figure 141. We consider several sample divisions: First, we split by the lepton flavor of the semitauonic and semileptonic signal, fitting $\mathcal{R}(D^{(*)})$ separately in the reconstructed electron and muon channels. We also decouple the D^+ and D^{*+} fit channels, allowing for an independent extraction of $\mathcal{R}(D^+)$ and $\mathcal{R}(D^{*+})$ in the D^+ sample, while fitting only $\mathcal{R}(D^{*+})$ in the D^{*+} sample.

The limited number of D^* feed-down signal events results in a large anti-correlation. The obtained $\mathcal{R}(D^+)$ ($\mathcal{R}(D^{*+})$) values in the split datasets are consistent with the nominal result with p-values from 10%-78% (20%-88%).

Additional cross-checks involve splitting the data by the charge of the signal lepton and by its polar angle, where the latter is divided at $\cos \theta_\ell = 0.22$ to approximately halve the

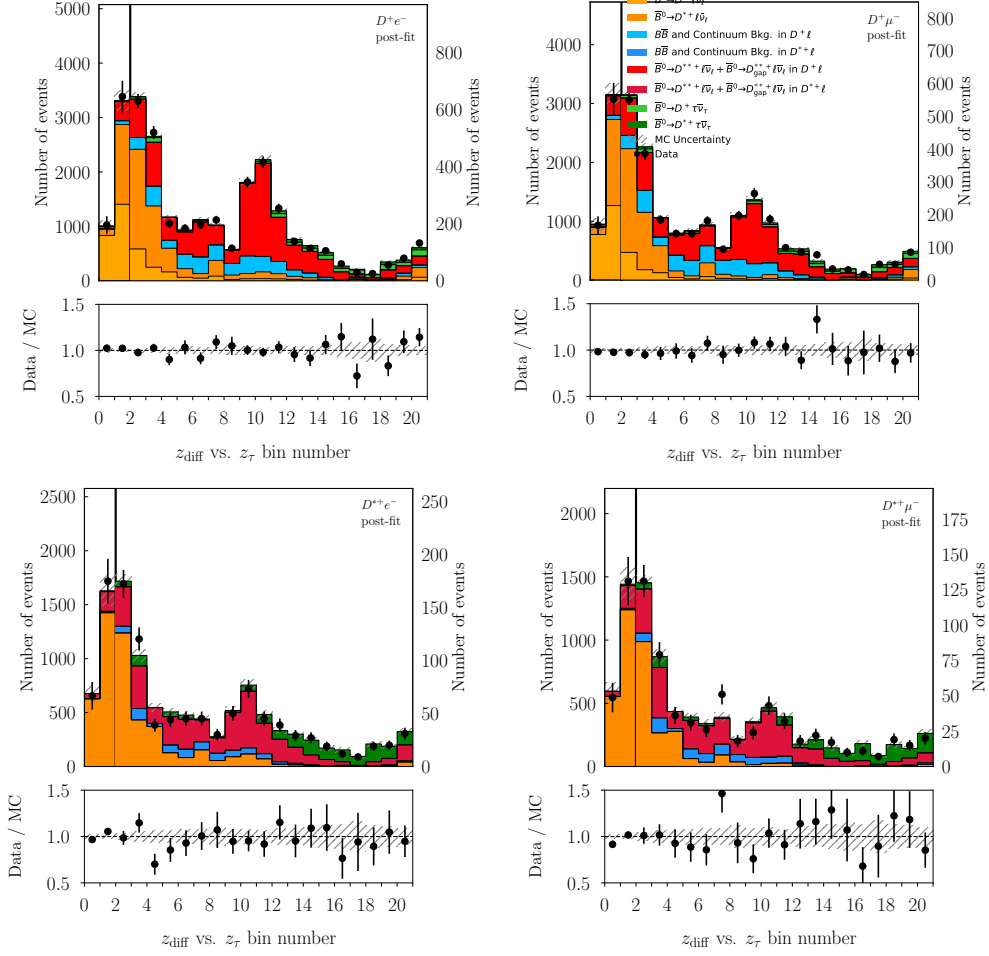


FIG. 132: The fitted classifier distributions for the D^+e^- , $D^+\mu^-$, $D^{*+}e^-$, and $D^{*+}\mu^-$ categories are shown. The hatched regions of the histograms correspond to the systematic uncertainties.

dataset. We also consider a separation based on the number of reconstructed tracks on the signal side (excluding the lepton), using a threshold of four tracks to ensure a meaningful division. Beyond that, the dataset is split according to different D meson reconstruction modes, selecting alternating modes based on their branching fractions. Furthermore, we perform a division of the dataset based on the decay modes of the tag-side B^0 : We divide the dataset by choosing only tag-side semileptonic decays with electrons and with muons, respectively. Second, we further categorize events by distinguishing between tag-side decays with D mesons and those with D^* mesons. We also test the stability using a more strict LID selection and find good agreement between the results.

The ratio of branching fractions of $\bar{B}^0 \rightarrow D^+\ell^-\bar{\nu}_\ell$ over $\bar{B}^0 \rightarrow D^{*+}\ell^-\bar{\nu}_\ell$ is 0.40 ± 0.01 and consistent with the prediction from Ref. [10] of 0.417 ± 0.012 within 1.1 standard deviations and the world average of Ref. [7] of 0.431 ± 0.014 within 1.8 standard deviations.

Table XXXVII summarises the results of the split sample fits by indicating the difference between each set of $\mathcal{R}(D^{(*)})$ values, a p value quantifying their compatibility computed by taking into consideration the covariance matrix from the fit as well as a p value corresponding to the combined $\mathcal{R}(D) - \mathcal{R}(D^*)$ result compared to the nominal fit results as described in Section 11.11.9.

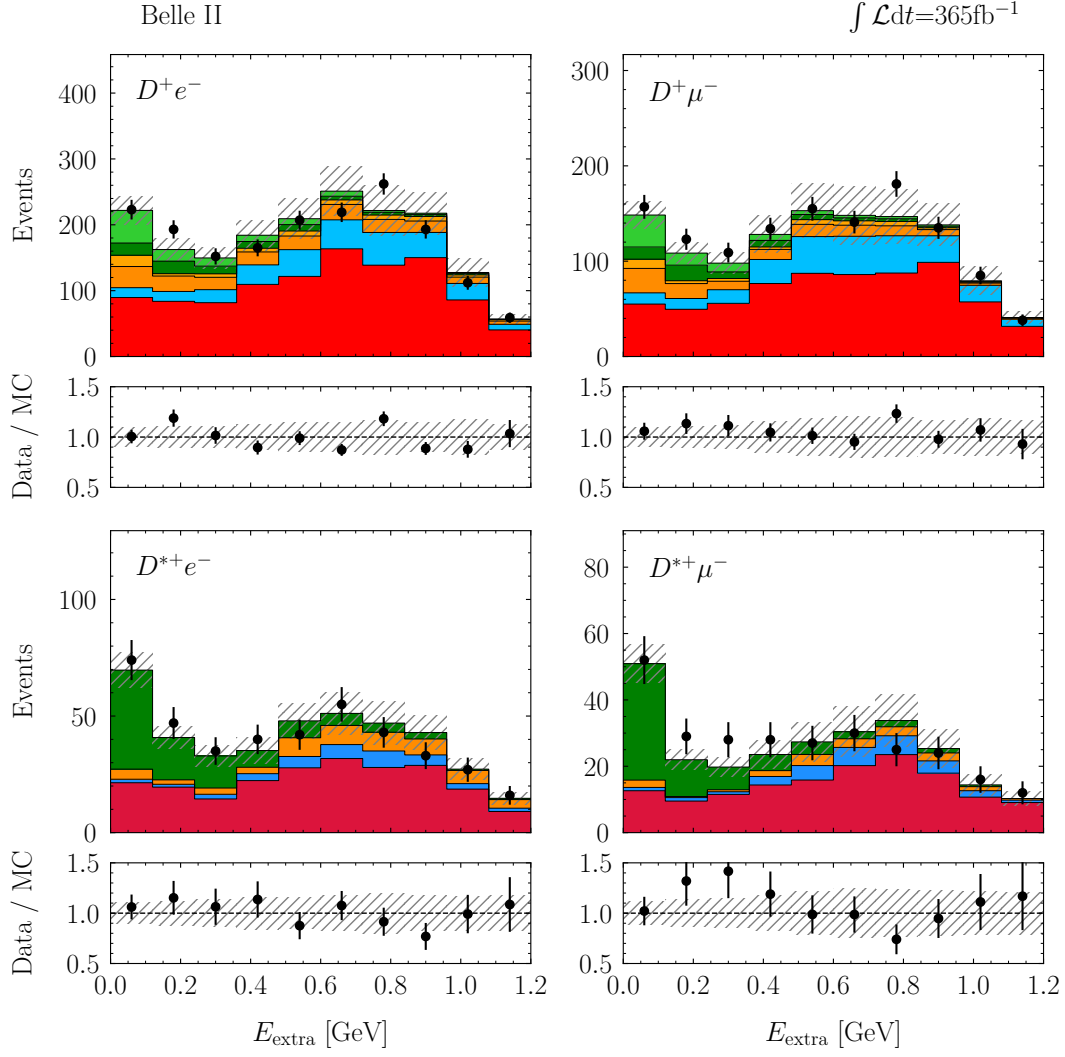


FIG. 133: Distributions of E_{extra} with post-fit scaling applied and selecting only signal enriched bins (5,6,10-20) for the four fit categories.

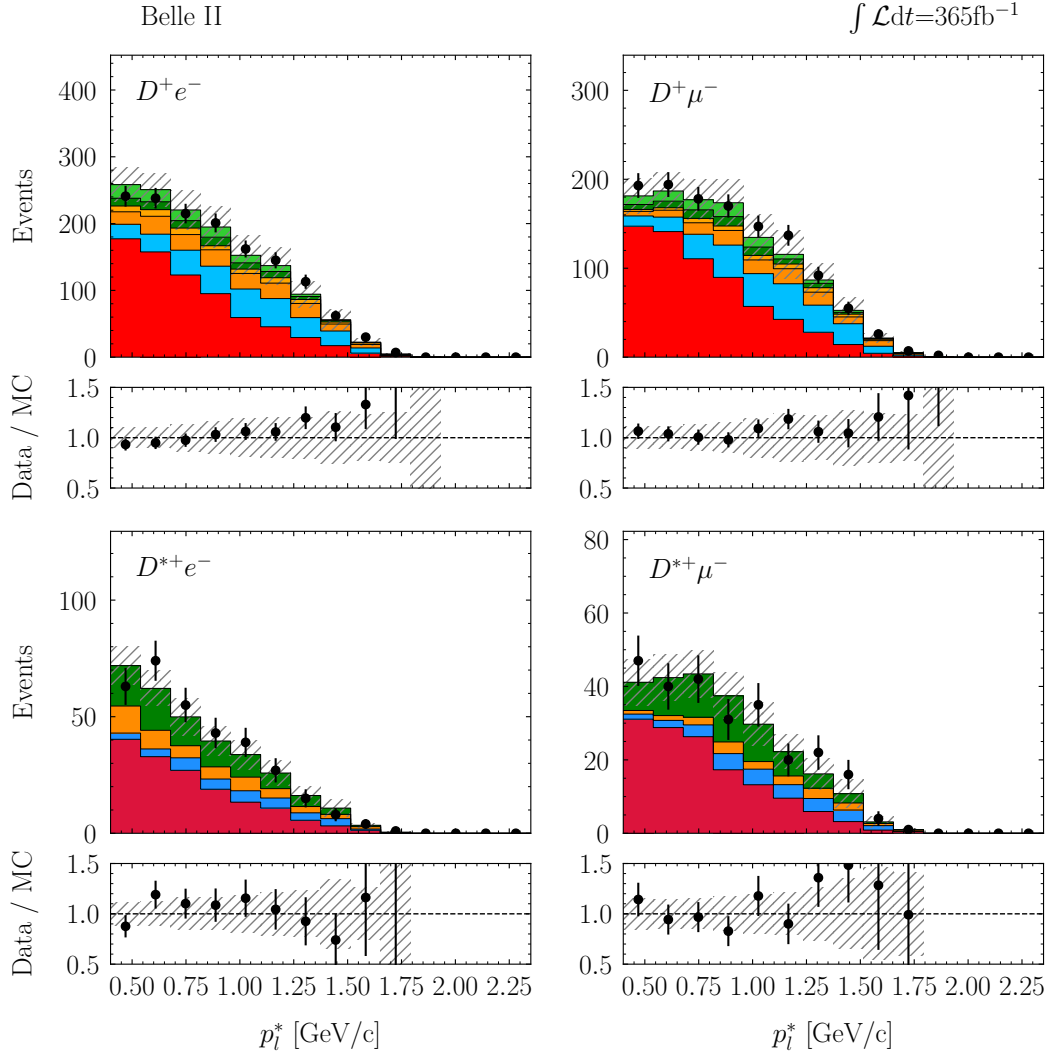


FIG. 134: Distributions of p_l^* with post-fit scaling applied and selecting only signal enriched bins (5,6,10-20) for the four fit categories.

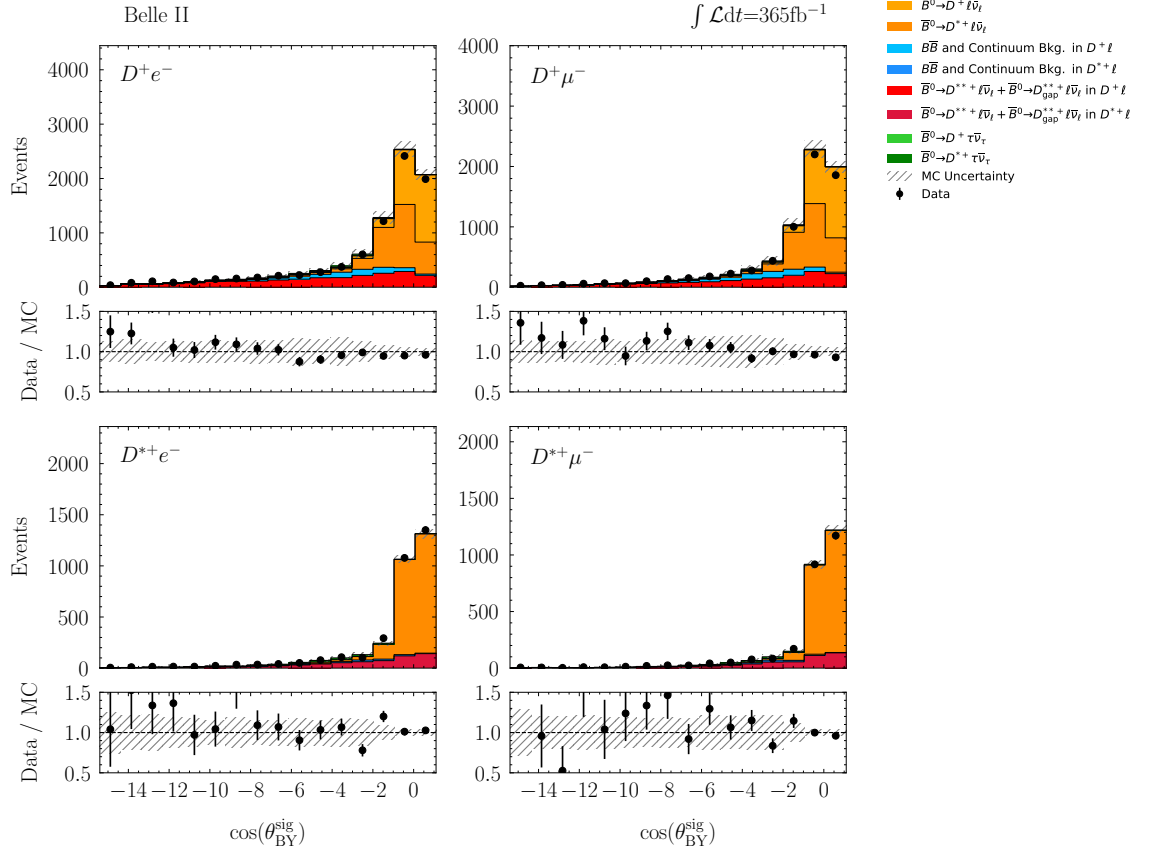


FIG. 135: Distribution of $\cos \theta_{BY}$ with best-fit scaling, including all systematic uncertainties for the four fit categories: $D^+ e^-$, $D^+ \mu^-$, $D^{*+} e^-$ and $D^{*+} \mu^-$.

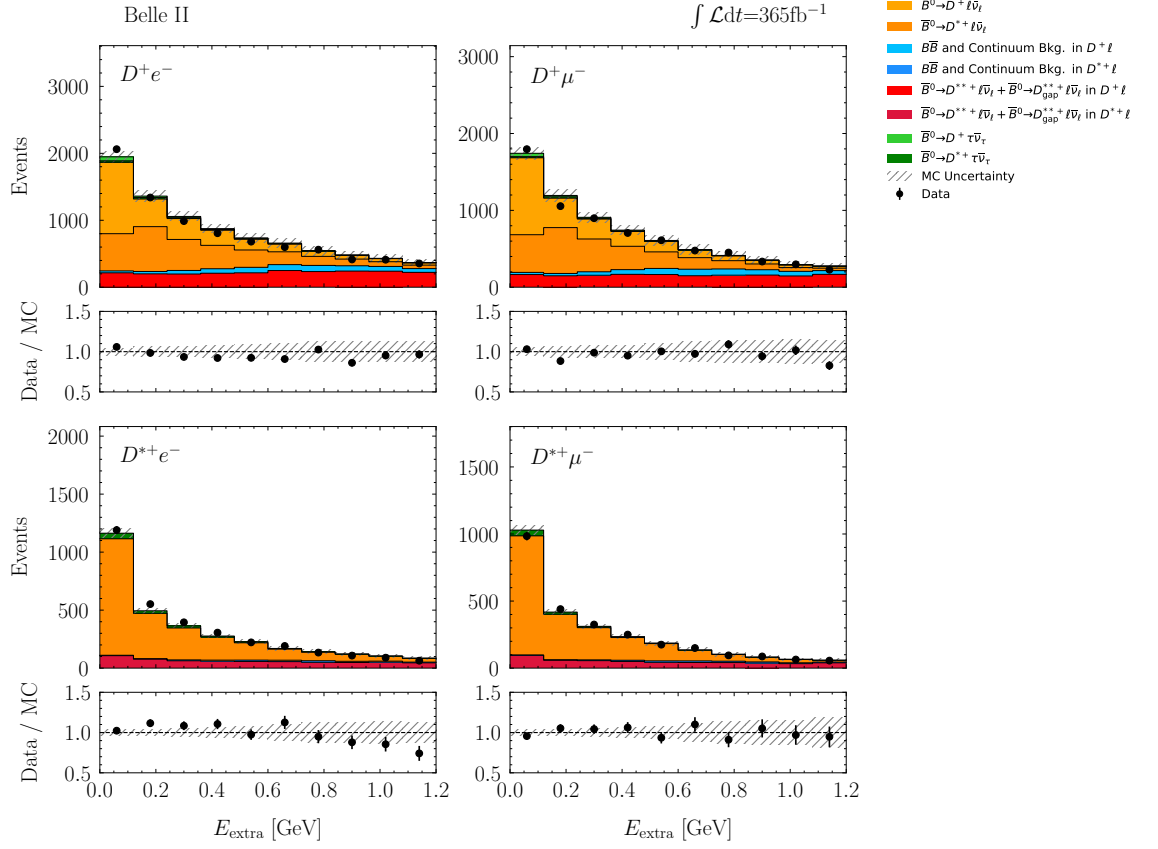


FIG. 136: Distribution of E_{extra} with best-fit scaling, including all systematic uncertainties for the four fit categories.

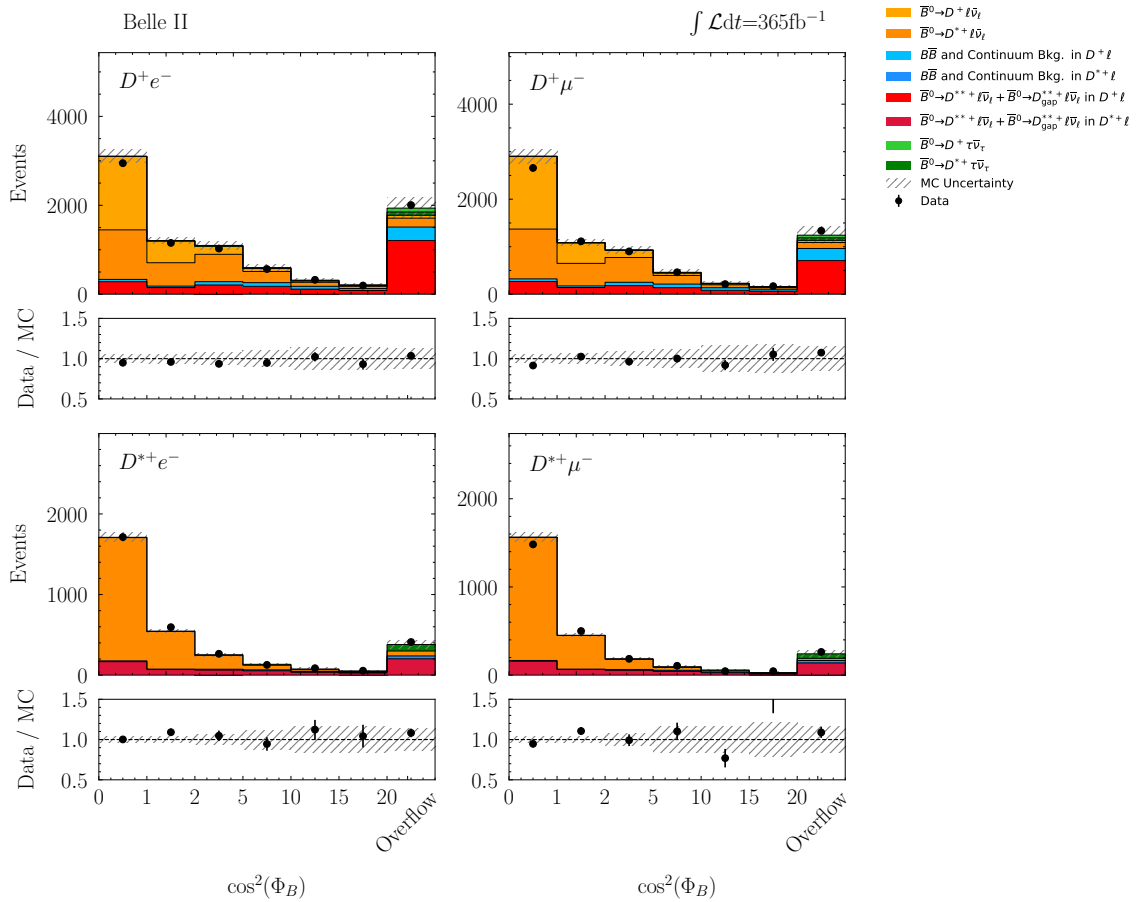


FIG. 137: Distribution of $\cos^2 \Phi_B$ with best-fit scaling, including all systematic uncertainties for the four fit categories. Note the use of non-uniform binning, including a large overflow bin, which accounts for the absence of significant structure in the region of high $\cos^2 \Phi_B$.

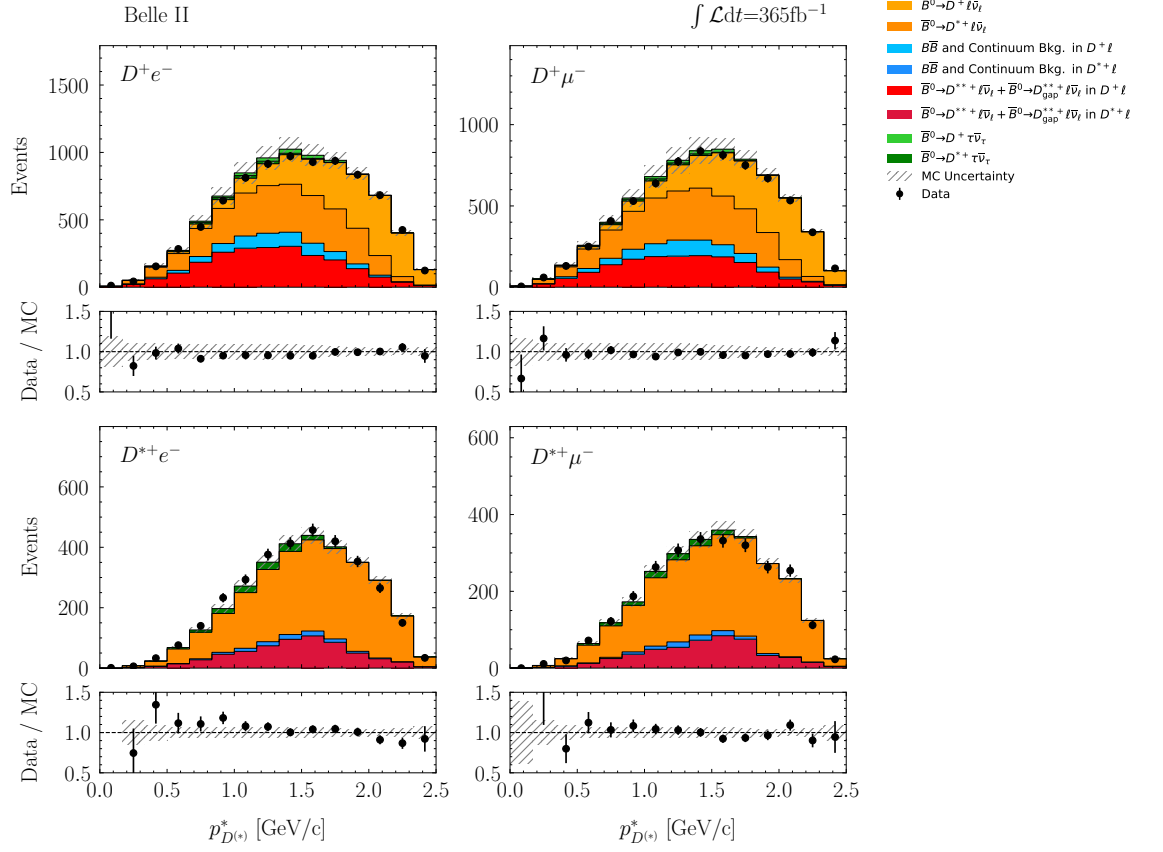


FIG. 138: Distribution of p_D^* with best-fit scaling, including all systematic uncertainties for the four fit categories.

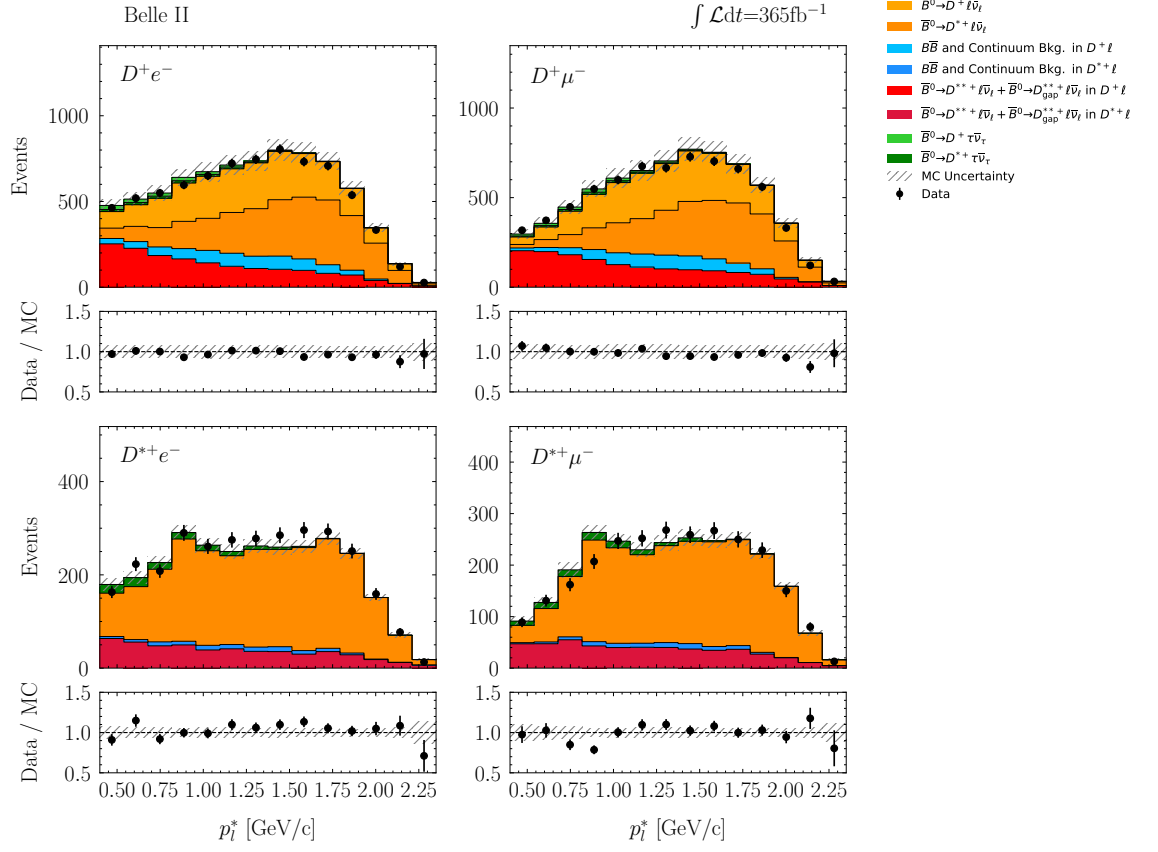
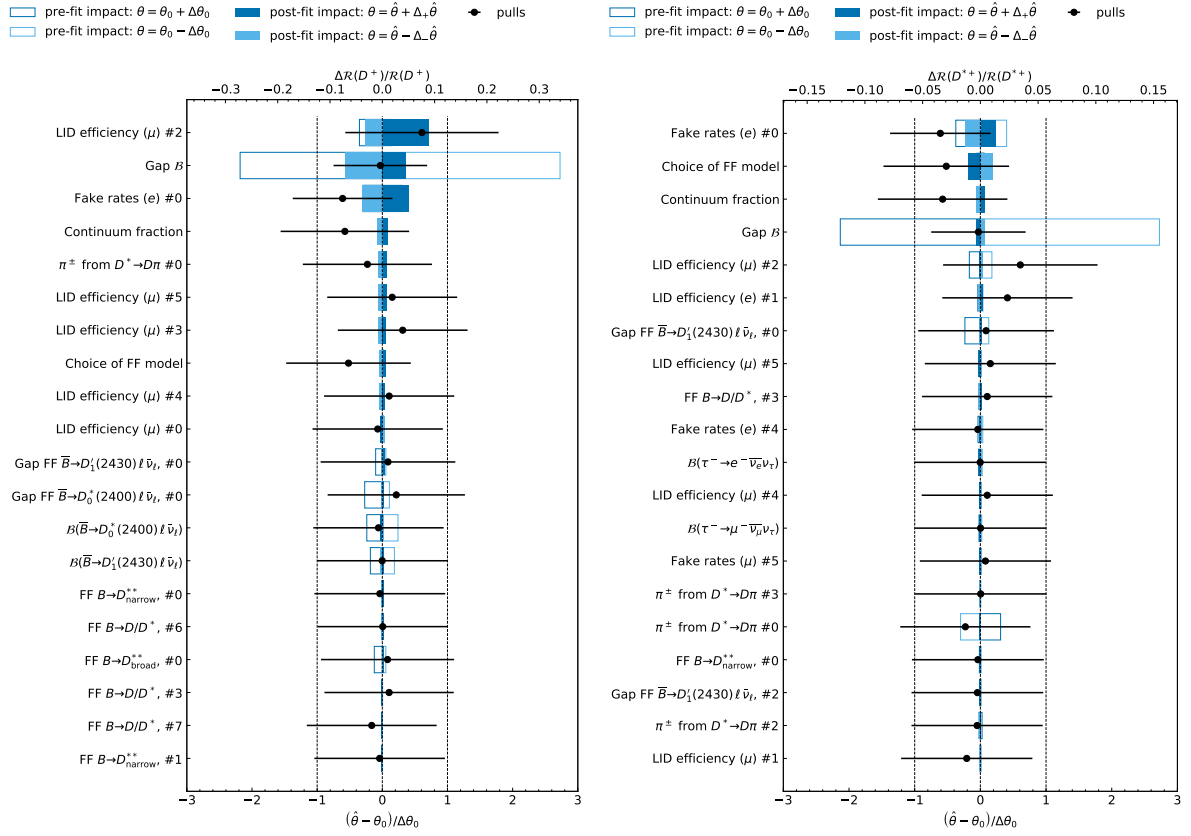


FIG. 139: Distribution of p_l^* with best-fit scaling, including all systematic uncertainties for the four fit categories.



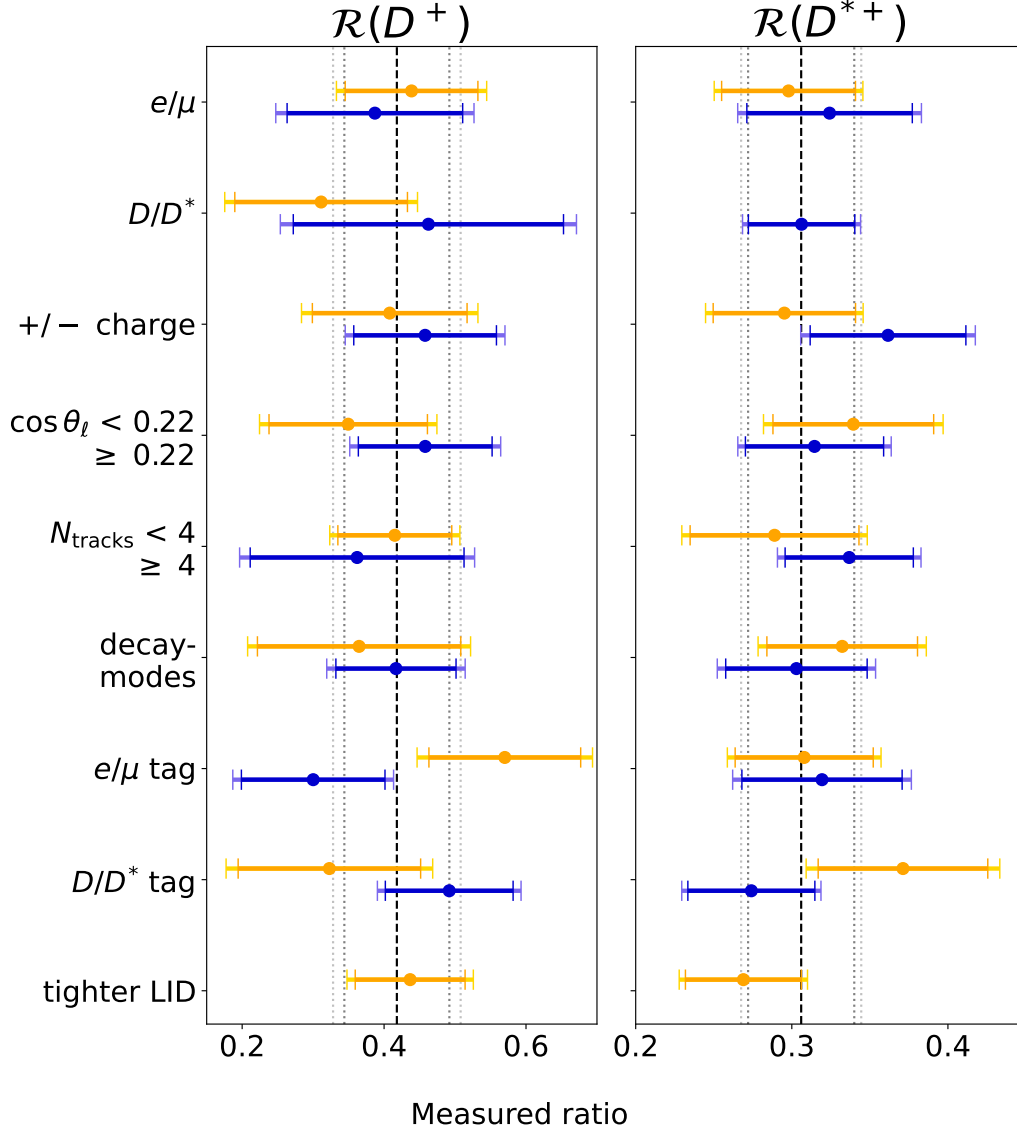


FIG. 141: Summary of the various stability checks. The central values of each sample split fit (blue and orange error bars), along with the sizes of the statistical and total uncertainties are shown (inner and outer ticks). The nominal results Eqs. 82 and 83 are also shown (dashed black line) with statistical and total error (inner and outer gray line). Left shows $\mathcal{R}(D)$, and on the right $\mathcal{R}(D^*)$.

Cut Criterion	$\mathcal{R}(D^{(*)})_{\text{cut}}$	$\mathcal{R}(D^{(*)})_{\text{anti-cut}}$	$\Delta\mathcal{R}(D^{(*)})$	χ^2	p	p_{comb}
e/μ	$0.439 \pm 0.094_{\text{stat}} \pm 0.050_{\text{syst}}$	$0.387 \pm 0.124_{\text{stat}} \pm 0.065_{\text{syst}}$	0.05 ± 0.17	0.09	0.76	0.92
D/D^*	$0.311 \pm 0.122_{\text{stat}} \pm 0.060_{\text{syst}}$	$0.462 \pm 0.190_{\text{stat}} \pm 0.085_{\text{syst}}$	-0.16 ± 0.21	0.54	0.46	0.41
$\pm \text{charge}$	$0.408 \pm 0.109_{\text{stat}} \pm 0.060_{\text{syst}}$	$0.458 \pm 0.101_{\text{stat}} \pm 0.051_{\text{syst}}$	0.05 ± 0.16	0.09	0.76	0.49
$\cos \theta_\ell < 0.22$	$0.349 \pm 0.112_{\text{stat}} \pm 0.056_{\text{syst}}$	$0.458 \pm 0.094_{\text{stat}} \pm 0.049_{\text{syst}}$	0.11 ± 0.16	0.46	0.5	0.7
$N_{\text{tracks}} < 4$	$0.415 \pm 0.080_{\text{stat}} \pm 0.044_{\text{syst}}$	$0.362 \pm 0.151_{\text{stat}} \pm 0.069_{\text{syst}}$	-0.05 ± 0.19	0.08	0.78	0.74
decaymodes	$0.365 \pm 0.143_{\text{stat}} \pm 0.064_{\text{syst}}$	$0.417 \pm 0.085_{\text{stat}} \pm 0.048_{\text{syst}}$	-0.05 ± 0.18	0.08	0.78	0.86
e/μ tag	$0.570 \pm 0.107_{\text{stat}} \pm 0.062_{\text{syst}}$	$0.300 \pm 0.101_{\text{stat}} \pm 0.051_{\text{syst}}$	0.27 ± 0.16	2.75	0.10	0.24
D/D^* tag	$0.323 \pm 0.129_{\text{stat}} \pm 0.068_{\text{syst}}$	$0.492 \pm 0.090_{\text{stat}} \pm 0.046_{\text{syst}}$	-0.17 ± 0.17	0.95	0.33	0.33
e/μ	$0.298 \pm 0.043_{\text{stat}} \pm 0.021_{\text{syst}}$	$0.324 \pm 0.053_{\text{stat}} \pm 0.026_{\text{syst}}$	-0.03 ± 0.07	0.13	0.72	0.92
D/D^*	—	$0.306 \pm 0.034_{\text{stat}} \pm 0.016_{\text{syst}}$	-0.16 ± 0.21	0.54	0.46	0.41
$\pm \text{charge}$	$0.295 \pm 0.046_{\text{stat}} \pm 0.022_{\text{syst}}$	$0.362 \pm 0.050_{\text{stat}} \pm 0.025_{\text{syst}}$	0.07 ± 0.08	0.78	0.38	0.49
$\cos \theta_\ell < 0.22$	$0.339 \pm 0.051_{\text{stat}} \pm 0.026_{\text{syst}}$	$0.315 \pm 0.044_{\text{stat}} \pm 0.021_{\text{syst}}$	-0.02 ± 0.08	0.11	0.74	0.7
$N_{\text{tracks}} < 4$	$0.289 \pm 0.054_{\text{stat}} \pm 0.025_{\text{syst}}$	$0.337 \pm 0.041_{\text{stat}} \pm 0.021_{\text{syst}}$	0.05 ± 0.07	0.41	0.52	0.74
decaymodes	$0.332 \pm 0.048_{\text{stat}} \pm 0.024_{\text{syst}}$	$0.303 \pm 0.045_{\text{stat}} \pm 0.023_{\text{syst}}$	0.03 ± 0.07	0.16	0.69	0.86
e/μ tag	$0.308 \pm 0.044_{\text{stat}} \pm 0.022_{\text{syst}}$	$0.319 \pm 0.051_{\text{stat}} \pm 0.025_{\text{syst}}$	-0.01 ± 0.08	0.02	0.88	0.24
D/D^* tag	$0.371 \pm 0.054_{\text{stat}} \pm 0.030_{\text{syst}}$	$0.274 \pm 0.041_{\text{stat}} \pm 0.018_{\text{syst}}$	0.10 ± 0.08	1.63	0.20	0.33

TABLE XXXVII: Results of the split sample fits for $\mathcal{R}(D^+)$ (above) and $\mathcal{R}(D^{*+})$ (below) indicating the difference between the two fitted values, a χ^2 value derived using the correlation matrix of the fit with corresponding p value as well as a p_{comb} value indicating the combined compatibility with the nominal results.

Split	Difference $\mathcal{R}(D)$	Pull $\mathcal{R}(D)$	Difference $\mathcal{R}(D^*)$	Pull $\mathcal{R}(D^*)$
e/μ	0.0517	0.33	-0.0263	-0.39
D/D^*	-0.1512	-0.67	-	-
$\pm \text{charge}$	-0.0499	-0.34	-0.0664	-0.98
$\cos \theta_\ell < 0.22 / \geq 0.22$	-0.1085	-0.74	0.0249	0.37
$N_{\text{signal side tracks}} < 4 / \geq 4$	0.0531	0.31	-0.0478	-0.70
decaymodes	-0.0520	-0.31	0.0293	0.44

TABLE XXXVIII: Statistical pulls defined as $\Delta\mathcal{R}(D^{(*)})/\sqrt{\sigma_{\text{stat1}}^2 + \sigma_{\text{stat2}}^2}$ between the two splits.

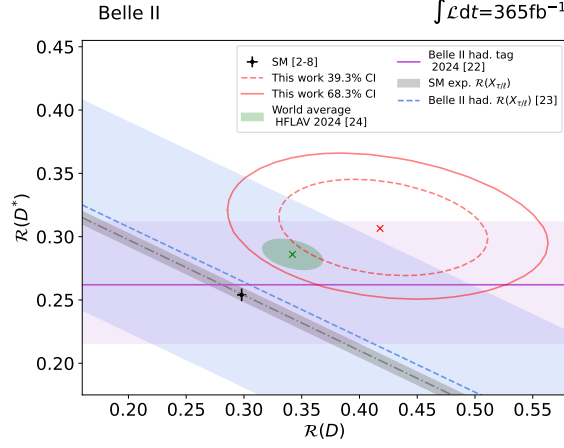


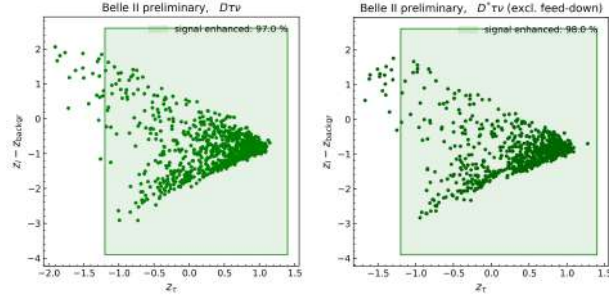
FIG. 142: The measured $\mathcal{R}(D^+)$ and $\mathcal{R}(D^{*+})$ ratios are shown (red marker) with the corresponding 68.3% CI (red solid contour), 39.3% CI (red dashed contour), and compared to the SM prediction (black marker), the world average from Ref. [7] (green ellipse) are shown. The $\mathcal{R}(D^*)$ measurement of Belle II, which uses hadronic modes to reconstruct B_{tag} candidates (dark violet band), and the constraint on $\mathcal{R}(D)$ and $\mathcal{R}(D^*)$ from the inclusive $\mathcal{R}(X_{\tau/\ell})$ (blue band) are shown.

13. SUMMARY

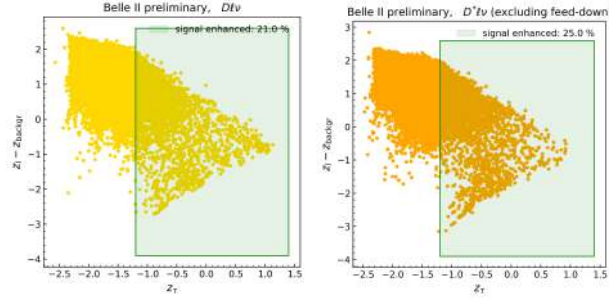
We report measurements of the ratios of $\mathcal{R}(D^+)$ and $\mathcal{R}(D^{*+})$ and test the predictions of lepton-flavor-universality of the SM. For this we analyzed a 365 fb^{-1} data sample, recorded by the Belle II experiment between 2019-2022. Signal events are selected by first reconstructing the companion B meson in semileptonic modes using a hierarchical approach. The signal side is analyzed using a multi-class multivariate approach, combining the discriminating power of five variables. The selected events are analyzed using a likelihood fit and we determine

$$\begin{aligned}\mathcal{R}(D^+) &= 0.418 \pm 0.074 \text{ (stat)} \pm 0.051 \text{ (syst)}, \\ \mathcal{R}(D^{*+}) &= 0.306 \pm 0.034 \text{ (stat)} \pm 0.018 \text{ (syst)},\end{aligned}$$

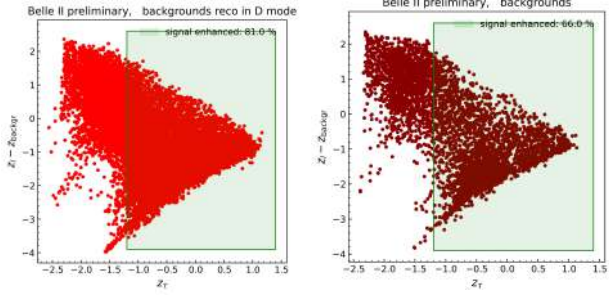
consistent with the SM expectation within 1.7 standard deviations. Figure 142 shows the 2D confidence intervals (CI) and compares this result with the SM expectation and the Belle II measurements Refs. [3, 4], which analyzed an orthogonal data set. Further the world average of Ref. [7] is shown with which the presented measurement is consistent within 0.6 standard deviations. The uncertainty of the measurements of the ratios are dominated by statistical uncertainties and the largest systematic uncertainty is the limited simulated sample size used to determine efficiencies, train the multi-class classification algorithm, and template shapes.



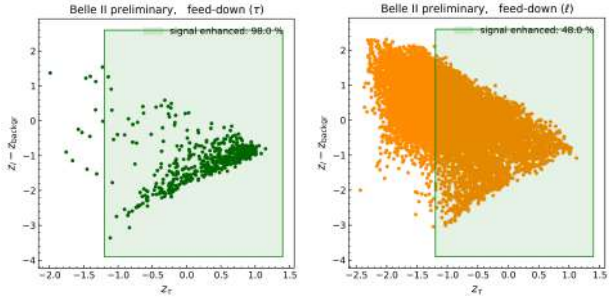
(a)



(b)

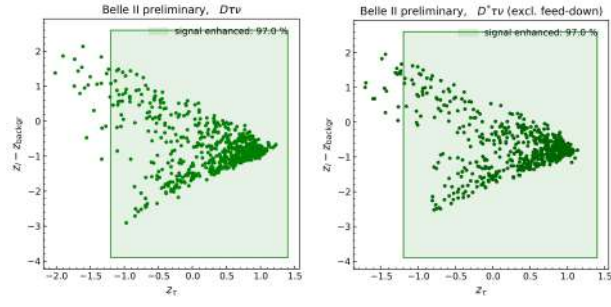


(c)

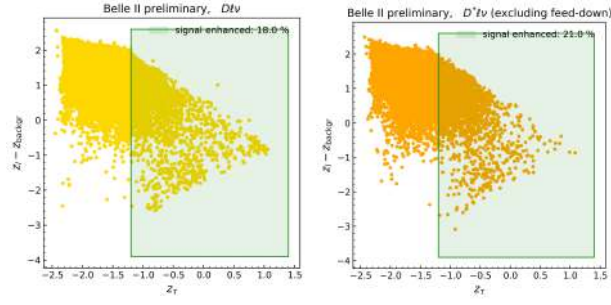


(d)

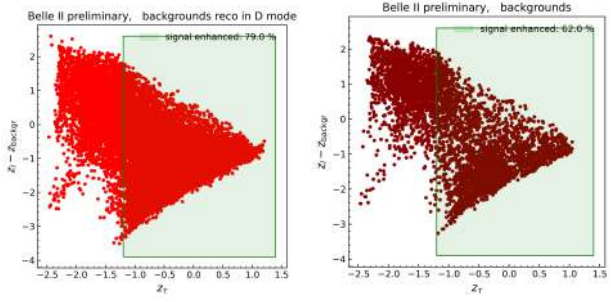
FIG. 143: 2D space of z_τ and $z_l - z_{\text{backgr}}$, showing the six fit categories (while keeping the respective feed-down components separate) in the electron mode, highlighting the definition of the **signal enhanced** region in the green. The legend provides information on how large the fraction of events is in a specific sideband w.r.t. the total number of events within the respective category.



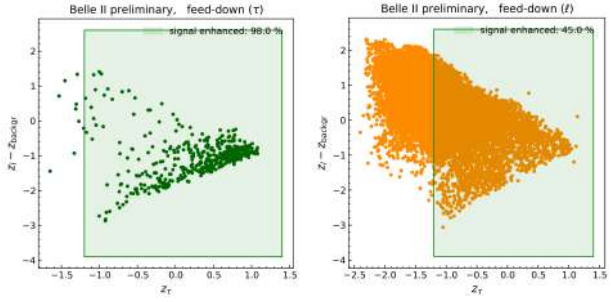
(a)



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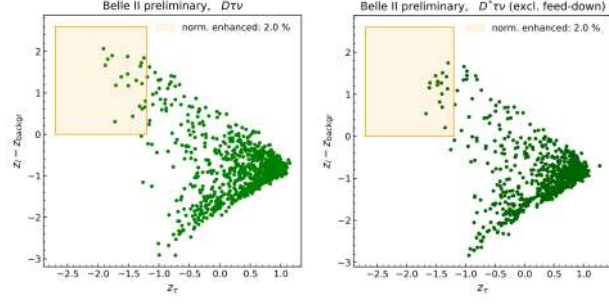


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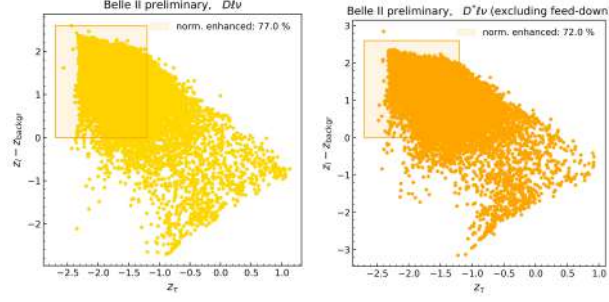


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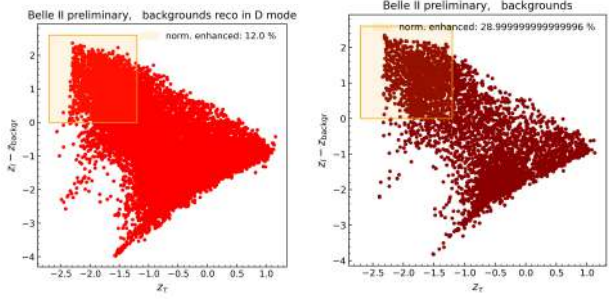
FIG. 144: Same as Fig.143, but for the muon mode



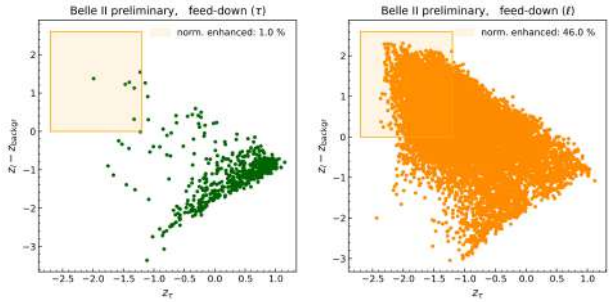
(a)



(b)

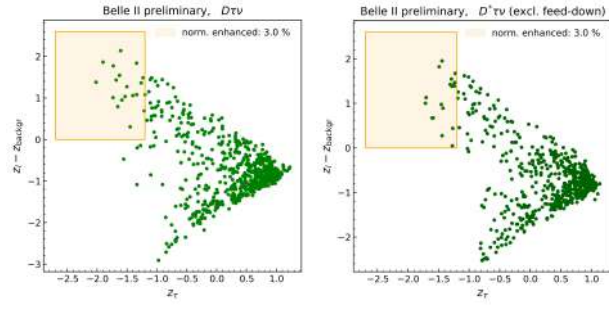


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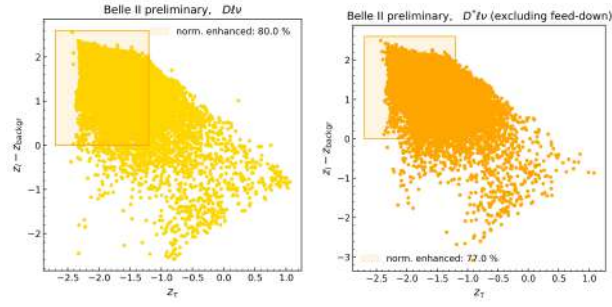


(d)

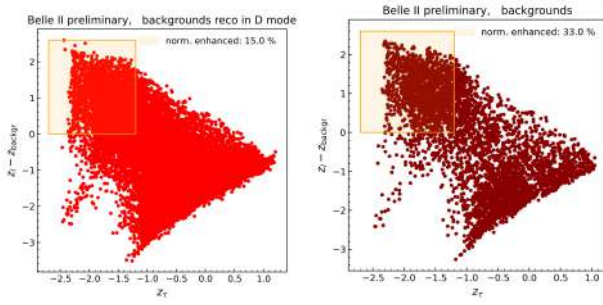
FIG. 145: 2D space of z_τ and $z_l - z_{\text{backgr}}$, showing the six fit categories (while keeping the respective feed-down components separate) in the electron mode, highlighting the definition of the **normalisation enhanced** region in the orange. The legend provides information on how large the fraction of events is in a specific sideband w.r.t. the total number of events within the respective category.



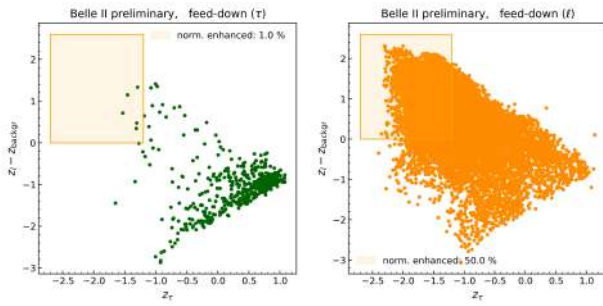
(a)



(b)

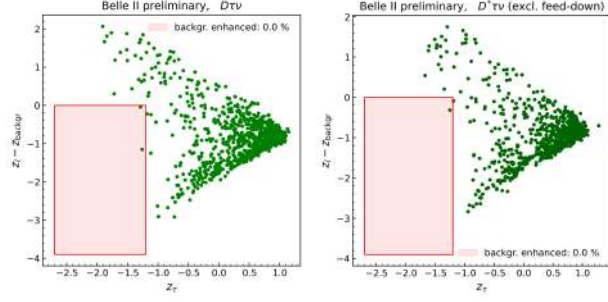


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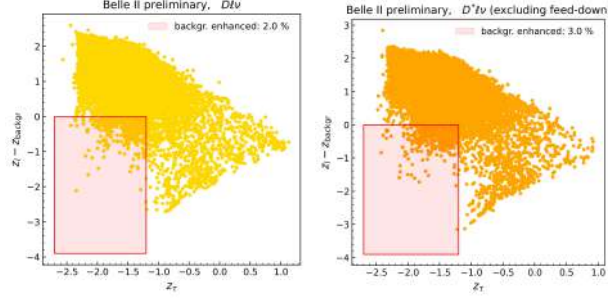


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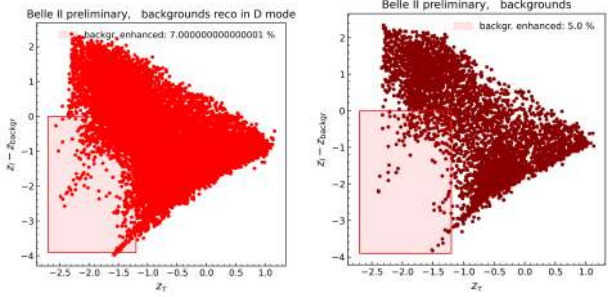
FIG. 146: Same as Fig.145, but for the muon mode



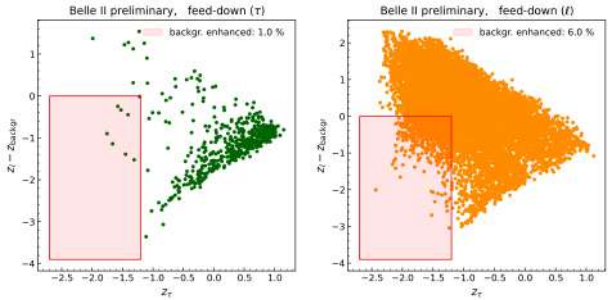
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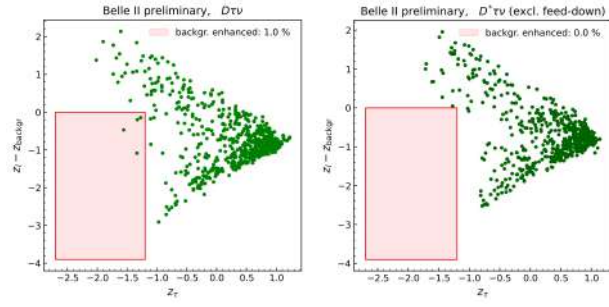


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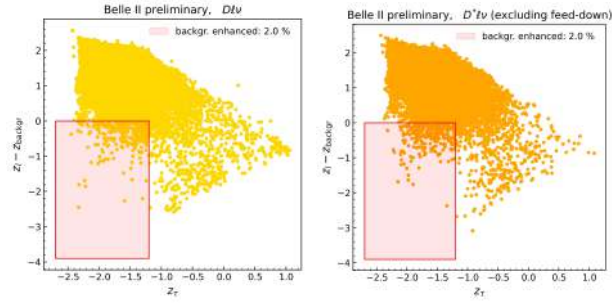


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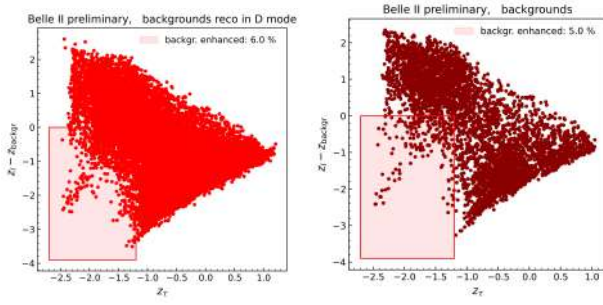
FIG. 147: 2D space of z_τ and $z_l - z_{\text{backgr}}$, showing the six fit categories (while keeping the respective feed-down components separate) in the electron mode, highlighting the definition of the **background enhanced** region in the red. The legend provides information on how large the fraction of events is in a specific sideband w.r.t. the total number of events within the respective category.



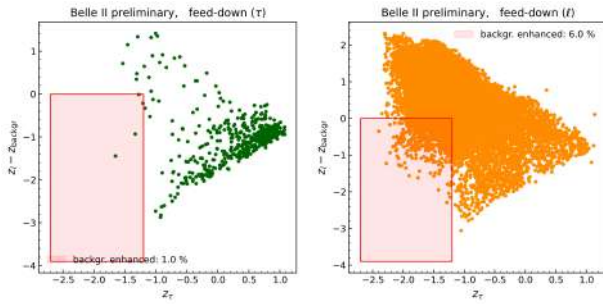
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(b)

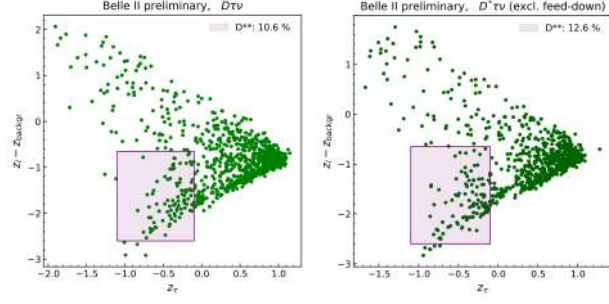


(c)

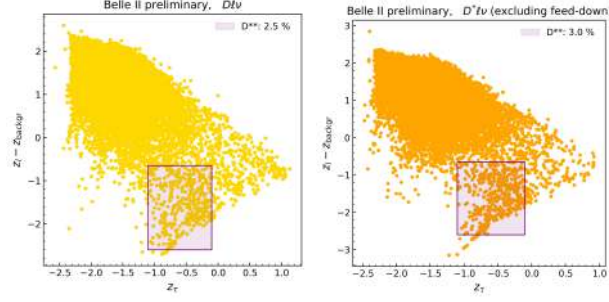


(d)

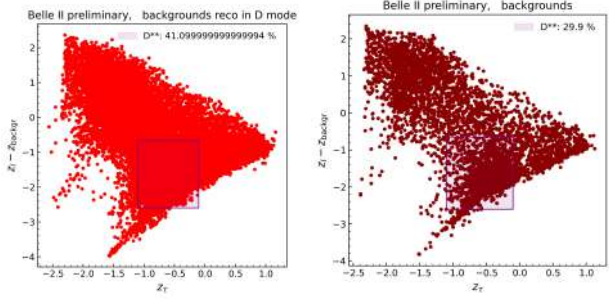
FIG. 148: Same as Fig.147, but for the muon mode



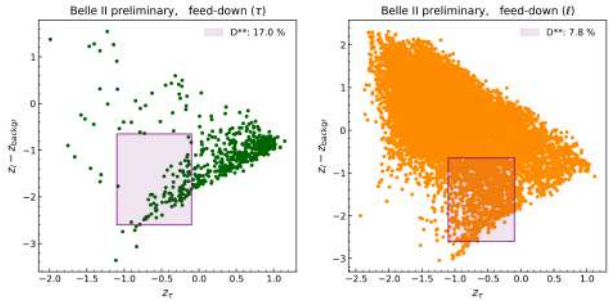
(a)



(b)

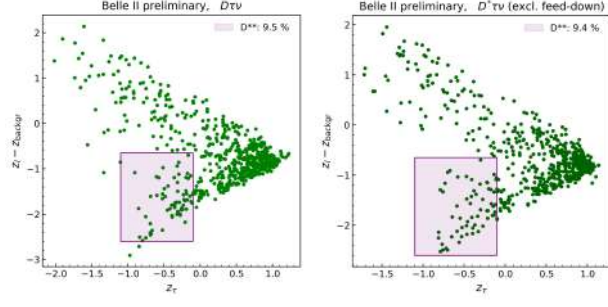


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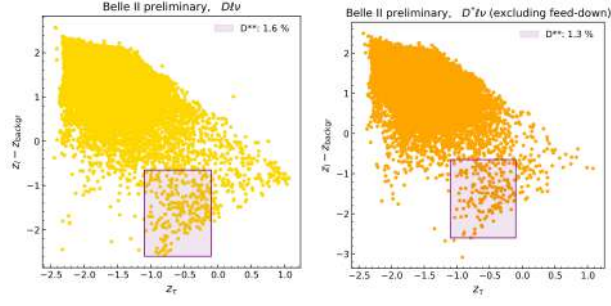


(d)

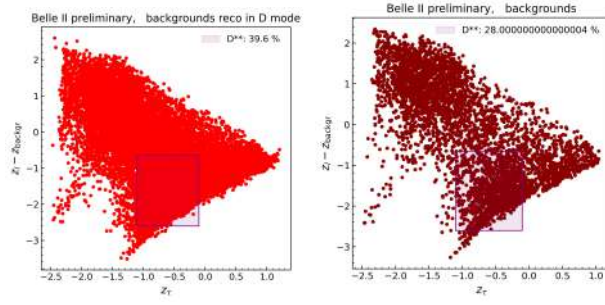
FIG. 149: 2D space of z_τ and $z_l - z_{\text{backgr}}$, showing the six fit categories (while keeping the respective feed-down components separate) in the electron mode, highlighting the definition of the **D**enhanced** region in the purple. The legend provides information on how large the fraction of events is in a specific sideband w.r.t. the total number of events within the respective category.



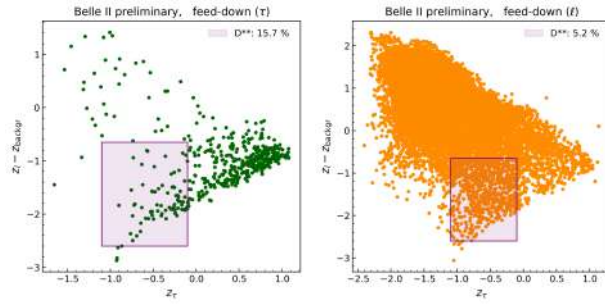
(a)



(b)

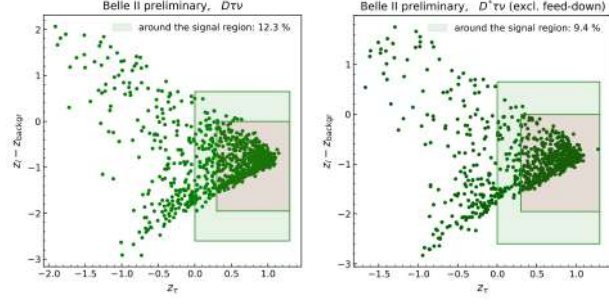


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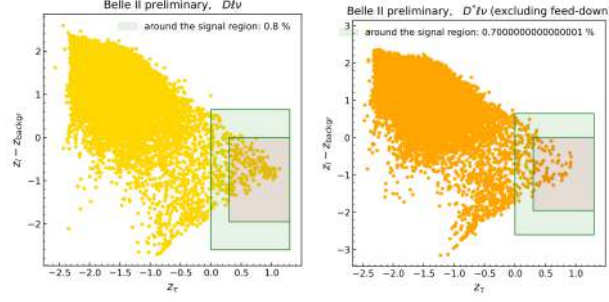


(d)

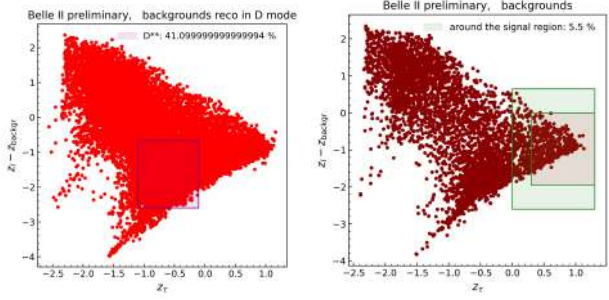
FIG. 150: Same as Fig.149, but for the muon mode



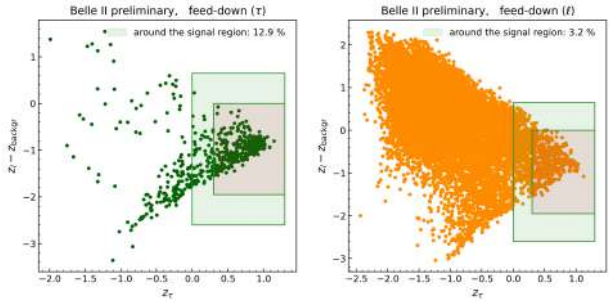
(a)



(b)

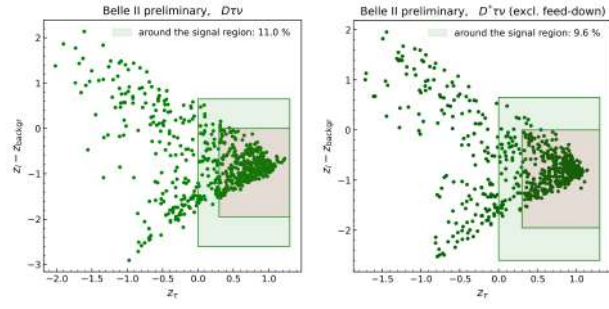


(c)

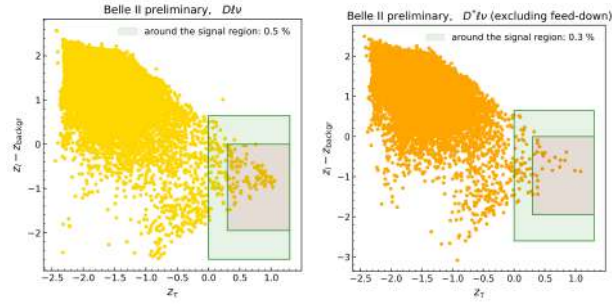


(d)

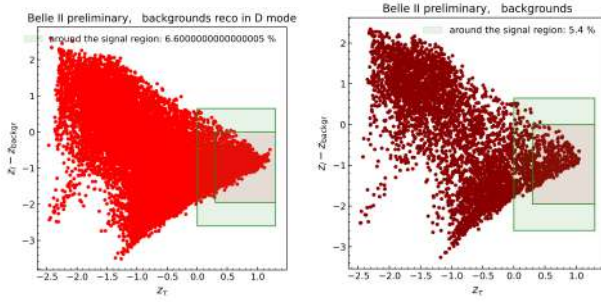
FIG. 151: 2D space of z_τ and $z_l - z_{\text{backgr}}$, showing the six fit categories (while keeping the respective feed-down components separate) in the electron mode, highlighting the definition of the region **around the signal region** in the dark green with the signal region in red. The legend provides information on how large the fraction of events is in a specific sideband w.r.t. the total number of events within the respective category.



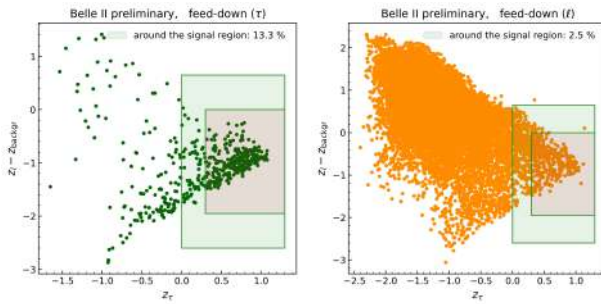
(a)



(b)



(c)



(d)

FIG. 152: Same as Fig.151, but for the muon mode

Appendix B: 2D fit variable distributions for different categories of events

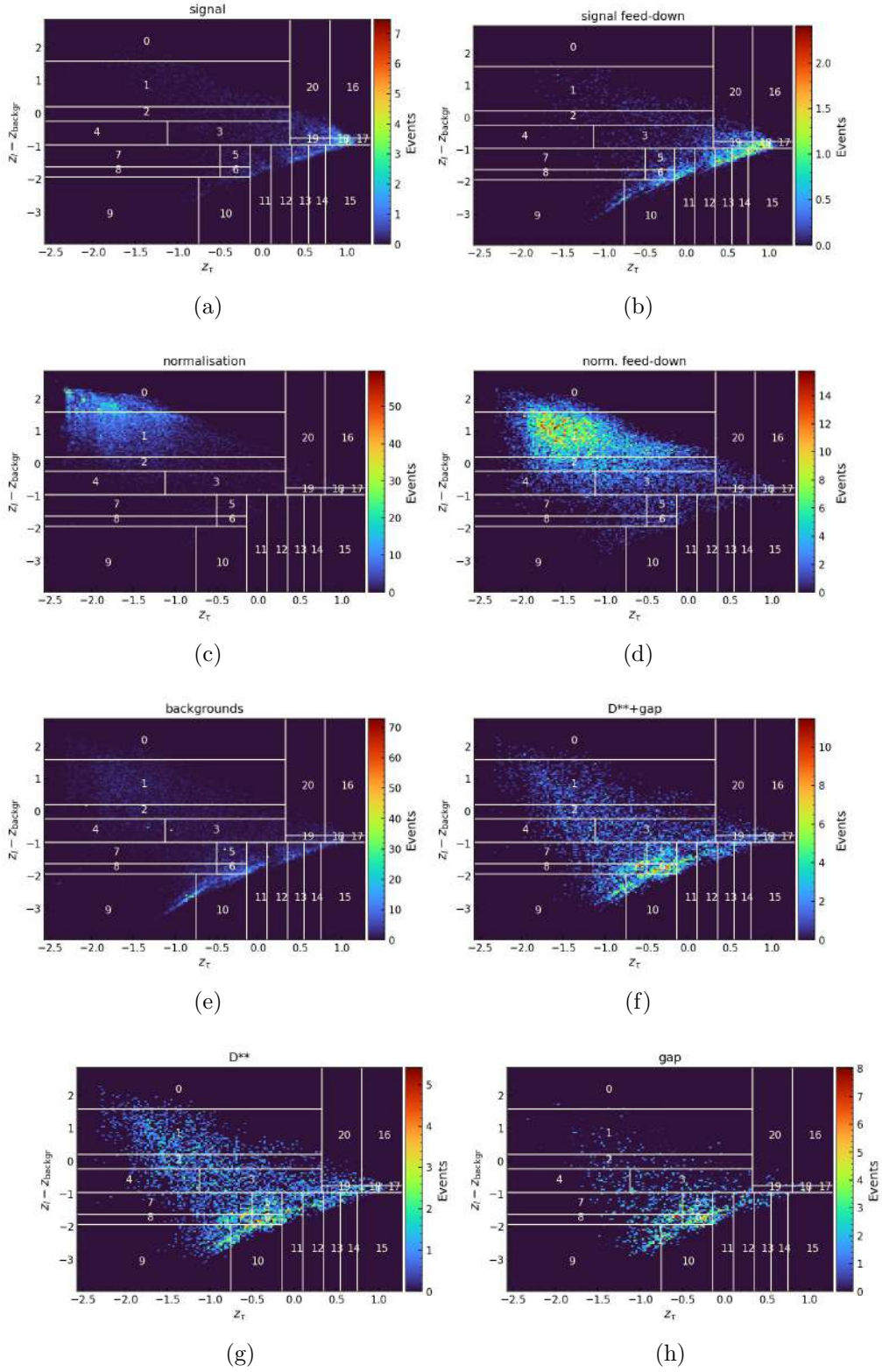


FIG. 153: Same as Fig.25, but for the $D\mu$ mode.

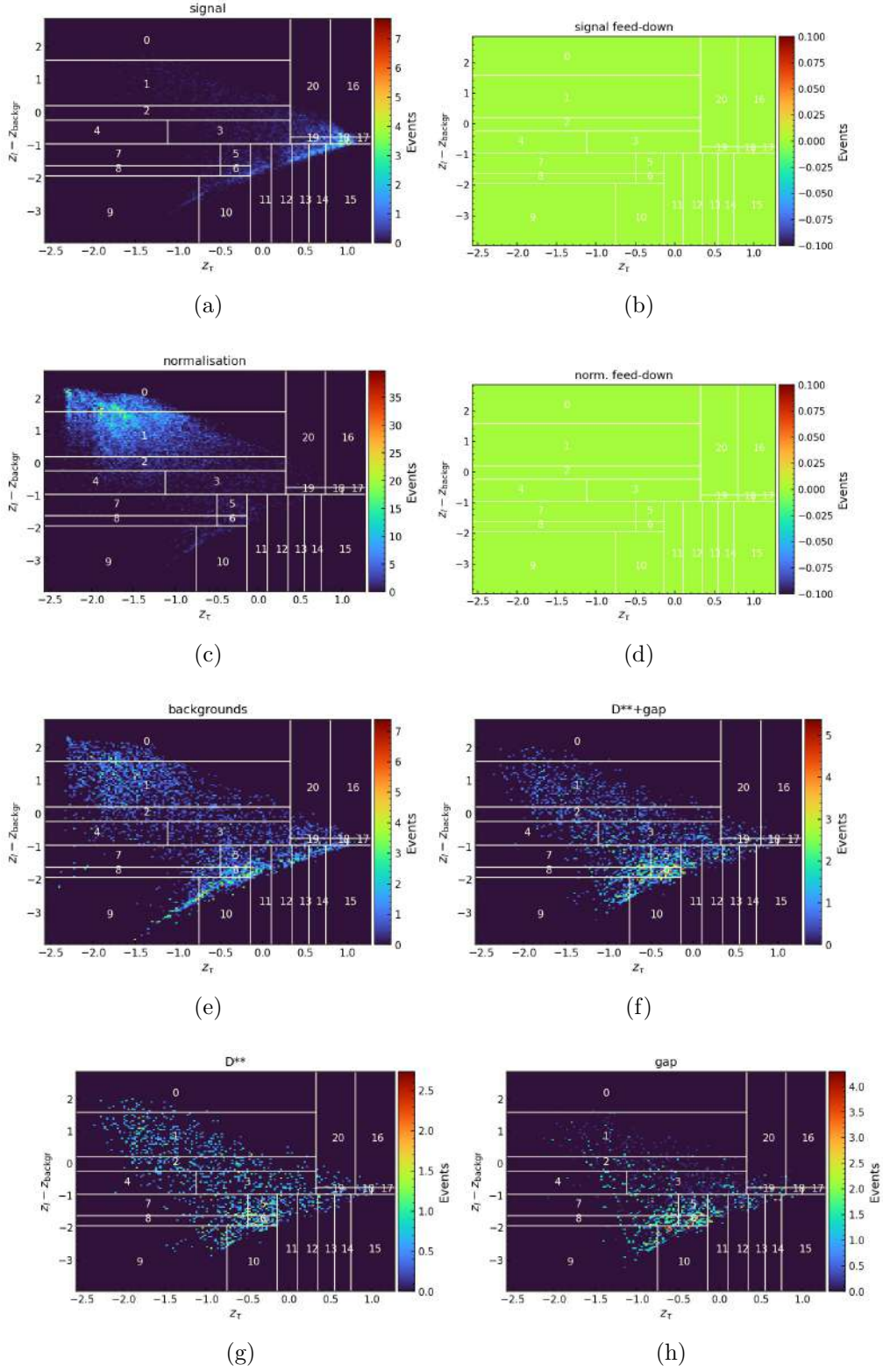


FIG. 154: Same as Fig.25, but for the D^*e mode.

1971 [1] private communication Alessandro Gaz.

1972 [2] Belle II Decay File. June 2020. <https://stash.desy.de/projects/B2/repos/software/>

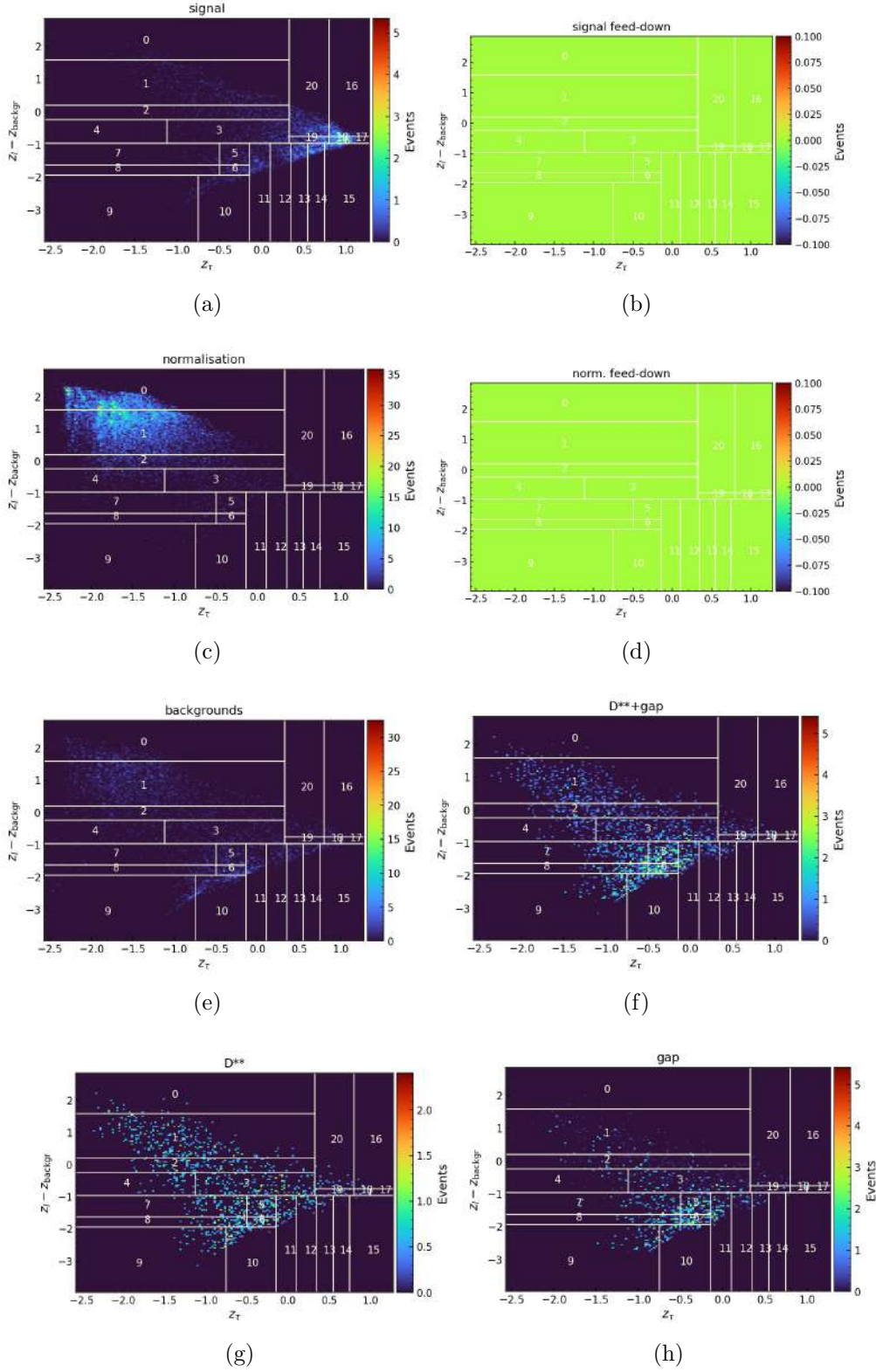


FIG. 155: Same as Fig.25, but for the $D^*\mu$ mode.

[browse/decfiles/dec/DECAY_BELLE2.DEC](#).

- [3] I. Adachi et al. A test of lepton flavor universality with a measurement of $R(D^*)$ using hadronic B tagging at the Belle II experiment. 1 2024.

- [4] I. Adachi et al. First Measurement of $R(X\tau/\ell)$ as an Inclusive Test of the $b \rightarrow c\tau\nu$ Anomaly. *Phys. Rev. Lett.*, 132(21):211804, 2024.
- [5] Y. Amhis, Sw. Banerjee, E. Ben-Haim, F. U. Bernlochner, M. Bona, A. Bozek, C. Bozzi, J. Brodzicka, M. Chrzaszcz, J. Dingfelder, and et al. Averages of b -hadron, c -hadron, and τ -lepton properties as of 2018. *The European Physical Journal C*, 81(3), Mar 2021. <http://dx.doi.org/10.1140/epjc/s10052-020-8156-7>.
- [6] Steve Baker and Robert D. Cousins. Clarification of the Use of Chi Square and Likelihood Functions in Fits to Histograms. *Nucl. Instrum. Meth.*, 221:437–442, 1984.
- [7] Swagato Banerjee et al. Averages of b -hadron, c -hadron, and τ -lepton properties as of 2023. 11 2024.
- [8] F. Bernlochner, J. Dingfelder, M. Hohmann, H. Junkerkalefeld, P. Lewis, M. Milesi, P. Rocchetti, P. Urquijo, and P. Rocchetti. Measurement of the ratio $r(x)$ of inclusive $b \rightarrow x\tau\nu$ to $b \rightarrow x\ell\nu$ decays at belle ii. Sep 2021.
- [9] Florian U. Bernlochner, Sebastian Duell, Zoltan Ligeti, Michele Papucci, and Dean J. Robinson. Das ist der hammer: consistent new physics interpretations of semileptonic decays. *Eur. Phys. J. C*, 80:883, 2020.
- [10] Florian U. Bernlochner, Zoltan Ligeti, Michele Papucci, Markus T. Prim, Dean J. Robinson, and Chenglu Xiong. Constrained second-order power corrections in hqet. *Physical Review D*, 106(9), November 2022.
- [11] Florian U. Bernlochner, Zoltan Ligeti, Michele Papucci, Markus T. Prim, Dean J. Robinson, and Chenglu Xiong. Constrained second-order power corrections in HQET: $R(D^{(*)})$, $|V_{cb}|$, and new physics. *Phys. Rev. D*, 106(9):096015, 2022.
- [12] Florian U. Bernlochner, Zoltan Ligeti, Michele Papucci, and Dean J. Robinson. Combined analysis of semileptonic B decays to D and D^* : $\mathcal{R}(D^{(*)})$, $|V_{cb}|$, and new physics. *Physical Review D*, 95(11), jun 2017. <https://doi.org/10.1103/PhysRevD.95.115008>.
- [13] Florian U. Bernlochner, Zoltan Ligeti, Michele Papucci, and Dean J. Robinson. Combined analysis of semileptonic B decays to D and D^* : $\mathcal{R}(D^{(*)})$, $|V_{cb}|$, and new physics. *Phys. Rev. D*, 95(11):115008, 2017. [Erratum: *Phys. Rev. D* 97, 059902 (2018)].
- [14] Dante Bigi and Paolo Gambino. Revisiting $B \rightarrow D\ell\nu$. *Phys. Rev. D*, 94(9):094008, 2016.
- [15] Bigi, Dante and Gambino, Paolo. Revisiting $B \rightarrow D\ell\nu$. *Phys. Rev. D*, 94:094008, Nov 2016. <https://link.aps.org/doi/10.1103/PhysRevD.94.094008>.
- [16] Marzia Bordone, Martin Jung, and Danny van Dyk. Theory determination of $\bar{B} \rightarrow D^{(*)}\ell^-\bar{\nu}$ form factors at $\mathcal{O}(1/m_c^2)$. *Eur. Phys. J. C*, 80(2):74, 2020.
- [17] Claudia Cecchi, Guglielmo De Nardo, Elisa Manoni, and Mario Merola. B counting measurement of the run 1 belle ii data sample. Mar 2023.
- [18] ”Chaoyi Lyu and Florian Bernlochner and Petar Rados”. ”Determination of relative slow pion tracking efficiency”. Jun 2022.
- [19] Kyle Cranmer, George Lewis, Lorenzo Moneta, Akira Shibata, and Wouter Verkerke. HistFactory: A tool for creating statistical models for use with RooFit and RooStats. Technical report, New York U., New York, 2012.
- [20] Dante Bigi and Paolo Gambino and Stefan Schacht. $\mathcal{R}(D^*)$, $|V_{cb}|$, and the Heavy Quark Symmetry relations between form factors. *Journal of High Energy Physics*, 2017(11), Nov 2017. [https://doi.org/10.1007/JHEP11\(2017\)290](https://doi.org/10.1007/JHEP11(2017)290).
- [21] R. Aaij et al. Measurement of the ratio of the $b^0 \rightarrow d^{*-}\tau^+\nu_\tau$ and $b^0 \rightarrow d^{*-}\mu^+\nu_\mu$ branching fractions using three-prong τ -lepton decays. *Physical Review Letters*, 120(17), apr 2018. <https://doi.org/10.1103/PhysRevLett.120.171802>.
- [22] R. Aaij et al. Test of lepton flavor universality by the measurement of the $B^0 \rightarrow D^{*-}\tau^+\nu_\tau$ branching fraction using three-prong τ decays. *Physical Review D*, 97(7), apr 2018. <https://doi.org/10.1103/PhysRevD.97.072002>.

//doi.org/10.1103/PhysRevD.97.072013.

- [23] Florian Bernlochner and Michael Eliachevitch and Michael Hedges and Peter Lewis. Tests of lepton universality in angular observables of hadronically tagged $B \rightarrow D^* \ell \nu$ decays at Belle II. May 2022.
- [24] Florian U. Bernlochner and Stephan Duell and Zoltan Ligeti and Michele Papucci and Dean J. Robinson. Das ist der HAMMER: consistent new physics interpretations of semileptonic decays. *The European Physical Journal C*, 80(9), sep 2020. <https://doi.org/10.1140/EPJc%2Fs10052-020-8304-0>.
- [25] G. Caria et al. Measurement of $\mathcal{R}(d)$ and $\mathcal{R}(d^*)$ with a semileptonic tagging method. *Physical Review Letters*, 124(16), apr 2020. <https://doi.org/10.1103/PhysRevLett.124.161803>.
- [26] Paolo Gambino, Martin Jung, and Stefan Schacht. The V_{cb} puzzle: An update. *Phys. Lett. B*, 795:386–390, 2019.
- [27] Gill, M. Measurement of $B \rightarrow D$ Form Factors in the Semileptonic Decay $B^0 \rightarrow D^* \ell \nu$ at BABAR. 6 2018. <https://www.osti.gov/biblio/1442697>.
- [28] R. Glattauer et al. Measurement of the decay $B \rightarrow D \ell \nu_\ell$ in fully reconstructed events and determination of the Cabibbo-Kobayashi-Maskawa matrix element $|V_{cb}|$. *Phys. Rev. D*, 93(3):032006, 2016.
- [29] "Philip Grace and Paul Jackson". "measurement of $|v_{ub}|/|v_{cb}|$ with sl fei tagging in early phase iii data". May 2022. "belle2-note-ph-2022-019".
- [30] LeptonID group and Belle II Collaboration. Muon and electron identification efficiencies and hadron-lepton mis-identification rates at Belle II for Moriond 2021. Mar 2021. <https://docs.belle2.org/record/2318?ln=en>.
- [31] Charles R Harris, K Jarrod Millman, Stéfan J van der Walt, Ralf Gommers, Pauli Virtanen, David Cournapeau, Eric Wieser, Julian Taylor, Sebastian Berg, Nathaniel J Smith, Robert Kern, Matti Picus, Stephan Hoyer, Marten H van Kerkwijk, Matthew Brett, Allan Haldane, Jaime F del Río, Mark Wiebe, Pearu Peterson, Pierre Gérard-Marchant, Kevin Sheppard, Tyler Reddy, Warren Weckesser, Hameer Abbasi, Christoph Gohlke, and Travis E Oliphant. Array programming with NumPy. *Nature*, 585(7825):357–362, 2020.
- [32] Lukas Heinrich, Matthew Feickert, and Giordon Stark. pyhf: Pure python implementation of histfactory with tensors and autograd.
- [33] Hirose, S. et al. Measurement of the τ lepton polarization and $R(D^*)$ in the decay $\bar{B} \rightarrow D^* \tau^- \bar{\nu}_\tau$ with one-prong hadronic τ decays at Belle. *Phys. Rev. D*, 97:012004, Jan 2018. <https://link.aps.org/doi/10.1103/PhysRevD.97.012004>.
- [34] Sneha Jaiswal, Soumitra Nandi, and Sunando Kumar Patra. Extraction of $|V_{cb}|$ from $B \rightarrow D^{(*)} \ell \nu_\ell$ and the Standard Model predictions of $R(D^{(*)})$. *JHEP*, 12:060, 2017.
- [35] Kazuki Kojima, Qi-Dong Zhou, Kodai Matsuoka, Toru Iijima, Katsuro Nakamura, Koji Hara, Taichiro Koga, Racha Cheaib, and Hannah Marie Wakeling. Measurement of $r(d^*)$ with the hadronic fei tagging method using ichep 2022 dataset. May 2022.
- [36] T. Kuhr, C. Pulvermacher, M. Ritter, T. Hauth, and N. Braun. The Belle II Core Software. *Computing and Software for Big Science*, 3(1), Nov 2018. <http://dx.doi.org/10.1007/s41781-018-0017-9>.
- [37] Lees, J. P. et al. Evidence for an Excess of $\bar{B} \rightarrow D^{(*)} \tau^- \bar{\nu}_\tau$ Decays. *Phys. Rev. Lett.*, 109:101802, Sep 2012. <https://link.aps.org/doi/10.1103/PhysRevLett.109.101802>.
- [38] Lees, J. P. et al. Measurement of an excess of $\bar{B} \rightarrow D^{(*)} \tau^- \bar{\nu}_\tau$ decays and implications for charged Higgs bosons. *Phys. Rev. D*, 88:072012, Oct 2013. <https://link.aps.org/doi/10.1103/PhysRevD.88.072012>.

- [39] G. Martinelli, S. Simula, and L. Vittorio. $|V_{cb}|$ and $R(D)^{(*)}$ using lattice QCD and unitarity. *Phys. Rev. D*, 105(3):034503, 2022.
- [40] L. Moneta, K. Cranmer, D. Baskakov, G. Stark, and W. K. Tung. iminuit: A Python interface to Minuit2. 2011.
- [41] In order to solve the tension between the inclusive and exclusive branching fractions of semileptonic B decays, in our generic MC simulation, there are decays into final states like $D\pi\pi\ell\nu$, such that the second daughter of the B would not be a lepton, but a pion. By using the dedicated D^{**} signal MC, these modes are excluded in any case.
- [42] In principle, we also allow for $B^0 \rightarrow [D^{*+} \rightarrow D^+\gamma](\ell/\tau)\nu_{\ell/\tau}$ events, however, the amount of those events is negligibly small.
- [43] If there is any signal event in the respective sideband, it is added to the normalisation category, as those two are assumed to have the same scale factors.
- [44] Due to the background subtraction performed in this analysis, the FEI efficiency attributed to the backgrounds will not matter for the final fit result.
- [45] When splitting between electrons and muons the fractions of each category are almost the same.
- [46] The latter is also possible in pyhf but not used for the systematic uncertainties in our case.
- [47] Olaf Behnke et al. Data Analysis in High Energy Physics. Mar 2013.
- [48] Peter Lewis. Everything you ever wanted to know about the Full Event Interpretation. Dec 2022. https://docs.google.com/presentation/d/1q6ZW0n_GDx_keqiJzXtMw7Fo8Ihva15TSxxlZSkaNmM/edit#slide=id.g573c983e85_0_0.
- [49] R. Aaij et al. Measurement of the Ratio of Branching Fractions $\mathcal{B}(\bar{B}^0 \rightarrow D^{*-}\tau^+\bar{\nu}_\tau)/\mathcal{B}(\bar{B}^0 \rightarrow D^{*-}\mu^+\bar{\nu}_\mu)$. *Physical Review Letters*, 115(11), sep 2015. <https://doi.org/10.1103/2Fphysrevlett.115.111803>.
- [50] Philip Grace Racha Cheaib, Trevor Shillington. Standard WG Skims. <https://confluence.desy.de/display/Bi/Standard+WG+Skims>.
- [51] Anders Ryd, David Lange, Natalia Kuznetsova, Sophie Versille, Marcello Rotondo, David P. Kirkby, Frank K. Wuerthwein, and Akimasa Ishikawa. EvtGen: A Monte Carlo Generator for B-Physics. 5 2005.
- [52] S. Hirose et al. Measurement of the τ Lepton Polarization and $\mathcal{R}(D^*)$ in the Decay $\bar{B} \rightarrow D^*\tau\bar{\nu}$. *Physical Review Letters*, 118(21), may 2017. <https://doi.org/10.1103/2Fphysrevlett.118.211801>.
- [53] Sneha Jaiswal and Soumitra Nandi and Sunando Kumar Patra. Extraction of $|V_{cb}|$ from $B \rightarrow D(*)\ell\nu_\ell$ and the Standard Model predictions of $R(D^*)$. *Journal of High Energy Physics*, 2017(12), dec 2017. <https://doi.org/10.1007/2Fjhep12%282017%29060>.
- [54] William Sutcliffe. BinFit. <https://stash.desy.de/users/sutclw/repos/binfit/browse>.
- [55] E. Waheed et al. Measurement of the CKM matrix element $|V_{cb}|$ from $B^0 \rightarrow D^{*-}\ell^+\nu_\ell$ at Belle. *Phys. Rev. D*, 100(5):052007, 2019. [Erratum: Phys.Rev.D 103, 079901 (2021)].